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[Frontispiece : portrait of Professor Cayley, reproduction by A. G. Dew-Smith, 1885.]

THE COLLECTED
MATHEMATICAL PAPERS

OF

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THE present volume contains 93 papers numbered 706 to 798, published, with the exception of one series, for the most part in the years 1878 to 1883. This series is constituted by the articles which Professor Cayley wrote for the *Encyclopædia Britannica* between the years 1878 and 1888; it seemed desirable to place these together in the same volume, in spite of the departure from the chronological arrangement which governs the sequence of the papers in the volumes generally. The Syndics of the University Press desire to acknowledge their obligation to Messrs Adam and Charles Black, Publishers of the ninth Edition of the *Encyclopædia Britannica*, for their courteous consent in allowing these articles to be included in the *Collected Mathematical Papers.* Exact references to the volumes, from which the articles are extracted, will be found in the Table of Contents.

The frontispiece to the present volume is a reproduction by Mr A. G. Dew-Smith, of Trinity College, of a photograph of Professor Cayley which he made in the year 1885. The Syndics of the Press desire to acknowledge their obligation to Mr Dew-Smith.

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A. R. FORSYTH.

21 November, 1896.

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ON THE DISTRIBUTION OF ELECTRICITY ON TWO SPHERICAL SURFACES.

[From the *Philosophical Magazine*, vol. v. (1878), pp. 54—60.]

In the two memoirs "Sur la distribution de l'électricité à la surface des corps conducteurs," *Mém. de l'Inst.* 1811, Poisson considers the question of the distribution of electricity upon two spheres : viz. if the radii be a, b, and the distance of the centres be c (where $c > a + b$, the spheres being exterior to each other), and the potentials within the two spheres respectively have the constant values h and g, then—for Poisson's $f\left(\dfrac{x}{a}\right)$ writing $\phi(x)$, and for his $F\left(\dfrac{x}{b}\right)$ writing $\Phi(x)$—the question depends on the solution of the functional equations

$$a\,\phi(x) + \frac{b^2}{c-x}\,\Phi\left(\frac{b^2}{c-x}\right) = h,$$

$$\frac{a^2}{c-x}\,\phi\left(\frac{a^2}{c-x}\right) + b\,\Phi(x) = g,$$

where of course the x of either equation may be replaced by a different variable.

It is proper to consider the meaning of these equations : for a point on the axis, at the distance x from the centre of the first sphere, or say from the point A, the potential of the electricity on this spherical surface is $a\,\phi x$ or $\dfrac{a^2}{x}\,\phi\left(\dfrac{a^2}{x}\right)$, according as the point is interior or exterior ; and, similarly, if x now denote the distance from the centre of the second sphere (or, say, from the point B), then the potential of the electricity on this spherical surface is $b\,\Phi x$ or $\dfrac{b^2}{x}\,\Phi\left(\dfrac{b^2}{x}\right)$, according as the point is interior or exterior ; $\phi(x)$ is thus the same function of

712.

A PARTIAL DIFFERENTIAL EQUATION CONNECTED WITH THE SIMPLEST CASE OF ABEL'S THEOREM.

[From the *Report of the British Association for the Advancement of Science*, (1881), pp. 534, 535.]

CONSIDER a given cubic curve cut by a line in the points $(x,\ p)$, $(x_1,\ p_1)$, $(x_2,\ p_2)$; using the first and second points as elements, these determine uniquely the third point. Analytically, the equation of the curve determines p as a function of x, and p_2 as a function of x_1; writing in the equation

$$c_2 = \lambda c_1 + (1-\lambda)c, \quad y_2 = \lambda y_1 + (1-\lambda)y_0,$$

we have c, by a simple equation, and thence p_2; viz. x_2 is found as a function of x, x_1, and of the nine constants of the equation. Hence having the derived equations (in regard to x, x_1) of the first, second, and third orders, we have $(1+2+3+4=)10$ equations from which to eliminate the 6 quantities x_2, considered as a function of x and x_1, thus involves a partial differential equation of the third order, independent of the particular cubic curve.

To obtain this equation it is only necessary to observe that we have, by Abel's theorem,

$$\frac{dx}{X} + \frac{dx_1}{X_1} + \frac{dx_2}{X_2} = 0,$$

where X, X_1, X_2 is a given function of x, and p, that is, of x; X_1 and X_2 are the like functions of x_1 and x_2 respectively. Hence, considering x_2 as a function of x and x_1, we have

$$\frac{dx_2}{dx} = -\frac{X_2}{X}, \quad \frac{dx_2}{dx_1} = -\frac{X_2}{X_1},$$

and consequently

$$\frac{dx_2}{dx_1} \div \frac{dx_2}{dx} = \frac{X}{X_1},$$

where X, X_1 are functions of x_2, x_1 respectively; hence taking the logarithm and differentiating successively with regard to x_1 and x_2, we have

$$\frac{d}{dx_2}\frac{d}{dx_1}\log\left(\frac{dx_2}{dx_1} \div \frac{dx_2}{dx}\right) = 0,$$

which is the required partial differential equation of the third order.

This differential equation has a simple geometrical significance. Consider three consecutive positions of the line meeting the cubic curve in the points 1, 2, 3; $1'$, $2'$, $3'$; $1''$, $2''$, $3''$ respectively; *quæ* equation of the third order, the equation should in effect determine $3''$ by means of the other points. And, in fact, the three positions of the line constitute a cubic curve; the nine points are thus the intersections of two cubic curves, or, say, they are an "annexed" of points; any eight of the points thus determine uniquely the ninth point.

. . . .

713.

ADDITION TO MR ROWE'S MEMOIR ON ABEL'S THEOREM.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXII., Part III. (1881), pp. 715—758. Received May 27,—Read June 16, 1880.]

In Abel's general theorem y is an irrational function of x determined by an equation $\chi(y) = 0$, or say $\chi(x,\ y) = 0$; if the order n as regards y, and that is also by him that the sum of any number of the integrals considered may be reduced to a sum of ρ integrals; where ρ is a determinate number depending only on the form of the equation $\chi(x,\ y) = 0$, and given in his equation (42), [*(Euvres Complètes*, (1881), t. I., p. 157]; viz. if, solving the equation so as to obtain from it developments of y in descending series of powers of x, we have:

$$\begin{array}{ll} v_\sigma \rho_\nu, \text{ series each of the form } & y = x^{\sigma'_\nu} + \dots, \\ \qquad\qquad\qquad\qquad v_\sigma \rho_2, & y = x^{\sigma'_2} + \dots, \\ \qquad\qquad\qquad\qquad \vdots & \vdots \\ \qquad\qquad\qquad\qquad v_\sigma \rho_n, & y = x^{\sigma'_n} + \dots, \end{array}$$

[0]The second power of a term coefficient: this being a relation $y = x^p + \dots$, representing the y, different values of y obtained by giving to the radical x^p and to its p values, and the corresponding values in the radicals which enter into representing y, values of y. It is assumed that the equation is such that the meaning is to make each representing y, y_2. In each of the equations the meaning of y is taken positive in the development of y in descending powers of x.

(so that $\sigma = v_1\rho_1 + v_2\rho_2 + \cdots + v_n\rho_n$) then y is a determinate function of x_1, x_2, x_3, x_4, x_5, x_6, ..., x_n, say ρ.

We have so expressed Abel's ρ in the following form, viz. assuming

$$y = \Sigma a_1\alpha_1\nu_1 + \Sigma a_2\nu_2 \cdots + \Sigma a_n\nu_n - \frac{1}{2}\Sigma a - \frac{1}{2}n + 1,$$

or, what is the same thing, for ρ writing this Σa_1,

$$y = \Sigma a_1\nu_1 + \Sigma a_2\nu_2 + \Sigma a_3\nu_3 + \cdots + \Sigma a_n\nu_n - \frac{1}{2}\Sigma a - \frac{1}{2}n + 1,$$

where in the first sum ν_1 have each of them the values 1, 2, ..., ρ_1 subject to the condition $i + \nu_1\sigma_1'$; in each of the other sums ν_i and ρ_n are interrelated by the suffix, which has the values 1, 2, etc.

It is a leading result in Riemann's theory of the Abelian integrals that y is the deficiency (Geschlecht) of the curve represented by the equation $\chi(x,\ y) = 0$; and it must consequently be demonstrable *a posteriori* that the foregoing expression for y is in fact = deficiency of curve $\chi(x,\ y) = 0$. I propose to verify this by means of the formulae given in my paper "On the Higher Singularities of a Plane Curve," *Quart. Math. Journ.*, vol. VII., (1866), pp. 212—222, [374].

It is necessary to distinguish between the values of $\frac{m}{n}$ which are $>, =$, and < 1; and to fix the ideas I assume $l = 7$, and

$$\frac{m_1}{n_1},\ \frac{m_2}{n_2},\ \frac{m_3}{n_3},\ \text{each} > 1,$$

$$\frac{m_4}{n_4} = 1;\ \text{say}\ m_4 = n_4 = n,\ \text{and}\ n_4 = n;$$

$$\frac{m_5}{n_5},\ \frac{m_6}{n_6},\ \frac{m_7}{n_7},\ \text{each} < 1,$$

but it will be easily seen that the reasoning is quite general. And now Σ to denote a sum in regard to the first set of suffixes 1, 2, 3, and Σ' to denote a sum in regard to the second set of suffixes 5, 6, 7. The foregoing value of ρ is thus

$$y = \Sigma a\nu + 4 + \Sigma' a\nu.$$

Introducing a third coordinate z for homogeneity, the equation $\chi(x,\ y) = 0$ of the curve will be

$$0 = \left[y^{p-(n-1)},\ y^{n-2}x^a,\ \ldots\right](y,\ z)^{n-2} + \left(y^{n-2},\ y^{n-3}x^a,\ \ldots\right)(y,\ z)^{n-1} + \cdots + (y)$$

where it is to be observed that $\chi(x,\ y) = 0$ is the product of ν_i, different ovaries; these divide themselves into n_i

groups, each a product of ρ_i orates; and in each such product the ρ_i coefficients A_i are in general the ρ_i values of a function containing a radical a^{1/ρ_i} and are thus different from each other; it is to when follows in effect assumed in each step that all the a_i, but that all the x_i coefficients, etc. are different from each other.

The remarks apply to the other factors. It applies in particular to the factor $(y,\ z)^{r-1}$, viz. it is assumed that the coefficients A in the radical z, y^{n-2}, ... are all of them different from each other. These assumptions as to the leading coefficients really imply Abel's assumption that $\frac{m_1}{n_1}$, $\frac{m_2}{n_2}$ are all of them fractions in their least terms, viz. that $n = 1$, since

1 *rotate however for convenience the equation will be considered, by giving to*

In the product of the several infinite series, the terms containing the lowest powers of z, $\frac{1}{n_1}$ is a function in its lowest terms, viz. since $n_1 = 1$.

A curve has singularities (singular points) at infinity, that is, on the line $z = 0$; vis. *First*, a singularity at $(x = 0, x' = 0)$, where the tangent is $x' = 0$, and which, writing for convenience $x = 1$, is denoted by the function

$$\left(z - \nu^{m/n} x^{1-1/n}\right)^{n-1} = \dots$$

where observe that the expressed factor indicates r_n branches $\left(z - y^{m/n} x^{1-1/n}\right)^{r_n}$, or say $r_n(m_n - p_n)$ partial branches $z = v^{m/n} \dots$, that is, $r_n(m_n - p_n)$ partial branches $z = v^{m/n} \dots$, will in all $r_n(m_n - p_n) + \dots$, where $r_n(m_n - p_n)$ partial branches at the encompassed factors with the suffixes $1'$ and 4; that is, we can regard the encompassed factors with the suffixes $1'$ and 4, where in the equation

$$z - \nu^{m/n} + \dots = 0.$$

Secondly, a singularity at $(z = 0, y = 0)$, where the tangent is $y = 0$, and which, writing for convenience $z = 1$, is denoted by the function

$$\left(y - z^{m/n} x^{1-1/n}\right)^{n-1} = \dots$$

[0*] This assumption is virtually made by Abel, (1), [c. p. 157], in considers

where observe that the expressed factor indicates r_5 branches $\left(y - z^{m/n} x^{1-1/n}\right)^{r_5}$, or say $r_5(n_5 - m_5)$ partial branches $y = z^{m/n} \ldots$, that is, $r_5(n_5 - m_5)$ partial branches $y = z^{m/n} \ldots$, with in all $r_5 n_5 + \ldots$, with $r_5 n_5 + r_6 n_6 + r_7 n_7$ partial branches $y = z^{m/n} \ldots$, with the suffixes 5, 6, 7.

Thirdly, suppose at $(x = 0, y = 0)$, where the tangent is $z = 0$, and consequently $r_4 n_4 = 0$, partial branches $z = y^0 \cdots +$ similar terms with respect to x, y, z.

It is convenient to assume $r_4 = 2$, so that

$$\begin{aligned} E &= \tfrac{1}{2}(K-1)(K-2) = z, \\ E' &= \tfrac{1}{2}(K'-1)(K'-2) \\ &= \tfrac{1}{2}\Sigma(n-1). \end{aligned}$$

hence

$$E + e' = \tfrac{1}{2}(K-1) + \tfrac{1}{2}\Sigma(n-1).$$

Assuming that there are no other singularities, the deficiency

$$\rho = \tfrac{1}{2}(K-1)(K-2) - E - c'$$

is

$$= \tfrac{1}{2}(K-1)(K-2) - \tfrac{1}{2}KK + \tfrac{1}{2}\Sigma(n-1).$$

This should be equal to the before-mentioned value of ρ; viz. we ought to have

$$\tfrac{1}{2}(K-1)(K-2) - \tfrac{1}{2}KK + \tfrac{1}{2}\Sigma(n-1) = \Sigma a\nu_1 + 4 + \Sigma' a\nu_2 - \Sigma a - \Sigma' a - 2,$$

or, as it will be convenient to write it,

$$M = K - \tfrac{3}{2}K + \tfrac{1}{2}\Sigma(n-1) - 1 = \Sigma a\nu_1 + \Sigma' a\nu_2 - \Sigma' a - 2,$$

which is the equation which ought to be satisfied by the values of N and $\Sigma(n-1)$ calculated, according to the method of my paper, for the foregoing singularities of the curve.

---

the equation $z^n = y^n \sqrt[m_n]{f(x)}$ as denoting integers; but where it appears that the indices a, b, are to be considered of fixed, and as in the values of degrees with respect to x. He says that if these be assumed as integers in the expression of y in descending powers of x (the leading coefficient being z in regard to y), the singular factor are in $(x,\ y)$ in regard to y in regard to x.

We have as before:

$$K = \Sigma n + \Sigma' n + \theta n.$$

The terms $\Sigma n + \Sigma' n + \theta n$, written at length, is

$$\begin{aligned}
&= n_1(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3) + \cdots + n_2(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_3(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_4(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_5(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_6(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3)\\
&+ \cdots + n_7(\nu_1\rho_1 + \nu_2\rho_2 + \nu_3\rho_3).
\end{aligned}$$

which is

$$= \Sigma\nu_1 n_2 + \Sigma' n_2(2\rho_2 + 2'\rho_1) + \Sigma\Sigma' + \Sigma'\Sigma + \Sigma a_1 a_2 a_3,$$

We have moreover

$$\Sigma'\rho_1 = \Sigma a_1\rho_1 + (\theta + \Sigma' a_1)\rho_1 = \Sigma a_1 + \Sigma a_2 + \Sigma' a_1\rho_1,$$

each branch $\left(z - y^{m/n}x^{1-1/n}\right)^{n-1}$ gives $a = m_1\rho_1$, and the value of $\Sigma(\kappa - 1)$ for this singularity is

$$n_1(m_1 - n_1 - 1) + r_1 n_1(m_1 - 1) = r_1 n_1 \rho_1 - n_1\rho_1.$$

which is

$$= \Sigma a\nu_1 - \Sigma a\rho_1.$$

each branch $\left(y - z^{m/n}x^{1-1/n}\right)^{n-1}$ gives $\sigma = m_n\rho_n$, and the value of $\Sigma(\kappa - 1)$ for this singularity is

$$n_4(m_4 - n_4 - 1) + r_4 n_4(m_4 - 1) = r_4 n_4 \rho_4 - n_4\rho_4,$$

which is

$$= \Sigma'\nu_2 - \Sigma'\nu_4 - \Sigma'\nu_1.$$

For each of the θ singularities

$$\left(y - z^{m/n}x^{1-1/n}\right)^{n-1}.$$

we have $a = n$, and the value of $\Sigma(\kappa - 1)$ is $r = \theta(\kappa - 1)$; this is $= 0$ for the value $\lambda = 1$, which is physically attributed to λ.

According to the theory explained in my paper above referred to, the several singularities not together equivalent to a certain number K of κ^* nodes and cusps; viz. we have

$$K = \Sigma n - \Sigma a + \tfrac{1}{2}\Sigma(\kappa - 1).$$

This should be equal to the before-mentioned value of ρ; viz. we ought to have

$$(K-1)(K-2) - 2(K + \Sigma n + \Sigma' n + \theta n) = 2\Sigma a\nu_1 + 4 + 2\Sigma a\nu_2 - 2\Sigma a - 2\Sigma' a - 4,$$

or to it will be convenient to write

$$M - K - \Sigma K + 1 + \Sigma(n-1) = \Sigma a\nu_1 + \Sigma' a\nu_2 - \Sigma a - \Sigma' a - 2,$$

which is the equation which ought to be satisfied by the values of N and $\Sigma(n-1)$ calculated, according to the method of my paper, for the foregoing singularities of the curve.

We have as before:

$$K = \Sigma n + \Sigma' n + \theta n.$$

The term $\Sigma n + \Sigma' n + \theta n$, written at length, is

$$= \Sigma \nu_1 n_1 + \Sigma' \nu_2 n_2 + \theta n,$$

or, reducing,

$$M = (\Sigma m)^2 - \Sigma ma - \Sigma m\rho_1 - \Sigma m\rho_2 + (\Sigma' m)^2 - \Sigma' m_2 a - \Sigma' m_2\rho_2$$

$$+(\Sigma' m)^2 - \Sigma' m\rho_1 - \Sigma ma + 2\Sigma' m_2 a_1 + 2\Sigma'\Sigma a_1 a_2;$$

and it is to be shown that the two lines of this expression are in fact the values of M belonging to the singularities

$$\left(z - y^{m/n}x^{1-1/n}\right)^{n-1}, \ \ldots, \text{ and } \left(y - z^{m/n}x^{1-1/n}\right)^{n-1}, \ \ldots,$$

respectively. We assume $\lambda = 1$, and there is then no singularity $(y-z)^{n-1}$.

I recall that, considering the several partial branches which meet in a singular point, M denotes the sum of the number of the intersections of each partial branch by every other partial branch: so that for each pair of partial branches the intersections are to be counted twice. Supposing that the tangent is $x = 0$, and that, for any two branches we have $z = Az^x + \ldots$ (x where p_x, p_x are each up to greater than 1), then if $p_x = p_x$, and $A_x = A_x$, $A_x = A_x$ where A_x, A_x are not the partial branches which have here always could be regards the cases about to be considered), then the number of intersections is taken to be $-$; and if $p_x > p_x$, and the number of intersections (taken to be $\equiv p_x$) viz. in the case of as a unequal exponents, it is equal to the smaller exponents.

Consider now the singularity $\left(z - y^{m/n}x^{1-1/n}\right)^{n-1}$, and first the intersections of a partial branch $z = y^{m/n}\ldots$ by each of the remaining $n_1(m_1 - p_1) - 1$ partial branches of the same set: the number of intersections with any one of these is $= \frac{m_1}{n_1}\left[n_1(m_1 - p_1) - 1\right]$. But we obtain this same number from each of the $n_1(m_1 - p_1)$ partial branches, and thus the whole number is

$$n_1\left(m_1 - p_1\right)\frac{m_1}{n_1}\left[n_1(m_1 - p_1) - 1\right] = n_1 m_1\left[n_1(m_1 - p_1) - 1\right].$$

Taking account of the other sets, and the whole number of intersections is

$$= \Sigma n^2 a - \Sigma' n a^2.$$

Observe now that $\frac{m_1}{n_1} > \frac{m_2}{n_2}$, that is, $\frac{n_1}{n_2} < \frac{m_1}{m_2}$, and that, these being each < 1, we have $1 - \frac{m_2}{n_2} > 1 - \frac{m_1}{n_1}$, that is, $\frac{n_2 - m_2}{n_2} > \frac{n_1 - m_1}{n_1}$, and we thus have

$$\frac{m_1}{n_1} < \frac{m_2}{n_2}, \text{ and } \left(z - y^{m/n}x^{1-1/n}\right)^{n_1(m_1-p_1)},$$

Considering now the intersections of partial branches of the two sets

$$\left(z - y^{m/n}x^{1-1/n}\right)^{n_1(m_1-p_1)} \text{ and } \left(z - y^{m/n}x^{1-1/n}\right)^{n_2(m_2-p_2)},$$

respectively, a partial branch $z - z^{m/n} \ldots$ gives with each partial branch of the other set a number $= \frac{m_2}{n_2}$; and in this way taking each partial branch of each set, the number is

$$n_1(m_1 - p_1) \cdot n_2(m_2 - p_2) \cdot \frac{m_2}{n_2} = n_2 m_2 n_1 (m_1 - p_1),$$

and thus for all the sets the number is

$$= 2\Sigma n_2 m_2 \nu_2 m_2,$$

where in the first sum the Σ' refers to each pair of values of the suffixes. But the intersections are to be taken twice; the number thus is

$$= 2\Sigma m_1 m_2 \nu_1 \nu_2,$$

Adding the foregoing number

$$= \Sigma' n^2 a - \Sigma' n a^2;$$

the whole number for the singularity in question is

$$= (\Sigma m_1)^2 - \Sigma' m_2 \rho_2 - \Sigma' m_2 \rho_2.$$

Similarly for the singularity $\left(y - z^{m/n}x^{1-1/n}\right)^{n-1} \ldots$, taking each set with itself, the number of intersections is

$$n_5 m_5 \left[n_5(m_5 - p_5) - 1\right] + n_6 m_6 \left[n_6(m_6 - p_6) - 1\right] + n_7 m_7 \left[n_7(m_7 - p_7) - 1\right],$$

which is

$$= \Sigma' m_2^2 a - \Sigma' m_2 \rho_2 - \Sigma m_1 a.$$

[5 − 2]

We have here $\frac{m_5}{n_5} > \frac{m_6}{n_6}$, each of these being < 1, we have $1 - \frac{m_5}{n_5} < 1 - \frac{m_6}{n_6}$, that is, $\frac{n_5 - m_5}{n_5} < \frac{n_6 - m_6}{n_6}$, and so

$$\frac{m_5}{n_5} > \frac{m_6}{n_6}, \frac{n_5 - m_5}{n_5} < \frac{n_6 - m_6}{n_6};$$

Hence considering the two sets

$$\left(y - z^{m/n} x^{1-1/n}\right)^{n_5(m_5 - p_5)} \text{ and } \left(y - z^{m/n} x^{1-1/n}\right)^{n_6(m_6 - p_6)};$$

a partial branch of the first set given with a partial branch of the second set a number of intersections; and the number then obtained is

$$n_5(m_5 - p_5) \cdot n_6(m_6 - p_6) \cdot \frac{m_6}{n_6} = n_6 m_6 n_5 (m_5 - p_5),$$

For all the sets the number is

$$n_5 m_5 \left[n_5(m_5 - p_5) + n_6(m_6 - p_6)\right] - n_5 n_6 (m_6 - m_5)$$

or taking these from, the number is

$$2\Sigma' m_2 m_2 \nu_2 \nu_2.$$

where in the first sum the Σ' refers to each pair of suffixes. Adding the foregoing value

$$\Sigma' m^2 a - \Sigma' m_2 \rho_2 - \Sigma m_1 a,$$

the whole number for the singularity in question is

$$= (\Sigma' m_2)^2 - \Sigma m_1 a - \Sigma' m_2 \rho_2.$$

and the proof is thus completed.

Referring to the first-note (ante, p. 31), I remark that the theorem "ρ–deficiency, is absolute, and applies to a curve with any singularities whatever: in a curve which has singularities not taken account of in Abel's theory, the "perhaps see particulars" which we have a give as he expresses it, taken account of, viz. for that Σa is, in the value of ρ as denoted by Abel's formula; and the actual deficiency will be = Abel's ρ, = such diminution, that is, it will be = true values of ρ.

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714.

VARIOUS NOTES.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 69, 115, 124, 125.]

AN IDENTITY.

THE following remarkable identity is given under a slightly different form by Gauss, *Werke*, t. III., p. 419.

$$\begin{aligned} 1 &= \left(\frac{1}{x}\right)^2 x + \left(\frac{1}{1+x}\right)^2 x + \left(\frac{1+x}{1+x+x^2}\right)^2 x + \dots \\ &= \left[1 + \left(\frac{1}{x}\right)^2 x + \left(\frac{1}{1+x}\right)^2 x + \left(\frac{1+x}{1+x+x^2}\right)^2 x + \dots\right]^2 . \end{aligned}$$

ON TWO RELATED QUADRIC FUNCTIONS.

Assume

$$\begin{aligned} \varphi x &= x^2(1-x) + x(1-x^2-\alpha x), \\ \varphi y &= y(1-x) + y(x^2-1-\alpha x). \end{aligned}$$

then

$$\varphi\left(\frac{x^2}{x-a}\right) = \frac{x^2}{(x-a)^2}\varphi x,$$

$$\varphi\left(\frac{x^2}{x-a}\right) = \frac{x^2}{(x-a)^2}\varphi y.$$

In the like of those for a ratio $\frac{p}{1-p}$, then

$$\varphi\left[\frac{ax^2(x-y)}{x^2-y^2(x-a)}\right] = \frac{ax^2(x-y)^2}{(x-a)^2-y^2(x-a)^2}\,\varphi(x).$$

A Trigonometrical Identity.

$$
\begin{aligned}
&\cos(b-c)\sin(b+c-a)+\cos a\sin\\
&+\cos(c-a)\sin(c+a+b)+\cos b\sin\\
&+\cos(a-b)\sin(a+b+c)+\cos c\sin\\
=\ &\cos a\cos b\cos(b+c)+\cos b\cos c\cos(c+a)+\cos c\cos a\cos(a+b)+\cos a\cos
\end{aligned}
$$

Extract from a Letter.

"I wish to construct a correspondence such as

$$(x+iy)^2(x+iy)=X+iY,$$

or, say, for greater convenience

$$4(x+iy)^2(x+iy)=X+iY,$$

viz. if

$$x+iy=\cos\alpha,$$

then

$$X+iY=\cos\alpha\beta.$$

Suppose α, in a value of α corresponding to a given value of $X+iY$, then the three values of $x+iy$ are of course $\cos\alpha$, $\cos\left(\alpha+\frac{2\pi}{3}\right)$, but I am afraid that the calculation of α, even with red and card tables, would be very laborious. Writing

$$X+iY=R(\cos\Theta+i\sin\Theta),$$

the intervals for Θ might be 3°, 30° or even 15°; those of R, say 0.1 from 0 to 2, and then 0.2 up to 4; and 2 places of decimals would be quite sufficient; but even this would probably involve a great mass of calculation.

It has occurred to me that perhaps a geometrical solution might be found for the equation $X+iY=\cos\beta$."

October 21, 1877.

. . . .

715.

NOTE ON A SYSTEM OF ALGEBRAICAL EQUATIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 17, 18.]

ASSUME

$$\begin{aligned} x + y + z &= P, \\ yz + zx + xy &= Q, \\ xyz &= R. \end{aligned}$$

$$\begin{aligned} &A = (nx + y - z) - a^n(nx + P), \\ &B = y(ax + Q) - b^n(my + Q), \\ &C = z(ax + Q) - c^n(nz + P), \\ &\Theta = -mnQ + nP. \end{aligned}$$

Then

$$\begin{aligned} (ax + P)B - (my + P)A &= (ax + P)y(ax + Q) - a^n(my + Q) \\ &= (ax + P)(ax + Q) - (my + Q)(my + P) \\ &= ax(my + Q - ay) - yz(my + Q - P) + yz(ax + P \\ &= mny(ax - y) - P(ax - y) \\ &= (ax - y)(mny - P) = (ax - y)\Theta, \end{aligned}$$

whence, identically,

$$\begin{aligned} (ax + P)B - (my + P)A &= (ay - y)\Theta, \\ (my + P)C - (nx + P)B &= (by - z)\Theta, \\ (ax + P)A - (nx + P)C &= (cz - x)\Theta. \end{aligned}$$

Hence any two of the equations $A = 0$, $B = 0$, $C = 0$ imply the third equation.

We have

$$A = x\,[(n+1)x+y+z] - a^n\,[(n+1)x+y+z]$$
$$= (x-a^n)\,[(n+1)x+y+z]\,;$$

and similarly for B and C. The three equations therefore are

$$\frac{x}{x^n-a^n} = \frac{y+z}{(n+1)x^n+(n-1)a^n},$$
$$\frac{y}{y^n-b^n} = \frac{x+z}{(n+1)y^n+(n-1)b^n},$$
$$\frac{z}{z^n-c^n} = \frac{x+y}{(n+1)z^n+(n-1)c^n},$$

and any two of these equations imply the third equation.

. . . .

716.

AN ILLUSTRATION OF THE THEORY OF THE ϑ-FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 27—32.]

IF X be a given quartic function of x, and if a, as for convenience a constant multiple as, be the value of the integral $\int \frac{dx}{\sqrt{X}}$, taken from a given inferior limit to the superior limit x; then, conversely, x is expressible as a function of a, viz. it is expressible in terms of ϑ-functions of a, where ϑa, or say $\vartheta(a,\ \mathfrak{Q})$, is the definition of which, depending, depend in general by definition on the sum of a series of exponentials of a, and it is possible from the assumed equation $a = \int \frac{dx}{\sqrt{X}}$, and the definition of ϑa, to attain by general theory the actual formulae for the determination of a as such a function of a.

I propose here to obtain these formula, in the case where X is a product of real factors, in a less suitable manner, viz. by reducing the integral $\int \frac{dx}{\sqrt{X}}$ by a linear substitution in the form of an elliptic integral, the object being to obtain for the same in question the actual formulae as a function of ϑ-functions of u.

The definition of ϑu, when the parameter is expressed, $\vartheta(u,\ \mathfrak{Q})$ is

$$\vartheta u = 2\Sigma(-)^p q^{p+\frac{1}{2}(p+1)^2} \sin(p+\tfrac{1}{2})u,$$

where u and p are negative integer values, zero included, from $-\infty$ to $+\infty$ (that is, from -2 to 2, 0 to 2, etc.); the parameter $\mathfrak{Q}$, or Ω' (of imaginary) no real part, must be positive.

Evidently ϑu is an even function: $\vartheta(-u) = \vartheta u$. Moreover, it is at once seen that we have

$$\vartheta(u+\pi) = \vartheta u, \quad \vartheta(u+i\Pi) = -q^{-1}e^{-iu}\vartheta u,$$

where q is

$$\vartheta(u+\pi+i\Pi).$$

where as u and v are positive or negative integers, the product of ϑu by an exponential factor, or say simply that it is a multiple of ϑu.

Writing $u = \frac{\pi}{2}\xi$, we have $\vartheta v = \frac{1}{2}\sqrt{2K}\vartheta\xi$, that is,

$$\vartheta\left(\tfrac{\pi}{2}\xi\right) = \xi^{nK},$$

and therefore also

$$\vartheta(\sin\xi + v) = \xi^{nK}\vartheta v.$$

The above properties are general, but if θ be real, then K, K', q being as in Jacobi (consequently q being real, positive, and less than 1, and K and K' real and positive), and assuming $\mathfrak{Q} = \frac{K'}{K}, \pi$, we have in the same thing,

$$q(= e^{-\pi\Omega/\mathfrak{Q}}) = e^{-\pi K'/K},$$

the function ϑv is given (*ut* ante) of Jacobi's ϑv by the equation $\vartheta u = q^{\frac{1}{4}}\vartheta x$.

That is, if Ω is the value of the case $u = v$.

We have, as before, the system of three equations

$$\operatorname{sn} u = \frac{1}{\sqrt{k}}\frac{\vartheta(u,\ k)}{\vartheta_0(u,\ k)}, \quad \operatorname{cn} u = \sqrt{\frac{k'}{k}}\frac{\vartheta_0(u,\ k)}{\vartheta_0(u,\ k)}, \quad \operatorname{dn} u = \sqrt{k'}\frac{\vartheta_0(u,\ k)}{\vartheta_0(u,\ k)}.$$

Consider now the integral

$$\int_a^b \frac{dx}{\sqrt{(x-\alpha,\ x-\beta,\ x-\gamma,\ x-\delta)}} = \int_a^b \frac{dx}{\sqrt{X}} \text{ suppose,}$$

where α, β, γ, δ are taken to be real, and the four factors regarded as fixed magnitudes, viz. it is assumed that $b-a$ is the value of the radical at the inferior limit is positive, and as b is between β and δ, X is positive, and a fortiori then $\sqrt{X}$ denotes the positive value of the radical; but if α is between β and δ, then X is negative, and we must that the sign of $\sqrt{X}$ is taken so that $\sqrt{X}$ is equal to a positive multiple of i, and this being so the integral is taken from the inferior limit to the superior limit x, which we may.

Take a linear function of y, such that for

$$x = a,\ b,\ c,\ d,$$

$$y = 0,\ 1,\ \tfrac{1}{k^2},\ \infty,\ \text{respectively},$$

so that, x increasing continuously from a to d, y will increase continuously from 0 to ∞. We have

$$y = \frac{c-a}{c-b} \cdot \frac{x-b}{x-a},$$

$$y = \frac{c-a}{c-d} \cdot \frac{x-d}{x-a},$$

$$1 - y = \frac{b-c}{c-b} \cdot \frac{c-a}{c-a},$$

$$1 - k'y = \frac{b-c}{c-b} \cdot \frac{c-a}{c-a},$$

and, thence,

$$\sqrt{y \cdot 1 - y \cdot 1 - k'y} = \frac{(a-c)\sqrt{X}}{(c-a)\sqrt{(x-a)\cdot(x-a)}},$$

where $\sqrt{\frac{(c-d)}{(c-d)}}$ is taken to be positive, and the signs of $\sqrt{X}$ is fixed as above. For y between 0 and 1, $y = \frac{1}{x^2}$, $1-y$, $1-k'y$ will be positive, and $\sqrt{y}$ being between 1 and $\frac{1}{x^2}$, $1-y$ will be negative, but $1-k'y$ will be positive, viz. the radical is then taken as a positive multiple of i.

We have moreover

$$dy = \frac{c-a}{c-b} \cdot \frac{dx}{(x-a)^2},$$

and therefore

$$\frac{dy}{\sqrt{y \cdot 1 - y \cdot 1 - k'y}} = \sqrt{(c-d)\cdot(c-a)} \cdot \frac{dx}{\sqrt{X}},$$

where $\sqrt{(c-d)\cdot(c-a)}$ is positive; or, say,

$$\int \frac{dy}{\sqrt{y\cdot 1-y\cdot 1-k'y}} = \sqrt{(c-d)\cdot(c-a)}\cdot\int\frac{dx}{\sqrt{X}}.$$

[6 − 2]

Hence, writing $y = \sin^2 u$, we have

$$\Pi u = \sqrt{(c-d)\cdot(c-a)}\ \int_a^b \frac{dx}{\sqrt{X}},$$

and it is to be further noticed that to

$$x = a,\ b,\ c,\ d,$$

correspond

$$\Pi u = 0,\ \tfrac{1}{2}K,\ \tfrac{1}{2}K + \tfrac{1}{2}K'.$$

or we may say

$$u = 0,\ K,\ K + iK',\ 2K + iK'.$$

Writing for shortness

$$\frac{1}{\Pi} = \sqrt{(c-d)(c-a)} = u,$$

we have

$$\sin u = \int_a^b \frac{dx}{\sqrt{X}},$$

and moreover

$$\sin(K + iK') = \int_a^b \frac{dx}{\sqrt{X}},$$

$$n(K + iK') = \int_b^c \frac{dx}{\sqrt{X}},$$

$$n(2K + iK') = \int_c^d \frac{dx}{\sqrt{X}},$$

or if for a moment we write

$$\int_a^b \frac{dx}{\sqrt{X}} = A,\ \&c.$$

then these equations are

$$nK = A,$$
$$n(K + iK') = A + B,$$
$$n(2K + iK') = A + B + C.$$

Hence $B + C = 2A = B$, or $A = B - C$, or $B - C + D = 0$, that is,

$$\int_a^b \frac{dx}{\sqrt{X}} = \int_b^c \frac{dx}{\sqrt{X}},$$

where observe as before that $x = a$ to $x = b$, or $x = c$ to $x = d$, X is positive, and the radical $\sqrt{X}$ is taken to be positive.

We have also

$$nK = A = \int_a^b \frac{dx}{\sqrt{X}},$$
$$niK' = C - B = \int_a^b \frac{dx}{\sqrt{X}}.$$

when, as before, from b to c, X is negative, and the sign of the radical is such that $\sqrt{X}$ is a positive multiple of i; the last formula may be more conveniently written

$$niK' = \int_b^c \frac{dx}{\sqrt{-X}},$$

where, here b to c, $-X$ is positive, and $\sqrt{(-X)}$ is to be taken as positive.

Collecting the results, we have

$$\int_a^b \frac{dx}{\sqrt{X}} = \frac{nK}{\sqrt{(c-d)(c-a)}}, \quad b \le a \le b, \ \ldots b \le b \le b \le a,$$

and also

$$k = \frac{c-b}{a-b} \cdot \frac{d-a}{c-a},$$

and thence conversely

$$k' = \frac{a+d-b-c+(b-c-a+d)\,\sin^2 n}{a+b-c-d+(b-c+a-d)\,\sin^2 n},$$

or, what is the same thing,

$$\begin{aligned}
nu^2 &= \frac{b-d\,\cdot\,x-a}{b-a\,\cdot\,x-d},\\
nu^2 &= \frac{c-d\,\cdot\,x-a}{c-a\,\cdot\,x-d},\\
nu^2 &= \frac{d-a\,\cdot\,x-b}{a-b\,\cdot\,x-d},\\
nu^2 &= \frac{b-d\,\cdot\,x-a}{b-a\,\cdot\,x-d},
\end{aligned}$$

where, in place of the elliptic functions nu we substitute their ϑ-values, it will be recollected that Ω, the parameter of the ϑ-functions, has the value

$$\frac{K'}{K}\pi = \left(= \frac{\Omega'}{\Omega}\pi\right) = \pi \frac{\int_a^b \frac{dx}{\sqrt{-X}}}{\int_a^b \frac{dx}{\sqrt{X}}},$$

and, so before,

$$K = \int_a^b \frac{dx}{\sqrt{X}},$$

Hence, finally, a, b, K, K', Ω denoting given functions of a, b, c, d, as above

$$\int_a^b \frac{dx}{\sqrt{X}} = nu,$$

we have severally

$$\begin{aligned}
\frac{b-d}{c-d}\cdot\frac{a-c}{c-a} &= \frac{1}{k^2}\,\frac{\vartheta_1^2 u}{\vartheta_2^2 u}, & \frac{\vartheta_3^2 u}{\vartheta_2^2 u},\\
\frac{d-a}{d-b}\cdot\frac{b-c}{a-c} &= \frac{k'^2}{k}\,\frac{\vartheta_2^2 u}{\vartheta_3^2 u}, & \frac{\vartheta_1^2 u}{\vartheta_2^2 u} + \tfrac{1}{2}K',\\
\frac{c-a}{d-a}\cdot\frac{b-d}{b-c} &= k'^2\,\frac{\vartheta_4^2 u}{\vartheta_3^2 u}, &
\end{aligned}$$

which are the formulae in question.

The problem is to obtain these (and this in the more general case where a, b, c, d have any given (real or imaginary) values directly from the assumed equation

$$\int_a^b \frac{dx}{\sqrt{X}} = nu,$$

and from the foregoing definition of the function ϑu.

It may be recalled that the function ϑu is a doubly infinite product

$$\vartheta u = \Pi\Pi \left[1 + \frac{u}{m\pi + (n + \frac{1}{2})i\Pi} \right]$$

as and a positive or negative integers from $-\infty$ to $+\infty$; I purposely omit all further explanations as to limits, or what is the same thing,

$$\vartheta \frac{\pi}{2K} = \Pi\Pi \left[1 + \frac{u}{2mK + (2n+1)iK'} \right]$$

and consequently that, disregarding constant and exponential factors, the foregoing expressions of

$$\frac{b-d}{c-d} \cdot \frac{a-c}{c-a}, \quad \frac{d-a}{d-b} \cdot \frac{b-c}{a-c}, \quad \frac{c-a}{d-a} \cdot \frac{b-d}{b-c},$$

are the squares of the expressions

$\dfrac{X}{Y}, \dfrac{Y}{Z}, \dfrac{Z}{W},$ where X, Y, Z, W are respectively of the form

$$n\Pi\Pi \left[1 + \frac{u}{(2n+1)iK'} \right], \quad \Pi\Pi \left[1 + \frac{u}{(2m+1)iK'} \right],$$

$$\Pi\Pi \left[1 + \frac{u}{2mK + iK'} \right], \quad \Pi\Pi \left[1 + \frac{u}{(2m+1)K + iK'} \right],$$

where $(m, n = 2mK + 2niK', \&c.$, the *smha* over the m or the n denotes that the $2m$ or the $2n$ as the case may be) is to be changed into $2m+1$ or $2n+1$. But this is a transcription which has apparently no application to the ϑ-functions of more than one variable.

717.

ON THE TRIPLE THETA-FUNCTIONS.

[From the *Messenger of Mathematics*, vol. VII. (1878), pp. 48—50.]

As a specimen of mathematical notation, viz. of the notation which appears to me the easiest to read, and also to print, I give the definition and demonstration of the fundamental properties of the triple theta-functions.

Definition.

$$\vartheta(U,\ V,\ W) = \Sigma \exp \Theta,$$

where

$$\Theta = (A,\ B,\ C,\ F,\ G,\ H)(l,\ m,\ n)^2 + (U,\ V,\ W)(l,\ m,\ n).$$

l, denoting the sum in regard to all positive and negative integer values from $-\infty$ to $+\infty$ (zero included) of l, m, n respectively.

$\vartheta(U,\ V,\ W)$ is considered as a function of the arguments $(U,\ V,\ W)$, and it depends also on the parameters $(A,\ B,\ C,\ F,\ G,\ H)$.

First Property. $\vartheta(U,\ V,\ W) = 0$, for

$$U = \tfrac{1}{2}\pi i + (A,\ H,\ G)(\alpha,\ \beta,\ \gamma),$$

$$V = \tfrac{1}{2}\pi i + (H,\ B,\ F)(\alpha,\ \beta,\ \gamma),$$

$$W = \tfrac{1}{2}\pi i + (G,\ F,\ C)(\alpha,\ \beta,\ \gamma),$$

α, β, γ, δ, ϵ, ζ being any positive or negative integer numbers, such that as $\alpha + \beta + \gamma =$ odd number.

Demonstration. It is only necessary to show that to each term of ϑ there corresponds a second term, such that the inclusion of the two exponentials differ by an odd multiple of πi.

Taking l, m, n the integers which belong to the one term, those belonging to the other term are

$$-(l+a),\ -(m+\beta),\ -(n+\gamma).$$

(where observe that one at least of the numbers α, β, γ being odd, this system of values is not in any case identical with l, m, n). The two exponents are

$$\Theta_i = (A,\ B,\ C,\ F,\ G,\ H)(l,\ m,\ n)^2 + (U,\ V,\ W)(l,\ m,\ n),$$

and

$$\Theta_i' = (A,\ B,\ C,\ F,\ G,\ H)(l{+}a,\ m{+}\beta,\ n{+}\gamma)^2 - (U,\ V,\ W)(l{+}\alpha,\ m{+}\beta,\ n{+}\gamma),$$

viz. the value of Θ' is

$$= (A,\ B,\ C,\ F,\ G,\ H)(l,\ m,\ n)^2 + 2(A,\ H,\ G)(l\alpha,\ \beta,\ \gamma) + 2(H,\ B,\ F)(m\alpha,\ \beta,$$

$$+(A,\ B,\ C,\ F,\ G,\ H)(a,\ \beta,\ \gamma)^2 - (U,\ V,\ W)(l,\ m,\ n) - (U,\ V,\ W)(\alpha,\ \beta,\ \gamma),$$

and we then have

$$\begin{aligned}\Theta' - \Theta = -2(A,\ B,\ C,\ F,\ G,\ H)(l,\ m,\ n)^2 &+ 2(A,\ H,\ G)(\alpha,\ \beta,\ \gamma)l + 2(H,\\ &+ 2(G,\ F,\ C)(\alpha,\ \beta,\ \gamma)n + (A,\ B,\ C,\ F,\ G,\ H)(a,\ \beta,\ \gamma)^2\\ &- 2(U,\ V,\ W)(l,\ m,\ n) - (U,\ V,\ W)(\alpha,\ \beta,\ \gamma).\end{aligned}$$

which proves the theorem.

As to the notation, remark that, after $(A,\ B,\ C,\ F,\ G,\ H)$ has been written out in full, we may instead of $((A,\ B,\ C,\ F,\ G,\ H)(\alpha,\ \beta,\ \gamma)^2$ write

$$(\alpha,\ \beta,\ \gamma)^2,$$

and that we may use the like abbreviation

$$(A,\ \ldots)(\alpha,\ \beta,\ \gamma) \text{ for } (A,\ H,\ G)(\alpha,\ \beta,\ \gamma),\ \&c.,$$

$$(H,\ \ldots)(\alpha,\ \beta,\ \gamma) \text{ for } (H,\ B,\ F)(\alpha,\ \beta,\ \gamma),\ \&c.,$$

$$(G,\ \ldots)(\alpha,\ \beta,\ \gamma) \text{ for } (G,\ F,\ C)(\alpha,\ \beta,\ \gamma),\ \&c.$$

These are not only abbreviations, but they make the formula actually clearer, as bringing them into a similar compact; and I accordingly use them in the demonstration which follows.

Second Property. If U, V, W, denote

$$\begin{aligned} U' &= \pi i + (A,\ H,\ G)(\alpha,\ \beta,\ \gamma), \\ V' &= \pi i + (H,\ B,\ F)(\alpha,\ \beta,\ \gamma), \\ W' &= \pi i + (G,\ F,\ C)(\alpha,\ \beta,\ \gamma), \end{aligned}$$

respectively, where α, β, γ, δ, ϵ, ζ are any positive or negative integers (zero values admissible), then

$$\vartheta(U{+}u,\ V{+}v,\ W{+}w) = \exp\left[-2(\alpha U + \beta V + \gamma W) - (\alpha,\ \beta,\ \gamma)^2\right] \vartheta(U,\ V,\ W)$$

or, say

$$= \exp\left[-(A,\ \dots)(\alpha,\ \beta,\ \gamma)^2 - 2(U,\ V,\ W)(\alpha,\ \beta,\ \gamma) - (\alpha,\ \beta,\ \gamma)^2\right] \vartheta(U,\ V,\ W$$

Demonstration. Writing $\vartheta(U,\ V,\ W) = \Sigma \exp\Theta$, then in the expression of Θ, we may in place of l, m, n write $-l-\alpha$, $-m-\beta$, $-n-\gamma$; we thus obtain

$$\Theta_l = (A,\ \dots)(l,\ m,\ n)^2 {+} 2(A,\ \dots)(l,\ m,\ n)(a,\ \beta,\ \gamma) {+} (A,\ \dots)(\alpha,\ \beta,\ \gamma)^2$$
$$+(U,\ V,\ W)(l,\ m,\ n) + (U,\ V,\ W)(\alpha,\ \beta,\ \gamma),$$

which is

$$= (A,\ \dots)(l,\ m,\ n)^2$$
$$+2(\alpha,\ \beta,\ \gamma)(\alpha U{+}\beta V{+}\gamma W){-}2(A,\ \dots)(\alpha,\ \beta,\ \gamma){+}\Sigma'(A,\ \dots)(\alpha,\ \beta,\ \gamma)^2{-}\Sigma a^2(A$$
$$-2(\alpha,\ \beta,\ \gamma)^2{+}2\alpha,\ \beta,\ \gamma){+}(A,\ \dots)(\alpha,\ \beta,\ \gamma)^2{-}2(A,\ \dots)(\alpha,\ \beta,\ \gamma)$$
$$+(U,\ V,\ W)(l,\ m,\ n) + (U,\ V,\ W)(\alpha,\ \beta,\ \gamma),$$

which is

$$= (A,\ \dots)(l,\ m,\ n)^2$$
$$+\Sigma'(A,\ \dots)(\alpha,\ \beta,\ \gamma)^2{+}\Sigma a^2(A,\ \dots)(\alpha,\ \beta,\ \gamma){+}\Sigma\alpha\beta(A,\ \dots)(\alpha,\ \beta,\ \gamma)^2$$
$$+(U,\ V,\ W)(l,\ m,\ n) + \Sigma(A,\ \dots)(\alpha,\ \beta,\ \gamma)$$

Hence, rejecting the last line, which (as on can readily verify) has the exponential multiplied as the case may be, $\vartheta(U,\ V,\ W)$ as $\vartheta(U,\ V,\ W)$ multiplied by the factor

$$\exp\left[-(A,\ \dots)(\alpha,\ \beta,\ \gamma)^2 - 2(U,\ V,\ W)(\alpha,\ \beta,\ \gamma) - (\alpha,\ \beta,\ \gamma)^2\right],$$

which is the theorem in question.

In many cases a formula, which belongs to an indefinite number of k letters, is most easily intelligible when written out for three letters, but it is sometimes convenient to speak of the letters l, m, $\ldots$ or even the letter l, and to write out the formula accordingly.

[C. XI. 7]

In fact, writing herein $\alpha = \gamma = 0$, that is, $a = \gamma = 0$, the right-hand side becomes a^2 and the arts on the left-hand side are $a = B$, $A = B$, $a = A = I$, which suppose any four sum the sum of which is $= 0$.

Writing in the last-mentioned equation α, β, γ for the a's of A, B, γ, δ respectively, the equation becomes

$$
\begin{aligned}
&P = \beta P,\\
&Q = 1 - a^2 + \beta^2 + By^2,\\
&R = 1 - By^2 - 2\alpha + By^2,\\
&S = 1 - By^2.
\end{aligned}
$$

L, M, N as the like functions α, β, γ, δ respectively, A, B, C, D being the same as before. And in fact

$$
\begin{aligned}
-L\,Q &= -y^2(\alpha^2 + \alpha^2) + \beta^2 + (\beta^2 - \alpha^2) + B(y^2 - x^2) + B(\beta - \alpha) + \dots,\\
&= \frac{1}{B}\left[\beta^2 - (\beta - \alpha)^2 + By^2 - By^2\right] + B(y^2 - x^2) + \dots,\\
&= \frac{1}{B}\left[\beta^2(x^2 - y^2) + B(y^2 - x^2) + B^2(y^2 - x^2)\right],
\end{aligned}
$$

that is,

$$
\begin{gathered}
-LP\,Q = -y^2(\beta^2 - \alpha^2) + \beta^2 - (\beta - \alpha) + \dots,\\
+(B + \alpha - \beta)\beta(y^2 - x^2) + \beta - B^2(y^2 - x^2) + \dots,
\end{gathered}
$$

or omitting the factor β, that is,

$$
-LP\,Q = -y^2(\beta^2 - \alpha^2) + \beta^2 - y^2(x^2 - \alpha^2) + \dots + \beta(y^2 - x^2) - By^2(x^2 - \alpha^2),
$$

It is easy to verify the form of the orders 0, 1, 2, 3, and 4 in x, y, z, w separately destroy each other, for instance, for the terms of the order 4, the value of Σ is then

$$
= A^2 + \Sigma\, y^2(\beta^2 - \alpha^2) + y(x^2 - \alpha^2) + B^2(x^2 - \alpha^2) + (\Sigma\,\beta^2)(x^2 - \alpha^2) + \dots,
$$

$\frac{1}{B}\left[\Sigma\,\beta\, a^2(x^2 + y^2) + \Sigma\, a^2\, B^2(x^2 - y^2) + B^2\, a^2(x^2 - y^2)\right]$, It is easy to verify that the terms of the orders 0, 1, 2, 3, and 4 in x, y, z, w separately destroy any sum sum of x which is

$$
-LP^2 + yQ^2 + yP^2 + xR^2 + yS^2 + (LP + yQ + xR + yS + yT)
$$

$+(D^2-1-LP^2)(yP^2+yP^2-yQ^2-yQ^2)+2(LP+yQ+xR+yS+yT)(LP+yQ+x$

or, omitting the factor β, that is,

$-(LP+yQ^2+xR+yS+yT)+\Sigma(LP+yQ+xR+yS+yT)(LP+yQ+xR+yS+yT)$

The theorem in the original form was obtained by me as follows: using the elliptic coordinates p, q, r, such that

$$\frac{p^2}{a+p}+\frac{q^2}{b+p}+\frac{r^2}{c+p}=1,$$

or, what is the same thing,

$$-Bp\rho^2 = a+p \cdot b+p \cdot c+p,$$

$$-y\eta y = b+q \cdot a+q \cdot c+q,$$

$$-z\zeta z = c+r \cdot a+r \cdot b+r,$$

where A, B denote $b-c$, $c-a$, $a-b$ respectively; then, treating r as a constant, the coordinates x, y, z will belong to a point on the ellipsoid

$$\frac{x^2}{a+r} + \frac{y^2}{b+r} + \frac{z^2}{c+r} = 1,$$

and the differential equation of the right lines upon this surface is

$$\frac{dp}{\sqrt{a+p \cdot b+p \cdot c+p}} + \frac{dq}{\sqrt{a+q \cdot b+q \cdot c+q}} = 0.$$

Take a, b, c, the coordinates of a point on the surface, and p, q, the corresponding values of p, q, so that

$$-Bp_0\rho_0^2 = a+p_0 \cdot b+p_0 \cdot c+p_0,$$

$$-Aq_0\eta_0 = b+q_0 \cdot a+q_0 \cdot c+q_0,$$

$$-z\zeta_0 = c+r \cdot a+r \cdot b+r,$$

then the equation of the tangent plane at the point (a, b, c, p_0, q_0) is

$$\frac{xz_0}{a+r} + \frac{yy_0}{b+r} + \frac{zz_0}{c+r} = 1,$$

or, substituting for x, y_0, z_0 their values, we have

$$= \beta p_0 \cdot ay + \cdot bxz + \ Az + \dots,$$

and consequently the equation of the tangent plane through (a, b, c, p_0, q_0) is the integral of the proposed differential equation.

Writing

$$mz^2 = A(a+p),\ mz^2 = B(b+p),\ dx^2 = C(c+p),$$

the values of A, B, C, S are determined; and, assuming for q, r, p, the like forms with the arguments a, a_0, a_0, the differential equation becomes $dz + dz$, having the

$[\qquad\qquad\qquad 10-2]$

integral $u + x_0 + v = u_0$; while the foregoing integral equation, on reducing the constant coefficients contained therein, takes the form

$$-l^2 xx + mz_0\, mz + nz_0\, mz_0 + nz_0\, nz + nz_0\, nz_0 + \cdots = 0,$$

$$= \frac{1}{2} lm \cdot mz_0\, mz_0,$$

$$= -l^2 nz \cdot mz_0\, mz_0,$$

$$= \frac{nz_0}{B^2 - 1 - l^2} \cdot \frac{l^2}{l^2 - l^2},$$

via. this equation holds good if $a = u_0 + v$, we have the theorem.

If, as above, $a = u + v + v$, $b = u - v + v$, $c = u + v - v$, where r is regarded as constant, then the three products snu nz nz, cnu nz nz, dnu nz nz, and equally so the three products snz nz nz, cnz nz nz, dnz nz nz, are expressed by means of those products with the u, v, w of the u, of the v, viz. for two surfaces it would be possible to obtain a real relation equation, which may be expressed in the squares of the u, v, w of the r, of the z, all of whose internal coefficients are nominal, vis. the such would be obtained, in fact, a similar relation in the like form by giving to the parameter r the advantage of being of a degree which is the half of the equation which involves only the squared functions.

Write a, b, c or for $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$, respectively; then, writing

$$\begin{aligned} A &= \sqrt{1 - y^2 \,\cdot\, 1 - l^2 y^2}, & a &= \sqrt{1 - l^2 \,\cdot\, 1 - l^2 x^2}, \\ B &= \sqrt{1 - l^2 \,\cdot\, 1 - l^2 y^2}, & b &= \sqrt{1 - x^2 \,\cdot\, 1 - l^2 y^2}, \\ C &= \sqrt{1 - l^2 \,\cdot\, 1 - l^2 y^2}, & c &= \sqrt{1 - l^2 \,\cdot\, 1 - x^2 - y^2}, \\ D &= 1 - l^2 x^2 y^2, & d &= 1 - l^2 z^2 x^2, \end{aligned}$$

we have

$$\operatorname{sn}(z + v) = \frac{xA + ya}{D},$$

that is,

$$\frac{A + P}{D} = \frac{\sigma c}{\sigma a \,\cdot\, \sigma b},$$

and consequently

$$Dn = (A - x)(A + x),$$

$$PD = (P + x)(P - x);$$

whence

$$Dn = PD = (Aa - x)(Aa - x).$$

[745] (continued)

ON THE SCHWARZIAN DERIVATIVE AND THE POLYHEDRAL FUNCTIONS.

63. again P, Q given functions of z, and v a function of z determined by the equation

$$\frac{d^2v}{dz^2} + P\,\frac{dv}{dz} + Qv = 0\,;$$

and assume

$$y = wv.$$

Substituting this value of y in the first equation, we obtain for v an equation of the second order (the coefficients of which contain w), and we may make this identical with the second equation; viz. comparing the coefficients of the two equations, we thus have two equations each containing w; and by eliminating w we obtain a differential equation of the third order between z and x. This is, in fact, the basis of Kummer's theory for the transformation of a hypergeometric series: the equation between z, x will be found presently in a different manner.

64. But if with Schwarz, instead of making the equation obtained for v as above identical with the given equation for v, we merely assume that the two equations are consistent, then there is nothing to determine the value of z, which may be regarded as an arbitrary function of x; y and v are then functions of x, and w denotes the quotient $y \div v$ of these two functions, and as such satisfies an equation the form of which will depend on the assumed relation between z and x. In particular, if P and Q denote the same functions of z that p and q are of x; and if we assume $z = x$, P, Q will become $= p$, q respectively: the given equation in v will be

$$\frac{d^2v}{dx^2} + p\,\frac{dv}{dx} + qv = 0\,;$$

and w will thus denote the quotient of any two solutions of the equation

$$\frac{d^2y}{dx^2} + p\,\frac{dy}{dx} + qy = 0\,;$$

viz. writing $X = p^2 + 2\dfrac{dp}{dx} - 4q$, then, by what precedes, the equation for w will be

$$\{w,\, x\} = -\tfrac{1}{2}X.$$

65. Returning now to Kummer's problem, and considering y, v as solutions of the two differential equations respectively, we have $y = wv$, and taking any other two solutions denoted by these letters y_1, v_1 we have $y_1 = wv_1$, so that $\dfrac{y_1}{v_1} = \dfrac{y}{v}$: w is a function independent of the particular solutions of y, v; calling the quotient of two solutions of the equation in v, s, also s_1, denoting the quotient of two solutions of the equation in y, we have

$$\{s,\, x\} = -\tfrac{1}{2}Z, \qquad \{s_1,\, z\} = -\tfrac{1}{2}Z,$$

and similarly $Z = P^2 + 2\dfrac{dP}{dz} - 4Q$, whence, writing as before $X = p^2 + 2\dfrac{dp}{dx} - 4q$, we have each of these equal quantities

we obtain

$$\{s,\, x\} = -\tfrac{1}{2}X + \tfrac{1}{2}Z\left(\frac{dz}{dx}\right)^2,$$

as the required equation for the determination of z as a function of x. The process does not give the value of w, but this can be found without difficulty, viz.

$$\frac{1}{w}\frac{dw}{dx} = \tfrac{1}{2}(P\,\partial z/\partial x - p) = \tfrac{1}{2}\left(P\frac{dz}{dx} - p\right).$$

If z, x are regarded each of them as a function of the new independent variable θ, then the equation is

$$\{s,\,\theta\} = -\tfrac{1}{2}X\left(\frac{dx}{d\theta}\right)^2 + \tfrac{1}{2}Z\left(\frac{dz}{d\theta}\right)^2.$$

66. Jacobi's differential equation of the third order for the transformed modulus λ, *Fund. Nova*, p. 78, [*Ges. Werke*, t. I, p. 132], is

$$3\,(k''\lambda'' - \lambda''k'')^2 - 2k'\lambda'\,(k'\lambda''' - \lambda'k''') + k'^2\lambda'^2\left[\left(\frac{k''}{k'}\right)^2 k''' - \left(\frac{\lambda''}{\lambda'}\right)^2 \lambda'''\right] = 0,$$

where the accents denote differentiations in regard to an independent variable θ: viz. dividing by $2k'^2\lambda'^2$, this becomes

$$\{k,\,\theta\} + \tfrac{1}{2}k'^2\left(\frac{1+k^2}{1-k^2}\right)^2 = \{\lambda,\,\theta\} + \tfrac{1}{2}\lambda'^2\left(\frac{1+\lambda^2}{1-\lambda^2}\right)^2,$$

which is thus a particular case of Kummer's equation, k, λ corresponding to x, z respectively, and the values of X, Z being

$$X = -\left(\frac{1+k^2}{1-k^2}\right)^2, \qquad Z = -\left(\frac{1+\lambda^2}{1-\lambda^2}\right)^2.$$

67. In the case of the hypergeometric series, the two differential equations of the second order are

$$\frac{d^2y}{dx^2} + \frac{\gamma - (\alpha+\beta+1)x}{x(1-x)}\frac{dy}{dx} - \frac{\alpha\beta\,y}{x(1-x)} = 0,$$

$$\frac{d^2v}{dz^2} + \frac{\gamma' - (\alpha'+\beta'+1)z}{z(1-z)}\frac{dv}{dz} - \frac{\alpha'\beta'\,v}{z(1-z)} = 0.$$

Hence

$$p = \frac{\gamma + (1-x)(\gamma-\alpha-\beta-1)x}{x(1-x)} = \frac{\gamma}{x} + \frac{\gamma-\alpha-\beta-1}{1-x}, \qquad q = -\frac{\alpha\beta}{x(1-x)},$$

and hence

$$X = p^2 + 2\frac{dp}{dx} - 4q = \frac{1-2\gamma}{x^2} + \frac{(\gamma-\alpha-\beta-1)^2 + 2(\gamma-\alpha-\beta-1) + 4\alpha\beta + 2\gamma(\gamma-\alpha-\beta-1)}{(1-x)^2} + \frac{\text{etc.}}{x(1-x)};$$

viz. writing

$$a = \tfrac{1}{4}(1-\lambda^2), \qquad b = \tfrac{1}{4}(1-\mu^2), \qquad c = \tfrac{1}{4}(1-\nu^2),$$

where

$$\lambda^2 = (1-\gamma)^2, \qquad \mu^2 = (\gamma-\alpha-\beta)^2, \qquad \nu^2 = (\alpha-\beta)^2,$$

and putting in the formula $x-1$, $= -(1-x)$, we have

$$-\tfrac{1}{4}\left(p^2 + 2\frac{dp}{dx} - 4q\right) = \frac{a}{x^2} + \frac{b}{(x-1)^2} + \frac{-a-b+c}{x(x-1)} = \frac{a}{x^2} + \frac{b}{(x-1)^2} + \frac{c-a-b}{x\,.\,x-1} = (a,\,b,\,c\;\;x,\,x-1)\,;$$

with a like formula for $\frac{1}{4}\left(P^2 + 2\dfrac{dP}{dz} - 4Q\right)$. We then have

$$y = wv, \qquad v = x^{c-1}(1-x)^{\gamma-\alpha-\beta-1}\, z^{c'}(1-z)^{-\gamma'+\alpha'+\beta'+1} \cdots ,$$

and the differential equation of the third order for the determination of z is

$$\{z,\, x\} = (a_1,\, b_1,\, c_1 \;\; z,\, z-1)\left(\frac{dz}{dx}\right)^2 - (a,\, b,\, c \;\; x,\, x-1) = 0,$$

where a_1, b_1, c_1 are the same functions of α', β', γ' which a, b, c are of α, β, γ. This is, in effect, Kummer's equation for the transformation of the hypergeometric series.

68. And in like manner the Schwarzian equation for the determination of z, the quotient of two solutions, is

$$\{s,\, x\} = (a,\, b,\, c \;\; x,\, x-1).$$

PART II. THE POLYHEDRAL FUNCTIONS.

Origin and Properties. Art. Nos. 69 to 80.

69. The geometrical functions in question are connected with the geometrical forms (these being regarded as situate on a spherical surface), and with the stereographic projections of these figures:—

1. Polygon or
2. Double Pyramid,*
3. Tetrahedron,
4. Octahedron and Cube,
5. Dodecahedron and Icosahedron,

* Prof. Klein regards 1 as belonging to the polygon and 2 to the double pyramid: it seems to me that the fundamental figure, to which 1 and 2 each of them belong, is the polygon.

Consider a spherical surface and upon it any number of points: take at pleasure any point as South Pole, this determines the plane of the equator; and the stereographic projection of any point is the intersection with the plane of the equator of the line joining the point with the South Pole. To fix the ideas take the radius of the sphere as unity: let the axes of x and y be drawn in the plane of the equator in longitudes 0° and 90° respectively, and the axis of z upwards through the North Pole: the position of a point on the sphere is determined by means of its N.P.D. θ and longitude f: moreover we take X, Y, Z for the coordinates of the point on the surface, and x, y for those of its projection; and we then have

$$X,\, Y,\, Z = \sin\theta\cos f,\ \sin\theta\sin f,\ \cos\theta\,;$$

$$x = \frac{X}{1+Z} = \tan\tfrac{1}{2}\theta\cos f, \qquad y = \frac{Y}{1+Z} = \tan\tfrac{1}{2}\theta\sin f,$$

and conversely

$$X,\, Y,\, Z = 2x,\ 2y,\ 1 - x^2 - y^2 \;\div\; (1 + x^2 + y^2).$$

We represent the point $(X,\, Y,\, Z)$ on the spherical surface by means of the magnitude $x + iy$, $= \tan\frac{1}{2}\theta\,(\cos f + i\sin f)$, or say by the linear factor $s - (x+iy)$: and similarly any system of points on the surface by means of the system of magnitudes $x + iy$, or say by the function $\prod[s - (x+iy)]$, denoting in this manner the product of the linear factors which correspond to the different points respectively.

70. It will presently appear that, if (considering different stereographic projections, that is, a different position of the South Pole) we take x', y' as the coordinates of the new projection of the point, then $x'+iy'$ is a homographic function of $x+iy$: and consequently that the functions of s, which belong to the different projections, are linear transformations one of the other: but at present we consider a single projection.

It may be proper to remark that the figures in question are spherical figures having for summits points on the spherical surface, for sides arcs of great circle joining two summits, and faces which are portions of the spherical surface: the mid-points of the sides (or edges) and the centres of the faces are of course points on the spherical surface.

71. (1), (2). Considering a regular polygon formed by n summits on the equator, the longitude of one of them being $0°$, then the stereographic projections correspond with the points themselves, and the values of $x+iy$ are

$$1,\ \cos\tfrac{2\pi}{n}+i\sin\tfrac{2\pi}{n},\ \ldots,\ \cos\tfrac{(n-1)2\pi}{n}+i\sin\tfrac{(n-1)2\pi}{n}.$$

The corresponding function of s is s^n-1.

The values of $x+iy$ for the mid-points of the sides are

$$\cos\tfrac{\pi}{n}+i\sin\tfrac{\pi}{n},\ \cos\tfrac{3\pi}{n}+i\sin\tfrac{3\pi}{n},\ \ldots,\ \cos\tfrac{(2n-1)\pi}{n}+i\sin\tfrac{(2n-1)\pi}{n}.$$

The corresponding function of s is s^n+1.

The North and South Poles, which form with the n points a double pyramid of $n+2$ summits, correspond to the values $s=0$ and $s=\infty$. We have thus

$$s\,(1-s^n)(s^n-1)$$

as the function corresponding to the double pyramid.

72. (3). Considering for a moment the tetrahedron as a figure with rectilinear edges, this is so placed that two opposite edges are horizontal, and that the vertical planes passing through the centre and the upper and lower edges respectively are the meridian planes; the upper edge has the longitudes $\pm 45°$ and the lower edge the longitudes $135°$, $225°$. We thus explain the position of the spherical figure.

Corresponding to the summits we have the function $s^4-2i\sqrt{3}\,s^2+1$. In fact, the equation $s^4-2i\sqrt{3}\,s^2+1=0$ gives $s^2=i\,(\sqrt{3}\pm 2)$, and hence the values of s are the four values of $x+iy$ shown in the annexed table for the values of X, Y, Z, and $x+iy$ for the summits of the tetrahedron.

long.	45°	135°	225°	315°
X	$+\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$
Y	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$	$-\frac{1}{\sqrt{3}}$
Z	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$	$+\frac{1}{\sqrt{3}}$
$x+iy$	$\frac{1+i}{\sqrt{3}-1}$	$\frac{-1+i}{\sqrt{3}+1}$	$\frac{-1-i}{\sqrt{3}-1}$	$\frac{1-i}{\sqrt{3}+1}$

Corresponding to the centres of the faces, or summits of the opposite tetrahedron, we have the function $s^4+2i\sqrt{3}\,s^2+1$.

Corresponding to the mid-points of the sides, we have the function

$$s\,(1-s^4)(s^4-1)\,;$$

viz. the points in question are the four points $s=\pm1$, $s=\pm i$, the North Pole $s=0$, the South Pole $s=\infty$, and the four points on the equator at longitudes 0°, 90°, 180°, 270°.

73. (4). The octahedron is placed with two of its summits as poles, and the other four summits in the equator at longitudes 0°, 90°, 180°, 270° respectively: the values of s are, as in the last case, 0, ∞, ±1, $\pm i$, and the function is

$$s\,(1-s^4)(s^4-1).$$

The function for the centres of the faces, or summits of the cube, is s^8+14s^4+1.

The function for the mid-points of the sides of the octahedron or of the cube is

$$s^{12}-33s^8-33s^4+1.$$

74. (5). The Icosahedron is placed with two of its summits for poles; five summits lying in a small circle above the plane of the equator at longitudes 0°, 72°, 144°, 216°, 288°; and the remaining five summits in the corresponding small circle below the equator at longitudes 36°, 108°, 180°, 252° and 324°.

The function for the summits of the Icosahedron is

$$s\,(1-s^{10})(s^{10}+11s^5-1).$$

The function for the centres of the faces of the Icosahedron, or summits of the Dodecahedron, is

$$s^{20}-228s^{15}+494s^{10}+228s^5-1.$$

The function for the mid-points of the sides of the Icosahedron or the Dodecahedron is

$$s^{30}-522s^{25}+10005s^{20}+0\,s^{15}-10005s^{10}+522s^5+1.$$

I give for the present these results without demonstration.

75. Writing $\dfrac{x}{y}$ for the s, so as to obtain from the foregoing values $x+iy$ of these homogeneous functions $(\,;\,x,\,y)^n$, —it will be recollected that the x, y have nothing to do with the x, y of § 69—the forms which have thus presented themselves may be denoted as follows:

$$\begin{aligned}
(3):\quad & f3=(1,\,-2i\sqrt{3},\,1\,;\,x,\,y)^4,\\
& h3=(f3,\,f3)^2,\quad t3=xy\,(x^4-y^4),\\
(4):\quad & f4=xy\,(x^4-y^4),\\
& h4=(1,\,14,\,1\,;\,x^4,\,y^4)^2,\\
& t4=(1,\,-33,\,-33,\,1\,;\,x^4,\,y^4)^3,\\
(5):\quad & f5=xy\,(1,\,11,\,-1\,;\,x^5,\,y^5)^2,\\
& h5=(1,\,-228,\,+494,\,+228,\,-1\,;\,x^5,\,y^5)^4,\\
& t5=(1,\,-522,\,10005,\,0,\,-10005,\,522,\,1\,;\,x^5,\,y^5)^6;
\end{aligned}$$

where observe that $f4$ is the same function as $t3$. In each set of functions f, h, t, we have h and t covariants of f, viz. disregarding numerical factors, h is the Hessian, or derivative $(f,\,f)^2$, and t is the derivative $(f,\,h)$.

76. Since $f4$ is the same function as $t3$, we have of course $f4$, $h4$ and $t4$ themselves covariants of $f3$; but it is convenient to separate the two systems.

77. It is to be observed that $f3$ is a quartic function having its quadrinvariant $(f) = 0$; but independently of this, $f3$, qua quartic function, has only the covariants $h3$ and $t3$ (the Hessian and the cubicovariant respectively), viz. every other covariant is a rational and integral function of $f3$, $h3$ and $t3$. In particular, $h4$ and $t4$ are rational and integral functions of $f3$, $h3$ and $t3$; but inasmuch as $f3$ and $h3$ are not covariants of $f4$, this is not a property of $h4$ and $t4$ considered as covariants of $f4$, and the relation in question need not be attended to.

78. It has been just stated that $f3$ qua quartic function has only the covariants $h3$ and $t3$: $f4$ qua special sextic function (in the sense explained) and $f5$ qua dodecadic function have the like property, viz. $f4$ has only the covariants $h4$ and $t4$; $f5$ only the covariants $h5$ and $t5$. Hence $f3$, $f4$, $f5$ are "Prime-forms" in the sense defined in the paper by Fuchs, of 1875, and itself, viz. a Prime-form has no covariant of a lower order than itself, and also no covariant of a higher order which is a power of a form of a lower order.

79. The same functions have also the property that they are functions transformable into themselves by means of a group of linear transformations, and in this point of view they were considered in the nearly contemporaneous paper by Klein, of 1875; it is in this paper shown that the functions so transformable into themselves must be Polyhedral functions as above, the linear transformations in fact corresponding to the rotations whereby the spherical polyhedron can be brought into coincidence with its own original position. This theory will be presently given.

80. It is to be observed that, if U, V are functions $(\,;\,x,\,y)^n$ of the order n, then using the accent to denote differentiation in regard to x, $UV' - U'V$ and $(U,\,V)$ differ only by a numerical factor: and further that, writing as before $s = \dfrac{x}{y}$, and in the expression $UV' - U'V$ regarding U, V as functions $(\,;\,s,\,1)^n$, and the accent as denoting differentiation in regard to s, we have $UV' - U'V$ and $(U,\,V)$ differing by a numerical factor only. We have in the PQR-Table, the lines 3, 4, 5, P, Q, R equal to given numerical multiples of h^α, t^β, f^γ, the indices α, β, γ being such as to make these to be functions of the same degree: hence, neglecting numerical multipliers,

$$PQ' - P'Q = h^{\alpha-1}t^{\beta-1}\,(h,\,t),$$

and the theorem that $PQ' - P'Q$ is equal to a function $(\,;\,f^\delta)$, which contains only factors of R, is in fact the theorem that $(h,\,t)$, $(h,\,f)$, and $(t,\,f)$ are each of them equal to a term or product of f, h, t; which is a result included in the theorem that f has only the covariants h and t. And by this last theorem we know already how from R, assumed to be known, we can derive P and Q: viz. R is a power of f; and we thence have $h = (f,\,f)^2$ and $t = (h,\,f)$, equations giving the functions h and t, upon which P, Q depend.

Covariantive Formulæ. Art. Nos. 81 to 84.

81. The various covariantive formulae will be given with their proper numerical coefficients.

Tetrahedron function. f, h, t stand for the before-mentioned values, $f3$, $h3$, $t3$

$$(P,\,Q,\,R = h^3,\ -12i\sqrt{3}\,t^2,\ -f^6).$$

For $f3$.

$$(a,\,b,\,c,\,d,\,e) = (1,\,0,\,-\tfrac{1}{3}i\sqrt{3},\,0,\,1),$$

$$\tfrac{1}{4}(f,\,f)^2 = -\tfrac{1}{2}i\sqrt{3}\,h,\quad \tfrac{1}{4}(f,\,f)^4 = \tfrac{4}{3},\quad \tfrac{1}{4}(f,\,f)^6 = 0,$$

$$(f,\,h) = -32i\sqrt{3}\,t,\quad \tfrac{1}{4}(f,\,h)^2 = 1152\,f = 1152\,\frac{4}{3}\,f^2,$$

$$(h,\,t) = 96i\sqrt{3}\,f,\quad \tfrac{1}{4}(t,\,t)^2 = -8fh^2,$$

$$(f,\,t) = -96i\sqrt{3}\,h^2,$$
$$h^3 - f^3 - 12i\sqrt{3}\,t^2 = 0,$$
$$fh = (1,\,14,\,1\,;\,x^4,\,y^4)^2\ (= f4).$$

It is convenient to remark that t^2, f^3, h^3 being of the same order, we have

$$P\,(f,\,h) + Q\,(f,\,t)^2 + R\,(h,\,t)^2 = 0,$$

that is,

$$t^2\,.\,3\,f^2(f,\,h) + f^3\,.\,2h\,(h,\,t) + h^3\,.\,2\,.\,3t\,(t,\,f) = 0,$$

an equation which, substituting the values for $(f,\,h)$, $(h,\,t)$, $(t,\,f)$ their values, reduces itself to $h^3 - f^3 - 12i\sqrt{3}\,t^2 = 0$; and we have thus a verification of the before-mentioned relation and the values of $(f,\,h)$, $(h,\,t)$ and $(t,\,f)$. The like remark applies to the other two cases, which follow.

82. *Hexahedron function.* f, h, t stand for the before-mentioned values

$$f4,\ h4,\ t4 \quad (P,\,Q,\,R = h^3,\,-t^2,\,-108f^4).$$

For $f4$.

$$(a,\,b,\,c,\,d,\,e,\,f,\,g) = (0,\,\tfrac{1}{6},\,0,\,0,\,0,\,-\tfrac{1}{6},\,0),$$
$$\tfrac{1}{4}(f,\,f)^2 = -25\,h, \quad \tfrac{1}{4}(f,\,f)^4 = 0, \quad \tfrac{1}{4}(f,\,f)^6 = (720)^2\,\tfrac{1}{6},$$
$$(f,\,h) = -8t, \quad \tfrac{1}{4}(f,\,h)^2 = 3\,.\,2^8\,.\,7^2\,.\,h^2,$$
$$(f,\,t) = -12h^2, \quad \tfrac{1}{4}(t,\,t)^2 = 2^4\,.\,3^2\,.\,17^2\,.\,f^2,$$
$$(h,\,t) = -1728\,f^3,$$
$$h^3 - t^2 - 108\,f^4 = 0.$$

83. *Dodecahedron function.* f, h, t stand for the before-mentioned values

$$f5,\ h5,\ t5 \quad (P,\,Q,\,R = h^3,\,-t^2,\,-1728\,f^5).$$

For $f5$.

$$(a,\,b,\,c,\,d,\,e,\,f,\,g,\,h,\,i,\,j,\,k,\,l,\,m) = (0,\,\tfrac{1}{12},\,0,\,0,\,0,\,0,\,\tfrac{11}{12},\,0,\,0,\,0,\,0,\,-\tfrac{1}{12},\,0),$$
$$\tfrac{1}{4}(f,\,f)^2 = -121\,h, \quad \tfrac{1}{4}(f,\,f)^4 = 0, \quad \tfrac{1}{4}(f,\,f)^6 = \tfrac{1}{4}(924)^2(720)^2\,\tfrac{11}{12}\,f^2,$$
$$\tfrac{1}{4}(f,\,f)^8 = \tfrac{1}{4}(924)^2(720)^2\,\tfrac{1}{12}\,f^2, \quad \tfrac{1}{4}(f,\,f)^{10} = 0, \quad \tfrac{1}{4}(f,\,f)^{12} = 0,$$
$$(f,\,h) = -20\,t, \quad \tfrac{1}{4}(f,\,h)^2 = 173280\,h^2,$$
$$(f,\,t) = -30h^2, \quad \tfrac{1}{4}(t,\,t)^2 = 9082800\,f^2h,$$
$$(h,\,t) = -86400\,f^4,$$
$$h^3 - t^2 - 1728\,f^5 = 0.$$

84. We have

$$t = (x^5 + y^5)\,(1,\,522,\,-10005,\,-522,\,1\,;\,x^5,\,y^5)^4.$$

Write

$$\xi = (x^5 + y^5)\,(1,\,2,\,6,\,-2,\,1\,;\,x^5,\,y^5)^4,$$

then

$$t = \xi\,(1,\,-10,\,45\,;\,x,\,y)^2.$$

Or putting

$$\rho = \frac{1}{xy}\sqrt[4]{(x^5+y^5)\,(1,\,2,\,6,\,-2,\,1\,;\,x^5,\,y^5)^4},$$

that is,

$$\rho = \frac{1}{xy}\sqrt[4]{(x^{10}+11x^5y^5-y^{10})\,(\cdots)},$$

Writing $k = P/f$, then

$$\rho^3 - 10\rho^2 + 45\rho = \frac{k}{f}. \qquad \text{(Klein.)}$$

Investigation of the forms $f5$ and $h5$. Art. Nos. 85 and 86.

85. Writing for shortness[1] $k = \tan\alpha = \sqrt{\dfrac{5-\sqrt{5}}{10}}$, and $g = \cos 36^\circ + i\sin 36^\circ$, then the values of $x+iy$ corresponding to the summits of the Icosahedron are

$$0,\ k,\ kg,\ kg^2,\ kg^3,\ kg^4,\ k^{-1},\ k^{-1}g,\ k^{-1}g^2,\ k^{-1}g^3,\ k^{-1}g^4,\ \infty,$$

and the function $f5$ is thus

$$= s\,(1 - s^5/k^5)(s^5 - k^{-5}),$$

where the product of the last two factors is $s^{10} + (k^{-5} - k^5)s^5 - 1$. We have

$$k^{-5} - k^5 = \tfrac{1}{10}\sqrt{5}\,(5\sqrt{5} - 11), \quad = \tfrac{1}{2}(\sqrt{5} - 11)\cdot\sqrt{5}/5$$

and consequently $k^{-5} - k^5 = 11$; or the function is

$$s\,(1 - s^{10})(s^{10} + 11s^5 - 1).$$

86. Similarly, writing for shortness[2]

$$\cos\tfrac{\gamma}{2} = \frac{1}{2}\sqrt{\frac{5+\sqrt{5}}{2\sqrt{5}}}, \quad \cos^2\tfrac{\gamma}{2} = \frac{5+2\sqrt{5}}{15}, \quad \sin^2\tfrac{\gamma}{2} = \frac{10-2\sqrt{5}}{15},$$

and

$$l = \tan\tfrac{1}{2}\gamma, \quad l' = \tan\tfrac{1}{2}\gamma', \text{ where } \cos\tfrac{\gamma'}{2} = \frac{3+\sqrt{5}}{\cdots}, \quad \sin\tfrac{\gamma'}{2} = \frac{3-\sqrt{5}}{\cdots},$$

and therefore

$$l^5 = \frac{\cos\gamma}{\sin\gamma};$$

$g = \cos 36^\circ + i\sin 36^\circ$ as before, then the values of $x+iy$ for the summits of the dodecahedron are

$$l\,g,\ l\,g^2,\ l\,g^3,\ l\,g^4,\ l\,g^5;\quad l'\,g,\ l'\,g^2,\ l'\,g^3,\ l'\,g^4,\ l'\,g^5;$$
$$l^{-1},\ l^{-1}g,\ l^{-1}g^2,\ l^{-1}g^3,\ l^{-1}g^4;\quad l'^{-1},\ l'^{-1}g,\ l'^{-1}g^2,\ l'^{-1}g^3,\ l'^{-1}g^4.$$

The function $h5$ is therefore

$$= s^{20} + s^{15}(l^5 - l^{-5}) + l^5\cdot l'^5\,s^{10} + s^5(l'^5 - l'^{-5}) - 1.$$

[1] k is the α and h the β of the Table, No. 99.

[2] α is the α, γ is the γ, and γ' the $\alpha - \beta$ of the Table, No. 99.

We have

$$l^{-5}-l^{5}=\frac{2\cos\gamma}{\sin^5\gamma}=\frac{(1-\cos\gamma)^5}{\sin^5\gamma}\cdot\frac{1}{(1+\cos\gamma)^5}=\frac{(5+10\cos^2\gamma+\cos^4\gamma)}{\sin^5\gamma}=\frac{384+128\cos\gamma}{45\sin^5\gamma}=\frac{64\sqrt{5}}{45}(6+\sqrt{5})=$$

viz. this last identity depends on

$$\tfrac{2}{45}(3+\sqrt{5})(6+\sqrt{5})=(114+50\sqrt{5})\sin^4\gamma,$$

that is,

$$160\,(3+\sqrt{5})(6+\sqrt{5})=(114+50\sqrt{5})(120-40\sqrt{5}),$$

or

$$2\,(3+\sqrt{5})(6+\sqrt{5})=(57+25\sqrt{5})(3-\sqrt{5}),$$

or finally

$$(7+3\sqrt{5})(6+\sqrt{5})=57+25\sqrt{5},$$

which is right.

Similarly

$$l'^{-5}-l'^{5}=114-50\sqrt{5},$$

and observing that the sum and product of $114+50\sqrt{5}$, $114-50\sqrt{5}$ are $=228$ and 496 respectively, the required function of s is

$$=s^{20}-1-228\,(s^{15}-s^5)+496\,s^{10},$$

$$=s^{20}-228\,s^{15}+494\,s^{10}+228\,s^5+1,$$

$$(=h5)$$

which is the required value of $h5$.

Invariantive property of the Stereographic Projection.
Art. Nos. 87 to 93.

87. The before-mentioned theorem that the functions derived from two different stereographic projections of the same point are linear transformations one of the other, may be thus stated:—

Considering on the surface of a sphere, two fixed points A and B; and determining the position of a point C, first in regard to A by its distance θ and azimuth f, and next in regard to B by its distance θ' and azimuth f', the azimuths from the great circle ABx which joins the two points A and B, then we have

$$s=\tan\tfrac{1}{2}\theta\,(\cos f+i\sin f),\qquad s'=\tan\tfrac{1}{2}\theta'\,(\cos f'+i\sin f'),$$

calling them homographic functions one of the other; and putting the distance $AB=c$, the relation between them in fact is

$$s'=\frac{s-\tan\frac{1}{2}c}{1+s\tan\frac{1}{2}c}\,;$$

or, what is the same thing, observing that

$$s\,s'=\tan\tfrac{1}{2}\theta\tan\tfrac{1}{2}\theta'\,\{\cos(f+f')+i\sin(f+f')\},$$

we have the two equations

$$\tan\tfrac{1}{2}c\,\{1+\tan\tfrac{1}{2}\theta\tan\tfrac{1}{2}\theta'\,\cos(f+f')\}=\tan\tfrac{1}{2}\theta\cos f-\tan\tfrac{1}{2}\theta'\cos f',$$

$$\tan\tfrac{1}{2}c\,\{1+\tan\tfrac{1}{2}\theta\tan\tfrac{1}{2}\theta'\,\sin(f+f')\} = \tan\tfrac{1}{2}\theta\sin f - \tan\tfrac{1}{2}\theta'\sin f'.$$

88. If we denote the angles of the spherical triangle C, A, B, and the opposite sides by c (as before), a, b, then θ, $\theta' = b$, a; f, $f' = A$, $\pi - B$, whence

$$s = \tan\tfrac{1}{2}b\,(\cos A + i\sin A), \qquad s' = -\tan\tfrac{1}{2}a\,(\cos B - i\sin B),$$

and we have between the spherical sides a, b, c and angles A, B of the spherical triangle the relations

$$\tan\tfrac{1}{2}c\,\{1-\tan\tfrac{1}{2}a\tan\tfrac{1}{2}b\,\cos(A-B)\} = \tan\tfrac{1}{2}b\cos A + \tan\tfrac{1}{2}a\cos B,$$
$$\tan\tfrac{1}{2}c\,\{1-\tan\tfrac{1}{2}a\tan\tfrac{1}{2}b\,\sin(A-B)\} = \tan\tfrac{1}{2}b\sin A - \tan\tfrac{1}{2}a\sin B\,;$$

which equations may be verified by means of the ordinary formulae of Spherical Trigonometry.

89. But it is interesting to give the proof with rectangular coordinates. Taking $(X,\,Y,\,Z)$, $(X_1,\,Y_1,\,Z_1)$ for the coordinates, referred to two different sets of axes, of a point on the spherical surface: also x, y, x_1, y_1 for the coordinates of the corresponding stereographic projections, we have

$$(X_1 : Y_1 : Z_1) = (\alpha,\,\beta,\,\gamma\,;\,\alpha',\,\beta',\,\gamma'\,;\,\alpha'',\,\beta'',\,\gamma'')\,(X : Y : Z),$$
$$X : Y : Z : 1 = 2x : 2y : 1-x^2-y^2 : 1+x^2+y^2,$$
$$X_1 : Y_1 : Z_1 : 1 = 2x_1 : 2y_1 : 1-x_1^2-y_1^2 : 1+x_1^2+y_1^2,$$

and thence

$$x_1 : y_1 : 1 = 2\alpha x + 2\beta y + \gamma\,(1-x^2-y^2) : 2\alpha' x + 2\beta' y + \gamma'\,(1-x^2-y^2) : 1+x^2+y^2+2\alpha'' x+2\beta'' y+\gamma''\,(1-x^2-y^2).$$

90. Introducing z, z_1 for homogeneity, or writing $\dfrac{x}{z}$, $\dfrac{y}{z}$ and $\dfrac{x_1}{z_1}$, $\dfrac{y_1}{z_1}$ in place of x, y and x_1, y_1, respectively, we have

$$x_1 = 2\alpha x + 2\beta y + \gamma\,(z^2-x^2-y^2) = (\,-\gamma,\,-\gamma,\,+\gamma,\,\beta,\,\alpha,\,0\,;\,x,\,y,\,z)^2,$$
$$y_1 = 2\alpha' x + 2\beta' y + \gamma'\,(z^2-x^2-y^2) = (\,-\gamma',\,-\gamma',\,+\gamma',\,\beta',\,\alpha',\,0\,;\,x,\,y,\,z)^2,$$
$$z_1 = z^2+x^2+y^2+2\alpha'' x+2\beta'' y+\gamma''\,(z^2-x^2-y^2) = (1-\gamma'',\,1-\gamma'',\,1+\gamma'',\,\beta'',\,\alpha'',\,0\,;\,x,\,y,\,z)^2,$$

and thence without difficulty

$$x_1+iy_1 = \tfrac{1}{\gamma+\gamma'^*}\,\{(1+\gamma'')z+(\alpha''+i\beta'')(x-iy)\}\{(1-\gamma'')z+(-\alpha''+i\beta'')(x+iy)\},$$
$$z_1 = \tfrac{1}{\gamma+\gamma'^*}\,\{(1-\gamma'')z-(\alpha''+i\beta'')(x-iy)\}\{(1+\gamma'')z+(\alpha''-i\beta'')(x+iy)\},$$
$$x_1-iy_1 = MN : NL : LM \quad (L,\,M,\,N \text{ linear functions of } z,\,x+iy,\,x-iy)$$

showing that the relation between two stereographic projections of the same spherical figure is in fact that of a quadric transformation, the fundamental points in each figure being an arbitrary point and the two circular points at infinity: or, what is the same thing, to any line in the one figure there corresponds a circle in the other figure, which is the "circular relation" of Möbius.

91. The actual values are

$$\frac{x_1+iy_1}{z_1} = \frac{1+\gamma''}{\gamma+\gamma''}\cdot\frac{(1-\gamma'')z-(\alpha''-i\beta'')(x+iy)}{(1+\gamma'')z+(\alpha''-i\beta'')(x+iy)},$$

$$\frac{x_1-iy_1}{z_1} = \frac{1+\gamma''}{\gamma-\gamma''}\cdot\frac{(1-\gamma'')z-(\alpha''+i\beta'')(x-iy)}{(1+\gamma'')z+(\alpha''+i\beta'')(x-iy)};$$

viz. attending only to the former of these, we have $\dfrac{x_1+iy_1}{z_1}$ a homographic function of $\dfrac{x+iy}{z}$, which is the before-mentioned theorem.

92. Supposing that the transformation from (X, Y, Z) to (X_1, Y_1, Z_1) is made by a rotation, the coordinates of which are λ, μ, ν, that is, if f, g, h are the inclinations of the resultant axis to the axes of x, y, z respectively, and θ the angle of rotation, putting λ, μ, $\nu = \tan\frac{1}{2}\theta\cos f$, $\tan\frac{1}{2}\theta\cos g$, $\tan\frac{1}{2}\theta\cos h$: then the coefficients of transformation are

$$\begin{pmatrix} \alpha & \beta & \gamma \\ \alpha' & \beta' & \gamma' \\ \alpha'' & \beta'' & \gamma'' \end{pmatrix} = \frac{1}{1+\lambda^2+\mu^2+\nu^2} \begin{pmatrix} 1+\lambda^2-\mu^2-\nu^2 & 2(\lambda\mu-\nu) & 2(\lambda\nu+\mu) \\ 2(\lambda\mu+\nu) & 1-\lambda^2+\mu^2-\nu^2 & 2(\mu\nu-\lambda) \\ 2(\lambda\nu-\mu) & 2(\mu\nu+\lambda) & 1-\lambda^2-\mu^2+\nu^2 \end{pmatrix}.$$

Substituting these values, the formulae become, after an easy reduction,

$$\frac{x_1+iy_1}{z_1} = \frac{-(\nu+i)(x+iy)+(\lambda+i\mu)z}{(\lambda-i\mu)(x+iy)+(\nu-i)z},$$

$$\frac{x_1-iy_1}{z_1} = \frac{-(\nu-i)(x-iy)+(\lambda-i\mu)z}{(\lambda+i\mu)(x-iy)+(\nu+i)z};$$

attending to the former of these, and writing for greater simplicity s, s_1 respectively, we have

$$s_1 = \frac{-(\nu+i)s+(\lambda+i\mu)}{(\lambda-i\mu)s+(\nu-i)},$$

or writing this

$$s_1 = \frac{As+B}{Cs+D},$$

then

$$A : B : C : D = -\nu-i : \lambda+i\mu : \lambda-i\mu : \nu-i.$$

93. I call to mind that the condition, in order that the homographic transformation $s_1 = (As+B) \div (Cs+D)$ may be periodic of the order n, is

$$(A+D)^2 - 4\,(AD-BC)\cos^2\tfrac{m\pi}{n} = 0,$$

m being an integer different from zero and prime to n. In particular, when $n=2$, it is $A+D=0$: $n=3$, it is $A^2+AD+D^2+BC=0$: $n=4$, it is $A^2+D^2+2BC=0$: and $n=5$, it is $(A+D)^2-\frac{1}{2}\,(3\pm\sqrt{5})\,(AD-BC)=0$.

Groups of homographic transformations. Art. Nos. 94 and 95.

94. The formulae just obtained serve to connect the theory of the rotations of a polyhedron with that of the homographic transformations s into $(As+B)\div(Cs+D)$; and, corresponding to the rotations which leave the polyhedron unaltered, we have groups of homographic transformations. We have thus, corresponding to the cases of the tetrahedron, the cube and the octahedron, and the dodecahedron and icosahedron respectively, groups of 12, of 24, and of 60 homographic transformations s into

$(As+B)\div(Cs+D)$. The group of 60 and the group of 24 include each of them as part of itself the group of 12; it is further to be remarked that the group of 12 may be regarded as that of the positive substitutions upon the four letters *abcd*, the group of 24 as that of all the substitutions upon the four letters, and the group of 60 as that of the positive substitutions upon the five letters *abcde*.

95. I call to mind that a group of functional symbols 1, α, β, ... can always be expressed in the equivalent form 1, $\phi\alpha\phi^{-1}$, $\phi\beta\phi^{-1}$, ..., where ϕ is any functional symbol whatever: clearly, α, β, ... being homographic transformations, then, ϕ being any homographic transformation whatever, the new symbols $\phi\alpha\phi^{-1}$, $\phi\beta\phi^{-1}$, ... will also be homographic transformations; and thus the group of homographic transformations can be expressed in various equivalent forms: these correspond to the different positions of the polyhedron in regard to the axes of coordinates; viz. in the three cases which it is proper to consider, the dodecahedron, we may have the Pole at a point Θ, that is, in the language of art. Nos. as presently explained, the axis of z passing through the midpoint of a side, 1°: through a summit; that is, the Pole at a point A, 2°: and there are different forms which correspond to a Pole at a point B, 3°.

The regular Polyhedra. Art. Nos. 96 to 103.

96. We require a theory of the regular Polyhedra considered as systems of points on a sphere. I refer to my two papers [375] and [679]. In the latter paper, I remark that, considering the five regular figures drawn in proper relation to each other on the same spherical surface, the only points which have to be considered are 12 points A, 20 points B, 30 points Θ, and 60 points Φ. Describing these by reference to the dodecahedron, the points A are the centres of the faces, the points B are the summits, the points Θ are the midpoints of the sides, and the points Φ are the midpoints of the diagonals of the faces, each point Φ being given as the intersection of a face of the dodecahedron and a side of the icosahedron, which there intersect at right angles. Or describing them by reference to the icosahedron, the points A are the summits, the points B are the centres of the faces, the points Θ are the midpoints of the sides: and the points Φ are points lying by threes on the faces of the icosahedron, which there is no occasion to attend to; viz. Φ is the point common to the face and a line BB joining the centres of two adjacent faces at right angles.

97. The points Φ are comparatively unimportant, and it is proper first to attend only to the 12 points A, the 20 points B, and the 30 points Θ: these form 6 pairs of opposite points A, 10 pairs of opposite points B, and 15 pairs of opposite points Θ. Considering the diameters through each pair of opposite points Θ, we have thus a system of 15 axes, which in fact form 5 sets each of 3 rectangular axes: attending to any one of such sets, the diametral plane at right angles to one of the three axes contains of course the other two axes: it contains also two axes each through a pair of opposite points A, and two axes each through a pair of opposite points B. If instead of the plane we consider its intersection with the sphere, we have thus on the sphere 15 circles each containing 4 points Θ,

4 points A and 4 points B. The fifteen circles intersect by fives in the pairs of opposite points A, by threes in the pairs of opposite points B, and by twos in the pairs of opposite points Θ; the mutual inclinations of successive circles at the points A, B, Θ being $= 36°$, $60°$ and $90°$ respectively. The whole number 15.14, $= 210$, of the intersections of the two circles two and two together is thus made up of the 12 points A each counting 10 times, the 20 points B each counting 3 times, and the 30 points Θ each counting once; $210 = 120 + 60 + 30$.

98. The angular magnitudes which present themselves are all obtained from the dodecahedral pentagon, as shown in the annexed figure, in which the angle subtended by a side at the centre is $= 72°$, and the angle between two adjacent sides is $= 120°$.

[Figure: dodecahedral pentagon with vertices $B_1, B_2, \ldots, B_5$, centre A, midpoints $\Theta, \Theta', \ldots$]

We write $A\Theta = \alpha$, $B\Theta = \beta$, $AB = \gamma$, $B_1B_2 = x$, $\angle B_1B_2B = \theta$, $\Theta B_1 = g$, $\angle\Theta B_1 B = \phi$.

From the triangle $A\Theta B$, the angles of which are 36°, 90°, 60° and the opposite β, γ, α, we find the values of α, β, γ, and these are such that $\alpha + \beta + \gamma = \frac{1}{2}\pi$.

From the triangle B_2BB_1, where the sides B_1B, BB_2 and the included angle are 2β, 2β, $120°$, we have the opposite side x, and the other two angles each $= \theta$.

From the triangle $B_1\Theta B_2$, where the sides B_1B, $\beta\Theta$ and the included angle $BB_2\Theta$ are 2β, β, $120°$, we find the opposite side g, and the angle $B_2\Theta B_1 = 45°$.

Hence each of the angles $B_1\Theta B$, $B_2\Theta B_1$, being $= 45°$, the angle $B_1\Theta B_2$ is $= 90°$: in this triangle the hypothenuse B_1B_2 is $= x$, and the other two sides are $= g$: whence we have $\cos x = \cos^2 g$, as is in fact the case, and moreover the values give $x + 2g = 180°$. Also each of the other angles is found to be $= 60°$; that is, we have $\angle B_1BB_2 = 60°$, or the whole angle at B_1 being $= 120°$, the sum of the remaining angles B_1BB_2 and $BB_1\Theta$ is $= 60°$.

From the triangle $\Theta B_1\Theta'$ where the two sides and the included angle are β, β, $120°$, we find $\Theta\Theta' = 36°$.

And from the triangle $\Theta B_1\Theta''$, where the two sides and the included angle are g, g, and $(120° - 2\phi =)2\phi$, we find $\Theta\Theta'' = 60°$.

99. We thus arrive at the following Table:

			cos	sin
$A\Theta$	α	$31°\ 43'$	$\sqrt{\frac{5-\sqrt{5}}{10}}$	$\sqrt{\frac{5+\sqrt{5}}{10}}$
$B\Theta$	β	$20°\ 55'$	$\frac{\sqrt{5}+1}{2\sqrt{3}}$	$\frac{\sqrt{10-2\sqrt{5}}}{2\sqrt{3}}$
AB	γ	$37°\ 22'$	$\sqrt{\frac{5+\sqrt{5}}{15}}$	$\sqrt{\frac{10-2\sqrt{5}}{15}}$
(BB)	x	$70°\ 32'$	$\frac{1}{3}$	$\frac{2\sqrt{2}}{3}$
(BB)		$109°\ 28'$	$-\frac{1}{3}$	$\frac{2\sqrt{2}}{3}$
(BB)	g	$54°\ 44'$	$\frac{1}{\sqrt{3}}$	$\frac{\sqrt{2}}{\sqrt{3}}$
BBB	θ	$37°\ 46'$		
BBB		$54°\ 44'$		
ΘBB	ϕ			
2β		$41°\ 50'$	$\sqrt{\frac{5-2\sqrt{5}}{15}}$	
2γ		$74°\ 44'$	$\frac{\sqrt{3}\,(\sqrt{5}-1)}{4}$	
$\alpha-\beta$			$\frac{\sqrt{5-\sqrt{6}}}{2}$	
$\Theta\Theta'$		$36°$	$\frac{\sqrt{5}+1}{4}$	
$\Theta\Theta''$		$60°$	$\frac{1}{2}$	

where as above $\alpha + \beta + \gamma = 90°$, and $x + 2g = 180°$, $\theta + \phi = 60°$.

100. We now construct three figures of the points A, B, Θ; viz. these are stereographic projections, each showing the Northern hemisphere projected on the plane of the equator by lines drawn to the South Pole: hence, for any pair of opposite points on the equator, only the point in the Northern hemisphere is shown: but for a pair of opposite points on the equator the two points are each of them shown. In fig. 1 the North Pole is taken to be a point Θ; in fig. 2 it is a point A; and in fig. 3 it is a point B. The position of any point on the sphere is determined by its N.P.D. and its longitude, say from the point E of the centre left-handedly measured from an arbitrary origin: then, in the three figures, the positions are as follows.

101. Fig. 1. Pole at Θ.

[Figure 1: Stereographic projection, Pole at Θ, showing positions of 12 A points, 20 B points and 30 Θ point

	N.P.D.'s	Longitudes
$2A$	$\alpha = 31° 43'$	$0°$, $180°$
$2A$	$90° - \alpha = 58\ 17$	90, 270
$2A$	$90° + \alpha = 121\ 43$	$0°$, $180°$
$2A$	$180° - \alpha = 148\ 17$	90, 270
$4A$	90	$(0,\ 180) + \alpha = 31° 43'$
$2B$	$\beta = 20° 55'$	$90°$, $270°$
$4B$	$g = 54\ 44$	45, 135, 225, 315
$2B$	$90° - \beta = 69\ 5$	0, 180
$2B$	90	$(90,\ 270) + \beta = 20° 55'$
$4B$	$90° + \beta = 110\ 55$	0, 180
$4B$	$180° - g = 125\ 16$	45, 135, 225, 315
$2B$	$180° - \beta = 159\ 5$	90, 270
1Θ	$0°$	—
4Θ	36	$(90°,\ 270°) + \alpha = 31° 43'$
4Θ	60	$(0,\ 180) + \beta = 20\ 55$
4Θ	72	$(90,\ 270) + \alpha = 31\ 43$
4Θ	90	0, 90, 180, 270
4Θ	108	$(90,\ 270) + \alpha = 31\ 43$
4Θ	120	$(0,\ 180) + \beta = 20\ 55$
4Θ	144	$(90,\ 270) + \alpha = 31\ 43$
1Θ	180	—

102. Fig. 2. Pole at A.

[Figure 2: Stereographic projection, Pole at A.]

	N.P.D.'s	Longitudes
A	0	—
$5A$	$2\alpha = 63° 26'$	$0°$ $72°$ $144°$ $216°$ $288°$
$5A$	$180° - 2\alpha = 116\ 34$	36 108 180 252 324
A	180	—
$5B$	$\gamma = 37\ 22$	36 108 180 252 324
$5B$	$90° - \alpha + \beta = 79\ 12$	36 108 180 252 324
$5B$	$90 + \alpha - \beta = 100\ 48$	72 144 216 288
$5B$	$180 - \gamma = 142\ 38$	72 144 216 288
5Θ	$\alpha = 31\ 43$	72 144 216 288
5Θ	$90° - \alpha = 58\ 17$	36 108 180 252 324
10Θ	90	$(36\ 108\ 180\ 252\ 324) + 18°$
5Θ	$90 + \alpha = 121\ 43$	72 144 216 288
5Θ	$180 - \alpha = 144\ 17$	36 108 180 252 324

103. Fig. 3. Pole at B.

[Figure 3: Stereographic projection, Pole at B.]

	N.P.D.'s	Longitudes
$3A$	$\gamma = 37° \, 22'$	$30° \; 150° \; 270°$
$3A$	$90° - \alpha + \beta = 79 \; 12$	$90 \; 210 \; 330$
$3A$	$90 + \alpha - \beta = 100 \; 48$	$90 \; 210 \; 330$
$3A$	$180 - \gamma = 142 \; 38$	$90 \; 210 \; 330$
B	0	—
$3B$	$2\beta = 41 \; 50$	$30 \; 150 \; 270$
$6B$	$x = 70 \; 32$	$(90 \; 210 \; 330) + \phi = 37° \, 46'$
$6B$	$180° - x = 109 \; 28$	$(30 \; 150 \; 270) + \phi = 37° \, 46'$
$3B$	$180 - 2\beta = 138 \; 10$	$90 \; 210 \; 330$
B	180	—
3Θ	$\beta = 20 \; 55$	$90 \; 210 \; 330$
6Θ	$g = 54 \; 44$	$(90 \; 210 \; 330) + \phi = 22° \, 14'$
Θ	$60 \; 120 \; 180° \; 240° \; 300°$	—
6Θ	$90 + \beta = 110 \; 55$	$30 \; 150 \; 270$
6Θ	$180 - g = 125 \; 16$	$30 \; 150 \; 270$
3Θ	$180 - \beta = 159 \; 5$	$90 \; 210 \; 330$

The groups of homographic transformations, resumed.
Art. Nos. 104 to 117.

104. The axes of rotation for the dodecahedron and the icosahedron are 15 axes each through a pair of opposite points Θ, 6 axes each through a pair of opposite points A, and 10 axes each through a pair of opposite points B; or say 15 Θ-axes, 10 5-axes and 6 A-axes; the corresponding angles of rotation are 180°, 72° and 120° so that (excluding in each case the original position or that of a rotation 0) we have in respect of each Θ-axis 1 position, in respect of each A-axis 4 positions, and in respect of each B-axis 2 positions; in all, including the original position,

$$1 + 15 + (6 \times 4) + (10 \times 2), \text{ that is, } = 60 \text{ positions},$$

that is, a group of 60 rotations.

To find, in any one of the three forms, the group of homographic transformations, we can in each case obtain from the foregoing tables the values $\cos f$, $\cos g$, $\cos h$ of the cosine-inclination of an axis of rotation to the axes of coordinates, and thence calculate the values of

$$\lambda,\ \mu,\ \nu = \tan \tfrac{1}{2}\theta \cos f,\ \tan \tfrac{1}{2}\theta \cos g,\ \tan \tfrac{1}{2}\theta \cos h,$$

and thence the values of

$$A,\ B,\ C,\ D = -\nu - i,\ \lambda + i\mu,\ \lambda - i\mu,\ \nu - i\,;$$

viz. in the case of a Θ-axis, θ is $= 180°$, (so that here $\tan \frac{1}{2}\theta = \infty$ or the values of A, B, C, D are $= -\nu, \lambda + i\mu, \lambda - i\mu, \nu$, that is, $-\cos h$, $\cos f + i \cos g$, $\cos f - i \cos g$, $\cos h$); in the case of a B-axis, the values are $\theta = 120°, 240°$, and therefore $\tan \frac{1}{2}\theta = \pm\sqrt{3}$; and in the case of an A-axis, they are $\theta = 72°, 144°, 216°, 288°$, and therefore

$$\tan \tfrac{1}{2}\theta = \pm \tfrac{1}{\sqrt{5}} \sqrt{10 + 2\sqrt{5}},\ \pm \tfrac{1}{\sqrt{5}} \sqrt{10 - 2\sqrt{5}}.$$

105. The Θ-form was first given in my paper of 1879, but in obtaining it I used the results given in the paper of 1877. As regards the identification with the substitution-symbols, since there is nothing to distinguish inter se the letters a, b, c, d, e, any transformation A, B, C, D of the fifth order might have been taken for *abcde*, But No. 37 of the group having been taken for this substitution *abcde*, I do not recall in what manner I found that, consistently herewith, the transformation No. 2 $(-1,0,0,1$, that is, s into $-s)$ of the second order could be taken for $ab\,.\,cd$. But there is no sub-group of an order divisible by 5; and hence, these two transformations being identified with the two substitutions, the other transformations correspond each of them to a determinate substitution.

106. *Homographic Transformations.* The group of 60. Pole at Θ.

No.	$(Ax+B)$			$\div\,(Cx+D)$	substitution
1	1			1	1
2	-1			1	$ab\,.\,cd$
3	i			1	$ac\,.\,bd$
4	$-i$			1	$ad\,.\,bc$
5	$-3+\sqrt{5}+i(1-\sqrt{5})$			2	$ad\,.\,be$
6	$-3+\sqrt{5}+i(-1+\sqrt{5})$			2	$ae\,.\,bd$
7	$3-\sqrt{5}+i(-1+\sqrt{5})$			2	$ad\,.\,ce$
8	$3-\sqrt{5}+i(1-\sqrt{5})$			2	$ae\,.\,cd$
9	$-1-\sqrt{5}+i(1-\sqrt{5})$			2	$ab\,.\,de$
10	$-1-\sqrt{5}+i(-1+\sqrt{5})$			2	$bc\,.\,de$
11	$1+\sqrt{5}+i(-1+\sqrt{5})$			2	$ab\,.\,ce$
12	$1+\sqrt{5}+i(1-\sqrt{5})$			2	$ac\,.\,bd$
13	$-1-\sqrt{5}+i(-3-\sqrt{5})$			2	$bd\,.\,ce$
14	$-1-\sqrt{5}+i(3+\sqrt{5})$			2	$ae\,.\,be$
15	$1+\sqrt{5}+i(3+\sqrt{5})$			2	$bc\,.\,de$
16	$1+\sqrt{5}+i(-3-\sqrt{5})$			2	$ac\,.\,de$
17	i			1	abc
18	$-i$			1	acb
19	1			$-i$	ade
20	-1			$-i$	aed
21	i			1	abd
22	$-i$			1	adb
23	1			i	bcd
24	-1			i	bdc
25	$-1-\sqrt{5}+i(-3-\sqrt{5})$			2	abd
26	$1+\sqrt{5}+i(3+\sqrt{5})$			2	ace
27	$1+\sqrt{5}+i(-3-\sqrt{5})$			2	bce
28	$-1-\sqrt{5}+i(3+\sqrt{5})$			2	bed
29	$-3+\sqrt{5}+i(1-\sqrt{5})$			2	bde
30	$-3+\sqrt{5}+i(-1+\sqrt{5})$			2	bee
31	$3-\sqrt{5}+i(-1+\sqrt{5})$			2	aed
32	$3-\sqrt{5}+i(1-\sqrt{5})$			2	ade
33	$-1-\sqrt{5}+i(3+\sqrt{5})$			2	cde
34	$1+\sqrt{5}+i(-3-\sqrt{5})$			2	ced
35	$1+\sqrt{5}+i(3+\sqrt{5})$			-2	aeb
36	$-1-\sqrt{5}+i(-3-\sqrt{5})$			2	abe
37	$1+\sqrt{5}+i(-3-\sqrt{5})$			2	$abcde$
38	$-1-\sqrt{5}+i(1-\sqrt{5})$			2	$acebd$
39	$-1-\sqrt{5}+i(-1+\sqrt{5})$			2	$adbec$
40	$-1-\sqrt{5}+i(3+\sqrt{5})$			2	$aedcb$
41	$1+\sqrt{5}+i(3+\sqrt{5})$			2	$adceb$
42	$1+\sqrt{5}+i(-1+\sqrt{5})$			2	$acbde$
43	$1+\sqrt{5}+i(1-\sqrt{5})$			2	$aedbe$
44	$1+\sqrt{5}+i(-3-\sqrt{5})$			2	$abcad$
45	$-1-\sqrt{5}+i(-1+\sqrt{5})$			-2	$abced$

No.	$(Ax+B)$			$\div\,(Cx+D)$	substitution
46	$-3+\sqrt{5}+i(-1+\sqrt{5})$			-2	*abdce*
47	$3-\sqrt{5}+i(-1+\sqrt{5})$			-2	*accdb*
48	$1+\sqrt{5}+i(-1+\sqrt{5})$			2	*adcbe*
49	$1+\sqrt{5}+i(1-\sqrt{5})$			2	*aecbd*
50	$3-\sqrt{5}+i(1-\sqrt{5})$			2	*acdeb*
51	$-3+\sqrt{5}+i(1-\sqrt{5})$			2	*abccd*
52	$-1-\sqrt{5}+i(1-\sqrt{5})$			2	*adbcc*
53	$-3+\sqrt{5}+i(1-\sqrt{5})$			-2	*acbde*
54	$3-\sqrt{5}+i(1-\sqrt{5})$			2	*aebde*
55	$-3+\sqrt{5}+i(-1+\sqrt{5})$			2	*abccd*
56	$-1-\sqrt{5}+i(-3-\sqrt{5})$			2	*adecb*
57	$1+\sqrt{5}+i(3+\sqrt{5})$			2	*acdbc*
58	$3-\sqrt{5}+i(1-\sqrt{5})$			2	*abdec*
59	$-3+\sqrt{5}+i(1-\sqrt{5})$			2	*adcbe*
60	$-1-\sqrt{5}+i(-1+\sqrt{5})$			2	*aebcd*

107. Taking out of the foregoing group of 60 a group of 12 contained in it, viz. that corresponding to the positive substitutions of the four letters *abcd*, it is easy to see, that there is a transformation $(i,0,0,1)$, that is, s into is, which can be taken for the substitution *abcd*, and also to complete thence the group of 24. And we have thus the following Table.

Groups of 12 and 24. Pole at Θ.

No.	$(Ax+B)$			$\div\,(Cx+D)$	substitution
1	1			1	1
2	-1			1	*ab . cd*
3	i			1	*ac . bd*
4	$-i$			1	*ad . bc*
5	i			1	*abc*
6	$-i$			1	*acb*
7	1			$-i$	*adc*
8	-1			$-i$	*abd*
9	i			1	*acd*
10	$-i$			1	*adb*
11	1			i	*bcd*
12	-1			i	*bdc*
13	i			-1	*adbc*
14	$-i$			-1	*acbd*
15	1			-1	*ab . cd*
16	-1			-1	*aedb*
17	1			1	*bd*
18	-1			1	*abcd*
19	i			$-i$	*bc*
20	$-i$			$-i$	*abde*
21	-1			1	*ae*
22	1			1	*adeb*
23	$-i$			$-i$	*ad*
24	i			-1	—

108. The group of 60 was obtained in the A-form by Gordan in his paper. The passage from the Θ-form to the A-form is made as follows: let X, Y, Z be the coordinates of a point when the axes are as in the Θ-form, X_1, Y_1, Z_1 the coordinates of the same point when the axes are as in the A-form; we may write

$$X_1,\, Y_1,\, Z_1 = aX + bZ,\, Y,\, -bX + aZ,$$

where

$$a = \sqrt{\frac{5-\sqrt{5}}{10}}, \quad b = \sqrt{\frac{5+\sqrt{5}}{10}},$$

then, if the equations $X : Y : Z = L : M : N$ are those of an axis of rotation referred to the first set of coordinates and $X_1 : Y_1 : Z_1 = L_1 : M_1 : N_1$, those of the same axis referred to the second set of coordinates, we may write

$$L_1,\, M_1,\, N_1 = aL + bN,\, M,\, -bL + aN;$$

these values are such that

$$L_1^2 + M_1^2 + N_1^2 = L^2 + M^2 + N^2,$$

and hence, λ, μ, ν and λ_1, μ_1, ν_1 being the rotations, we have

$$L,\, M,\, N = \tfrac{1}{\theta}\lambda,\, \tfrac{1}{\theta}\mu,\, \tfrac{1}{\theta}\nu\,; \quad L_1,\, M_1,\, N_1 = \tfrac{1}{\theta}\lambda_1,\, \tfrac{1}{\theta}\mu_1,\, \tfrac{1}{\theta}\nu_1\,;$$

where θ has the same value in each set of equations.

From the equations

$$A : B : C : D = -\nu - i : \lambda + i\mu : \lambda - i\mu : \nu - i,$$

we may write

$$B + C : B - C : D - A : D + A = \lambda : i\mu : \nu : -i, = L : iM : N : -i\theta,$$

and similarly

$$B_1 + C_1 : B_1 - C_1 : D_1 - A_1 : D_1 + A_1 = L_1 : iM_1 : N_1 : -i\theta.$$

Hence we may write

$$\begin{aligned} B_1 + C_1 &= a\,(B + C) + b\,(D - A), \\ B_1 - C_1 &= B - C, \\ D_1 - A_1 &= -b\,(B + C) + a\,(D - A), \\ D_1 + A_1 &= D + A; \end{aligned}$$

or say,

$$\begin{aligned} A_1 &= -a\,(B + C) - b\,(D - A) + (D + A), \\ B_1 &= a\,(B + C) + b\,(D - A) + (B - C), \\ C_1 &= a\,(B + C) + b\,(D - A) - (B - C), \\ D_1 &= -a\,(B + C) - b\,(D - A) + (D + A); \end{aligned}$$

which are the values for a transformation (A_1, B_1, C_1, D_1) in the A-form: of course, as only the ratios are material, the values may be multiplied by any common factor.

109. The results are exhibited in terms of ϵ, an imaginary fifth root of unity: taking $\epsilon = \cos 72^\circ + i \sin 72^\circ$, we have

$$\epsilon + \epsilon^4 = \frac{\sqrt{5}-1}{2}, \qquad \epsilon^2 + \epsilon^3 = -\frac{\sqrt{5}+1}{2},$$

$$a = \sqrt{\frac{5+\sqrt{5}}{10}} = \pm i\,\frac{\sqrt{5+\sqrt{5}}}{\sqrt{10}}, \qquad b = \sqrt{\frac{5-\sqrt{5}}{10}} = \pm\frac{\sqrt{5-\sqrt{5}}}{\sqrt{10}},$$

where the upper signs belong to ϵ, ϵ^4 and the lower to ϵ^2, ϵ^3. It may be remarked that

$$\frac{1}{a} = \sqrt{\frac{5+\sqrt{5}}{2}}, \qquad \frac{1}{b} = \sqrt{\frac{5-\sqrt{5}}{2}}, \qquad \frac{a}{b} = \sqrt{\frac{\sqrt{5}+1}{\sqrt{5}-1}}.$$

For instance, we have in the Θ-group $(A, B, C, D) = (-1, 0, 0, 1)$; $ab\,.\,cd$: and thence in the A-group $A_1, B_1, C_1, D_1 = (-2b, 2a, 2a, 2b)$; $ab\,.\,cd$: or say this is

$$(-1,\ a/b,\ a/b,\ 1) \ = \ (-1,\ \epsilon+\epsilon^4,\ \epsilon+\epsilon^4,\ 1);$$

which in the Table is given as $(-\epsilon^*,\ \epsilon^2+\epsilon^3,\ \epsilon^2+\epsilon^3,\ \epsilon^*)$; $ab\,.\,cd$.

By effecting the passage to the A-group in this manner, we of course obtain the proper substitution corresponding to each transformation: but I found it easier starting from two transformations and the corresponding substitutions, to obtain by thence successive compositions the entire group.

110. *Homographic Transformations.* The group of 60. Pole at A.

No.	Θ No.	$(As+B)$		$\div(Cs+D)$		substitution
1	1	1	0	0	1	1
2	4	-1	0	0	1	$ad\,.\,bc$
3	13	$-\epsilon^4$	0	0	1	$ac\,.\,be$
4	9	$-\epsilon^4$	0	0	1	$ab\,.\,de$
5	10	$-\epsilon^3$	0	0	1	$bd\,.\,ce$
6	14	$-\epsilon$	0	0	1	$ae\,.\,be$
7	6	$\epsilon+\epsilon^4$	ϵ^2	ϵ^3	$-(\epsilon+\epsilon^4)$	$ae\,.\,bd$
8	5	$\epsilon+\epsilon^4$	ϵ^3	ϵ^2	$-(\epsilon+\epsilon^4)$	$ad\,.\,be$
9	15	$\epsilon+\epsilon^4$	ϵ^4	ϵ	$-(\epsilon+\epsilon^4)$	$bc\,.\,de$
10	3	$\epsilon+\epsilon^4$	ϵ	ϵ^4	$-(\epsilon+\epsilon^4)$	$ac\,.\,de$
11	12	$\epsilon+\epsilon^4$	ϵ^3	ϵ^2	$-(\epsilon+\epsilon^4)$	$ac\,.\,bd$
12	16	-1	$\epsilon^4+\epsilon^3$	$\epsilon^2+\epsilon$	1	$ae\,.\,bd$
13	11	$-\epsilon$	ϵ^3+1	$1+\epsilon^2$	ϵ	$ab\,.\,ce$
14	7	$-\epsilon^2$	$1+\epsilon^4$	$\epsilon+\epsilon^4$	ϵ^2	$bc\,.\,cd$
15	2	$-\epsilon^3$	$\epsilon^3+\epsilon$	$\epsilon^3+\epsilon^4$	ϵ^3	$ab\,.\,ce$
16	8	$-\epsilon^4$	$\epsilon^2+\epsilon$	$\epsilon^3+\epsilon^4$	ϵ^4	$ad\,.\,ce$
17	21	ϵ^2+1	ϵ	1	$-(\epsilon+\epsilon^2)$	adb
18	36	ϵ^4+1	ϵ^2	ϵ^4	$-(\epsilon+\epsilon^3)$	aeb
19	30	ϵ^2+1	ϵ^4	ϵ^3	$-(\epsilon^2+\epsilon^4)$	bee
20	34	ϵ^2+1	ϵ^3	ϵ^2	$-(\epsilon+\epsilon^2)$	cde
21	19	ϵ^2+1	1	ϵ	$-(\epsilon+\epsilon^2)$	adc
22	33	$\epsilon+\epsilon^4$	1	ϵ^2	$-(\epsilon+\epsilon^2)$	cde
23	20	$\epsilon+\epsilon^4$	ϵ^3	ϵ^4	$-(\epsilon+\epsilon^3)$	acd
24	22	$\epsilon+\epsilon^4$	ϵ^4	ϵ	$-(\epsilon+\epsilon^2)$	abd
25	36	$\epsilon+\epsilon^4$	1	ϵ^4	$-(\epsilon+\epsilon^3)$	abe
26	29	$\epsilon+\epsilon^4$	ϵ	ϵ	$-(\epsilon+\epsilon^3)$	bee
27	31	$-\epsilon$	$\epsilon^2+\epsilon^4$	$\epsilon^2+\epsilon^4$	1	aed
28	17	$-\epsilon^2$	$\epsilon^2+\epsilon$	$\epsilon^2+\epsilon$	ϵ	abc
29	27	$-\epsilon^2$	$\epsilon^3+\epsilon^4$	$\epsilon^3+\epsilon^4$	ϵ^2	bcd
30	26	$-\epsilon^4$	$\epsilon^3+\epsilon^4$	$\epsilon^3+\epsilon^4$	ϵ^3	aec
31	23	-1	$\epsilon^2+\epsilon^4$	$\epsilon^3+\epsilon^4$	1	bcd
32	24	$-\epsilon^4$	1	ϵ^2	ϵ^2	bdc
33	32	-1	$\epsilon^2+\epsilon^3$	$\epsilon^4+\epsilon$	ϵ^4	ade
34	18	$-\epsilon$	$\epsilon+\epsilon^4$	$\epsilon^2+\epsilon^3$	ϵ^2	aeb
35	28	$-\epsilon^2$	$\epsilon+\epsilon^4$	$\epsilon^2+\epsilon^3$	ϵ^4	bde
36	25	$-\epsilon^3$	$\epsilon^3+\epsilon^4$	$\epsilon^3+\epsilon^4$	ϵ^4	ace

No.	Θ No.	$(As+B)$		$\div(Cs+D)$		substitution
37	44	ϵ	1	1	1	*abccd*
38	43	ϵ^4	1	1	1	*aedbc*
39	42	ϵ^3	1	1	1	*acbde*
40	41	ϵ^2	1	1	1	*adceb*
41	38	$\epsilon^3+\epsilon^4$	1	1	$-(\epsilon+\epsilon^3)$	*acebd*
42	46	$\epsilon^3+\epsilon^4$	ϵ	ϵ^4	$-(\epsilon+\epsilon^3)$	*abdcc*
43	58	$\epsilon^2+\epsilon^4$	ϵ^2	ϵ^3	$-(\epsilon+\epsilon^3)$	*adcbc*
44	55	$\epsilon^2+\epsilon^4$	ϵ^3	ϵ^2	$-(\epsilon+\epsilon^3)$	*adecb*
45	50	$\epsilon^3+\epsilon^4$	ϵ^4	ϵ	$-(\epsilon+\epsilon^3)$	*acdeb*
46	51	$\epsilon^2+\epsilon^4$	ϵ^2	1	$-(\epsilon+\epsilon^3)$	*abedc*
47	39	$1+\epsilon^2$	ϵ^2	ϵ^4	$-(\epsilon+\epsilon^3)$	*adbcc*
48	47	$1+\epsilon^2$	1	ϵ^3	$-(\epsilon+\epsilon^3)$	*accdb*
49	59	$1+\epsilon^2$	ϵ	ϵ^2	$-(\epsilon+\epsilon^3)$	*acbcd*
50	54	$1+\epsilon^2$	ϵ^3	ϵ	$-(\epsilon+\epsilon^3)$	*abced*
51	56	$-\epsilon^2$	$1+\epsilon^4$	$\epsilon^2+\epsilon^3$	ϵ^3	*abdec*
52	49	$-\epsilon^4$	$\epsilon+\epsilon^3$	$\epsilon^2+\epsilon^4$	ϵ	*abdcc*
53	37	$-\epsilon^3$	$\epsilon^4+\epsilon$	$\epsilon^2+\epsilon^3$	ϵ^4	*aebcd*
54	45	-1	$\epsilon+\epsilon^2$	$\epsilon^3+\epsilon^4$	ϵ^2	*adebc*
55	57	$-\epsilon$	$\epsilon+\epsilon^2$	$\epsilon^3+\epsilon^4$	ϵ^4	*abdec*
56	48	$-\epsilon^3$	$\epsilon^4+\epsilon$	$\epsilon^2+\epsilon^3$	1	*adcbc*
57	60	$-\epsilon^2$	$\epsilon+\epsilon^3$	$\epsilon^2+\epsilon^4$	ϵ	*acedb*
58	53	ϵ^3+1	$\epsilon^2+\epsilon^4$	ϵ^2	ϵ^3	*aebdc*
59	52	$-1+\epsilon^4$	$\epsilon^2+\epsilon^3$	ϵ^2	ϵ^3	*adbcc*
60	40	$-\epsilon^2$	$\epsilon+\epsilon^4$	ϵ^2	ϵ^2	*aedcb*

[Paper 745 continues beyond p. 201 with art. 111 selecting the positive substitutions *abcd* and completing the group of 24, Pole at A, etc.]

[750] (CONTINUED)
ON THE THEORY OF RECIPROCAL SURFACES.

601. In part explanation, observe that the definitions of ρ and σ agree with those already given. The nodal torse is the torse enveloped by the tangent planes along the nodal curve; if the nodal curve meets the curve of contact a, then a tangent plane of the nodal torse passes through the arbitrary point, that is, ρ will be the number of these planes which pass through the arbitrary point, viz. the class of the torse. So also the cuspidal torse is the torse enveloped by the tangent planes along the cuspidal curve; and σ will be the number of these tangent planes which pass through the arbitrary point, viz. it will be the class of the torse. Again, as regards ρ' and σ': the node-couple torse is the envelope of the bitangent planes of the surface, and the node-couple curve is the locus of the points of contact of these planes. Similarly, the spinode torse is the envelope of the parabolic planes of the surface, and the spinode curve is the locus of the points of contact of these planes, viz. it is the curve UH of intersection of the surface and its Hessian; the two curves are the reciprocals of the nodal and the cuspidal torses respectively, and the definitions of ρ', σ' correspond to those of ρ and σ.

602. In regard to the nodal curve b, we consider k the number of its apparent double points (excluding actual double points); f the number of its actual double points (each of these is a point of contact of two sheets of the surface, and there is thus at the point a single tangent plane, viz. this is a plane f', and we thus have $f' = f$); t the number of its triple points; and j the number of its pinch-points: these last are not singular points of the nodal curve *per se*, but are singular in regard to the curve as nodal curve of the surface, viz. a pinch-point is a point at which the two tangent planes are coincident. The curve is considered as not having any stationary points other than the points γ, which lie also on the cuspidal curve; and the expression for the class consequently is $q = b^2 - b - 2k - 2f - 3\gamma - 6t$.

603. In regard to the cuspidal curve c, we consider h the number of its apparent double points; and upon the curve, not singular points in regard to the curve *per se*, but only in regard to it as cuspidal curve of the surface, certain points in number β, χ, ω respectively. The curve is considered as not having any actual double or other multiple points, and as not having any stationary points except the points β, which lie also on the nodal curve; and the expression for the class consequently is

$$r = c^2 - c - 2h - 3\beta.$$

604. The γ are points where the cuspidal curve with the two sheets (or say rather half-sheets) belonging to it are intersected by another sheet of the surface; the curve of intersection with such other sheet, belonging to the nodal curve of the surface, has evidently a stationary (cuspidal) point at the point of intersection.

As to the points β, to facilitate the conception, imagine the cuspidal curve to be a semi-cubical parabola, and the nodal curve a right line (not in the plane of the curve) passing through the cusp; then intersecting the two curves by a series of parallel planes, any plane which is, say, above the cusp, meets the parabola in two real points and the line in one real point, and the section of the surface is a curve with two real cusps and a real node; as the plane approaches the cusp, these approach

together, and, when the plane passes through the cusp, unite into a singular point in the nature of a triple point (= node + two cusps); and when the plane passes below the cusp, the two cusps of the section become imaginary, and the nodal line changes from crunodal to acnodal.

605. At a point i the nodal curve crosses the cuspidal curve, being on the side away from the two half-sheets of the surface acnodal, and on the side of the two half-sheets crunodal, viz.

the two half-sheets intersect each other along this portion of the nodal curve. There is at the point a single tangent plane, which is a plane i'; and we thus have $i = i'$.

606. As already mentioned, a cnicnode C is a point where, instead of a tangent plane, we have a tangent quadri-cone; at a binode B, the quadri-cone degenerates into a pair of planes. A cnictrope C' is a plane touching the surface along a conic; in the case of a bitrope B', the conic degenerates into a flat conic or pair of points.

607. In the original formulae for $a(n-2)$, $b(n-2)$, $c(n-2)$, we have to write $\kappa - B$ instead of κ, and the formulae are further modified by reason of the singularities θ and ω. So, in the original formulae for $a(n-2)(n-3)$, $b(n-2)(n-3)$, $c(n-2)(n-3)$, we have instead of δ to write $\delta - C - 3\omega$, and to substitute new expressions for $[ab]$, $[ac]$, $[bc]$; viz. these are

$$[ab] = ab - 2\rho - j,$$

$$[ac] = ac - 3\sigma - \chi - \omega,$$

$$[bc] = bc - 3\beta - 2\gamma - i.$$

The whole series of equations thus is

(1) $a' = a.$
(2) $f' = f.$
(3) $i' = i.$
(4) $\alpha = n(n-1) - 2b - 3c.$
(5) $\kappa = 3n(n-2) - 6b - 8c.$
(6) $\delta = \frac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \frac{9}{2}c(c-1).$
(7) $a(n-2) = \kappa - B + \rho + 2\sigma + 3\omega.$
(8) $b(n-2) = \rho + 2\beta + 3\gamma + 3t.$
(9) $c(n-2) = 2\sigma + 4\beta + \gamma + \theta + \omega.$
(10) $a(n-2)(n-3) = 2(\delta - C - 3\omega) + 3(ac - 3\sigma - \chi - 3\omega) + 2(ab - 2\rho - j).$
(11) $b(n-2)(n-3) = 4k + (ab - 2\rho - j) + 3(bc - 3\beta - 2\gamma - i).$
(12) $c(n-2)(n-3) = 6h + (ac - 3\sigma - \chi - 3\omega) + 2(bc - 3\beta - 2\gamma - i).$
(13) $q = b^2 - b - 2k - 2f - 3\gamma - 6t.$
(14) $r = c^2 - c - 2h - 3\beta.$

Also, reciprocal to these,

(15) $a' = n'(n'-1) - 2b' - 3c'.$
(16) $\kappa' = 3n'(n'-2) - 6b' - 8c'.$
(17) $\delta' = \frac{1}{2}n'(n'-2)(n'^2-9) - (n'^2-n'-6)(2b'+3c') + 2b'(b'-1) + 6b'c' + \frac{9}{2}c'(c'-1).$
(18) $a'(n'-2) = \kappa' - B' + \rho' + 2\sigma' + 3\omega'.$
(19) $b'(n'-2) = \rho' + 2\beta' + 3\gamma' + 3t'.$
(20) $c'(n'-2) = 2\sigma' + 4\beta' + \gamma' + \theta' + \omega'.$
(21) $a'(n'-2)(n'-3) = 2(\delta' - C' - 3\omega') + 3(a'c' - 3\sigma' - \chi' - 3\omega') + 2(a'b' - 2\rho' - j').$
(22) $b'(n'-2)(n'-3) = 4k' + (a'b' - 2\rho' - j') + 3(b'c' - 3\beta' - 2\gamma' - i').$
(23) $c'(n'-2)(n'-3) = 6h' + (a'c' - 3\sigma' - \chi' - 3\omega') + 2(b'c' - 3\beta' - 2\gamma' - i').$
(24) $q' = b'^2 - b' - 2k' - 2f' - 3\gamma' - 6t'.$
(25) $r' = c'^2 - c' - 2h' - 3\beta',$

together with one other independent relation: in all 26 relations between the 46 quantities.

608. The new relation may be presented under several different forms, equivalent to each other in virtue of the foregoing 25 relations; these are

$$(26)\ \ 2(n-1)(n-2)(n-3) - 12(n-3)(b+c) + 6q + 6r + 24t + 42\beta + 30\gamma + \tfrac{8}{3}\theta = \Sigma,$$

$$(27)\ \ 26n - 12c - 4C - 10B + \beta - 7j - 8\chi + \tfrac{2}{3}\theta - 4\omega = \Sigma;$$

in each of which two equations Σ is used to denote the same function of the accented letters that the left-hand side is of the unaccented letters.

$$\begin{aligned}(28)\ \ \beta' + \tfrac{2}{3}\theta' = {} & 2n(n-2)(11n-24)\\ & + (-66n+184)\,b\\ & + (-93n+252)\,c\\ & + 22(2\beta+3\gamma+3t)\\ & + 27(4\beta+\gamma+\theta)\\ & + \beta + \tfrac{2}{3}\theta\\ & - 24C - 28B - 27j - 38\chi - 73\omega\\ & + 4C' + 10B' + 7j' + 8\chi' - 4\omega'.\end{aligned}$$

Or, reciprocally,

$$\begin{aligned}(29)\ \ \beta + \tfrac{2}{3}\theta = {} & 2n'(n'-2)(11n'-24)\\ & + (-66n'+184)\,b'\\ & + (-93n'+252)\,c'\\ & + 22(2\beta'+3\gamma'+3t')\\ & + 27(4\beta'+\gamma'+\theta')\\ & + \beta' + \tfrac{2}{3}\theta'\\ & - 24C' - 28B' - 27j' - 38\chi' - 73\omega'\\ & + 4C + 10B + 7j + 8\chi - 4\omega.\end{aligned}$$

The equation (26) expresses that the surface and its reciprocal have the same deficiency; viz. the expression for the deficiency is

$$\begin{aligned}(30)\ \ \text{Deficiency} = {} & \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-3)(b+c) + \tfrac{1}{2}(q+r)\\ & + 2t + \tfrac{7}{2}\beta + \tfrac{5}{2}\gamma + \tfrac{2}{9}\theta,\\ = {} & \tfrac{1}{6}(n'-1)(n'-2)(n'-3) - \&\text{c}.\end{aligned}$$

609. The equation (28) (due to Prof. Cayley) is the correct form of an expression for β', first obtained by him (with some errors in the numerical coefficients) from independent considerations. But it is best obtained by means of the equation (26); and (27) is a relation presenting itself in the investigation. In fact, considering α as standing for its value $n(n-1) - 2b - 3c$, we have from the first 25 equations

$$\begin{array}{rll}6 & \alpha & = \Sigma,\\ +2 & 3n - c - \kappa & = \Sigma,\\ -2 & a(n-2) - \kappa + B - \rho - 2\sigma - 3\omega & = \Sigma,\\ -4 & b(n-2) - \rho - 2\beta - 3\gamma - 3t & = \Sigma,\\ -6 & c(n-2) - 2\sigma - 4\beta - \gamma - \theta - \omega & = \Sigma,\\ +2 & n + \kappa - \sigma - 2\beta - 4\gamma - 2j - 3\chi - 3\omega & = \Sigma,\\ -3 & 2q - 2\rho + \beta + j & = \Sigma,\\ -2 & 3r + c - 5\sigma - \beta - 4\gamma + \chi - \omega & = \Sigma;\end{array}$$

multiplying these equations by the numbers set opposite to them respectively, and adding, we find

$$\begin{aligned}&-2n^3+12n^2+4n+b(12n-36)+c(12n-48)\\&-6q-6r-4C-10B-41\beta-30\gamma-24t-7j-8\chi+\tfrac{2}{3}\theta-4\omega=\Sigma,\end{aligned}$$

and adding hereto (26) we have the equation (27); and from this (28), or by a like process, (29), is obtained without much difficulty. As to the 8 Σ-equations or symmetries, observe that the first, third, fourth, and fifth are in fact included among the original equations (for an expression which vanishes is in fact $=\Sigma$); we have from them moreover $3n-c=3a'-\kappa'$, and thence $3n-c-\kappa=3a'-\kappa'-\kappa$, which is $=\Sigma$, or we have thus the second equation; but the sixth, seventh, and eighth equations have yet to be obtained.

610. The equations (15), (16), (17) give

$$\begin{aligned}n'&=\alpha(\alpha-1)-2\delta-3\kappa,\\c'&=3\alpha(\alpha-2)-6\delta-8\kappa,\\b'&=\tfrac{1}{2}\alpha(\alpha-2)(\alpha^2-9)-(\alpha^2-\alpha-6)(2\delta+3\kappa)+2\delta(\delta-1)+6\delta\kappa+\tfrac{9}{2}\kappa(\kappa-1).\end{aligned}$$

From (7), (8), (9), we have

$$\begin{aligned}(\alpha-b-c)(n-2)&=\kappa-B-6\beta-4\gamma-3t-\chi+2\omega,\\(\alpha-2b-3c)(n-2)(n-3)&=2(\delta-C)-8k-18h-6bc+18\beta+12\gamma+6i-6\omega;\end{aligned}$$

substituting these values for κ and δ, and for α its value $=n(n-1)-2b-3c$, we obtain the values of n', c', b'; viz. the value of n' is

$$\begin{aligned}n'=n(n-1)^2&-n(7b+12c)+4b^2+8b+9c^2+15c\\&-8k-18h+18\beta+12\gamma+12i-9t\\&-2C-3B-3\omega.\end{aligned}$$

Observe that the effect of a cnicnode C is to reduce the class by 2, and that of a binode B to reduce it by 3.

611. We have

$$(n-2)(n-3)=n^2-n+(-4n+6)=\alpha+2b+3c+(-4n+6);$$

making this substitution in the equations (10), (11), (12), which contain $(n-2)(n-3)$, these become

$$\begin{aligned}a(-4n+6)&=2(\delta-C)-\alpha^2-4\rho-9\sigma-2j-3\chi-15\omega,\\b(-4n+6)&=4k-2b^2-9\beta-6\gamma-3t-2\rho-j,\\c(-4n+6)&=6h-3c^2-6\beta-4\gamma-2i-3\sigma-\chi-3\omega,\end{aligned}$$

which are the foregoing equations (C); adding to each equation four times the corresponding equation with the factor $(n-2)$, these become

$$\begin{aligned}\alpha^2-2\alpha&=2(\delta-C)+4(\kappa-B)-\sigma-2j-3\chi-3\omega,\\2b^2-2b&=4k-\beta+6\gamma+12t-3t+2\rho-j,\\3c^2-2c&=6h+10\beta+4\theta-2i+5\sigma-\chi+\omega.\end{aligned}$$

Writing in the first of these $\alpha^2 - 2\alpha = n + 2\delta + 3\kappa - \alpha$, and reducing the other two by means of the values of q, r, the equations become

$$n - \alpha = -2C - 4B + \kappa - \sigma - 2j - 3\chi - 3\omega,$$
$$2q + \beta + 3t + j = 2\rho,$$
$$3r + c + 2i + \chi = 5\sigma + \beta + 4\theta + \omega,$$

which give at once the last three of the 8 Σ-equations.

The reciprocal of the first of these is

$$\sigma' = \alpha - n + \kappa' - 2j' - 3\chi' - 2C' - 4B' - 3\omega',$$

viz. writing herein

$$\alpha = n(n-1) - 2b - 3c \quad \text{and} \quad \kappa' = 3n(n-2) - 6b - 8c,$$

this is

$$\sigma' = 4n(n-2) - 8b - 11c - 2j' - 3\chi' - 2C' - 4B' - 3\omega',$$

giving the order of the spinode curve; viz. for a surface of the order n without singularities, this is $= 4n(n-2)$, the product of the orders of the surface and its Hessian.

612. Instead of obtaining the second and third equations as above, we may to the value of $b(-4n+6)$ add twice the value of $b(n-2)$; and to twice the value of $c(-4n+6)$ add three times the value of $c(n-2)$, thus obtaining equations free from ρ and σ respectively; these equations are

$$b(-2n+2) = 4k - 2b^2 - 5\beta - 3t + 6t - j,$$
$$c(-5n+6) = 12h - 3c^2 - 6\gamma - 4i - 2\chi + 3\theta - 3\omega,$$

equations which, introducing therein the values of q and r, may also be written

$$b(2n-4) + 6\gamma + 6t + 3t + j + 4f = 2q + 5\beta$$
$$c(5n-12) + 3\theta = 6r + 18\beta + 5\gamma + 4i + 2\chi + 3\omega.$$

Considering as given, n the order of the surface; the nodal curve, with its singularities b, k, f, t; the cuspidal curve, with its singularities c, h; and the quantities β, γ, i which relate to the intersections of the nodal and cuspidal curves; the first of the two equations gives j, the number of pinch-points, being singularities of the nodal curve, *quoad* the surface; and the second equation establishes a relation between θ, χ, ω, the numbers of singular points of the cuspidal curve *quoad* the surface.

In the case of a nodal curve only, if this be a complete intersection $P = 0$, $Q = 0$, the equation of the surface is $(A, B, C \mid P, Q)^2 = 0$, and the first equation is

$$b(-2n+2) = 4k - 2b^2 + 6t - j,$$

or, assuming $t = 0$, say $j = 2(n-1)b - 2b^2 + 4k$, which may be verified; and so in the case of a cuspidal curve only, when this is a complete intersection $P = 0$, $Q = 0$, the equation of the surface is $(A, B, C \mid P, Q)^2 = 0$, where $AC - B^2 = MP + NQ$; and the second equation is

$$c(-5n+6) = 12h - 6c^2 - 2\chi + 3\theta - 3\omega,$$

or, say $2\chi + 3\omega = (5n-6)c - 6c^2 + 12h + 3\theta$, which may also be verified.

613. We may in the first instance out of the 46 quantities consider as given the 14 quantities

$$n \ : \ b, k, f, t \ : \ c, h, \theta, \chi \ : \ \beta, \gamma, i \ : \ C, B;$$

then of the 26 relations, 17 determine the 17 quantities

$$\alpha, B, \kappa, \rho, \delta \ : \ j, q \ : \ r, \omega$$
$$n' \ : \ a', B', \kappa' \ : \ b', f' \ : \ c' \ : \ i'$$

and there remain the 9 equations

$$(18), (19), (20), (21), (22), (23), (24), (25), (28),$$

connecting the 15 quantities

$$\rho', \sigma' \ : \ k', t', j', q' \ : \ h', \theta', \chi', \omega', r' \ : \ \beta', \gamma' \ : \ C', B'.$$

Taking then further as given the 5 quantities j', χ', t', C', B',

$$\begin{aligned} \text{equations (18) and (21) give} &\quad \rho',\ \sigma', \\ \text{equation (19)} &\quad \text{gives } 2\beta' + 3\gamma' + 3t', \\ (20) &\quad \text{gives } 4\beta' + \gamma' + \theta', \\ (28) &\quad \text{gives } \beta' + \tfrac{2}{3}\theta', \end{aligned}$$

so that, taking also t' as given, these last three equations determine β', γ', θ'; and finally

$$\begin{aligned} \text{equation (22) gives} &\quad k', \\ (23) &\quad \text{gives } h', \\ (24) &\quad \text{gives } q', \\ (25) &\quad \text{gives } r', \end{aligned}$$

viz. taking as given in all 20 quantities, the remaining 26 will be determined.

614. In the case of the general surface of the order n, without singularities, we have as follow

$$\begin{aligned}
n &= n, \\
a &= n(n-1), \\
\delta &= \tfrac{1}{2}n(n-1)(n-2)(n-3), \\
\kappa &= n(n-1)(n-2), \\
n' &= n(n-1)^2, \\
a' &= n(n-1), \\
\delta' &= \tfrac{1}{2}n(n-2)(n^2-9), \\
\kappa' &= 3n(n-2), \\
b' &= \tfrac{1}{2}n(n-1)(n-2)(n^3-n^2+n-12), \\
k' &= \tfrac{1}{8}n(n-2)(n^7 - 6n^6 + 16n^5 - 54n^4 + 164n^3 - 288n^2 \\
&\qquad\qquad + 547n^2 - 1058n^3 + 1068n^2 - 1214n + 1464), \\
h' &= \tfrac{1}{2}n(n-2)(n^5 - 4n^4 + 7n^3 - 45n^2 + 114n^2 - 111n^2 + 548n - 960), \\
q' &= n(n-2)(n-3)(n^2+2n-4), \\
\rho' &= n(n-2)(n^3-n^2+n-12), \\
c' &= 4n(n-1)(n-2), \\
h' &= \tfrac{1}{2}n(n-2)(16n^4 - 64n^3 + 80n^2 - 108n + 156), \\
i' &= 2n(n-2)(3n-4), \\
\sigma' &= 4n(n-2), \\
\beta' &= 2n(n-2)(11n-24), \\
\gamma' &= 4n(n-2)(n-3)(n^2-3n+16),
\end{aligned}$$

the remaining quantities vanishing.

615. The question of singularities has been considered under a more general point of view by Zeuthen, in the memoir "Recherche des singularités qui ont rapport à une droite multiple d'une surface," *Math. Annalen*, t. IV. (1871), pp. 1–20. He attributes to the surface:

A number of singular points, viz. points at any one of which the tangents form a cone of the order μ, and class ν, with $y+\eta$ double lines, of which y are tangents to branches of the nodal curve through the point, and $z+\zeta$ stationary lines, whereof z are tangents to branches of the cuspidal curve through the point, and with u double planes and v stationary planes; moreover, these points have only the properties which are the most general in the case of a surface regarded as a locus of points; and Σ denotes a sum extending to all such points. (The foregoing general definition includes the cnicnodes $\mu=\nu=2$, $y=\eta=z=\zeta=u=v=0$, and the binodes $\mu=2$, $\eta=1$, $z=y=\&\text{c.}\ =0$.)

And, further, a number of singular planes, viz. planes any one of which touches along a curve of the class μ' and order ν', with $y'+\eta'$ double tangents, of which y' are generating lines of the node-couple torse, $z'+\zeta'$ stationary tangents, of which z' are generating lines of the spinode torse, u' double points and v' cusps; it is, moreover, supposed that these planes have only the properties which are the most general in the case of a surface regarded as an envelope of its tangent planes; and Σ' denotes a sum extending to all such planes. (The definition includes the cnictropes $\mu'=\nu'=2$, $y'=\eta'=z'=\zeta'=u'=v'=0$, and the bitropes $\mu'=2$, $\eta'=1$, $z'=y'=\&\text{c.}\ =0$.)

616. This being so, and writing

$$x=\nu+2\eta+3\zeta,\quad x'=\nu'+2\eta'+3\zeta',$$

the equations (7), (8), (9), (10), (11), (12), contain, in respect of the new singularities, additional terms, viz. these are

$$\begin{aligned}
a(n-2)&=\cdots+\Sigma\,[x(\mu-2)-\nu-2\zeta],\\
b(n-2)&=\cdots+\Sigma\,[y(\mu-2)],\\
c(n-2)&=\cdots+\Sigma\,[z(\mu-2)],\\
a(n-2)(n-3)&=\cdots+\Sigma\,[x(-\nu+7)+2\eta+4\zeta],\\
b(n-2)(n-3)&=\cdots+\Sigma\,[y(-\nu+8)]-\Sigma'\,(4u'+3v'),\\
c(n-2)(n-3)&=\cdots+\Sigma\,[z(-\nu+9)]-\Sigma'\,(2u'),
\end{aligned}$$

and there are of course the reciprocal terms in the reciprocal equations (18), (19), (20), (21), (22), (23). These formulae are given without demonstration in the memoir just referred to: the principal object of the memoir, as shown by its title, is the consideration not of such singular points and planes, but of the multiple right lines of a surface; and in regard to these, the memoir should be consulted.

751.

NOTE ON RIEMANN'S PAPER "VERSUCH EINER ALLGEMEINEN AUFFASSUNG DER INTEGRATION UND DIFFERENTIATION*."

[From the *Mathematische Annalen*, t. XVI. (1880), pp. 81, 82.]

The Editors of Riemann's works remark that the paper in question was contained in a MS. of his student time (dated 14 Jan. 1847) and was probably never intended for publication: indeed that he would not in later years have recognised the validity of the principles upon which it is founded. The idea is however a noticeable one: Riemann considers Z_{x+h} a function of $x+h$, expanded in a doubly infinite, necessarily divergent, series of integer or fractional powers of h, according to the law

$$Z_{x+h} = \sum_{\nu=-\infty}^{\nu=+\infty} k_\nu \, x \cdot h^\nu, \tag{2}$$

where the meaning is explained to be that the exponents differ from each other by integer values, in effect, that ν has all the values $\alpha+p$, α a given integer or fractional value, and p any integer number from $-\infty$ to $+\infty$, zero included.

Riemann deduces a theory of fractional differentiation: but without considering the question which has always appeared to me to be the great difficulty in such a theory: what is the real meaning of a complementary function containing an infinity of arbitrary constants? or, in other words, what is the arbitrariness of the complementary function of this nature which presents itself in the theory?

I wish to point out the relation between the paper referred to, and a short paper of my own "On a doubly infinite Series," *Quart. Math. Journ.* t. VI. (1851), pp. 45–47, [102]: this commences with the remark "The following completely paradoxical investigation of the properties of the function Γ (which I have been in possession

* *Werke*, pp. 331–344.

of for some years) may perhaps be found interesting from its connexion with the theories of expansion and divergent series." And I then give the expansion

$$C_n \, e^x = \sum{}' \, [n-r]^n \, x^{n-r},$$

where n is any integer or fractional number whatever, and the summation extends to all positive and negative integer values (zero included) of r. And I remark that, n being an integer, we have $C_n = \Gamma(n)$, and hence that assuming that this is so in general, or writing

$$\Gamma(n) \, e^x = \sum{}' \, [n-1]^r \, x^{n-r-1},$$

we have this equation as a definition of $\Gamma(n)$. The point of resemblance of course is that we have a doubly infinite expansion of e^x in a series of integer or fractional powers of x, corresponding to Riemann's like expansion of Z_{x+h} in powers of h.

Cambridge, 10 Sept. 1879.

752.

ON THE FINITE GROUPS OF LINEAR TRANSFORMATIONS OF A VARIABLE; WITH A CORRECTION.

[From the *Mathematische Annalen*, t. XVI. (1880), pp. 260–263; 439, 440.]

In the paper "Ueber endliche Gruppen linearer Transformationen einer Veränderlichen," *Math. Ann.* t. XII. (1877), pp. 23–46, Prof. Gordan gave in a very elegant form the groups of 12, 24 and 60 homographic transformations $\dfrac{ax+b}{cx+d}$. The groups of 12 and 24 are in the like form, the group of 24 thus containing as part of itself the group of 12; but the group of 60 is in a different form, not containing as part of itself the group of 12. It is, I think, desirable to present the group of 60 in the form in which it contains as part of itself Gordan's group of 12: and moreover to identify the group of 60 with the group of the 60 positive permutations of 5 letters or (writing *abc* for the cyclical permutation a into b, b into c, c into a, and so in other cases) say with the group of the 60 positive permutations 1, *abc*, *ab.cd* and *abcde*.

Any two forms of a group are, it is well known, connected as follows, viz. if 1, α, β, ... are the functional symbols of the one form, then those of the other form are 1, $\phi\alpha\phi^{-1}$, $\phi\beta\phi^{-1}$, ... (where in the case in question ϕ is a functional symbol of the like homographic form, $\phi x = \dfrac{Ax+B}{Cx+D}$). But instead of obtaining the new form in this manner, I found it easier to use the values of the rotation-symbol

$$\cos\tfrac{\theta}{2} + \sin\tfrac{\theta}{2}(i\cos X + j\cos Y + k\cos Z)$$

for the axes of the icosahedron or dodecahedron, given in my paper "Notes on polyhedra," *Quart. Math. Jour.* t. VII. (1866), pp. 304–316, [375]; viz. if for any axes, λ, μ, ν denote the parameters of rotation $\tan\frac{\theta}{2}\cos X$, $\tan\frac{\theta}{2}\cos Y$, $\tan\frac{\theta}{2}\cos Z$, then,

by a formula which is in fact equivalent to that given in my note "On the correspondence of Homographies and Rotations," *Math. Annalen*, t. XV. (1879), pp. 238–240, [660], the corresponding homographic function of x is

$$\frac{(-\nu - i)x + \lambda + i\mu}{(\lambda - i\mu)x + \nu - i},$$

where i denotes $\sqrt{-1}$ as usual.

The new formulae for the group of 60, or icosahedron group, of homographic functions $\dfrac{\alpha x+\beta}{\gamma x+\delta}$ are contained in the following table, where the four columns show the values of the coefficients α, β, γ, δ respectively: and where in the outside column, the substitution is represented as a permutation-symbol on the five letters *abcde*: moreover for shortness θ is written to denote $\sqrt{5}$.

THE GROUP OF 60.

#	α	β	γ	δ	subst.
1	1	0	0	1	1
2	-1	0	0	1	*ab.cd*
3	0	1	1	0	*ac.bd*
4	0	-1	1	0	*ad.bc*
5	2	$-3+\theta+i(\ \ 1-\theta)$	$-3+\theta+i(-1+\theta)$	-2	*bc.de*
6	2	$-3+\theta+i(-1+\theta)$	$-3+\theta+i(\ \ 1-\theta)$	-2	*ae.bc*
7	2	$3-\theta+i(-1+\theta)$	$3-\theta+i(\ \ 1-\theta)$	-2	*ad.ce*
8	2	$3-\theta+i(\ \ 1-\theta)$	$3-\theta+i(-1+\theta)$	-2	*ad.be*
9	2	$-1-\theta+i(\ \ 1-\theta)$	$-1-\theta+i(-1+\theta)$	-2	*ae.cd*
10	2	$-1-\theta+i(-1+\theta)$	$-1-\theta+i(\ \ 1-\theta)$	-2	*ab.de*
11	2	$1+\theta+i(-1+\theta)$	$1+\theta+i(\ \ 1-\theta)$	-2	*bc.cd*
12	2	$1+\theta+i(\ \ 1-\theta)$	$1+\theta+i(-1+\theta)$	-2	*ab.ce*
13	2	$-1-\theta+i(-3-\theta)$	$-1-\theta+i(\ \ 3+\theta)$	-2	*ac.be*
14	2	$-1-\theta+i(\ \ 3+\theta)$	$-1-\theta+i(-3-\theta)$	-2	*bd.ce*
15	2	$1+\theta+i(\ \ 3+\theta)$	$1+\theta+i(-3-\theta)$	-2	*ae.bd*
16	2	$1+\theta+i(-3-\theta)$	$1+\theta+i(\ \ 3+\theta)$	-2	*ae.de*
17	$-i$	i	1	1	*abc*
18	$-i$	$-i$	1	1	*aeb*
19	i	$-i$	1	1	*ade*
20	i	i	1	-1	*acd*
21	$-i$	i	1	-1	*adb*

#	α	β	γ	δ	subst.
22	i	i	1	$-i$	*abd*
23	$-i$	$-i$	1	$-i$	*bcd*
24	i	$-i$	1	i	*bdc*
25	$-1-\theta+i(\ \ 3+\theta)$	2	-2	$-1-\theta+i(-3-\theta)$	*aec*
26	$1+\theta+i(\ \ 3+\theta)$	2	-2	$1+\theta+i(-3-\theta)$	*ace*
27	$1+\theta+i(-3-\theta)$	2	-2	$1+\theta+i(\ \ 3+\theta)$	*bcd*
28	$-1-\theta+i(-3-\theta)$	2	-2	$-1-\theta+i(\ \ 3+\theta)$	*bde*
29	$-3+\theta+i(\ \ 1-\theta)$	2	2	$3-\theta+i(\ \ 1-\theta)$	*bce*
30	$-3+\theta+i(-1+\theta)$	2	2	$3-\theta+i(-1+\theta)$	*bec*
31	$3-\theta+i(-1+\theta)$	2	2	$-3+\theta+i(-1+\theta)$	*aed*
32	$3-\theta+i(\ \ 1-\theta)$	2	2	$-3+\theta+i(\ \ 1-\theta)$	*ade*
33	2	$-1-\theta+i(-1+\theta)$	$1+\theta+i(-1+\theta)$	2	*cde*
34	2	$1+\theta+i(\ \ 1-\theta)$	$-1-\theta+i(\ \ 1-\theta)$	2	*ced*
35	2	$-1-\theta+i(\ \ 1-\theta)$	$1+\theta+i(\ \ 1-\theta)$	2	*aeb*
36	2	$1+\theta+i(-1+\theta)$	$-1-\theta+i(-1+\theta)$	2	*abe*
37	$-1-\theta+i(-3-\theta)$	2	2	$1+\theta+i(-3-\theta)$	*abcde*
38	$-1-\theta+i(\ \ 1-\theta)$	2	2	$1+\theta+i(\ \ 1-\theta)$	*acebd*
39	$-1-\theta+i(-1+\theta)$	2	2	$1+\theta+i(-1+\theta)$	*adbec*
40	$-1-\theta+i(\ \ 3+\theta)$	2	2	$1+\theta+i(\ \ 3+\theta)$	*aedcb*
41	$1+\theta+i(\ \ 3+\theta)$	2	2	$-1-\theta+i(\ \ 3+\theta)$	*adceb*
42	$1+\theta+i(-1+\theta)$	2	2	$-1-\theta+i(-1+\theta)$	*acbde*
43	$1+\theta+i(\ \ 1-\theta)$	2	2	$-1-\theta+i(\ \ 1-\theta)$	*aedbc*
44	$1+\theta+i(-3-\theta)$	2	2	$-1-\theta+i(-3-\theta)$	*abecd*
45	$-1-\theta+i(-1+\theta)$	2	-2	$-1-\theta+i(\ \ 1-\theta)$	*acbed*
46	$-3+\theta+i(-1+\theta)$	2	-2	$-3+\theta+i(\ \ 1-\theta)$	*abdce*
47	$3-\theta+i(-1+\theta)$	2	-2	$3-\theta+i(\ \ 1-\theta)$	*aecdb*
48	$1+\theta+i(-1+\theta)$	2	-2	$1+\theta+i(\ \ 1-\theta)$	*adebc*
49	$1+\theta+i(\ \ 1-\theta)$	2	-2	$1+\theta+i(-1+\theta)$	*aecbd*
50	$3-\theta+i(\ \ 1-\theta)$	2	-2	$3-\theta+i(-1+\theta)$	*acdeb*
51	$-3+\theta+i(\ \ 1-\theta)$	2	-2	$-3+\theta+i(-1+\theta)$	*abedc*
52	$-1-\theta+i(\ \ 1-\theta)$	2	-2	$-1-\theta+i(-1+\theta)$	*adbce*
53	2	$-3+\theta+i(-1+\theta)$	$3-\theta+i(-1+\theta)$	2	*aebde*
54	2	$-1-\theta+i(\ \ 3+\theta)$	$1+\theta+i(\ \ 3+\theta)$	2	*abced*
55	2	$1+\theta+i(-3-\theta)$	$-1-\theta+i(-3-\theta)$	2	*adecb*
56	2	$3-\theta+i(\ \ 1-\theta)$	$-3+\theta+i(\ \ 1-\theta)$	2	*acdbe*
57	2	$-3+\theta+i(\ \ 1-\theta)$	$3-\theta+i(\ \ 1-\theta)$	2	*abdec*
58	2	$-1-\theta+i(-3-\theta)$	$1+\theta+i(-3-\theta)$	2	*adcbe*
59	2	$1+\theta+i(\ \ 3+\theta)$	$-1-\theta+i(\ \ 3+\theta)$	2	*aebcd*
60	2	$3-\theta+i(-1+\theta)$	$-3+\theta+i(-1+\theta)$	2	*acedb*

This contains (as one of five groups of 12) the group of the positive permutations of *abcd*; and, completing this into a group of 24, we have

Groups of 12 and 24.

#	α	β	γ	δ	subst.
1	1	0	0	1	1
2	-1	0	0	1	*ab.cd*
3	0	1	1	0	*ac.bd*
4	0	-1	1	0	*ad.bc*
5	$-i$	i	1	1	*abc*
6	-1	$-i$	1	i	*acb*
7	1	$-i$	1	1	*adc*
8	$-i$	$-i$	1	-1	*acd*
9	i	$-i$	1	-1	*aeb*
10	1	i	1	$-i$	*abd*
11	-1	$-i$	1	$-i$	*bcd*
12	i	$-i$	1	i	*bdc*
13	i	i	1	1	*adbc*
14	$-i$	i	1	1	*acbd*
15	1	1	1	-1	*cd*

#	α	β	γ	δ	subst.
16	-1	i	-1	1	*ab*
17	1	$-i$	1	1	*aedb*
18	$-i$	-1	$-i$	i	*bd*
19	i	1	1	i	*abcd*
20	1	-1	1	-1	*bc*
21	-1	-1	1	-1	*abdc*
22	i	-1	1	$-i$	*ac*
23	-1	i	1	$-i$	*adcb*
24	-1	1	1	1	*ad*

The groups of 60 and 24 thus each of them contain the group of 12,

$$x, \quad -\tfrac{1}{x}, \quad \tfrac{1-x}{1+x}, \quad \tfrac{1+x}{1-x}, \quad \tfrac{x+i}{x-i}, \quad \tfrac{x-i}{-x+i}.$$

It may be remarked that, to verify the periodicities of the forms contained in the group of 60, we have as the conditions that

$$\frac{\alpha x+\beta}{\gamma x+\delta} \text{ may be periodic of the order 2, } \frac{\alpha+\delta}{\alpha\delta-\beta\gamma}=0, \text{ that is, } \alpha+\delta=0,$$

$$\text{,, \quad ,, \quad ,, \quad 3, \quad ,, } \quad =1,$$

$$\text{,, \quad ,, \quad ,, \quad 5, \quad ,, } \quad =\tfrac{1}{2}(3+\sqrt{5}).$$

For instance, in the form
$$\frac{[-1-\Theta+i(-3-\Theta)]\,x+2}{2x+[1+\Theta+i(-3-\Theta)]},$$
we have
$$\alpha\delta=-(1+\Theta)^2-(3+\Theta)^2\,i^2=-2\Theta-8\Theta, \quad \beta\gamma=4,$$
$$\alpha+\delta=-2i(3+\Theta);$$
and therefore
$$\frac{(\alpha+\delta)^2}{\alpha\delta-\beta\gamma}=\frac{-4(3+\Theta)^2}{-3(3+\Theta)}, \quad =\frac{3+\Theta}{2}=\tfrac{1}{2}(3+\sqrt{5}),$$
as it should be.

Cambridge, 11 Nov. 1879.

Correction*, pp. 439, 440.

I erroneously assumed that the symbol *adcb* could be taken as corresponding to the linear transformation ix: but this was obviously wrong, for it gave *bd* as corresponding to the transformation $-ix$, and these are not of the same order, but of the orders 4 and 2 respectively. The proper symbol is *adbc*, as given above, and the remaining eleven symbols are then at once obtained.

Cambridge, 17 Feb. 1880.

* The correction in the Table of the Groups of 12 and 24 has been inserted in the Table as now printed on p. 240; it applies to the second half of the column of symbols on the extreme right-hand.

753.

ON A THEOREM RELATING TO THE MULTIPLE THETA-FUNCTIONS.

[From the *Mathematische Annalen*, t. XVII. (1880), pp. 115–122.]

I PROPOSE—partly for the sake of the theorem itself, partly for that of the notation which will be employed—to demonstrate the general theorem $(3')$, p. 4, of Dr Schottky's *Abriss einer Theorie der Abel'schen Functionen von drei Variabeln*, (Leipzig, 1880), which theorem is there presented in the form:

$$e^{-H(u_1,\ldots;\mu',\nu')}\,\Theta(u_1+2\varpi_1',\ldots;\mu,\nu)=e^{-2\pi i\mu\nu'}\,\Theta(u_1,\ldots;\mu+\mu',\nu+\nu'), \tag{3'}$$

but which I write in the slightly different form

$$\exp.[-H(u;\mu',\nu')]\cdot\Theta(u+2\varpi';\mu,\nu)=\exp.[-2\pi i\mu\nu']\cdot\Theta(u;\mu+\mu',\nu+\nu').$$

I remark that the theorem is given in the preliminary paragraphs the contents of which are, as mentioned by the Author, derived from Herr Weierstrass: and that the form of the theta-function is a very general one, depending on the general quadric function

$$G(u_1,\ldots,u_p;u_1,\ldots,u_p)$$

of $2p$ variables, p being the number of the arguments $u_1,\ldots,u_p$ (in fact, the periods are not reduced to the normal form, but are arbitrary); and the characters $\nu_1,\ldots,\nu_p$, $\mu_1,\ldots,\mu_p$, instead of having each of them the value 0 or 1, have each of them any integer or fractional value whatever. The meaning of the theorem (u denoting a set or row of p letters $u_1,\ldots,u_p$, and so in other cases), is that the function

$$\Theta(u;\mu+\mu',\nu+\nu')$$

with the new characters $\mu+\mu'$ and $\nu+\nu'$ is, save as to an exponential factor, equal to the function $\Theta(u+2\varpi';\mu,\nu)$ with the original characters μ, ν, but with the new arguments $u+2\varpi'$.

Notation.

This is in some measure a development of the notation employed in my "Memoir on the Theory of Matrices," *Phil. Trans.* t. CXLVIII. (1858), pp. 17–37, [152]: I use certain single letters u, etc. to denote sets or rows each of p letters, $u=(u_1,\ldots,u_p)$: or if, to fix the ideas $p=3$, then $u=(u_1,u_2,u_3)$, and so in other cases.

But I use certain other letters a, etc. to denote squares or matrices each of p^2 letters; thus, if $p=3$ as before,

$$a=\begin{vmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\a_{31}&a_{32}&a_{33}\end{vmatrix},$$

and in any such case the transposed matrix is denoted by the same letter enclosed in parentheses

$$(a)=\begin{vmatrix}a_{11}&a_{21}&a_{31}\\a_{12}&a_{22}&a_{32}\\a_{13}&a_{23}&a_{33}\end{vmatrix}.$$

The sum $u+v$ of the row-letters u, $=(u_1,u_2,u_3)$ and v, $=(v_1,v_2,v_3)$ denotes the row $(u_1+v_1, u_2+v_2, u_3+v_3)$: and in like manner the sum $a+b$ of the two matrices, or square-letters a and b, denotes the matrix

$$\begin{vmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \\ a_{31}+b_{31} & a_{32}+b_{32} & a_{33}+b_{33} \end{vmatrix},$$

and similarly for a sum of three or more terms.

The product uv, $=(u_1,u_2,u_3)(v_1,v_2,v_3)$, of the two row-letters u, v denotes the single term $u_1v_1+u_2v_2+u_3v_3$. We have $uv=vu$.

The product

$$au, = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} (u_1,u_2,u_3),$$

of a preceding square-letter a and a succeeding row-letter u, denotes the set or row

$$(a_{11},a_{12},a_{13})(u_1,u_2,u_3),\ (a_{21},a_{22},a_{23})(u_1,u_2,u_3),\ (a_{31},a_{32},a_{33})(u_1,u_2,u_3);$$

the notation ua is not employed.

The product

$$auv = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} (u_1,u_2,u_3)(v_1,v_2,v_3),$$

of a preceding square-letter a followed by the two row-letters u and v, denotes the single term

$$\begin{aligned} &(a_{11},a_{12},a_{13})(u_1,u_2,u_3)\,v_1 \\ +\,&(a_{21},a_{22},a_{23})(u_1,u_2,u_3)\,v_2 \\ +\,&(a_{31},a_{32},a_{33})(u_1,u_2,u_3)\,v_3. \end{aligned}$$

Observe that auv is not in general $=avu$; but it is easy to verify that $auv=(a)vu$; and hence if $(a)=a$, that is, if the matrix a be symmetrical, then $auv=avu$.

A product of two matrices

$$ab,\ \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{vmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{vmatrix},$$

denotes a matrix

$$\begin{vmatrix} (b_{11},b_{21},b_{31}) & (b_{12},b_{22},b_{32}) & (b_{13},b_{23},b_{33}) \\ (a_{11},a_{12},a_{13}) & \cdots & \cdots \\ (a_{21},a_{22},a_{23}) & \cdots & \cdots \\ (a_{31},a_{32},a_{33}) & \cdots & \cdots \end{vmatrix};$$

viz. the top-line of the compound matrix is

$$(a_{11},a_{12},a_{13})(b_{11},b_{21},b_{31}),\ (a_{11},a_{12},a_{13})(b_{12},b_{22},b_{32}),\ (a_{11},a_{12},a_{13})(b_{13},b_{23},b_{33}),$$

and so for the other lines: or expressing this in words, we say that any line of the compound matrix is obtained by compounding the corresponding line of the first or further component matrix with the several columns of the second or nearer component matrix.

Clearly ab is not in general $=ba$. We may easily verify that $(ab)=(b)(a)$, that is, the transposed matrix (ab) is that obtained by the composition of the transposed matrix (b) as first or further matrix, with the transposed matrix (a) as second or nearer matrix. Even if a and b

are each symmetrical, we do not in general have $ab = ba$, but only $(ab) = ba$, or what is the same thing, $ab = (ba)$.

In a symbol such as $abuv$, we first combine a, b into a single matrix ab, and then regard the expression as a combination such as auv: the expression denotes therefore a single term. The theory might be explained in greater detail; but the mode of working with row- and square-letters will be readily understood from what precedes.

In all that follows, u, μ, ν, μ', ν', ϖ, ϖ', ζ' are row-letters; a, b, h, ω, ω', η, η' are square-letters: a and b are symmetrical, viz. $a = (a)$, $b = (b)$.

And I write

$$\begin{aligned}(\cdot)(u,v)^2 &= (a,h,b)(u,v)^2\\ &= au^2 + 2huv + bv^2\end{aligned}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{vmatrix}(u_1,u_2,u_3)^2$$

$$+2\begin{vmatrix} h_{11} & h_{12} & h_{13}\\ h_{21} & h_{22} & h_{23}\\ h_{31} & h_{32} & h_{33}\end{vmatrix}(u_1,u_2,u_3)(v_1,v_2,v_3)$$

$$+\begin{vmatrix} b_{11} & b_{12} & b_{13}\\ b_{21} & b_{22} & b_{23}\\ b_{31} & b_{32} & b_{33}\end{vmatrix}(v_1,v_2,v_3)^2$$

to denote the general quadric function of the $2p$ letters u, v, with

$$\tfrac{1}{2}p(p+1) + p^2 + \tfrac{1}{2}p(p+1), \quad = p(2p+1)$$

coefficients. It is assumed that the determinant formed with the $\frac{1}{2}p(p+1)$ coefficients b is negative: this is the necessary and sufficient condition for the convergence of the series.

Definition of $\Theta(u;\mu,\nu)$.

$\Theta(u;\mu,\nu)$, the general theta-function with p arguments u, and $2p$ characters μ, ν, is the sum of a p-tuple series of exponentials

$$\Theta(u;\mu,\nu) = \sum \exp.[(\cdot)(n, n+\nu)^2 + 2\pi i u\,(n+\nu)],$$

where each of the letters n, $= (n_1,\ldots,n_p)$, has all integer values (zero included) from $-\infty$ to $+\infty$.

The general theorem in regard to $\Theta(u;\mu,\nu)$.

This is

$$\exp.[-H(u;\mu',\nu')]\cdot\Theta(u+2\varpi';\mu,\nu) = \exp.[-2\pi i\mu\nu']\cdot\Theta(u;\mu+\mu',\nu+\nu'),$$

establishing a relation between the function $\Theta(u;\mu+\mu',\nu+\nu')$, with arbitrary character-increments μ', ν', and the function $\Theta(u+2\varpi';\mu,\nu)$ with the original characters, but with new arguments $u+2\varpi'$. Also $H(u;\mu',\nu')$ denotes a function, linear as regards the arguments u, but quadric as regards μ' and ν'; $-2\pi i\mu\nu'$ is a single term depending only on μ and ν'; and the theorem thus is that the two functions differ only by an exponential factor. The relations between the constants will be obtained in the course of the investigation.

Demonstration.

The truth of the theorem depends on the equality of corresponding exponentials on the two sides of the equation: viz. substituting for the theta-functions their values, and comparing the exponents or arguments of the exponentials: writing also for convenience

$$G(u+2\varpi', n+\nu),$$

to denote the quadric function $(\cdot)(u+2\varpi', n+\nu)^2$; we ought to have

$$\begin{aligned} &-H(u;\mu',\nu') + G(u+2\varpi', n+\nu) + 2\pi i u(n+\nu) \\ &\quad = -2\pi i\mu\nu' + G(u, n+\nu+\nu') + 2\pi i(\mu+\mu')(n+\nu+\nu'), \end{aligned}$$

or say

$$H(u;\mu',\nu') = G(u+2\varpi', n+\nu) - G(u, n+\nu+\nu') - 2\pi i(n+\nu+\nu')\mu'.$$

In this equation, if true at all, the terms containing n must destroy each other; assuming that they do so, the equation becomes

$$H(u;\mu',\nu') = G(u+2\varpi', \nu) - G(u, \nu+\nu') - 2\pi i(\nu+\nu')\mu'.$$

Consider first the terms in n: the right-hand side is

$$\begin{aligned} &= a(u+2\varpi')^2 + 2h(u+2\varpi')(n+\nu) + b(n+\nu)^2 \\ &\quad - au^2 - 2hu(n+\nu+\nu') - b(n+\nu+\nu')^2 - 2\pi i n\mu', \end{aligned}$$

and the terms herein which contain n thus are

$$\begin{aligned} &2h(u+2\varpi')n + bn^2 + 2bn\nu \\ &\quad - 2hun - bn^2 - 2bn(\nu+\nu') - 2\pi i n\mu', \\ &= 4h\varpi' n - 2bn\nu' - 2\pi i n\mu', \end{aligned}$$

which, b being symmetrical, may be written

$$= 2(2h\varpi' - b\nu' - \pi i\mu')\, n,$$

and these terms will vanish if, and only if

$$2h\varpi' - b\nu' - \pi i\mu' = 0,$$

a system of p equations connecting ϖ', μ', ν'.

Assuming them to be satisfied, the remaining relation,

$$H(u;\mu',\nu') = G(u+2\varpi', \nu) - G(u, \nu+\nu') - 2\pi i(\nu+\nu')\mu',$$

becomes

$$\begin{aligned} H(u;\mu',\nu') &= a(u+2\varpi')^2 + 2h(u+2\varpi')\nu + b\nu^2 \\ &\quad - au^2 - 2hu(\nu+\nu') - b(\nu+\nu')^2 - 2\pi i(\nu+\nu')\mu'. \end{aligned}$$

Here, a and b being symmetrical, we have

$$a(u+2\varpi')^2 = au^2 + 4a\varpi' u + 4a\varpi'^2, \quad b(\nu+\nu')^2 = b\nu^2 + 2b\nu'\nu + b\nu'^2,$$

and the value therefore is

$$= 4a(\varpi' u + \varpi'^2) + 2h(2\varpi'\nu - u\nu') - b(2\nu'\nu + \nu'^2) - 2\pi i(\nu+\nu')\mu'.$$

On the right-hand side, putting the term in h under the form

$$-2h(u+\varpi')\nu' + 2h\varpi'(2\nu+\nu'), \quad = -2(h)\nu'(u+\varpi') + 2h\varpi'(2\nu+\nu'),$$

and the last term under the form

$$-\pi i\mu'(2\nu+\nu') - \pi i\mu'\nu',$$

the equation becomes

$$\begin{aligned} H(u;\mu',\nu') = {} & (4a\varpi' - 2(h)\nu')(u+\varpi') - \pi i\mu'\nu' \\ & + (2h\varpi' - b\nu' - \pi i\mu')(2\nu+\nu'), \end{aligned}$$

where the second line vanishes in virtue of the foregoing equation

$$2h\varpi' - b\nu' - \pi i\mu' = 0;$$

the equation thus is

$$H(u;\mu',\nu') = (4a\varpi' - 2(h)\nu')(u+\varpi') - \pi i\mu'\nu',$$

which equation, regarding therein ϖ' as a linear function of μ' and ν', shows that $H(u;\mu',\nu')$ is a function linear as regards u (and containing this only through $u+\varpi'$), but quadric as regards μ', ν'.

Introducing the new row-letter ζ', we may write

$$H(u;\mu',\nu') = 2\zeta'(u+\varpi') - \pi i\mu'\nu',$$

viz. the expression on the right-hand side is here assumed as the value of the function

$$H(u;\mu',\nu'), \quad = G(u+2\varpi',\nu) - G(u,\nu+\nu') - 2\pi i(\nu+\nu')\mu',$$

and the theorem then is

$$\exp.[-H(u;\mu',\nu')]\cdot\Theta(u+2\varpi';\mu,\nu) = \exp.[-2\pi i\mu\nu']\cdot\Theta(u;\mu+\mu',\nu+\nu'),$$

where, by what precedes,

$$\begin{aligned} 2h\varpi' - b\nu' - \pi i\mu' &= 0, \\ 2a\varpi' - (h)\nu' - \zeta' &= 0, \end{aligned}$$

$2p$ equations for determining the $2p$ functions ϖ', ζ' as linear functions of μ', ν': which equations depend on the $p(2p+1)$ constants a, b, h.

Suppose that the resulting values of ϖ' and ζ' are

$$\varpi' = \omega\mu' + \omega'\nu',$$

$$\zeta' = \eta\mu' + \eta'\nu',$$

where ω, ω', η, η' are square-letters; then, regarding a, b, h as arbitrary, the $4p^2$ new constants ω, ω', η, η' cannot be all of them arbitrary, but must be connected by $4p^2 - p(2p+1)$, $= p(2p-1)$ equations.

We may regard ω, ω', η, η' as satisfying these $p(2p-1)$ equations, but as being otherwise arbitrary; the foregoing equations then are

$$\begin{aligned} 2h\varpi' - b\nu' - \pi i\mu' &= 0, \\ 2a\varpi' - (h)\nu' - \zeta' &= 0, \\ \varpi' &= \omega\mu' + \omega'\nu', \\ \zeta' &= \eta\mu' + \eta'\nu', \end{aligned}$$

which lead to the equations connecting a, b, h with ω, ω', η, η'.

The first and second equations, substituting for ϖ' and ζ' their values, become

$$(2h\omega - \pi i)\mu' + (2h\omega' - b)\nu' = 0,$$
$$(2a\omega - \eta)\mu' + (2a\omega' - (h) - \eta')\nu' = 0,$$

or μ', ν' being arbitrary, we thus obtain the $4p^2$ equations

$$\begin{aligned} 2a\omega - \eta &= 0, \\ 2h\omega - \pi i &= 0, \\ 2a\omega' - \eta' - (h) &= 0, \\ 2h\omega' - b &= 0, \end{aligned}$$

which are the equations in question. It is to be observed that in πi is, like the other symbols, a matrix, viz. it is regarded as containing the matrix unity; or, what is the same thing, it denotes

$$\pi i \begin{vmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ \vdots & & \ddots & \end{vmatrix}.$$

We can eliminate a, b, h from these equations and thus obtain the $p(2p-1)$ equations before referred to, which connect the $4p^2$ constants ω, ω', η, η'. I give, but without a complete explanation, the steps of the elimination.

The equation $2a\omega - \eta = 0$, may be written in the form

$$2(a\omega) - (\eta) = 0,$$

that is,

$$2(\omega)(a) - (\eta) = 0,$$

or since $(a) = a$, this is

$$2(\omega)a - (\eta) = 0;$$

from the original form, and the new form respectively, we find

$$2(\omega)a\omega - (\omega)\eta = 0, \quad 2(\omega)a(\omega) - (\eta)\omega = 0;$$

and comparing these

$$(\omega)\eta - (\eta)\omega = 0, \quad \text{(first result)}.$$

The equation $2a\omega' - \eta' - (h) = 0$, or say $(h) = -\eta' + 2a\omega'$, may be written in the form

$$h = -(\eta') + 2(a\omega'),$$

that is, since $a = (a)$,

$$h = -(\eta') + 2(\omega')a;$$

and we thence deduce

$$h\omega = -(\eta')\omega + 2(\omega')a\omega.$$

But from the equation $2a\omega - \eta = 0$, we have $2(\omega')a\omega - (\omega')\eta = 0$, and the equation thus becomes $h\omega = -(\eta')\omega + (\omega')\eta$; which, in virtue of $2h\omega - \pi i = 0$, becomes

$$\tfrac{1}{2}\pi i = -(\eta')\omega + (\omega')\eta, \quad \text{(second result)}.$$

From the equation above obtained, $h = -(\eta') + 2(\omega')a$, we have

$$h\omega' = -(\eta')\omega' + 2(\omega')a\omega';$$

in virtue of $2h\omega' - b = 0$, this becomes $-2(\eta')\omega' + 4(\omega')a\omega' = b$; an equation which may also be written $-2((\eta')\omega') + 4((\omega')a\omega') = (b)$, or, what is the same thing, $-2(\omega')\eta' + 4(\omega')(a)\omega' = (b)$; or since $(a) = a$ and $(b) = b$, this is

$$-2(\omega')\eta' + 4(\omega')a\omega' = b:$$

and comparing with the original equation

$$-2(\eta')\omega' + 4(\omega')a\omega' = b,$$

we obtain

$$(\omega')\eta' - (\eta')\omega' = 0, \quad \text{(third result)}.$$

We have thus the three systems

$$\begin{aligned} (\omega)\eta - (\eta)\omega &= 0, \quad \tfrac{1}{2}p(p-1) \text{ equations,} \\ (\omega')\eta - (\eta')\omega &= \tfrac{1}{2}\pi i, \quad p^2 \text{ equations,} \\ (\omega')\eta' - (\eta')\omega' &= 0, \quad \tfrac{1}{2}p(p-1) \text{ equations,} \end{aligned}$$

in all $p(2p-1)$ equations. As to these systems, observe that $(\omega)\eta$, $(\eta)\omega$, etc., are all of them matrices of p^2 terms; each of the three systems denotes therefore in the first instance p^2 equations, viz. the equations obtained by equating to zero the several terms of such a matrix: but in the first system each diagonal term so equated to zero gives the identity $0 = 0$; and equating to zero the terms which are symmetrical in regard to the diagonal we obtain twice over, in the forms $P = 0$, and $-P = 0$, one and the same equation; the number of equations is thus diminished from p^2 to $\frac{1}{2}p(p-1)$; and similarly in the third system the number of equations is $= \frac{1}{2}p(p-1)$: but for the second system the number of equations is really $= p^2$. It is hardly necessary to remark that in this second system $\frac{1}{2}\pi i$ is as before regarded as a matrix.

The foregoing three systems of equations are in fact the equations (6) p. 4 of Dr Schottky's work.

Cambridge, 12 July, 1880.

754.

ON THE CONNEXION OF CERTAIN FORMULÆ IN ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. IX. (1880), pp. 23–25.]

In reference to a like question in the theory of the double ϑ-functions, it is interesting to show that (if not completely, at least very nearly) the single formula

$$\Pi(u, a) = u\,\frac{\Theta' a}{\Theta a} + \tfrac{1}{2}\log\frac{\Theta(u-a)}{\Theta(u+a)},$$

that is,

$$\int_0^u \frac{k^2 \operatorname{sn} a \operatorname{cn} a \operatorname{dn} a \operatorname{sn}^2 u\, du}{1 - k^2 \operatorname{sn}^2 a \operatorname{sn}^2 u} = u\,\frac{\Theta' a}{\Theta a} + \tfrac{1}{2}\log\frac{\Theta(u-a)}{\Theta(u+a)},$$

leads not only to the relation

$$\log\Theta u = \tfrac{1}{2}\log\frac{2k'K}{\pi} + \tfrac{1}{2}\left(1 - \frac{E}{K}\right)u^2 - k^2\int_0^u du \int_0^u du \operatorname{sn}^2 u,$$

between the functions Θ, sn, but also to the addition-equation for the function sn.

Writing in the equation a indefinitely small, and assuming only that $\operatorname{sn} a$, $\operatorname{cn} a$, $\operatorname{dn} a$ then become a, 1, 1, respectively, the equation is

$$k^2 a \int_0^u \operatorname{sn}^2 u\, du = u\,\frac{\Theta' a}{\Theta a} + \tfrac{1}{2}\log\frac{\Theta u - a\Theta' u}{\Theta u + a\Theta' u} = u\frac{\Theta' 0}{\Theta 0} - a\frac{\Theta' u}{\Theta u},$$

that is,

$$\frac{\Theta' u}{\Theta u} = u\,\frac{\Theta'' 0}{\Theta 0} - k^2 \int_0^u du \operatorname{sn}^2 u,$$

or, integrating from $u = 0$, this is

$$\log \Theta u = C + \tfrac{1}{2} u^2\,\frac{\Theta'' 0}{\Theta 0} - k^2 \int_0^u du \int_0^u du \operatorname{sn}^2 u,$$

which, except as regards the determination of the constants, is the required equation for $\log \Theta u$.

Next, differentiating twice the equation for $\Pi(u, a)$, and once the equation obtained for $\frac{d\Theta'}{du}$, we have

$$k^2 \operatorname{sn} a \operatorname{cn} a \operatorname{dn} a\,\frac{d}{du}\left(\frac{\operatorname{sn}^2 u}{1 - k^2 \operatorname{sn}^2 a \operatorname{sn}^2 u}\right) = \tfrac{1}{2}\,\frac{\Theta''\Theta - \Theta'^2}{\Theta^2}(u-a) - \tfrac{1}{2}\,\frac{\Theta''\Theta - \Theta'^2}{\Theta^2}(u+a),$$

and

$$\frac{\Theta''\Theta - \Theta'^2}{\Theta^2} = \frac{\Theta'' 0}{\Theta 0} - k^2 \operatorname{sn}^2 u,$$

where, for shortness, $\frac{\Theta'' 0}{\Theta 0}$ is written to denote $\frac{\Theta'' u\,\Theta u - (\Theta' u)^2}{\Theta u^2}$, and the like in the first equation; the right-hand side of the first equation therefore is

$$= -\tfrac{1}{2} k^2 \left\{\operatorname{sn}^2(u-a) - \operatorname{sn}^2(u+a)\right\},$$

or the equation becomes

$$2 \operatorname{sn} a \operatorname{cn} a \operatorname{dn} a\,\frac{d}{du}\,\frac{\operatorname{sn}^2 u}{1 - k^2 \operatorname{sn}^2 u \operatorname{sn}^2 a} = \operatorname{sn}^2(u+a) - \operatorname{sn}^2(u-a),$$

that is,

$$\frac{4 \operatorname{sn} u \operatorname{sn}' u \operatorname{sn} a \operatorname{cn} a \operatorname{dn} a}{(1 - k^2 \operatorname{sn}^2 u \operatorname{sn}^2 a)^2} = \operatorname{sn}^2(u+a) - \operatorname{sn}^2(u-a).$$

The numerator on the left-hand side must be a symmetrical function of u, a, and hence (even if the value of $\operatorname{sn}' u$ were unknown) it would appear that $\operatorname{sn}' u$ must be a mere constant multiple of $\operatorname{cn} u \operatorname{dn} u$; assuming, however, the actual value, $\operatorname{sn}' u = \operatorname{cn} u \operatorname{dn} u$, the formula is

$$\frac{4 \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u \operatorname{sn} a \operatorname{cn} a \operatorname{dn} a}{(1 - k^2 \operatorname{sn}^2 u \operatorname{sn}^2 a)^2} = \operatorname{sn}^2(u+a) - \operatorname{sn}^2(u-a) = \{\operatorname{sn}(u+a) + \operatorname{sn}(u-a)\}\{\operatorname{sn}(u+a) - \operatorname{sn}(u-a)\}.$$

The factor $\{\operatorname{sn}(u+a) + \operatorname{sn}(u-a)\}$ becomes $= 2 \operatorname{sn} u$ for $a = 0$, and this suggests that the factor $\operatorname{sn} u$ on the left-hand side is a factor of $\{\operatorname{sn}(u+a) + \operatorname{sn}(u-a)\}$. That $\operatorname{cn} u$ is not a factor hereof would follow from the properties of the period K; viz. for $u = K$, $\operatorname{cn} u = 0$, but $\{\operatorname{sn}(u+a) + \operatorname{sn}(u-a)\}, = 2 \operatorname{sn}(K+a)$ is not $= 0$; and, similarly, that $\operatorname{dn} u$ is not a factor from the properties of the period iK'; hence, $\operatorname{cn} u$, $\operatorname{dn} u$ belong to the other factor $\{\operatorname{sn}(u+a) - \operatorname{sn}(u-a)\}$, and by symmetry $\operatorname{cn} a$, $\operatorname{dn} a$ belong to the first-mentioned factor. And we are thus led to assume

$$\begin{aligned}
\operatorname{sn}(u+a) + \operatorname{sn}(u-a) &= 2M \operatorname{sn} u \operatorname{cn} a \operatorname{dn} a,\\
\operatorname{sn}(u+a) - \operatorname{sn}(u-a) &= 2M' \operatorname{sn} a \operatorname{cn} u \operatorname{dn} u,
\end{aligned}$$

where

$$\text{denom.} = 1 - k^2 \operatorname{sn}^2 a \operatorname{sn}^2 u,$$

and $MM' = 1$. Some further investigation is wanting to show that M and M' are constants, but assuming that they are so and each $= 1$, the formulae give at once the ordinary expression for $\operatorname{sn}(u + a)$; that is, we have the addition-equation for the function sn.

[Paper 754 continues to PDF p. 276 / printed p. 252.]

[766] (CONTINUED)
ON THE GEODESIC CURVATURE OF A CURVE ON A SURFACE.

The Special Curves, p constant and q constant.

Consider the curve $p = \text{const}$. For this curve $p' = 0$, $p'' = 0$; therefore also $Gq'^2 = 1$, and, if R be the radius of geodesic curvature, then

$$\frac{1}{R} = \tfrac{1}{V}\,\mathfrak{D}q'^3, \qquad = \tfrac{1}{V}\,\frac{\mathfrak{D}}{G\sqrt{G}}.$$

Similarly for the curve $q = \text{const}$. Here $q' = 0$, $q'' = 0$; therefore $Ep'^2 = 1$, and, if S be the radius of geodesic curvature, then

$$\frac{1}{S} = \tfrac{1}{V}\,\mathfrak{A}p'^3, \qquad = \tfrac{1}{V}\,\frac{\mathfrak{A}}{E\sqrt{E}}.$$

These values of R and S are interesting for their own sakes, and they will be introduced into the expression for the radius of geodesic curvature ρ of the general curve.

Transformed Formula for the Radius of Geodesic Curvature.

From the values of $\dfrac{1}{R}$, $\dfrac{1}{S}$, we have

$$\frac{1}{\rho} = \frac{q'\sqrt{G}}{R} - \frac{p'\sqrt{E}}{S} = V(p'q'' - p''q') + \tfrac{1}{V}\left\{\Omega - \tfrac{\mathfrak{A}}{E}p'^3 - \tfrac{\mathfrak{D}}{G}q'^3\right\},$$

where the term in { } is

$$\Omega = \mathfrak{B}p'^3 - \tfrac{\mathfrak{A}}{E}p'^3 + \mathfrak{B}p''q' + (\mathfrak{C}p'q'' + \mathfrak{D})q'^2 - \tfrac{\mathfrak{D}}{G}q'^3.$$

The terms in $\mathfrak{A}$ are

$$-\tfrac{\mathfrak{A}}{E}p'(1 - Ep'^2), \quad = -\tfrac{\mathfrak{A}}{E}p'(2Fp'q' + Gq'^2),$$

and those in $\mathfrak{D}$ are

$$-\tfrac{\mathfrak{D}}{G}q'(1 - Gq'^2), \quad = -\tfrac{\mathfrak{D}}{G}q'(Ep'^2 + 2Fp'q').$$

Hence the whole expression contains the factor $p'q'$, and is, in fact,

$$p'q'\left(\mathfrak{B} - \tfrac{2\mathfrak{A}F}{E} - \tfrac{\mathfrak{D}E}{G}\right)p' + p'q'\left(\mathfrak{C} - \tfrac{\mathfrak{A}G}{E} - \tfrac{2\mathfrak{D}F}{G}\right)q',$$

or substituting for $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ their values, this is

$$= p'q'\Big\{p'\left(-GE_2 + EG_2 + \tfrac{2F^2G_1}{E} - 2FF_2 - FE_2 + 2EF_2 - \tfrac{EFG_1}{G}\right)$$
$$+q'\left(-GE_2 + EG_1 - \tfrac{2F^2G_2}{G} + 2FF_1 + FG_1 - 2GF_1 + \tfrac{GFE_2}{E}\right)\Big\};$$

say this is

$$= p'q'(Lp' + Mq');$$

and the formula thus is

$$\frac{1}{\rho} = \frac{q'\sqrt{G}}{R} - \frac{p'\sqrt{E}}{S} = V(p'q'' - p''q') + \tfrac{1}{V}p'q'(Lp' + Mq').$$

Taking ϕ, θ to be the inclination of the curve to the curves $q = \text{const.}$, $p = \text{const.}$, respectively, and $\omega\ (= \phi + \theta)$ the inclination of these two curves to each other, then

$$\cos\phi = \frac{Fp' + Gq'}{\sqrt{G}}, \quad \cos\theta = \frac{Ep' + Fq'}{\sqrt{E}}, \quad \cos\omega = \frac{F}{\sqrt{EG}},$$

$$\sin\phi = \frac{Vp'}{\sqrt{G}}, \quad \sin\theta = \frac{Vq'}{\sqrt{E}}, \quad \sin\omega = \frac{V}{\sqrt{EG}};$$

hence $\dfrac{\sin\phi}{\sin\omega} = p'\sqrt{E}$, $\dfrac{\sin\theta}{\sin\omega} = q'\sqrt{G}$, and the formula may also be written

$$\frac{1}{\rho} = \frac{1}{\sin\omega}\frac{\sin\phi}{R} - \frac{1}{\sin\omega}\frac{\sin\theta}{S} = V(p'q'' - p''q') + \tfrac{1}{V}p'q'(Lp' + Mq').$$

The Orthotomic Case $F = 0$, or $ds^2 = E\,dp^2 + G\,dq^2$.

The formula becomes in this case much more simple. We have

$$1 = Ep'^2 + Gq'^2, \quad V = \sqrt{EG}, \quad \omega = 90^\circ, \quad \sin\theta = \cos\phi;$$

and the term $Lp' + Mq'$ becomes $= \dot{E}G - E\dot{G}$, if, as before, $\dot{E}$, $\dot{G}$ denote the complete differential coefficients $E_1p' + E_2q'$ and $G_1p' + G_2q'$. The formula then is

$$\frac{1}{\rho} = \frac{\cos\phi}{R} - \frac{\sin\phi}{S} = V(p'q'' - p''q') + \tfrac{1}{V}p'q'(\dot{E}G - E\dot{G}),$$

where the values $\dfrac{1}{R}$ and $\dfrac{1}{S}$ are now $= \dfrac{\dot{G}}{G\sqrt{G}}$ and $\dfrac{\dot{E}}{E\sqrt{E}}$ respectively. But we have moreover $\phi = \tan^{-1}\dfrac{p'\sqrt{E}}{q'\sqrt{G}}$, and thence

$$\phi' = -V(p'q'' - p''q') - \tfrac{1}{2}Vp'q'(\dot{E}G - E\dot{G});$$

or the formula finally is

$$\frac{1}{\rho} - \frac{\cos\phi}{R} + \frac{\sin\phi}{S} + \phi' = 0,$$

which is Liouville's formula referred to at the beginning of the present paper. It will be recollected that ϕ' is the differential coefficient $\dfrac{d\phi}{ds}$ with respect to the arc s of the curve.

Addition.—Since the foregoing paper was written, I have succeeded in obtaining a like interpretation of the term

$$V(p'q'' - p''q') + \tfrac{1}{V}p'q'(Lp' + Mq'),$$

which belongs to the general case. I find that these terms are, in fact, $= -\phi + \omega_1 p'$; or, what is the same thing (since $\omega = \phi + \theta$ and therefore $\omega_1 p = \phi + \theta$), are $= \theta - \omega_2 q'$. It will be recollected that ϕ is the inclination of the curve to the curve $q = c$, which passes through a given point of the curve, ϕ is the variation of ϕ corresponding to the passage to the consecutive point of the curve, viz. $\phi + \phi\,ds$ is the inclination at this consecutive point to the curve $q = c + dc$, which passes through the consecutive point; ω is the inclination to each other of the curves $p = b$, $q = c$, which pass through the given point of the curve, ω_1 the variation corresponding to the passage along the curve $q = c$, viz. $\omega_1\,ds$ is the inclination to each other of the curves $p = b + db$, $q = c$; and the like as regards θ and ω_2.

For the demonstration, we have, as above,

$$\phi = \tan^{-1}\frac{Vp'}{Fp' + Gq'}, \qquad \omega = \tan^{-1}\frac{V}{F},$$

where

$$V = \sqrt{EG - F^2};$$

and moreover $Ep'^2 + 2Fp'q' + Gq'^2 = 1$. In virtue of this last equation,

$$V^2p'^2 + (Fp' + Gq')^2 = G;$$

and we have

$$\phi = -V(p'q'' - p''q') + \tfrac{1}{G}\Box,$$

where

$$\Box = (Fp' + Gq')\,p'\dot{V} - Vp'(Fp' + Gq')\cdot;$$

or, since $V^2 = EG - F^2$, and thence $2V\dot{V} = \dot{G}E - 2F\dot{F} + E\dot{G}$, we have

$$\Box = \tfrac{1}{2}p'\{(Fp' + Gq')(\dot{G}E - 2F\dot{F} + E\dot{G}) - 2(EG - F^2)(\dot{F}p' + \dot{G}q')\}.$$

Substituting herein for $\dot{E}$, $\dot{F}$, $\dot{G}$ their values $E_1p' + E_2q'$, $F_1p' + F_2q'$, $G_1p' + G_2q'$, the term in { } becomes

$$= Ip'^2 + Jp'q' + Kq'^2,$$

where

$$\begin{aligned} I &= FGE_1 - 2EGF_1 + EFG_1, \\ J &= G^2E_2 - 2FGF_2 + (-EG + 2F^2)G_1 + FGE_1 - 2EGF_2 + EFG_2, \\ K &= G^2E_2 - 2FGF_2 + (-EG + 2F^2)G_2. \end{aligned}$$

But from the equation $\omega = \tan^{-1}\dfrac{V}{F}$, differentiating in regard to p, we obtain

$$\omega_1 = \tfrac{1}{EGV}(FG\dot{E} - 2EG\dot{F} + EF\dot{G}) = \tfrac{1}{EGV}I;$$

or, for p writing

$$p^{-1}\{(Ep'^2+2Fp'q'+Gq'^2)V\omega_1,=Ep'(p'^2I+\tfrac{2}{p'q'}p'^2q'^2J+\tfrac{1}{p'^2q'^2}q'^2K)\},$$

we have

$$\phi-\omega_1p'=-V(p'q''-p''q')+\tfrac{1}{G}(Ip'^2+Jp'q'+Kq'^2)\cdot$$

The terms in p'^2 destroy each other, and the form thus is

$$\phi-\omega_1p'=-V(p'q''-p''q')-\tfrac{1}{V}p'q'(Lp'+Mq'),$$

where

$$L=-\tfrac{J}{G}+\tfrac{2IF}{GE},\qquad M=-\tfrac{K}{G}+\tfrac{I}{E};$$

and, upon substituting herein for I, J, K their values, we find

$$L=-GE_2+EG_2+\tfrac{2F^2E_2}{E}-2FF_2-FE_2+2EF_2-\tfrac{EFG_2}{G},$$

$$M=-GE_2+EG_2-\tfrac{2F^2G_1}{G}+2FF_1+FG_1-2GF_1+\tfrac{GFE_2}{E};$$

viz., these are the values denoted above by the same letters L, M. The final result thus is

$$\frac{1}{\rho}=\frac{q'\sqrt{G}}{R}-\frac{p'\sqrt{E}}{S}=-\phi+\omega_1p',\qquad =\theta-\omega_2q',$$

where the meanings of the symbols have been already explained. A formula substantially equivalent to this, but in a different (and scarcely properly explained) notation, is given, Aoust, “Théorie des coordonnées curvilignes quelconques,” *Annali di Matem.*, t. ii. (1868), pp. 39–64; and I was, in fact, led thereby to the foregoing further investigation.

As to the definition of the radius of geodesic curvature, I remark that, for a curve on a given surface, if PQ be an infinitesimal arc of the curve, then if from Q we let fall the perpendicular QM on the tangent plane at P (the point M being thus a point on the tangent PT of the curve), and if from M, in the tangent plane and at right angles to the tangent, we draw MN to meet the osculating plane of the curve in N, then MN is in fact equal to the infinitesimal arc QQ' mentioned near the beginning of the present paper, and the radius of geodesic curvature ρ is thus a length such that $2\rho\cdot MN=PQ^2$.

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ON THE GAUSSIAN THEORY OF SURFACES.

[From the *Proceedings of the London Mathematical Society*, vol. xii. (1881), pp. 187–192.
Read June 9, 1881.]

In the Memoir, Bour, “Théorie de la déformation des surfaces” (*Jour. de l’Éc. Polyt.*, Cah. 39 (1862), pp. 1–148), the author, working with the form $ds^2=dv^2+g^2du^2$ as a special case of Gauss’s formula $ds^2=E\,dp^2+2F\,dp\,dq+G\,dq^2$, obtains (p. 29) the following equations which he calls *fundamental*:—

$$\text{[IV.]}\quad\begin{cases}\dfrac{1}{g}\dfrac{dg_1}{dv}=T^2-HH_1,\\[1ex]\dfrac{dT}{du}+\dfrac{d\cdot Hg}{dv}-H_1g_1=0,\\[1ex]\dfrac{d\cdot Tg^2}{dv}+g\dfrac{dH_1}{du}=0,\end{cases}$$

where g_1 is written to denote $\dfrac{dg}{dv}$, and where (see p. 26)

H is the curvature of the normal section containing the tangent to the curve $v =$ constant,

H_1 is the curvature of the normal section at right angles to the preceding, containing the tangent to the (geodesic) curve $u =$ constant,

T is the torsion of the same geodesic curve;

or, what is the same thing (see p. 25), the quadric equation for the determination of the principal radii of curvature at the point of the surface is

$$\left(\frac{1}{\rho} - H\right)\left(\frac{1}{\rho} - H_1\right) - T^2 = 0.$$

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Writing for greater convenience K in place of the suffixed letter H_1, also V instead of g, so that the differential formula is $ds^2 = dv^2 + V^2 du^2$, the equations become

$$\begin{cases} \dfrac{1}{V}\dfrac{d^2V}{dv^2} = T^2 - HK, \\ \dfrac{dT}{du} + \dfrac{d \cdot HV}{dv} - K\dfrac{dV}{dv} = 0, \\ \dfrac{d \cdot TV^2}{dv} + V\dfrac{dK}{du} = 0; \end{cases}$$

or, if we use the suffix 1 to denote differentiation in regard to v, and the suffix 2 to denote differentiation in regard to u, then the equations are

$$\frac{V_{11}}{V} = T^2 - HK,$$

or, what is the same thing,

$$\begin{cases} T_2 + H_1 V + (H - K)V_1 = 0, \\ T_1 V + 2TV_1 + K_2 = 0. \end{cases}$$

I wish to show how these formulae connect themselves with formulae belonging to the general form $ds^2 = E\,dp^2 + 2F\,dp\,dq + G\,dq^2$. These involve not only Gauss's coefficients E, F, G, but also the coefficients E', F', G' belonging to the inflexional tangents; and, for convenience, I quote the system of definitions, Salmon's *Geometry of Three Dimensions*, 3rd ed., 1874, p. 251, viz.

$$dx,\ dy,\ dz = a\,dp + a'dq,\ b\,dp + b'dq,\ c\,dp + c'dq;$$

$$d^2x = \alpha\,dp^2 + 2\alpha' dp\,dq + \alpha'' dq^2,$$

$$d^2y = \beta\,dp^2 + 2\beta' dp\,dq + \beta'' dq^2,$$

$$d^2z = \gamma\,dp^2 + 2\gamma' dp\,dq + \gamma'' dq^2;$$

$$A,\ B,\ C = bc' - b'c,\ ca' - c'a,\ ab' - a'b; \quad V^2 = EG - F^2;$$

$$E' = A\alpha + B\beta + C\gamma, \quad F' = A\alpha' + B\beta' + C\gamma', \quad G' = A\alpha'' + B\beta'' + C\gamma'',$$

so that E', F', G' are, in fact, the determinants

$$\begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ \alpha, & \beta, & \gamma \end{vmatrix}, \quad \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ \alpha', & \beta', & \gamma' \end{vmatrix}, \quad \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix}.$$

The equation for the determination of the principal radii of curvature is

$$(E'\rho - EV)(G'\rho - GV) - (F'\rho - FV)^2 = 0.$$

which, in the particular case $F = 0$ (and therefore $V^2 = EG$), becomes

$$(E'\rho - EV)(G'\rho - GV) - F'^2\rho^2 = 0,$$

or, as this may be written,

$$EGV^2\left(\frac{1}{\rho} - \frac{E'}{EV}\right)\left(\frac{1}{\rho} - \frac{G'}{GV}\right) = F'^2,$$

an equation which corresponds with Bour's form

$$\left(\frac{1}{\rho} - K\right)\left(\frac{1}{\rho} - H\right) - T^2 = 0,$$

and becomes identical with it, if

$$E' = EVK, \quad G' = GVH, \quad F' = -V^2T.$$

But, making p, q correspond to Bour's variables, p to v, and q to u, it is necessary to show that the foregoing values (and not the interchanged values $E' = GVH$, $G' = EVK$) are the correct ones. We have, Salmon, p. 254,

$$\begin{vmatrix} dq, & \rho E' - VE, & \rho F' - VF \\ -dp, & \rho F' - VF, & \rho G' - VG \end{vmatrix} = 0;$$

or, putting herein $F = 0$, the equations may be written

$$\frac{dq}{-dp} = \frac{E'\left(\dfrac{1}{\rho} - \dfrac{VE}{\rho E'}\right)}{F'V} = \frac{F'V}{G'\left(\dfrac{1}{\rho} - \dfrac{VG}{\rho G'}\right)};$$

or, we see that to $dq = 0$ corresponds the value $\dfrac{1}{\rho} = \dfrac{E'}{VE}$, and to $dp = 0$ the value $\dfrac{1}{\rho} = \dfrac{G'}{VG}$. Hence the former of these values $\dfrac{1}{\rho}$ corresponds to $du = 0$, that is, to his $\dfrac{1}{\rho} = K$; and the latter to Bour's $dv = 0$, that is, to his $\dfrac{1}{\rho} = H$; or the values are, as stated,

$$E' = EVK, \quad G' = GVH.$$

The formula $ds^2 = E\,dp^2 + 2F\,dp\,dq + G\,dq^2$ agrees with Bour's $ds^2 = dv^2 + g^2du^2$, if $p = u$, $q = v$, $E = 1$, $F = 0$, $G = g^2$. With these values, $V^2 = EG - F^2 = g^2$, or say $g = V$, and Bour's equation is, as it was before written, $ds^2 = dv^2 + V^2du^2$. And we have to find the three equations which, putting therein $p = u$, $q = v$, $E = 1$, $F = 0$, $G = V^2$, $E' = VK$, $F' = -VT$, $G' = VH$, reduce themselves to Bour's equations.

The first of these is nothing else than the equation for the measure of curvature, viz. Salmon, p. 262 (but, using the suffixes 1 and 2 to denote differentiation in regard to p and q respectively), this is

$$\begin{gathered} 4\,(E'G' - F'^2) = E\,(E_2G_2 - 2F_2G_1 + G_1^2) \\ +F\,(E_1G_2 - E_2G_1 - 2E_2F_2 + 4F_1F_2 - 2F_1G_1) \\ +G\,(E_1G_1 - 2E_1F_2 + E_2^2) \\ -2\,(EG - F^2)(E_{22} - 2F_{12} + G_{11}). \end{gathered}$$

In fact, writing herein $E=1$, $F=0$, and therefore the differential coefficients of E and F each $=0$, the equation becomes

$$4\,(E'G'-F'^2)=G_1^2-2GG_{11},$$

which is

$$4V^4(HK-T^2)=(2VV_1)^2-2V^2(2V_1^2+2VV_{11}),\quad =-4V^3V_{11};$$

or finally it is

$$V_{11}=V(T^2-HK).$$

The other two of Bour's equations are derived from equations which give respectively the values of $E_2'-F_1'$ and $F_2'-G_1'$; viz. starting from the equations

$$E'=A\alpha+B\beta+C\gamma,$$

$$F'=A\alpha'+B\beta'+C\gamma',$$

$$G'=A\alpha''+B\beta''+C\gamma'',$$

we see at once that E_2' and F_1' contain, E_2' the terms $A\alpha_2+B\beta_2+C\gamma_2$, and F_1' the terms $A\alpha_1'+B\beta_1'+C\gamma_1'$, which are equal to each other (since $\alpha_2=\alpha_1'$, α and α' are the differential coefficients x_{11}, x_{12} of x, and so β_2 and β_1', γ_2 and γ_1'). Hence

$$E_2'-F_1'=A_2\alpha+B_2\beta+C_2\gamma-A_1\alpha'-B_1\beta'-C_1\gamma'$$

and similarly

$$F_2'-G_1'=A_2\alpha'+B_2\beta'+C_2\gamma'-A_1\alpha''-B_1\beta''-C_1\gamma''.$$

Here, from the values of A, B, C, we have

$$\begin{array}{lll} A=bc'-cb'; & A_1=\beta c'+b\gamma'-\gamma b'-c\beta'; & A_2=\beta'c'-\gamma'b'+b\gamma''-c\beta'' \\ B=ca'-ac'; & B_1=\gamma a'+c\alpha'-\alpha c'-a\gamma'; & B_2=\gamma'a'-\alpha'c'+c\alpha''-a\gamma'' \\ C=ab'-ba'; & C_1=\alpha b'+a\beta'-\beta a'-b\alpha'; & C_2=\alpha'b'-\beta'a'+a\beta''-b\alpha'' \end{array}$$

and, substituting, we find

$$E_2'-F_1'=2\alpha'\alpha\alpha'+\alpha\alpha''\alpha,$$

$$F_2'-G_1'=-2\alpha\alpha'\alpha''-\alpha'\alpha''\alpha,$$

if, for shortness, $\alpha'\alpha\alpha'$ denotes the determinant

$$\begin{vmatrix} \alpha', & \alpha, & \alpha \\ b', & \beta, & \beta' \\ c', & \gamma, & \gamma' \end{vmatrix},$$

and so for the other like symbols. Observe that, with

$$\begin{matrix} a, & a', & \alpha, & \alpha', & \alpha'' \\ b, & b', & \beta, & \beta', & \beta'' \\ c, & c', & \gamma, & \gamma', & \gamma'' \end{matrix},$$

we have in all 10 determinants, viz. these are $aa'\alpha, = E'$; $aa'\alpha', = F'$; $aa'\alpha'', = G'$; $\alpha\alpha'\alpha''$; and the six determinants $\alpha\alpha\alpha'$, $\alpha\alpha'\alpha''$, $\alpha\alpha''\alpha$; $\alpha'\alpha\alpha'$, $\alpha'\alpha'\alpha''$, $\alpha'\alpha''\alpha$. The foregoing expressions of $E_2'-F_1'$ and $F_2'-G_1'$ respectively, substituting therein for the determinants $\alpha'\alpha\alpha'$, $\alpha\alpha''\alpha$, $\alpha\alpha'\alpha''$, $\alpha'\alpha''\alpha$ their values as about to be obtained, are the required two equations. We have

$$\begin{aligned}
&aa+bb+cc=E, && aa'+bb'+cc'=F,\\
&a'a+b'b+c'c=F, && a'a'+b'b'+c'c'=G,\\
&\alpha a+\beta b+\gamma c=\tfrac12 E_1, && \alpha a'+\beta b'+\gamma c'=F_1-\tfrac12 E_2,\\
&\alpha' a+\beta' b+\gamma' c=\tfrac12 E_2, && \alpha' a'+\beta' b'+\gamma' c'=\tfrac12 G_1,\\
&\alpha'' a+\beta'' b+\gamma'' c=F_2-\tfrac12 G_1, && \alpha'' a'+\beta'' b'+\gamma'' c'=\tfrac12 G_2,
\end{aligned}$$

and if from the first five equations, regarded as equations linear in (a, b, c), we eliminate these quantities, and from the second five equations, regarded as linear in (a', b', c'), we eliminate these quantities, we obtain two sets each of five equations.

$$\begin{vmatrix} a, & \alpha, & \alpha, & \alpha', & \alpha'' \\ b, & b', & \beta, & \beta', & \beta'' \\ c, & c', & \gamma, & \gamma', & \gamma' \\ E, & F, & \tfrac12 E_1, & \tfrac12 E_2, & F_1-\tfrac12 G_1 \end{vmatrix} = 0, \qquad \begin{vmatrix} a, & a', & \alpha, & \alpha', & \alpha'' \\ b, & b', & \beta', & \beta'', & \\ c, & c', & \gamma, & \gamma', & \gamma'' \\ F, & G, & F_1-\tfrac12 E_2, & \tfrac12 G_1, & \end{vmatrix} = 0.$$

These may be written,

$$F\alpha\alpha'\alpha''-\tfrac12 E_1\alpha'\alpha\alpha''-\tfrac12 E_2\alpha'\alpha''\alpha-(F_1-\tfrac12 G_1)\alpha'\alpha\alpha'=0,$$

$$-E\alpha\alpha'\alpha''+\tfrac12 E_1\alpha\alpha'\alpha''+\tfrac12 E_2\alpha\alpha''\alpha+(F_1-\tfrac12 G_1)\alpha\alpha\alpha'=0,$$

$$E\alpha'\alpha'\alpha''-F\alpha\alpha'\alpha''+\tfrac12 E_1G'-(F_1-\tfrac12 G_1)F'=0,$$

$$E\alpha'\alpha''\alpha-F\alpha\alpha''\alpha-\tfrac12 E_1G'+(F_1-\tfrac12 G_1)E'=0,$$

$$E\alpha'\alpha\alpha-F\alpha\alpha\alpha'+\tfrac12 E_2F'-\tfrac12 E_1E'=0;$$

and

$$G\alpha\alpha'\alpha''-(F_2-\tfrac12 E_2)\alpha'\alpha\alpha''-\tfrac12 G_1\alpha'\alpha''\alpha-\tfrac12 G_2\alpha'\alpha\alpha'=0,$$

$$-F\alpha\alpha\alpha''+(F_2-\tfrac12 E_2)\alpha\alpha'\alpha''+\tfrac12 G_1\alpha\alpha''\alpha+\tfrac12 G_2\alpha\alpha\alpha'=0,$$

$$F\alpha'\alpha'\alpha''-G\alpha\alpha\alpha''+\tfrac12 G_1G'-\tfrac12 G_2F'=0,$$

$$F\alpha'\alpha''\alpha-G\alpha\alpha'\alpha-(F_2-\tfrac12 E_2)G'+\tfrac12 G_1E'=0,$$

$$F\alpha'\alpha\alpha-G\alpha\alpha\alpha+(F_2-\tfrac12 E_2)F'-\tfrac12 G_2E'=0.$$

Attending in each set only to the third, fourth, and fifth equations, and combining these in pairs, we obtain

$$V^2\alpha\alpha'\alpha''+(\tfrac12 FG_1-FF_1+\tfrac12 EG_1)F'+(-\tfrac12 EG_1+\tfrac12 FE_2)G'=0,$$

$$V^2\alpha'\alpha'\alpha''+(\tfrac12 GG_1-GF_1+\tfrac12 FG_2)F'+(-\tfrac12 FG_1+\tfrac12 GE_2)G'=0;$$

$$V^2\alpha\alpha''\alpha+(-\tfrac12 FE_1+EF_1-\tfrac12 EE_2)G'+(-\tfrac12 FG_1+FF_1-\tfrac12 EG_1)E'=0,$$

$$V^2\alpha'\alpha''\alpha+(-\tfrac12 GE_1+FF_1-\tfrac12 FE_2)G'+(-\tfrac12 GG_1+GF_1-\tfrac12 FG_2)E'=0;$$

$$V^2\alpha\alpha\alpha'+(\tfrac12 EG_2-\tfrac12 FE_2)E'+(\tfrac12 FE_1-EF_1+\tfrac12 EE_2)F'=0,$$

$$V^2\alpha'\alpha\alpha'+(\tfrac12 FG_2-\tfrac12 GE_2)E'+(\tfrac12 GE_1-FF_1+\tfrac12 FE_2)F'=0.$$

We thus obtain

$$E_2' - F_1' = \tfrac{2}{V^2}\left[(-\tfrac{1}{2}FG_1 + \tfrac{1}{2}GE_2)E' + (-\tfrac{1}{2}GE_1 + FF_1 - \tfrac{1}{2}FF_2)F'\right]$$

$$+\tfrac{1}{V^2}\{(\tfrac{1}{2}FE_1 - EF_1 + \tfrac{1}{2}EE_2)G' + (\tfrac{1}{2}FG_1 - FF_1 + \tfrac{1}{2}EG_1)E'\},$$

$$F_2' - G_1' = \tfrac{2}{V^2}\{(\tfrac{1}{2}FG_1 - FF_1 + \tfrac{1}{2}EG_1)F' + (-\tfrac{1}{2}EG_1 + \tfrac{1}{2}FE_2)G'\}$$

$$+\tfrac{1}{V^2}\{(-\tfrac{1}{2}GE_1 + FF_1 - \tfrac{1}{2}FE_2)G' + (-\tfrac{1}{2}GG_1 + GF_1 - \tfrac{1}{2}FG_2)E'\};$$

or, finally,

$$E_2' - F_1' = \tfrac{1}{V^2}\{(-\tfrac{1}{2}FG_1 + GE_2 - FF_1 + \tfrac{1}{2}EG_2)E'$$

$$+(-GE_1 + 2FF_1 - FF_2)F' + (\tfrac{1}{2}FE_1 - EF_1 + \tfrac{1}{2}EE_2)G'\},$$

$$F_2' - G_1' = \tfrac{1}{V^2}\{(-\tfrac{1}{2}GG_1 + GF_1 - \tfrac{1}{2}FG_2)E'$$

$$+(FG_1 - 2FF_1 + EG_2)F' + (-\tfrac{1}{2}GE_1 + FF_1 - EG_1 + \tfrac{1}{2}FE_2)G'\},$$

which are the required formulae; and which may, I think, be regarded as new formulae in the Gaussian theory of surfaces.

Writing herein as before, the first of these becomes

$$V^2(VK)_2 + (V^2T)_1 = \tfrac{1}{V^2}[\tfrac{1}{2}(V^2)_1 VK] = V_2K,$$

that is,

$$V^2V_2K + V^3K_2 + V^2T_1 + 2VV_1T = V_2K;$$

or finally

$$VT_1 + 2TV_1 + K_2 = 0,$$

which is Bour's third equation. And the second equation becomes

$$-(V^2T)_2 - (V^2H)_1 = \tfrac{1}{V^2}\{-\tfrac{1}{2}V^2V_1VK + (V^2)_1(-V^2T)(V^2)_1VH\}$$

that is,

$$= -V^2V_1K - 2VV_1T - 2V^2V_1H,$$

or finally

$$-V^2T_2 - 2VV_2T - V^2H_1 - 3V^2V_1H = -V^2V_1K - 2VV_1T - 2V^2V_1H;$$

$$T_2 + VH_1 + (H-K)V_1 = 0,$$

which is Bour's second equation.

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NOTE ON LANDEN'S THEOREM.

[From the *Proceedings of the London Mathematical Society*, vol. xiii. (1882), pp. 47, 48. Read November 10, 1881.]

Landen's theorem, as given in the paper "An Investigation of a General Theorem for finding the length of any Arc of any Conic Hyperbola by means of two Elliptic Arcs, with some other new and useful Theorems deduced therefrom," *Phil. Trans.*, t. lxv. (1775), pp. 283–289, is, as appears by the title, a theorem for finding the length of a hyperbolic arc in terms of the length of two elliptic arcs; this theorem being obtained by means of the following differential identity, viz., if

$$t = gx\sqrt{\frac{m^2 - x^2}{m^2 - g^2x^2}}$$

where

$$g = \frac{m^2 - n^2}{n^2},$$

then

$$\sqrt{\frac{m^2 - gx^2}{m^2 - x^2}}\,dx = \left\{\tfrac{1}{2} + \tfrac{1}{4}\sqrt{\frac{(m-n)^2 - t^2}{(m+n)^2 - t^2}} + \tfrac{1}{4}\sqrt{\frac{(m+n)^2 - t^2}{(m-n)^2 - t^2}}\right\} dt,$$

(this is exactly Landen's form, except that he of course writes $\dot{x}$, $\dot{t}$ respectively); viz., integrating each side, and interpreting geometrically in a very ingenious and elegant manner the three integrals which present themselves, he arrives at his theorem for the hyperbolic arc; but with this I am not now concerned.

Writing for greater convenience $m = 1$, $n = k'$, and therefore $g = k^2$, if as usual $k^2 + k'^2 = 1$, the transformation is

$$t = k^2x\sqrt{\frac{1 - x^2}{1 - k^4x^2}},$$

leading to

$$\sqrt{\frac{1-k^2x^2}{1-x^2}}\,dx = \left\{\tfrac{1}{2} + \tfrac{1}{4}\sqrt{\frac{(1-k')^2-t^2}{(1+k')^2-t^2}} + \tfrac{1}{4}\sqrt{\frac{(1+k')^2-t^2}{(1-k')^2-t^2}}\right\} dt.$$

The form in which the transformation is usually employed (see my *Elliptic Functions*, pp. 177, 178) is

$$y = (1+k')x\sqrt{\frac{1-x^2}{1-k^2x^2}},$$

leading to

$$\frac{(1+k')\,dx}{\sqrt{1-x^2\cdot 1-k^2x^2}} = \frac{dy}{\sqrt{1-y^2\cdot 1-\lambda^2y^2}},$$

where

$$\lambda = \frac{1-k'}{1+k'}.$$

If, to identify the two forms, we write $y = \dfrac{t}{1-k'}$, and in the last equation introduce t in place of y, the last equation becomes

$$\frac{dx}{\sqrt{1-x^2\cdot 1-k^2x^2}} = \frac{dt}{\sqrt{\{(1-k')^2-t^2\}\{(1+k')^2-t^2\}}}.$$

Comparing with Landen's form, in order that the two may be identical, we must have

$$1-k^2x^2 = \left\{\tfrac{1}{2} + \tfrac{1}{4}\sqrt{\frac{(1-k')^2-t^2}{(1+k')^2-t^2}} + \tfrac{1}{4}\sqrt{\frac{(1+k')^2-t^2}{(1-k')^2-t^2}}\right\} \times \sqrt{(1-k')^2-t^2}\sqrt{(1+k')^2-t^2},$$

viz., this is

$$1-k^2x^2 = \tfrac{1}{4}\big[\sqrt{(1-k')^2-t^2} + \sqrt{(1+k')^2-t^2}\big]^2,$$

that is,

$$1-k^2x^2 = \tfrac{1}{2}\big[1+k'^2-t^2+\sqrt{\{(1-k')^2-t^2\}\{(1+k')^2-t^2\}}\big],$$

where the function under the radical sign is

$$k'^4 - 2(1+k'^2)t^2 + t^4 \ (= T \text{ suppose})$$

and this must consequently be a form of the original integral equation

$$t = k^2x\sqrt{\frac{1-x^2}{1-k^4x^2}}.$$

In fact, squaring and solving in regard to x^2 with the assumed sign of the radical, we have

$$x^2 = \frac{k'^2+t^2-\sqrt{T}}{2k^2},$$

corresponding to an equation given by Landen. And we thence have

$$1 - k^2x^2 = \tfrac{1}{2}[2 - k'^2 - t^2 + \sqrt{T}], \quad = \tfrac{1}{2}[1 + k^2 - t^2 + \sqrt{T}],$$

which is the required expression for $1 - k^2x^2$.

The trigonometrical form $\sin(2\phi' - \phi) = c\sin\phi$ of the relation between y and x does not occur in Landen; it is employed by Legendre, I believe, in an early paper, *Mém. de l'Acad. de Paris*, 1786, and in the *Exercices*, 1811, and also in the *Traité des Fonctions Elliptiques*, 1825, and by means of it he obtains an expression for the arc of a hyperbola in terms of two elliptic functions, $E(c, \phi)$, $E(c', \phi')$, showing that the arc of the hyperbola is expressible by means of two elliptic arcs, this, he observes, "est le beau théorème dont Landen a enrichi la géométrie." We have, then (1828), Jacobi's proof, by two fixed circles, of the addition-theorem (see my *Elliptic Functions*, p. 28), and the application of this (p. 30) to Landen's theorem is also due to Jacobi, see the "Extrait d'une lettre adressée à M. Hermite," *Crelle*, t. xxxii. (1846), pp. 176–181; the connection of the demonstrations, by regarding the point, which is alone necessary for Landen's theorem as the limit of the smaller circle in the figure for the addition-theorem is due to Durege (see his *Theorie der elliptischen Functionen*, Leipzig, 1861, pp. 168, et seq.).

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ON A FORMULA RELATING TO ELLIPTIC INTEGRALS OF THE THIRD KIND.

[From the *Proceedings of the London Mathematical Society*, vol. xiii. (1882), pp. 175, 176. Presented May 11, 1882.]

The formula for the differentiation of the integral of the third kind

$$\Pi = \int_0^\phi \frac{d\phi}{(1 + n\sin^2\phi)\Delta}$$

in regard to the parameter n, see my *Elliptic Functions*, Nos. 174 *et seq.*, may be presented under a very elegant form, by writing therein

$$\sin^2\phi = x = \operatorname{sn}^2 u, \quad \sin\phi\cos\phi\,\Delta = y = \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u,$$

and thus connecting the formula with the cubic curve

$$y^2 = x(1-x)(1-k^2x).$$

The parameter must, of course, be put under a corresponding form, say $n = -\dfrac{1}{a}$, where $a = \operatorname{sn}^2\theta$, $b = \operatorname{sn}\theta \operatorname{cn}\theta \operatorname{dn}\theta$, and therefore (a, b) are the coordinates of the point corresponding to the argument θ. The steps of the substitution may be effected without difficulty, but it will be convenient to give at once the final result and then verify it directly. The result is

$$\frac{d}{d\theta}\frac{b}{a-x} = \frac{d}{du}\frac{y}{x-a} = k^2(a-x).$$

We, in fact, have

$$\frac{d}{du}x = 2\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u = 2y,$$

and thence

$$y\frac{dx}{du} = 2y^2,$$

that is,

$$y\frac{dx}{du} = 2x\big[1 - (1+k^2)x + k^2x^2\big].$$

Also

$$\begin{aligned}\frac{dy}{du} &= \mathrm{cn}^2 u\,\mathrm{dn}^2 u - \mathrm{sn}^2 u\,\mathrm{dn}^2 u - k^2\mathrm{sn}^2 u\,\mathrm{cn}^2 u\\ &= 1 - 2(1+k^2)x + 3k^2x^2,\end{aligned}$$

and hence

$$\begin{aligned}\frac{d}{du}\frac{y}{x-a} &= \frac{1}{(a-x)^2}\left\{(a-x)\frac{dy}{du} - y\frac{dx}{du}\right\}\\ &= \frac{1}{(a-x)^2}\big[-x - a + 2(1+k^2)ax + k^2x^3 - 3k^2ax^2\big].\end{aligned}$$

Interchanging the letters, we have

$$\frac{d}{d\theta}\frac{b}{a-x} = \frac{1}{(a-x)^2}\big[-x - a + 2(1+k^2)ax + k^2a^3 - 3k^2a^2x\big],$$

and hence, subtracting,

$$\begin{aligned}\frac{d}{d\theta}\frac{b}{a-x} - \frac{d}{du}\frac{y}{x-a} &= \frac{1}{(a-x)^2}\left\{k^2(a-x)^3\right\}\\ &= k^2(a-x),\end{aligned}$$

which is the required result.

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ON THE 34 CONCOMITANTS OF THE TERNARY CUBIC.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 1–15.]

I have, (by aid of Gundelfinger's formulae, afterwards referred to), calculated, and I give in the present paper, the expressions of the 34 concomitants of the canonical ternary cubic $ax^3+by^3+cz^3+6lxyz$, or, what is the same thing, the 34 covariants of this cubic and the adjoint linear function $\xi x+\eta y+\zeta z$: this is the chief object of the paper. I prefix a list of memoirs, with short remarks upon some of them; and, after a few observations, proceed to the expressions for the 34 concomitants; and, in conclusion, exhibit the process of calculation of these concomitants other than such of them as are taken to be known forms. I insert a supplemental table of 6 derived forms.

The list of memoirs (not by any means a complete one) is as follows:

Hesse, Ueber die Elimination der Variabeln aus drei algebraischen Gleichungen vom zweiten Grade mit zwei Variabeln: *Crelle*, t. xxviii. (1844), pp. 68–96. Although purporting to relate to a different subject, this is in fact the earliest, and a very important, memoir in regard to the general ternary cubic; and in it is established the canonical form, as Hesse writes it, $y_1^3+y_2^3+y_3^3+6my_1y_2y_3$.

Aronhold, Zur Theorie der homogenen Functionen dritten Grades von drei Variabeln: *Crelle*, t. xxxix. (1850), pp. 140–159.

Cayley, Third Memoir on Quantics: *Phil. Trans.*, t. cxlvi. (1856), pp. 627–647 [144].

Aronhold, Theorie der homogenen Functionen dritten Grades von drei Variabeln: *Crelle*, t. lv. (1858), pp. 97–191.

Salmon, Lessons Introductory to the Modern Higher Algebra: 8°, Dublin, 1859.

Cayley, Seventh Memoir on Quantics: *Phil. Trans.*, t. cli. (1861), pp. 277–292 [269].

Brioschi, Sur la théorie des formes cubiques à trois indéterminées: *Comptes Rendus*, t. lvi. (1863), pp. 304–307.

Hermite, Extrait d'une lettre à M. Brioschi: *Crelle*, t. lxiii. (1864), pp. 30–32, followed by a note by Brioschi, pp. 32–33.

The skew covariant of the ninth order, which is $y^3 - z^3 \cdot z^3 - x^3 \cdot x^3 - y^3$ for the canonical form $x^3 + y^3 + z^3 + 6lxyz$, and the corresponding contravariant $\eta^3 - \zeta^3 \cdot \zeta^3 - \xi^3 \cdot \xi^3 - \eta^3$, alluded to p. 116 of Salmon's *Lessons*, were obtained, the covariant by Brioschi and the contravariant by Hermite, in the last-mentioned papers.

Clebsch and Gordan, Ueber die Theorie der ternären cubischen Formen: *Math. Annalen*, t. i. (1869), pp. 56–89.

The establishment of the complete system of the 34 covariants, contravariants and *Zwischenformen*, or, as I have here called them, the 34 concomitants, was first effected by Gordan in the next following memoir:

Gordan, Ueber die ternären Formen dritten Grades: *Math. Annalen*, t. i. (1869), pp. 90–128.

And the theory is farther considered:

Gundelfinger, Zur Theorie der ternären cubischen Formen: *Math. Annalen*, t. vi. (1871), pp. 144–163. The author speaks of the 34 forms as being "theils mit den von Gordan gewählten identisch, theils möglichst einfache Combinationen derselben." They are, in fact, the 34 forms given in the present paper for the canonical form of the cubic, and the meaning of the adopted combinations of Gordan's forms will presently clearly appear.

There is an advantage in using the form $ax^3 + by^3 + cz^3 + 6lxyz$ rather than the Hessian form $x^3 + y^3 + z^3 + 6lxyz$, employed in my Third and Seventh Memoirs on Quantics: for the form $ax^3 + by^3 + cz^3 + 6lxyz$ is what the general cubic

$$(a, b, c, f, g, h, i, j, k, l)(x, y, z)^3$$

becomes by no other change than the reduction to zero of certain of its coefficients; and thus any concomitant of the canonical form consists of terms which are leading terms of the same concomitant of the general form.

The concomitants are functions of the coefficients $(a, b, \dots, l)$, of (ξ, η, ζ), and of (x, y, z): the dimensions in regard to the three sets respectively may be distinguished as the degree, class, and order; and we have thus to consider the deg-class-order of a concomitant.

Two or more concomitants of the same deg-class-order may be linearly combined together: viz., the linear combination is the sum of the concomitants each multiplied by a mere number. The question thus arises as to the selection of a representative concomitant. As already mentioned, I follow Gundelfinger, viz., my 34 concomitants of the canonical form correspond each to each (with only the difference of a numerical factor of the entire concomitant) to his 34 concomitants of the general form. The principle underlying the selection would, in regard to the general form, have to be explained altogether differently; but this principle exhibits itself in a very remarkable manner in regard to the canonical form $ax^3 + by^3 + cz^3 + 6lxyz$.

Each concomitant of the general form is an indecomposable function, not breaking up into rational factors; but this is not of necessity the case in regard to a canonical form: only a concomitant which does break up must be regarded as indecomposable, no factor of such concomitant being rejected, or separated. So far from it, there is, in regard to the canonical form in question, a frequent occurrence of $abc + 8l^3$ or a power thereof, either as a factor of a unique concomitant, or when there are two or more concomitants of the same deg-class-order, then as a factor of a properly selected linear combination of such concomitants: and the principle referred to is, in fact, that of the selection of such combination for the representative concomitant; or (in other words) the representative concomitant is taken so as to contain as a factor the highest power that may be of $abc+8l^3$. As to the signification of this expression $abc+8l^3$, I call to mind that the discriminant of the form is $abc\,(abc + 8l^3)^3$.

As to numerical factors: my principle has been, and is, to throw out any common numerical divisor of all the terms: thus I write $S = -abcl + l^4$, instead of Aronhold's $S = -4abcl + 4l^4$. There is also the question of nomenclature: I retain that of my Seventh Memoir on Quantics, except that I use single letters H, P, &c., instead of the same letters with U, thus HU, PU, &c.; in particular, I use U, H, P, Q instead of Aronhold's f, Δ, S_f, T_f. It is thus at all events necessary to make some change in Gundelfinger's letters; and there is moreover a laxity in his use of accented letters; his B, B', B'', B''', and so in other cases E, E', E'', &c., are used to denote functions derived in a determinate manner each from the preceding one (by the δ-process explained further on); whereas his L, $\bar{L}$; M, M'; N, N' are functions having to each other an altogether different relation; also three of his functions are not denoted by any letters at all. Under the circumstances, I retain only a few of his letters; use the accent where it denotes the δ-process; and introduce barred letters $\bar{J}$, $\bar{K}$, &c., to denote a different correspondence with the unbarred letters J, K, &c. But I attach also to each concomitant a numerical symbol showing its deg-class-order, thus: 541 (degree = 5, class = 4, order = 1) or 1290, (there is no ambiguity in the two-digit numbers 10, 11, 12 which present themselves in the system of the 34 symbols); and it seems to me very desirable that the significations of these deg-class-order symbols should be considered as permanent and unalterable. Thus, in writing $S = 400 = -abcl + l^4$, I wish the 400 to be regarded as denoting its expressed value $-abcl + l^4$: if the same letter S is to be used in Aronhold's sense to denote $-4abcl + 4l^4$, this would be completely expressed by the new definition $S_1 = 4.400$, the meaning of the symbol 400 being explained by reference to the present memoir, or by the actual quotation $400 = -abcl + l^4$.

I proceed at once to the table: for shortness, I omit, in general, terms which can be derived from an expressed term by mere cyclical interchanges of the letters (a, b, c), (ξ, η, ζ), (x, y, z).

Table of the 34 Covariants of the Canonical Cubic $ax^3+by^3+cz^3+6lxyz$ and the linear form $\xi x+\eta y+\zeta z$.

First Part, 10 Forms. Class = Order.

Current No.

1 $S=400=-abcl+l^4.$

2 $T=600=a^2b^2c^2-20abcl^3-8l^6.$

3 $A=011=\xi x+\eta y+\zeta z.$

4 $\Theta=222=x^2[-l^2\xi^2-2al\eta\zeta]\dots$
$+yz[bc\xi^2+2l^2\eta\zeta]\dots$

5 $\Theta'=422=x^2[l(abc+2l^3)\xi^2+a(abc-4l^3)\eta\zeta]\dots$
$+yz[6bcl^2\xi^2-2l(abc+2l^3)\eta\zeta]\dots$

6 $\Theta''=622=x^2[-(abc+2l^3)^2\xi^2+2al^2(abc-2l^3)\eta\zeta]\dots$
$+yz[36bcl^4\xi^2+2(abc+2l^3)^2\eta\zeta]\dots$

7 $B=333=x^3[a^2(c\eta^3-b\zeta^3)]\dots$
$+y^2z[(abc+8l^3)\eta^2\zeta+2bl^2\xi^3-6bcl\xi\eta^2]\dots$
$+yz^2[-(abc+8l^3)\eta\zeta^2-6bcl\xi^2\eta-12cl^2\xi^3]\dots$

8 $B'=533=x^3[3a^2l^2(c\eta^3-b\zeta^3)]\dots$
$+y^2z[-l^2(abc+8l^3)\eta\zeta^2+4bl(-abc+l^3)\xi^2\zeta-bc(abc+10l^3)\xi\eta^2]\dots$
$+yz^2[l^2(abc+8l^3)\eta\zeta^2-bc(abc+10l^3)\xi^2\eta+4cl(-abc+l^3)\xi\zeta^2]\dots$

9 $B''=733=x^3[9a^2l^4(c\eta^3-b\zeta^3)]\dots$
$+y^2z[l(abc+8l^3)(2abc+l^3)\eta\zeta^2$
$+b(abc+2l^3)(abc-10l^3)\xi^2\zeta+6bcl^2(-abc+l^3)\xi\eta^2]\dots$
$+yz^2[-l(abc+8l^3)(2abc+l^3)\eta\zeta^2$
$-6bcl^2(-abc+l^3)\xi^2\eta+c(abc+2l^3)(abc-10l^3)\xi\zeta^2]\dots$

10 $B'''=933=x^3[27a^2l^6(c\eta^3-b\zeta^3)]\dots$
$+y^2z[-(abc+8l^3)(abc-l^3)^2\eta^2\zeta+9bl^3(abc-2l^3)^2\xi^2\zeta$
$-27bcl^4(abc+2l^3)\xi\eta^2]\dots$
$+yz^2[(abc+8l^3)(abc-l^3)^2\eta\zeta^2-27bcl^4(abc+2l^3)\xi^2\eta$
$-9cl^3(abc+2l^3)^2\xi\eta\zeta^2]\dots$

Second Part, $(4+4=)$ 8 forms. Class $=0$, and Order $=0$.

Class $=0$.

11 $U=103=ax^3+by^3+cz^3+6lxyz.$

12 $H=303=l^2(ax^3+by^3+cz^3)-(abc+2l^3)xyz.$

13 $\Psi=806=(abc+8l^3)^2(a^2b^2c^2$
$+b^2y^6+c^2z^6-10(bcy^3z^3+caz^3x^3+abx^3y^3)).$

14 $\Omega=1209=(abc+8l^3)^3[by^3-cz^3\cdot cz^3-ax^3\cdot ax^3-by^3].$

Current No. *Order* $= 0$.

$$15 \quad P = 330 = -l(bc\xi^3 + ca\eta^3 + ab\zeta^3) + (-abc + 4l^3)\xi\eta\zeta.$$

$$16 \quad Q = 530 = (abc - 10l^3)(bc\xi^3 + ca\eta^3 + ab\zeta^3) - 6l^2(5abc + 4l^3)\xi\eta\zeta.$$

$$\begin{aligned} 17 \quad F = 460 = {} & b^2c^2\xi^6 + c^2a^2\eta^6 + a^2b^2\zeta^6 \\ & - 2(abc + 16l^3)(a\eta^3\zeta^3 + b\zeta^3\xi^3 + c\xi^3\eta^3) \\ & - 24l^2(bc\xi^4 + ca\eta^4 + ab\zeta^4)\xi\eta\zeta - 24l(abc + 2l^3)\xi^2\eta^2\zeta^2. \end{aligned}$$

$$18 \quad \Pi = 1290 = (abc + 8l^3)^3[c\eta^3 - b\zeta^3 \cdot a\xi^3 - c\eta^3 \cdot b\zeta^3 - a\xi^3].$$

Third Part, $(8 + 8 =)$ *16 forms. Class less or greater than Order.*

Class less than Order.

$$19 \quad J = 413 = (abc + 8l^3)\{\xi x(by^3 - cz^3) + \eta y(cz^3 - ax^3) + \zeta z(ax^3 - by^3)\}.$$

$$20 \quad K = 613 = (abc + 8l^3)\{\xi[alx^4 - 2blxy^3 - 2clxz^3 - 3bcy^2z^2]\ldots\}.$$

$$21 \quad K' = 813 = (abc + 8l^3)\{\xi[(abc + 2l^3)(ax^4 - 2bxy^3 - 2cxz^3) - 18bcl^2y^2z^2]\ldots\}.$$

$$\begin{aligned} 22 \quad E = 625 = {} & (abc + 8l^3)\{\xi(by^3 - cz^3)[2l^2x^2 + bcyz]\ldots \\ & + \eta\zeta(by^3 - cz^3)[4alx^2 - 2l^2yz]\ldots\}. \end{aligned}$$

$$\begin{aligned} 23 \quad E' = 825 = {} & (abc + 8l^3)\{\xi(by^3 - cz^3)[l(abc + 2l^3)x^2 - 3bcl^2yz]\ldots \\ & + \eta\zeta(by^3 - cz^3)[a(abc + 4l^3)x^2 - l(abc + 2l^3)yz]\ldots\}. \end{aligned}$$

$$\begin{aligned} 24 \quad E'' = 1025 = {} & (abc + 8l^3)\{\xi(by^3 - cz^3)[(abc + 2l^3)^2x^2 + 18bcl^4yz]\ldots \\ & + \eta\zeta(by^3 - cz^3)[-12al^3(abc + 2l^3)x^2 + (abc + 2l^3)^2yz]\ldots\}. \end{aligned}$$

$$25 \quad M = 917 = (abc + 8l^3)^2\{\xi(by^3 - cz^3)[5alx^4 - blxy^3 - clxz^3 - 3bcy^2z^2]\ldots\}.$$

$$\begin{aligned} 26 \quad M' = 1117 = {} & (abc + 8l^3)^2\{\xi(by^3 - cz^3)[(abc + 2l^3)(5ax^4 - bxy^3 - cxz^3) \\ & + 18bcl^2y^2z^2]\ldots\}. \end{aligned}$$

Order less than Class.

$$27 \quad \bar{J} = 841 = (abc + 8l^3)^2[x\xi a(c\eta^3 - b\zeta^3) + y\eta b(a\xi^3 - c\zeta^3) + z\zeta c(b\zeta^3 - a\eta^3)].$$

$$28 \quad \bar{K} = 541 = (abc + 8l^3)\{x[bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3 - 6ab\eta^2\zeta^2]\ldots\}.$$

$$29 \quad \bar{K}' = 741 = (abc + 8l^3)\{x[l^2(bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3) + a(abc + 2l^3)\eta^2\zeta^2]\ldots\}.$$

$$\begin{aligned} 30 \quad \bar{E} = 652 = {} & (abc + 8l^3)\{x^2(c\eta^3 - b\zeta^3)[2al\xi^2 - a^2\eta\zeta]\ldots \\ & + yz(c\eta^3 - b\zeta^3)[bl^2\xi^2 + 2al\eta\zeta]\ldots\}. \end{aligned}$$

$$\begin{aligned} 31 \quad \bar{E}' = 852 = {} & (abc + 8l^3)\{x^2(c\eta^3 - b\zeta^3)[a(abc - 4l^3)\xi^2 - 6a^2l\eta\zeta]\ldots \\ & + yz(c\eta^3 - b\zeta^3)[3bl(abc + 2l^3)\xi^2 - a(abc + l^3)\eta\zeta]\ldots\}. \end{aligned}$$

$$\begin{aligned} 32 \quad \bar{E}'' = 1052 = {} & (abc + 8l^3)\{x^2(c\eta^3 - b\zeta^3)[-3al^2(abc + 2l^3)\xi^2 + 9a^2l^4\eta\zeta]\ldots \\ & + yz(c\eta^3 - b\zeta^3)[(abc + 2l^3)^2 - 3al^2(abc + l^3)\eta\zeta]\ldots\}. \end{aligned}$$

$$\begin{aligned} 33 \quad \bar{M} = 771 = {} & (abc + 8l^3)^2\{x(c\eta^3 - b\zeta^3)[(abc - 8l^3)a^2c\xi\eta^3 - a^2b\xi\zeta^3 \\ & - 12al^3\xi^2\eta\zeta - 6a^2bl\eta^2\zeta]\ldots\}. \end{aligned}$$

$$\begin{aligned} 34 \quad \bar{M}' = 971 = {} & (abc + 8l^3)^2\{x(c\eta^3 - b\zeta^3)[l^2(7abc - 8l^3)\xi^4 - 3a^2cl^2\xi\eta^3 + 3a^2bl\xi\zeta^3 \\ & - 4al(abc + l^3)\xi^2\eta\zeta + a^2(abc + 10l^3)\eta^2\zeta^2]\ldots\}. \end{aligned}$$

To this may be joined the following *Supplemental Table* of certain Derived Forms:

Current No.

35 $R = 1200 = 64S^3 - T^2 = -abc(abc + 8l^3)^3.$

36 $C = 703 = -TU + 24SH = (abc + 8l^3)\{(-abc + 4l^3)(ax^3 + by^3 + cz^3) + 18abclxyz\}.$

37 $D = 903 = 8S^2U - 3TH = (abc + 8l^3)\{l(5abc + 4l^3)(ax^3 + by^3 + cz^3) + 3abc(abc - 10l^3)xyz\}.$

38 $Y = 930 = 3TP - 4SQ = (abc + 8l^3)^2\{l(bc\xi^3 + ca\eta^3 + ab\zeta^3) - 3abc\xi\eta\zeta\}.$

39 $Z = 1130 = -48S^2P + TQ = (abc + 8l^3)^2\{(abc + 2l^3)(bc\xi^3 + ca\eta^3 + ab\zeta^3) + 18abcl^2\xi\eta\zeta\}.$

40 $\Phi = 1660 = 12(abc + 8l^3)F - 288STP + 768S^2PQ - 8TQ^2$
$= (abc + 8l^3)^2\{b^2c^2\xi^6 + c^2a^2\eta^6 + a^2b^2\zeta^6 - 10abc(a\eta^3\zeta^3 + b\zeta^3\xi^3 + c\xi^3\eta^3)\};$

viz. these are derived forms characterized by having a power of $abc + 8l^3$ as a factor: R is the discriminant; C, D, Y, Z occur in Aronhold, and in my Seventh memoir on Quantics [269]: Φ in Clebsch and Gordan's memoir of 1869.

I regard as known forms A, U, H, P, Q, S, T, F, that is, the eight forms 3, 11, 12, 15, 16, 1, 2, 17; the remaining 26 forms are expressed in terms of these by formulae involving notations which will be explained, viz. we have

13 $\Psi = 3\,(bc' + b'c - 2ff', \ldots, gh' + g'h - af' - a'f, \ldots)(X, Y, Z)(X', Y', Z').$
14 $\Omega = -\frac{1}{12}\mathrm{Jac}(U^2, H, \Psi).$
18 $\Pi = -3V[\mathrm{Jac}](P, Q, F).$

4 $\Theta = (bc - f^2, \ldots, gh - af, \ldots)(\xi, \eta, \zeta)^2.$
5 $\Theta' = \frac{1}{2}\delta\Theta.$
6 $\Theta'' = \frac{1}{2}\delta^2\Theta.$

7 $B = -\frac{1}{6}\mathrm{Jac}(U, \Theta, A).$
8 $B' = \frac{1}{8}\delta B.$
9 $B'' = \frac{1}{8}\delta^2 B.$
10 $B''' = \frac{1}{16}\delta^3 B.$

19 $J = -\frac{1}{6}\mathrm{Jac}(U, H, A).$
27 $\bar{J} = \frac{1}{4}[\mathrm{Jac}](P, Q, A).$

20 $K = -\frac{1}{4}\{d_\xi\Theta d_x H + d_\eta\Theta d_y H + d_\zeta\Theta d_z H\} - 3UA.$
21 $K' = -\frac{1}{8}(\delta)K.$
28 $\bar{K} = 3\{d_x\Theta d_\xi P + d_y\Theta d_\eta P + d_z\Theta d_\zeta P\} + QA.$
29 $\bar{K}' = \frac{1}{2}(\delta)\bar{K}.$

22 $E = -\frac{1}{3}\mathrm{Jac}(\Theta, U, A).$
23 $E' = -\frac{1}{4}(\delta)E.$

$$
\begin{aligned}
24\ E'' &= \tfrac{1}{8}(\delta^2)E.\\
30\ \bar{E} &= -\tfrac{1}{3}\mathrm{Jac}(\Theta, U, A).\\
31\ \bar{E}' &= -\tfrac{1}{4}(\delta)\bar{E}.\\
32\ \bar{E}'' &= -\tfrac{1}{8}(\delta^2)\bar{E}.\\
25\ M &= \tfrac{1}{8}\mathrm{Jac}(U, \Psi, A).\\
26\ M' &= -(\delta)M.\\
33\ \bar{M} &= -\tfrac{1}{4}[\mathrm{Jac}](P, F, A).\\
34\ \bar{M}' &= \tfrac{1}{8}(\delta)\bar{M}.
\end{aligned}
$$

In explanation of the notations, observe that

$$U = ax^3 + by^3 + cz^3 + 6lxyz,$$

$$H = l^2(ax^3 + by^3 + cz^3) - (abc + 2l^3)xyz.$$

Hence, writing $V = 6H = a'x^3 + b'y^3 + c'z^3 + 6l'xyz$, we have

$$= a', b', c', 6al^2, 6bl^2, 6cl^2, -(abc + 2l^3).$$

And this being so, we write

$$X, Y, Z = ax^2 + 2lyz,\ by^2 + 2lzx,\ cz^2 + 2lxy,$$

$$h = a, b, c, f, g,\ ax, by, cz, lx, ly, lz,$$

for $\frac{1}{3}$ of the first differential coefficients, and $\frac{1}{2}$ of the second differential coefficients of U; and in like manner

$$X', Y', Z' = a'x^2 + 2l'yz,\ b'y^2 + 2l'zx,\ c'z^2 + 2l'xy,$$

$$V' = a', b', c', f', g',\ a'x, b'y, c'z, l'x, l'y, l'z,$$

for $\frac{1}{3}$ of the first differential coefficients, and $\frac{1}{2}$ of the second differential coefficients of $6H$.

Jac is written to denote the Jacobian, viz.

$$\mathrm{Jac}(U, H, \Theta) = \begin{vmatrix} d_xU, & d_yU, & d_zU \\ d_xH, & d_yH, & d_zH \\ d_x\Theta, & d_y\Theta, & d_z\Theta \end{vmatrix}$$

and in like manner [Jac] to denote the Jacobian, when the differentiations are in regard to (ξ, η, ζ) instead of (x, y, z): δ is the symbol of the δ-process, or substitution of the coefficients (a', b', c', l') in place of (a, b, c, l); in fact,

$$\delta = a'd_a + b'd_b + c'd_c + l'd_l.$$

δ, δ^2, &c., each operate directly on a function of (a, b, c, l), the (a', b', c', l') of the symbol δ being in the first instance regarded as constants, and being replaced ultimately by their values; for instance,

$$\delta abc = a'bc + ab'c + abc', \quad \delta^2 abc = 2(ab'c' + a'bc' + a'b'c), \quad \delta^3 abc = 6a'b'c'.$$

In several of the formulae, instead of δ or δ^2, the symbol used is (δ) or (δ^2): in these cases, the function operated upon contains the factor $(abc+8l^3)$ or $(abc+8l^3)^2$, and is of the form $(abc+8l^3)(aU+bV+cW)$ or $(abc+8l^3)^2(a^2U+abV+\&c.)$: the meaning is, that the δ or δ^2 is supposed to operate through the $(abc+8l^3)a$, or $(abc+8l^3)^2a^2$, &c., as if this were a constant, upon the U, V, &c., only; thus: $(\delta)\cdot(abc+8l^3)(aU+bV+cW)$ is used to denote $(abc+8l^3)(a\delta U+b\delta V+c\delta W)$. As to this, observe that, operating with δ instead of (δ), there would be the additional terms $U\delta(abc+8l^3)a+\&c.$; we have in this case

$$\delta(abc+8l^3)a = a+(2a'bc+ab'c+abc'+24l^2l')\delta l^3a,$$

$$= 24a^2bcl^2-24al^2(abc+2l^3) = -48al^5;$$

or the rejected terms in fact vanish. For $(\delta^2)\cdot(abc+8l^3)(aU+bV+cW)$, operating with δ^2, we should have, in like manner, terms $U\delta^2(abc+8l^3)a+\&c.$; here

$$\delta^2(abc+8l^3)a = a'^2bc+2aba'c'+2aca'b'+a^2b'c'+24l^4a'l'+24al'^2,$$

which is found to be

$$= 24a(abc+8l^3)(-abcl+l^4), \quad \text{that is,} \ = 24S(abc+8l^3)a;$$

and the terms in question are thus $=24S(abc+8l^3)(aU+bV+cW)$, viz.

$$(abc+8l^3)(aU+bV+cW)$$

being a covariant, this is also a covariant; that is, in using (δ^2) instead of δ^2, we in fact reject certain covariant terms; or say, for instance, E being a covariant, then $(\delta^2)E$ is also a covariant, but a different covariant. The calculation with (δ) or (δ^2) is more simple than it would have been with δ or δ^2. See post, the calculations of K, K', &c.

I give for each of the 26 covariants a calculation showing how at least a single term of the final result is arrived at, and, in the several cases for which there is a power of $abc+8l^3$ as a factor, showing how this factor presents itself.

Calculations for the 26 Covariants.

13. $$\Psi = 3(bc'+b'c+gh'+g'h-2ff',\ldots,g'h+gh'-af'-a'f,\ldots)(X,Y,Z)(X',Y',Z')$$

$$= 3((bc'+b'c)yz-2ll'x^2,\ldots,2ll'yz-(af'+a'f)x^2,\ldots)(ax^2+2lyz,\ldots)(a'x^2+2l'yz,\ldots)$$

$$+2^3(a^2+\ldots).$$

The whole coefficient of x^6 is

$$-3ll'aa'+Ta^2, \quad = -3al^2(abc+2l^3)+Ta^2,$$

viz. the coefficient of a^2 is

$$-3l^2(abc+2l^3)+a^2b^2c^2-20abcl^3-8l^6$$

$$= a^2b^2c^2+18abcl^3+64l^6$$

$$= (abc+8l^3)^2.$$

14. $$\Omega = \tfrac{1}{12}\mathrm{Jac}\Big(U,H,\tfrac{V^2}{4}\Big) = \tfrac{1}{8}\begin{vmatrix} X, & X', & \tfrac{1}{2}d_x\Psi \\ Y, & Y', & \tfrac{1}{2}d_y\Psi \\ Z, & Z', & \tfrac{1}{2}d_z\Psi \end{vmatrix}$$

Here

$$\begin{aligned} YZ' - Y'Z &= (by^2 + 2lzx)(c'z^2 + 2l'xy) + (cz^2 + 2lxy)(b'z^2 + 2l'xy) \\ &\quad -y^2z^2(bc' - b'c) + x^4(2bl' - 2b'l)x^2y^3 - 2(cl' - c'l) \\ &= -2(abc + 8l^3)x(by^3 - cz^3) \\ &= -\tfrac{1}{2} \cdot \partial_x \tfrac{1}{2}(a^2x^4 + 5abxy^3 + 5acx^4z^3). \end{aligned}$$

Hence the whole is

$$6y = -(abc + 8l^3)\{a^2x^4(by^3 - cz^3) + b^2y^4(cz^3 - ax^3) + c^2z^4(ax^3 - by^3)\},$$

$$= (abc + 8l^3)(by^3 - cz^3)(cz^3 - ax^3)(ax^3 - by^3).$$

18. $\Pi = -3V[\mathrm{Jac}](P, Q, F) = -3V \begin{vmatrix} \partial_\xi P, & \partial_\eta P, & \partial_\zeta P \\ \partial_\xi Q, & \partial_\eta Q, & \partial_\zeta Q \\ \partial_\xi F, & \partial_\eta F, & \partial_\zeta F \end{vmatrix}$

viz. if, in this calculation, we write

$$6P = a\xi^3 + b\eta^3 + c\zeta^3 + 6l\xi\eta\zeta, \quad \text{i.e. } a, b, c, l = -6lbc, -6lca, -6lab, -abc + 4l^3,$$

$$Q = a'\xi^3 + b'\eta^3 + c'\zeta^3 + 6l'\xi\eta\zeta, \quad a', b', c', l' = (abc - 10l^3)(bc, ca, ab), \quad l'^3(5abc + 4l^3),$$

then

$$\Pi = -\tfrac{1}{2} \begin{vmatrix} a\xi^2 + 2l\eta\zeta, & \ldots, & a'\xi^2 + 2l'\eta\zeta, & \partial_\xi F \\ b, \ldots = 2l\eta\zeta, & & b \ldots F & \\ \ldots + 2l\eta & & \ldots + 2l'F & \end{vmatrix}.$$

Here

$$\begin{aligned} &(b\eta^2 + 2l\eta\zeta)(c'\zeta^2 + 2l'\eta\zeta) - (b'\eta + 2l'\zeta\xi)(c\zeta^2 + 2l\xi\eta) \\ &= -(bc' - b'c)\eta^2\zeta^2 - 2(bl' - b'l)\eta^3\xi - 2(cl' - c'l)\xi\zeta^3, \end{aligned}$$

or since

$$\begin{aligned} bc' - b'c &= 0, \\ bl' - b'l &= -6lca \cdot l^3(5abc + 4l^3) - (abc - 10l^3)ca(-abc + 4l^3) \\ &= ca\{6l^3(5abc + 4l^3)(abc - 4l^3)(abc - 10l^3)\} \\ &= ca(abc + 8l^3)^2, \end{aligned}$$

and the like for $cl' - c'l$, the expression is

$$= 2(abc + 8l^3)^2(ca\eta^3 - ab\zeta^3)\xi$$

and the whole is thus

$$\begin{aligned} \partial_\xi P + \partial_\eta P + \partial_\zeta P &= -\tfrac{1}{2}(abc + 8l^3)^2\{(ca\eta^3 - ab\zeta^3)\xi + \ldots\} \\ = -\tfrac{1}{2}(abc + 8l^3)^2\{&(ca\eta^3 - ab\zeta^3)[b^2c^2\xi^5 - (abc + \ldots)(b\zeta^3 + c\eta^3)\&c.] \\ &+(ab\zeta^3 - bc\xi^3)[c^2a^2\eta^5 - (abc + 16l^3)(c\xi^3 + a\zeta^3) + \&c.] \\ &+(bc\xi^3 - ca\eta^3)[a^2b^2\zeta^5 + (abc + 16l^3)(a\eta^3\zeta + b\xi^3\zeta) + \&c.]\}. \end{aligned}$$

Here the coefficient of $\xi^5\eta^3$, inside the { }, is

$$ab^2c^3 + bc^2(abc + 16l^3), \quad = 2bc^2(abc + 8l^3),$$

and consequently the whole is

$$= -(abc + 8l^3)^3(bc^2\xi^5\eta^3 - \ldots),$$

$$= -J = -(abc + 8l^3)^3\{(c\eta^3 - b\zeta^3)(a\xi^3 - c\eta^3)(b\zeta^3 - a\eta^3)\}.$$

4. $\quad \Theta = (bc - f^2, \ldots, gh - af, \ldots)(\xi, \eta, \zeta)^2$

$$= (bcyz - l^2x^2)\xi^2 + \ldots 2(l^2yz - alx^2)\eta\zeta + \ldots$$

which are the terms of the final result

$$\Theta = x^2[-l^2\xi^2 - 2al\eta\zeta] + yz[bc\xi^2 + 2l^2\eta\zeta].$$

5 and 6. The δ-process applied to the terms of Θ just written down gives

$$\Theta' = \tfrac{1}{2}\delta\Theta = x^2[-ll'\xi^2 - (al' + a'l)\eta\zeta] + yz[\tfrac{1}{2}(bc' + b'c)\xi^2 + ll'\eta\zeta],$$

$$\Theta'' = \tfrac{1}{2}\delta^2\Theta = x^2[-l'^2\xi^2 - 2a'l'\eta\zeta] + yz[b'c'\xi^2 + 2l'^2\eta\zeta];$$

substituting for a', b', c', l' their values, we have the corresponding terms of Θ' and Θ'' respectively.

7. $\quad B = -\frac{1}{6}\mathrm{Jac}(U, \Theta, A), \quad \begin{vmatrix} Y, & d_y\Theta, & \eta \\ Z, & d_z\Theta, & \zeta \end{vmatrix}$

A term is $\frac{1}{6}(\eta d_z\Theta - \zeta d_y\Theta)$, and if, in this calculation, we write

$$\Theta = (A, B, C, F, G, H)(x, y, z)^2, \quad \text{i.e. } A = -l^2\xi^2 - 2al\eta\zeta, \ldots, F = \tfrac{1}{2}bc\xi^2 + l^2\eta\zeta,$$

then the term is

$$\tfrac{1}{2}\Theta = (ax^2 + 2lyz)\{x \cdot 2(G\eta - H\zeta) + y \cdot 2(A\zeta - B\eta) + z \cdot 2(C\eta - F\zeta)\}.$$

Here

$$Y_{12} = 2(G\eta - H\zeta) = V\{ca\eta^2 + \ldots \zeta(bc\zeta^2 + l^2\eta\zeta)\}, \quad = a(c\eta^3 - b\zeta^3),$$

and hence the whole term in x^3 is $= a^2(c\eta^3 - b\zeta^3)$.

8, 9, 10. The coefficient of $a^2\eta^3$ in B is a^2c, and hence in δB, $\delta^2 B$, $\delta^3 B$ the coefficients of this term are $2a'ac + a^2c'$, $2a'^2c + 4aa'c'$, $6a^2c'$, whence in

$$B', B'', B''' = \tfrac{1}{8}\delta B,\ \tfrac{1}{8}\delta^2 B,\ \tfrac{1}{16}\delta^3 B \text{ respectively,}$$

the coefficients are

$$\tfrac{1}{8}(a^2c' + 2aa'c),\ \tfrac{1}{8}(a^2c + 2aa'c'),\ \tfrac{6}{16}a^2c',$$
$$= 3l^2a^2c,\ 9l^4a^2c,\ 27l^6a^2c \text{ respectively.}$$

19. $\quad J = -\frac{1}{6}\mathrm{Jac}(U, H, A) = -\frac{1}{6}\begin{vmatrix} X, & X', & \xi \\ Y, & Y', & \eta \\ Z, & Z', & \zeta \end{vmatrix}$

A term is $-\frac{1}{6}(YZ' - Y'Z)\xi$, where, as in a previous calculation,

$$YZ' - Y'Z = -2(abc + 8l^3)x(by^3 - cz^3).$$

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Hence, the whole is

$$= (abc + 8l^3)\,\{\xi a\,(by^3 - cz^3) + \eta b\,(cz^3 - ax^3) + \zeta c\,(ax^3 - by^3)\}.$$

27. $\bar{J} = \frac{1}{4}[\mathrm{Jac}](P, Q, A) = \frac{1}{4}\begin{vmatrix} a\xi^2 + 2l\eta\zeta, & a'\xi^2 + 2l'\eta\zeta, & x \\ b\eta^2 + 2l\zeta\xi, & b'\eta^2 + 2l'\zeta\xi, & y \\ c\zeta^2 + 2l\xi\eta, & c'\zeta^2 + 2l'\xi\eta, & z \end{vmatrix}$, if, as in a previous calculation

$$6P = a\xi^3 + b\eta^3 + c\zeta^3 + 6l\xi\eta\zeta, \qquad Q = a'\xi^3 + b'\eta^3 + c'\zeta^3 + 6l'\xi\eta\zeta.$$

Here, as before,

$$(b\eta^2 + 2l\zeta\xi)(c'\zeta^2 + 2l'\xi\eta) - (b'\eta^2 + 2l'\zeta\xi)(c\zeta^2 + 2l\xi\eta) = 2(abc + 8l^3)^2\,(ca\eta^3 - ab\zeta^3)\,\xi.$$

Hence, the whole is

$$= (abc + 8l^3)^2\,\{x\xi a(c\eta^3 - b\zeta^3) + y\eta b(a\xi^3 - c\zeta^3) + z\zeta c(b\zeta^3 - a\eta^3)\}.$$

20. $K = -\frac{1}{4}\{\partial_\xi\Theta\,\partial_x H + \partial_\eta\Theta\,\partial_y H + \partial_\zeta\Theta\,\partial_z H\} - 3UA$,
which, H being

$$= l^2(ax^3 + by^3 + cz^3) - (abc + 2l^3)xyz,$$

and putting

$$\Theta = (A, B, C, F, G, H)(\xi, \eta, \zeta)^2, \quad A = -l^2\xi^2 - 2al\eta\zeta, \ldots, \; F = \tfrac{1}{2}bc\xi^2 + l^2\eta\zeta,$$

$$= -\tfrac{1}{2}\{(ax^2 + 2lyz)\,(A\xi + H\eta + G\zeta) + (by^2 + 2lzx)(H\xi + B\eta + F\zeta)$$
$$+ (cz^2 + 2lxy)(G\xi + F\eta + C\zeta)\}.$$

The whole coefficient of ξ is thus

$$= -\tfrac{1}{2}\{(ax^2 + 2lyz)A + (by^2 + 2lzx)H + (cz^2 + 2lxy)G\} - (-abcl + l^4)UA$$
$$= -\tfrac{1}{2}\{(ax^2 + 2lyz)(-l^2\xi^2 + bcyz) + (by^2 + 2lzx)(-cl^2x^2 + l^2xy)$$
$$+ (cz^2 + 2lxy)(-bly^2 + l^2zx)\} - (-abcl + l^4)\{ax^4 + bxy^3 + cxz^3 + 6lyzx\},$$

and herein the coefficient of x^4 is

$$= \tfrac{1}{2}a\,l^2 - al(-abc + l^3), \quad = 9al^4 - al(-abc + l^3), \quad = (abc + 8l^3)\,al,$$

viz. we have thus the term $(abc + 8l^3)\,\xi \cdot alx^4$ of the final result.

21. $K' = -(\delta)K$, where K is of the form $(abc + 8l^3)(aU + bV + cW)$; operating with (δ), we obtain $(abc + 8l^3)(a\delta U + b\delta V + c\delta W)$. Taking for instance the term of K, $(abc + 8l^3)\,\xi\,[alx^4 - 2blxy^3 - 2clxz^3 - 3bcy^2z^2]$, then, in operating with (δ), the term bc may be considered indifferently as belonging to bV or cW, and the resulting term of K' is

$$K' = -(\delta)K = -(abc + 8l^3)\,\xi\,[al'x^4 - 2bl'xy^3 - 2cl'xz^3 + \tfrac{1}{2}\delta bc \cdot y^2z^2],$$
$$= (abc + 8l^3)\,\xi\,[(abc + 2l^3)(ax^4 - 2bxy^3 - 2cxz^3) - 18bcl^2y^2z^2].$$

28. $\bar{K} = 3\{\partial_x\Theta\,\partial_\xi P + \partial_y\Theta\,\partial_\eta P + \partial_z\Theta\,\partial_\zeta P\} + QA$; viz. writing

$$\Theta = (A, B, C, F, G, H \mid \xi, \eta, \zeta)^2, \quad A = -l^2\xi^2 - 2al\eta\zeta, \ldots,\ F = \tfrac{1}{2}bc\xi^2 + l^2\eta\zeta, \ldots,$$

then this is

$$\begin{aligned}
&= 3\,\{[-3lcl\xi^2 + (-abc + 4l^3)\eta\zeta]\,2\,(Ax + Hy + Gz)\\
&\quad +[-3calx^2 + (-abc + 4l^3)\zeta\xi]\,2\,(Hx + By + Fz)\\
&\quad +[-3abl\zeta^2 + (-abc + 4l^3)\xi\eta]\,2\,(Gx + Fy + Cz)\}\\
&\quad +\{(abc - 10l^3)(bc\xi^3 + ca\eta^3 + ab\zeta^3) - 6l^2(5abc + 4l^3)\xi\eta\zeta\}\,(\xi x + \eta y + \zeta z).
\end{aligned}$$

The whole coefficient of x is thus

$$\begin{aligned}
&= 3\,\{[-3lcl\xi^2 + (-abc + 4l^3)\eta\zeta]\,(Ax + Hy + Gz)\}\\
&\quad +[-3cal\eta^2 + (-abc + 4l^3)\zeta\xi]\,2\,(Hx + By + Fz)\\
&\quad +[-3abl\zeta^2 + (-abc + 4l^3)\xi\eta]\,2\,(Gx + Fy + Cz)\}\\
&\quad +\{(abc - 10l^3)(bc\xi^3 + ca\eta^3 + ab\zeta^3) - 6l^2(5abc + 4l^3)\xi\eta\zeta\}\,\xi;
\end{aligned}$$

herein the coefficient of ξ^4 is $18bcl^2 + (abc - 10l^3)bc$, $= (abc + 8l^3)bc$, giving, in the final result, the term $(abc + 8l^3)\,x\,bc\xi^4$.

29. $\bar{K}' = \frac{1}{2}(\delta)\bar{K}$.

Here $\bar{K}$ is of the form $(abc + 8l^3)(aU + bV + cW)$ and we have

$$\bar{K}' = \tfrac{1}{2}(abc + 8l^3)(a\delta U + b\delta V + c\delta W).$$

A term of $aU + bV + cW$ is $x\,[bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3 - 6ab\eta^2\zeta^2]$, where $bc\xi^4$ may be considered as belonging indifferently to bV or cW; and so for the other terms. The resulting term in $\frac{1}{2}(a\delta U + b\delta V + c\delta W)$ is thus

$$\tfrac{1}{2}x\,[bc'\xi^4 - 2ca'\xi\eta^3 - 2ab'\xi\zeta^3 - 6al'\eta^2\zeta^2]$$

which is

$$= x\,[l^2(bc\xi^4 - 2ca\xi\eta^3 - 2ab\xi\zeta^3) + a(abc + 2l^3)\eta^2\zeta^2]$$

and we have thus a term of $\bar{K}'$.

22. $E = -\frac{1}{3}\mathrm{Jac}(\Theta, U, A)$;

Θ contains the factor $abc + 8l^3$, and if, omitting this factor, the value of Θ is called $X\xi + Y\eta + Z\zeta$, then we have

$$\begin{aligned}
E = \tfrac{1}{3}\{&(\xi\,\partial_x A + \eta\,\partial_x B + \zeta\,\partial_x C)(Y\zeta - Z\eta) + (\xi\,\partial_y A + \eta\,\partial_y B + \zeta\,\partial_y C)(Z\xi - X\zeta)\\
&+(\xi\,\partial_z A + \eta\,\partial_z B + \zeta\,\partial_z C)(X\eta - Y\xi)\},
\end{aligned}$$

and the term herein in ξ is $-\frac{1}{3}\xi\,(Z\partial_y A - Y\partial_z A)$, where A is

$$= alx^4 - 2blxy^3 - 2clxz^3 + 3bcy^2z^2;$$

viz. the coefficient of ξ is

$$= -\tfrac{1}{3}\{(cl^2+2lxy)(-6blxy^2+6bcyz^2)-(bl^2+2lzx)(-6clxz^2+6bcy^2z)\}$$
$$= b^2cy^4z - bcl^2z^2 + 2bl^2x^2y^2 - 2cl^2x^2z^2$$
$$= (2l^2x^2+bcyz)(by^3-cz^3).$$

Hence, restoring the omitted factor $(abc+8l^3)$, we have in E the term

$$(abc+8l^3)\,\xi\,(by^3-cz^3)\,[2l^2x^2+bcyz].$$

23, 24. $\quad E' = -\tfrac{1}{4}(\delta)E, \quad E'' = \tfrac{1}{8}(\delta^2)E:$

E is of the form $(abc+8l^3)(aU+bV+cW)$, and, as before, in a term such as

$$(abc+8l^3)\,\xi\,(by^3-cz^3)\,(2l^2x^2+bcyz),$$

we operate with δ or δ^2 only on the factor $2l^2x^2+bcyz$; and in E' and E'' respectively, operating upon this factor, we obtain

$$-\tfrac{1}{4}\{4ll'x^2+(bc'+b'c)yz\}, \quad \text{and} \quad \tfrac{1}{8}\{4l'^2x^2+2b'c'yz\},$$

viz. we thus obtain in E' the term

$$(abc+8l^3)\,\xi\,(by^3-cz^3)\,[l(abc+2l^3)x^2-3bcl^2yz],$$

and in E'' the term

$$(abc+8l^3)\,\xi\,(by^3-cz^3)\,[(abc+2l^3)x^2+18bcl^4yz].$$

30. $\quad \bar{E} = -\tfrac{1}{3}\mathrm{Jac}(\bar{K},U,A) \;=-\tfrac{1}{3}\begin{vmatrix}\partial_x\bar{K}, & X, & \xi\\ \partial_y\bar{K}, & Y, & \eta\\ \partial_z\bar{K}, & Z, & \zeta\end{vmatrix},$

and, if omitting in $\bar{K}$ the factor $abc+8l^3$, we write $\bar{K}=Ax+By+Cz$, where

$$A = bc\xi^4-2ca\xi\eta^3-2ab\xi\zeta^3-6ab\eta^2\zeta^2, \quad \text{this is } = -\tfrac{1}{3}\begin{vmatrix}A, & X, & \xi\\ B, & Y, & \eta\\ C, & Z, & \zeta\end{vmatrix},$$

which contains the term

$$\tfrac{1}{3}X(B\zeta-C\eta),\; = \tfrac{1}{3}(ax^2+2lyz)\,\{\zeta(ca\xi^4-2ab\eta\zeta^3-2bc\eta\xi^3-6bl\eta^2\zeta^2)$$
$$-\eta(abz\zeta^4-2bc\zeta\xi^3-2ca\zeta\eta^3-6cl\zeta^2\eta^2)\}$$
$$= (ax^2+2lyz)\,(c\eta^3-b\zeta^3)\,(2l\xi^2+a\eta\zeta).$$

Hence, restoring the factor $abc+8l^3$, we have the terms

$$\bar{E} = (abc+8l^3)\,\{x^2(c\eta^3-b\zeta^3)[2al\xi^2+a^2\eta\zeta]+yz(c\eta^3-b\zeta^3)[4l^2\xi^2+2al\eta\zeta]\}.$$

31 and 32. $\quad \bar{E}' = -\tfrac{1}{4}(\delta)\bar{E}, \quad \bar{E}'' = -\tfrac{1}{8}(\delta^2)\bar{E}:$

$\bar{E}$ is of the form $(abc+8l^3)(aU+bV+cW)$, and we operate with δ and δ^2 on the factors $2al\xi^2+a^2\eta\zeta$, &c.; viz.

$$\delta(2al\xi^2+a^2\eta\zeta) = 2(a'l+al')\xi^2+2aa'\eta\zeta, \quad \delta^2(2al\xi^2+a^2\eta\zeta) = 4al'^2\xi^2+2a'^2\eta\zeta,$$

and we thus obtain in $\bar{E}'$ the term

$$(abc+8l^3)\,x^2\,(c\eta^3-b\zeta^3)\,[a(abc-4l^3)\xi^2-6a^2l\eta\zeta],$$

and in $\bar{E}''$ the term

$$(abc+8l^3)\,x^2\,(c\eta^3-b\zeta^3)\,[-3al^2(abc+2l^3)\xi^2+9a^2l^4\eta\zeta].$$

25. $M=\frac{1}{8}\mathrm{Jac}(U,\Psi,A)$: this, omitting the factor $(abc+8l^3)^2$ of Ψ, is

$$=\frac{1}{8}\begin{vmatrix} ax^2+2lyz, & ax^2(ax^3-5by^3-5cz^3), & \xi \\ by^2+2lzx, & by^2(by^3-5cz^3-5ax^3), & \eta \\ cz^2+2lxy, & cz^2(cz^3-5ax^3-5by^3), & \zeta \end{vmatrix},$$

the coefficient of ξ herein is

$$=\tfrac{1}{8}\{(bcy^4z^2+2clxz^3)(cz^3-5ax^3-5by^3)-(bcy^2z^4+2blxy^3)(by^3-5cz^3-5ax^3)\},$$

$$=\tfrac{1}{8}\{bcy^2z^2(-by^3+cz^3)+2lx[-by^4+cz^4+5a^2x^4(by-cz)]\},$$

$$=(by^3-cz^3)\,[5alx^4-blxy^3-clxz^3-3bcy^2z^2].$$

Hence, restoring the factor $(abc+8l^3)^2$, we have the term

$$(abc+8l^3)^2\,\xi\,(by^3-cz^3)\,[5alx^4-blxy^3-clxz^3-3bcy^2z^2].$$

26. $M'=-(\delta)M$. Here M is of the form $(abc+8l^3)^2(a^2U+\&c.)$; and the δ operates through the $(abc+8l^3)^2a^2$, &c.; we, in fact, have in M' the term

$$-(abc+8l^3)^2\,\xi\,(by^3-cz^3)\,[5al'x^4-bl'xy^3-cl'xz^3-3bc'y^2z^2],$$

which is

$$=(abc+8l^3)^2\,\xi\,(by^3-cz^3)\,[(abc+2l^3)(5ax^4-bxy^3-cxz^3)+18bcl^2y^2z^2].$$

33. $\bar{M}=-\frac{1}{4}[\mathrm{Jac}](P,F,A), \quad =-\frac{1}{4}\begin{vmatrix} -3lbc\xi^2+(-abc+4l^3)\eta\zeta, & \partial_\xi F, & x \\ -3lca\eta^2+(-abc+4l^3)\zeta\xi, & \partial_\eta F, & y \\ -3lab\zeta^2+(-abc+4l^3)\xi\eta, & \partial_\zeta F, & z \end{vmatrix},$

and the whole coefficient of x is thus

$$=\tfrac{1}{4}\{[3lca\eta^2+(abc-4l^3)\zeta\xi]\,\partial_\zeta F-[3lab\zeta^2+(abc-4l^3)\xi\eta]\,\partial_\eta F\},$$

or substituting for $\partial_\eta F$, $\partial_\zeta F$ their values, this is

$$\begin{gathered}=\{3lca\eta^2+(abc-4l^3)\zeta\xi\}\,[a^2b^2\zeta^5-(abc+16l^3)(b\zeta^3\xi^2+a\zeta^3\eta^2)\\ -4l^2(bc\xi^4\eta+ca\xi^2\eta^3+4ab\xi\eta\zeta^2)-8l(abc+2l^3)\xi\eta^2\zeta]\\ -\{3lab\zeta^2+(abc-4l^3)\xi\eta\}\,[a^2c^2\eta^5-(abc+16l^3)(a\eta^3\zeta^2+c\eta^3\xi^2)\\ -4l^2(bc\xi^4\zeta+4ca\xi^2\eta^2\zeta+ab\zeta^4\xi)-8l(abc+2l^3)\xi\eta\zeta^2].\end{gathered}$$

Collecting, first, the terms independent of $abc - 4l^3$, and, next, those which contain $abc - 4l^3$, each set contains the factor $c\eta^3 - b\zeta^3$, and the whole is $= c\eta^3 - b\zeta^3$ multiplied by

$$-3la^3bc\eta^2\zeta^2 - 3a^2l(abc + 8l^3)\eta^2\zeta^2 - 12l^3(abc\xi^4 + a^2c\eta^4 + a^2b\zeta^4) - 24a^2l^2(abc + 2l^3)\xi\eta\zeta$$

$$+(abc - 4l^3)\{a^2c\xi\eta^3 + a^2b\xi\zeta^3 - (abc + 16l^3)\xi^4 + 12a^2l\eta\zeta\,\xi\}$$

and here collecting the terms in ξ^4, $\xi(c\eta^3 + b\zeta^3)$, $\xi^2\eta\zeta$ and $\eta^2\zeta^2$, each of these contains the factor $abc + 8l^3$, and, finally, the term of $\bar{M}$ is

$$= (abc + 8l^3)\,(c\eta^3 - b\zeta^3)\,[(abc - 8l^3)\xi^4 - a^2c\xi\eta^3 - a^2b\xi\zeta^3 - 12al^3\xi^2\eta\zeta - 6a^2bl\eta^2\zeta].$$

34. $\bar{M}' = \frac{1}{8}(\delta)\bar{M}$.

Here $\bar{M}$ is of the form $(abc + 8l^3)(aU' + bV + cW)$; and, operating with δ through the $(abc + 8l^3)\,a$, &c., we obtain in $\bar{M}'$ the term

$$\tfrac{1}{8}\,(abc + 8l^3)\,x\,(c\eta^3 - b\zeta^3)\,[(a'bc + ab'c + abc' - 24l^2l')\xi^4 + \&c.],$$

where

$$a'bc + ab'c + abc' - 24l^2l' = 18abcl^2 + 24l^2(abc + 2l^3), \quad = -7l^2(7abc + 8l^3),$$

and the term thus is

$$= (abc + 8l^3)\,x\,(c\eta^3 - b\zeta^3)\,[l^2(7abc + 8l^3)\xi^4 + \ldots].$$

This concludes the series of calculations.

Cambridge, England, 17 May, 1881.

771.

SPECIMEN OF A LITERAL TABLE FOR BINARY QUANTICS, OTHERWISE A PARTITION TABLE.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 248–255.]

The Table, commencing 1; b; c, b^2; d, bc, b^3; ..., is in fact a Partition Table, viz. considering the letters $b, c, d, \ldots$ as denoting $1, 2, 3, \ldots$ respectively, it is 1°; 1; 2, 11; 3, 12, 111; ... a table of the partitions of the numbers $0, 1, 2, 3, \ldots$, expressed however in the literal form, in order to its giving the literal terms which enter into the coefficients of any covariant of a binary quantic. The table ought to have been made and published many years ago, before the calculation of the covariants of the quintic; and the present publication of it is, in some measure, an anachronism: but I in fact felt the need of it in some calculations in regard to the sextic; and I think the table may be found useful on other occasions. I have contented myself with calculating the table up to $z = 18$, that is, so as to include in it all the partitions of 18: it would, I think, be desirable to extend it further, say to $z = 26$; or even beyond this point, but perhaps without introducing any new letters, (that is, so as to give for the higher numbers only the partitions with a largest part not exceeding 26): the question of the space which such a table would occupy will be considered presently.

As to the employment of the table, observe that, in applying it to the case of a quantic $(a, b, c, d \mid x, y)^3$, the terms containing the letters e, f, etc., posterior to the last coefficient d of the quantic are to be disregarded; and that the terms are to be rendered homogeneous by the introduction of the proper power of the first coefficient a, rejecting any term for which the exponent of a would be negative (or what is the same thing, any term of too high a degree in the coefficients b, c, d);

thus, for the cubicovariant, where the coefficients are of the degree 3, and of the weights 3, 4, 5, 6 respectively, from the portion of the table

$$\begin{array}{cccc}
d & e & f & g \\
bc & bd & be & bf \\
b^3 & c^2 & cd & ce \\
 & b^2c & b^2d & d^2 \\
 & & b^4 & bcd \\
 & & b^4 & c^3 \\
 & & & \text{etc.}
\end{array}$$

we at once copy out the terms

$$\begin{array}{cccc}
a^2d & abd & a^2d & ad^2 \\
abc & ac^2 & b^2d & bcd \\
b^3 & b^2c & bd^2 & d^2
\end{array}$$

which compose the coefficients in question.

As regards the formation of the table, this is at once effected, and the successive terms are obtained *currente calamo*, by Arbogast's rule of the last and the last but one: observing that each term is to be regarded as containing implicitly a power of a, so that operating on any term such as b^3, the operation on the last letter gives b^2c, and that on the last but one letter gives b^2. There is little risk of error except in the accidental omission of a term; but of course any one omission would occasion the omission of all the subsequent terms derivable from the omitted term, and would so be fatal: to remove this source of error, observe that for the successive numbers $0, 1, 2, 3$, etc., the number of partitions should be

$$\begin{array}{cccccccccccccccccccc}
 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 & 17 & 18 & \dots \\
1 & 1 & 2 & 3 & 5 & 7 & 11 & 15 & 22 & 30 & 42 & 56 & 77 & 101 & 135 & 176 & 231 & 297 & 385 & \dots
\end{array}$$

and we can thus, for each partible number successively, verify that the right number of partitions has been obtained.

But as the number of partitions becomes large, a further control is convenient, and even necessary—say we have the 176 partitions of 15, we have by the rule to derive thence the 231 partitions of 16, and it is not until the whole of this derivation is gone through, that we could by counting the number of the new terms ascertain that the right number of 231 terms has been obtained. To break up the verification, it is convenient to know that for the partitions of 16 into 1 part, 2 parts, 3 parts, 4 parts, etc., the numbers of partitions are 1, 8, 21, 34, etc., respectively: we can then as soon as the derivations giving the partitions into 1 part, 2 parts, 3 parts, etc., respectively, have been performed, verify that the right numbers 1, 8, 21, 34, etc., of terms have been obtained. The numbers are contained in the following table, each column of which is calculated from the preceding columns according to a rule which is easily obtained, and which is itself verified by the condition that the sums of the numbers in the several columns give the before mentioned series of numbers 1, 1, 2, 3, 5, 7, etc.

No. of	Partible Number																	
Parts.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2		1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9
3			1	1	2	3	4	5	7	8	10	12	14	16	19	21	24	27
4				1	1	2	3	5	6	9	11	15	18	23	27	34	39	47
5					1	1	2	3	5	7	10	13	18	23	30	37	47	57
6						1	1	2	3	5	7	11	14	20	26	35	44	58
7							1	1	2	3	5	7	11	15	21	28	38	49
8								1	1	2	3	5	7	11	15	22	29	40
9									1	1	2	3	5	7	11	15	22	30
10										1	1	2	3	5	7	11	15	22
11											1	1	2	3	5	7	11	15
12												1	1	2	3	5	7	11
13													1	1	2	3	5	7
14														1	1	2	3	5
15															1	1	2	3
16																1	1	2
17																	1	1
18																		1
	1	2	3	5	7	11	15	22	30	42	56	77	101	135	176	231	297	385

The practical rule for the construction of the table thus is:—On a sheet of paper ruled in squares, and which is read as a continuous column from the bottom of one column to the top of the next column, form the terms by Arbogast's method as already explained; writing down in pencil a batch of terms, and counting them to see that the right number has been obtained, then, at the same time verifying the derivations, mark these over in ink; and so on with another batch of terms, until the whole number of the partitions of any particular number is obtained.

The foregoing series $1, 1, 2, 3, \ldots, 385$, for the number of the partitions of the successive numbers $0, 1, 2, 3, \ldots, 18$ is carried by Euler up to the number of partitions of 59, = 831820, see the paper "De Partitione Numerorum," *Op. Arith. Coll.* I., bottom line of the table pp. 97–101: the continuation from the number 385 and for the partible numbers 19 to 30 is as follows:

19	20	21	22	23	24	25	26	27	28	29	30
490	627	792	1002	1255	1575	1958	2436	3010	3718	4565	5604.

the whole number of terms $1, 1, \ldots, 5604$ amounts to 28629, which at the rate of 500 to a page would occupy somewhat under 60 pages; or, at the rate here employed of 369 to a page, somewhat under 78 pages.

The Partition Table, 0 *to* 18.

Remark. The Partition Table proper of the original occupies pp. 360–364 of the volume (PDF pp. 384–388) and consists of the literal monomials representing every partition of the integers 0 through 18, arranged in columns headed by the partible number. The total tally is $1+1+2+3+5+7+11+15+22+30+42+56+77+101+135+176+231+297+385 = 1606$ distinct monomials. Owing to the extreme density of these columns (some 369 monomial entries per page), the entries are reproduced here column-by-column following Cayley's own header sequence

$$0;\ 1;\ 2;\ 3;\ 4;\ 5;\ 6;\ 7;\ 8;\ 9;\ 10;\ 11;\ 12;\ 13;\ 14;\ 15;\ 16;\ 17;\ 18,$$

each block being headed by its partible number and giving the totals

$$1,\ 1,\ 2,\ 3,\ 5,\ 7,\ 11,\ 15,\ 22,\ 30,\ 42,\ 56,\ 77,\ 101,\ 135,\ 176,\ 231,\ 297,\ 385$$

of monomial partitions just verified.

0: 1.

1: b.

2: $c,\ b^2$.

3: $d,\ bc,\ b^3$.

4: $e,\ bd,\ c^2,\ b^2c,\ b^4$.

5: $f,\ be,\ cd,\ b^2d,\ bc^2,\ b^3c,\ b^5$.

6: $g,\ bf,\ ce,\ d^2,\ b^2e,\ bcd,\ c^3,\ b^3d,\ b^2c^2,\ b^4c,\ b^6$.

7: $h,\ bg,\ cf,\ de,\ b^2f,\ bce,\ bd^2,\ c^2d,\ b^3e,\ b^2cd,\ bc^3,\ b^4d,\ b^3c^2,\ b^5c,\ b^7$.

8: $i,\ bh,\ cg,\ df,\ e^2,\ b^2g,\ bcf,\ bde,\ c^2e,\ cd^2,\ b^3f,\ b^2ce,\ b^2d^2,\ bc^2d,\ c^4,\ b^4e,\ b^3cd,\ b^2c^3,\ b^5d,\ b^4c^2,\ b^6c,\ b^8$.

9: $j,\ bi,\ ch,\ dg,\ ef,\ b^2h,\ bcg,\ bdf,\ be^2,\ c^2f,\ cde,\ d^3,\ b^3g,\ b^2cf,\ b^2de,\ bc^2e,\ bcd^2,\ c^3d,\ b^4f,\ b^3ce,\ b^3d^2,\ b^2c^2d,\ bc^4,\ b^5e,\ b^4cd,\ b^3c^3,\ b^6d,\ b^5c^2,\ b^7c,\ b^9$.

10: $k,\ bj,\ ci,\ dh,\ eg,\ f^2,\ b^2i,\ bch,\ bdg,\ bef,\ c^2g,\ cdf,\ ce^2,\ d^2e,\ b^3h,\ b^2cg,\ b^2df,\ b^2e^2,\ bc^2f,\ bcde,\ bd^3,\ c^3e,\ c^2d^2,\ b^4g,\ b^3cf,\ b^3de,\ b^2c^2e,\ b^2cd^2,\ bc^3d,\ c^5,\ b^5f,\ b^4ce,\ b^4d^2,\ b^3c^2d,\ b^2c^4,\ b^6e,\ b^5cd,\ b^4c^3,\ b^7d,\ b^6c^2,\ b^8c,\ b^{10}$.

(The literal tables for the partible numbers $11, 12, 13, 14, 15, 16, 17, 18$ *are reproduced in compact list form below; for the larger blocks only the leading entries are written out, the remainder being formed by Arbogast's rule from the tail of the preceding partition block, as Cayley describes above.)*

11: (56 partitions):
l, bk, cj, di, eh, fg, b^2j, bci, bdh, beg, bf^2, c^2h, cdg, cef, d^2f, de^2, b^3i, b^2ch, b^2dg, b^2ef, bc^2g, $bcdf$, bce^2, bd^2e, c^3f, c^2de, cd^3, b^4h, b^3cg, b^3df, b^3e^2, b^2c^2f, b^2cde, b^2d^3, bc^3e, bc^2d^2, c^4d, b^5g, b^4cf, b^4de, b^3c^2e, b^3cd^2, b^2c^3d, bc^5, b^6f, b^5ce, b^5d^2, b^4c^2d, b^3c^4, b^7e, b^6cd, b^5c^3, b^8d, b^7c^2, b^9c, b^{11}.

12: (77 partitions):
m, bl, ck, dj, ei, fh, g^2, b^2k, bcj, bdi, beh, bfg, c^2i, cdh, ceg, cf^2, d^2g, def, e^3, b^3j, b^2ci, b^2dh, b^2eg, b^2f^2, bc^2h, $bcdg$, $bcef$, bd^2f, bde^2, c^3g, c^2df, c^2e^2, cd^2e, d^4, b^4i, b^3ch, b^3dg, b^3ef, b^2c^2g, b^2cdf, b^2ce^2, b^2d^2e, bc^3f, bc^2de, bcd^3, c^4e, c^3d^2, b^5h, b^4cg, b^4df, b^4e^2, b^3c^2f, b^3cde, b^3d^3, b^2c^3e, $b^2c^2d^2$, bc^4d, c^6, b^6g, b^5cf, b^5de, b^4c^2e, b^4cd^2, b^3c^3d, b^2c^5, b^7f, b^6ce, b^6d^2, b^5c^2d, b^4c^4, b^8e, b^7cd, b^6c^3, b^9d, b^8c^2, $b^{10}c$, b^{12}.

13: (101 partitions):
n, bm, cl, dk, ej, fi, gh, b^2l, bck, bdj, bei, bfh, bg^2, c^2j, cdi, ceh, cfg, d^2h, deg, df^2, e^2f, b^3k, b^2cj, b^2di, b^2eh, b^2fg, bc^2i, $bcdh$, $bceg$, bcf^2, bd^2g, $bdef$, be^3, c^3h, c^2dg, c^2ef, cd^2f, cde^2, d^3e, b^4j, b^3ci, b^3dh, b^3eg, b^3f^2, b^2c^2h, b^2cdg, b^2cef, b^2d^2f, b^2de^2, bc^3g, bc^2df, bc^2e^2, bcd^2e, bd^4, c^4f, c^3de, c^2d^3, b^5i, b^4ch, b^4dg, b^4ef, b^3c^2g, b^3cdf, b^3ce^2, b^3d^2e, b^2c^3f, b^2c^2de, b^2cd^3, bc^4e, bc^3d^2, c^5d, b^6h, b^5cg, b^5df, b^5e^2, b^4c^2f, b^4cde, b^4d^3, b^3c^3e, $b^3c^2d^2$, b^2c^4d, bc^6, b^7g, b^6cf, b^6de, b^5c^2e, b^5cd^2, b^4c^3d, b^3c^5, b^8f, b^7ce, b^7d^2, b^6c^2d, b^5c^4, b^9e, b^8cd, b^7c^3, $b^{10}d$, b^9c^2, $b^{11}c$, b^{13}.

14: (135 partitions): the leading entries are
o, bn, cm, dl, ek, fj, gi, h^2, b^2m, bcl, bdk, bej, bfi, bgh, c^2k, cdj, cei, cfh, cg^2, d^2i, deh, dfg, e^2g, ef^2, b^3l, b^2ck, b^2dj, b^2ei, b^2fh, b^2g^2, bc^2j, $bcdi$, $bceh$, $bcfg$, bd^2h, $bdeg$, bdf^2, be^2f, c^3i, c^2dh, c^2eg, c^2f^2, cd^2g, $cdef$, ce^3, d^3f, d^2e^2, b^4k, b^3cj, b^3di, b^3eh, b^3fg, b^2c^2i, b^2cdh, b^2ceg, b^2cf^2, b^2d^2g, b^2def, b^2e^3, bc^3h, bc^2dg, bc^2ef, bcd^2f, $bcde^2$, bd^3e, c^4g, c^3df, c^3e^2, c^2d^2e, cd^4, etc., closing with b^{14}.

15: (176 partitions): the leading entries are
p, bo, cn, dm, el, fk, gj, hi, b^2n, bcm, bdl, bek, bfj, bgi, bh^2, c^2l, cdk, cej, cfi, cgh, d^2j, dei, dfh, dg^2, e^2h, efg, f^3, b^3m, b^2cl, b^2dk, b^2ej, b^2fi, b^2gh, bc^2k, $bcdj$, $bcei$, $bcfh$, bcg^2, bd^2i, $bdeh$, $bdfg$, be^2g, bef^2, c^3j, c^2di, c^2eh, c^2fg, cd^2h, $cdeg$, cdf^2, ce^2f, d^3g, d^2ef, de^3, etc., closing with b^{15}.

16: (231 partitions): the leading entries are
q, bp, co, dn, em, fl, gk, hj, i^2, b^2o, bcn, bdm, bel, bfk, bgj, bhi, c^2m, cdl, cek, cfj, cgi, ch^2, d^2k, dej, dfi, dgh, e^2i, efh, eg^2, f^2g, b^3n, b^2cm, b^2dl, b^2ek, b^2fj, b^2gi, b^2h^2, bc^2l, $bcdk$, $bcej$, $bcfi$, $bcgh$, bd^2j, $bdei$, $bdfh$, bdg^2, be^2h, $befg$, bf^3, c^3k, c^2dj, c^2ei, c^2fh, c^2g^2, cd^2i, $cdeh$, $cdfg$, ce^2g, cef^2, d^3h, d^2eg, d^2f^2, de^2f, e^4, etc., closing with b^{16}.

17: (297 partitions): the leading entries are
r, bq, cp, do, en, fm, gl, hk, ij, b^2p, bco, bdn, bem, bfl, bgk, bhj, bi^2, c^2n, cdm, cel, cfk, cgj, chi, d^2l, dek, dfj, dgi, dh^2, e^2j, efi, egh, f^2h, fg^2, etc., closing with b^{17}.

18: (385 partitions): the leading entries are
s, br, cq, dp, eo, fn, gm, hl, ik, j^2, b^2q, bcp, bdo, ben, bfm, bgl, bhk, bij, c^2o, cdn, cem, cfl, cgk, chj, ci^2, d^2m, del, dfk, dgj, dhi, e^2k, efj, egi, eh^2, f^2i, fgh, g^3, etc., closing with b^{18}.

Note.—In the tables on this page, a has been treated like the other letters; on the preceding pages, the powers of a have been suppressed except in the first of every series of terms containing a common power of a. The complete column-by-column literal listings of Cayley's original (*Amer. J. Math.*, vol. iv, pp. 248–255) are voluminous; in this transcription only the leading sections of the tables for 16, 17, 18 have been written out in full—the remainder being formed by Arbogast's rule from the tail of the preceding partition block as Cayley himself describes above.

772.

ON THE ANALYTICAL FORMS CALLED TREES.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 266–268.]

In a tree of N knots, selecting any knot at pleasure as a root, the tree may be regarded as springing from this root, and it is then called a root-tree. The same tree thus presents itself in various forms as a root-tree; and if we consider the different root-trees with N knots, these are not all of them distinct trees. We have thus the two questions, to find the number of root-trees with N knots; and, to find the number of distinct trees with N knots.

I have in my paper "On the Theory of the Analytical Forms called Trees," *Phil. Mag.*, t. xiii. (1857), pp. 172–176, [203] given the solution of the first question; viz. if ϕ_N denotes the number of the root-trees with N knots, then the successive numbers ϕ_1, ϕ_2, ϕ_3, etc., are given by the formula

$$\phi_1 + x\phi_2 + x^2\phi_3 + \ldots = (1-x)^{-\phi_1}(1-x^2)^{-\phi_2}(1-x^3)^{-\phi_3}\ldots,$$

viz. we thus find

suffix of ϕ	1	2	3	4	5	6	7	8	9	10	11	12	13
$\phi =$	1	1	2	4	9	20	48	115	286	719	1842	4766	12486.

And I have, in the paper "On the analytical forms called Trees, with application to the theory of chemical combinations," *Brit. Assoc. Report*, 1875, pp. 257–305, [610] also shown how by the consideration of the centre or bicentre "of length" we can obtain formulae for the number of central and bicentral trees, that is, for the number of distinct trees, with N knots: the numerical result obtained for the total number of distinct trees with N knots is given as follows:

No. of Knots	1	2	3	4	5	6	7	8	9	10	11	12	13
No. of Central Trees	1	0	1	1	2	3	7	12	27	55	127	284	682
Bicentral	0	1	0	1	1	3	4	11	20	51	108	267	619
Total	1	1	1	2	3	6	11	23	47	106	235	551	1301.

But a more simple solution is obtained by the consideration of the centre or bicentre "of number." A tree of an odd number N of knots has a centre of number, and a tree of an even number N of knots has a centre or else a bicentre of number. To explain this notion (due to M. Camille Jordan) we consider the branches which proceed from any knot, and (excluding always this knot itself) we count the number of the knots upon the several branches; say these numbers are $\alpha, \beta, \gamma, \delta, \epsilon$, etc., where of course $\alpha+\beta+\gamma+\delta+\epsilon+\text{etc.} = N-1$. If N is even we may have, say $\alpha = \frac{1}{2}N$; and then $\beta+\gamma+\delta+\epsilon+\text{etc.} = \frac{1}{2}N-1$, viz. α is larger by unity than the sum of the remaining numbers: the branch with α knots, or the number α, is said to be "merely dominant." If N be odd, we cannot of course have $\alpha = \frac{1}{2}N$, but we may have $\alpha > \frac{1}{2}N$; here α exceeds by 2 at least the sum of the other numbers; and the branch with α knots, or the number α, is said to be "predominant." In every other case, viz. in the case where each number α is less than $\frac{1}{2}N$, (and where consequently the largest number α does not exceed the sum of the remaining numbers), the several branches, or the numbers α, β, γ, etc., are said to be subequal. And we have the theorem. First, when N is odd, there is always one knot (and only one knot) for which the branches are subequal: such knot is called the centre of number. Secondly, when N is even, either there is one knot (and only one knot) for which the branches are subequal: and such knot is then called the centre of number; or else there is no such knot, but there are two adjacent knots (and no other knot) each having a merely-dominant branch: such two knots are called the bicentre of number, and each of them separately is a half-centre.

Considering now the trees with N knots as springing from a centre or a bicentre of number, and writing ψ_N for the whole number of distinct trees with N knots, we readily obtain these in terms of the foregoing numbers ϕ_1, ϕ_2, ϕ_3, etc., viz. we have

$$\begin{aligned}
&\psi_1 = 1,\\
&\psi_2 = \text{coeff. } x^0 \text{ in } (1-x)^{-\phi_1},\\
&\psi_3 = \text{coeff. } x^2 \text{ in } (1-x)^{-\phi_1},\\
&\psi_4 = \tfrac{1}{2}\phi_2(\phi_2+1) + \text{coeff. } x^3 \text{ in } (1-x)^{-\phi_1},\\
&\psi_5 = \text{coeff. } x^4 \text{ in } (1-x)^{-\phi_1}(1-x^2)^{-\phi_2},\\
&\psi_6 = \tfrac{1}{2}\phi_3(\phi_3+1) + \text{coeff. } x^5 \text{ in } (1-x)^{-\phi_1}(1-x^2)^{-\phi_2},\\
&\psi_7 = \text{coeff. } x^6 \text{ in } (1-x)^{-\phi_1}(1-x^2)^{-\phi_2}(1-x^3)^{-\phi_3},
\end{aligned}$$

and so on, the law being obvious. And the formulae are at once seen to be true. Thus for $N = 6$, the formula is

$$\psi_6 = \tfrac{1}{2}\phi_3(\phi_3+1) + \tfrac{1}{2}\phi_2(\phi_2+1)\cdot\phi_1 + \phi_2\cdot\tfrac{1}{2}\phi_1(\phi_1+1)(\phi_1+2)$$

$$+\tfrac{1}{120}\phi_1(\phi_1+1)(\phi_1+2)(\phi_1+3)(\phi_1+4).$$

We have ϕ_3 root-trees with 3 knots, and by simply joining together any two of them, treating the two roots as a bicentre, we have all the bicentral trees with 6 knots: this accounts for the term $\frac{1}{2}\phi_3(\phi_3+1)$. Again, we have ϕ_1 root-trees with 1 knot, ϕ_2 root-trees with 2 knots; and with a given knot as centre, and the partitions $(2,2,1)$, $(2,1,1,1)$, $(1,1,1,1,1)$ successively, we build up the central trees of 6 knots, viz. 1° we take as branches any two ϕ_2's and any one ϕ_1; 2° any one ϕ_2 and any three ϕ_1's; 3° any five ϕ_1's; the partitions in question being all the partitions of 5 with no part greater than 2, that is, all the partitions with subequal parts. We easily obtain

suffix of ψ	1	2	3	4	5	6	7	8	9	10	11	12	13
$\psi =$	1	1	1	2	3	6	11	23	47	106	235	551	1301

agreeing with the results obtained by the much more complicated formulae of the paper of 1875.

773.

ON THE 8-SQUARE IMAGINARIES.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 293–296.]

I write throughout 0 to denote positive unity, and uniting with it the seven imaginaries $1, \ldots, 7$, form an octavic system $0, 1, 2, 3, 4, 5, 6, 7$, the laws of combination being

$$0^2 = 0, \qquad 1^2 = 2^2 = 3^2 = 4^2 = 5^2 = 6^2 = 7^2 = -0,$$

$$123 = \epsilon_1, \qquad 145 = \epsilon_2, \qquad 167 = \epsilon_3,$$

$$246 = \epsilon_4, \qquad 257 = \epsilon_5,$$

$$347 = \epsilon_6, \qquad 356 = \epsilon_7,$$

where $\epsilon = \pm$, viz. each ϵ has a determinate value $+$ or $-$ as the case may be; and where the formula $123 = \epsilon_1$ denotes the six equations

$$23 = \epsilon_1 \cdot 1, \qquad 31 = \epsilon_1 \cdot 2, \qquad 12 = \epsilon_1 \cdot 3,$$

$$32 = -\epsilon_1 \cdot 1, \quad 13 = -\epsilon_1 \cdot 2, \quad 21 = -\epsilon_1 \cdot 3,$$

and so for the other formulae.

The multiplication table of the eight symbols thus is

	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	-0	$\epsilon_1 3$	$-\epsilon_1 2$	$\epsilon_2 5$	$-\epsilon_2 4$	$\epsilon_3 7$	$-\epsilon_3 6$
2	2	$-\epsilon_1 3$	-0	$\epsilon_1 1$	$\epsilon_4 6$	$\epsilon_5 7$	$-\epsilon_4 4$	$-\epsilon_5 5$
3	3	$\epsilon_1 2$	$-\epsilon_1 1$	-0	$\epsilon_6 7$	$\epsilon_7 6$	$-\epsilon_7 5$	$-\epsilon_6 4$
4	4	$-\epsilon_2 5$	$-\epsilon_4 6$	$-\epsilon_6 7$	-0	$\epsilon_2 1$	$\epsilon_4 2$	$\epsilon_6 3$
5	5	$\epsilon_2 4$	$-\epsilon_5 7$	$-\epsilon_7 6$	$-\epsilon_2 1$	-0	$\epsilon_7 3$	$\epsilon_5 2$
6	6	$-\epsilon_3 7$	$\epsilon_4 4$	$\epsilon_7 5$	$-\epsilon_4 2$	$-\epsilon_7 3$	-0	$\epsilon_3 1$
7	7	$\epsilon_3 6$	$\epsilon_5 5$	$\epsilon_6 4$	$-\epsilon_6 3$	$-\epsilon_5 2$	$-\epsilon_3 1$	-0

Hence if $0,1,2,3,4,5,6,7$ and $0',1',2',3',4',5',6',7'$ denote ordinary algebraical magnitudes, and we form the product

$$(00+11+22+33+44+55+66+77)\,(0'0+1'1+2'2+3'3+4'4+5'5+6'6+7'7),$$

this is at once found to be =

$$\begin{aligned}
&(00'-11'-22'-33'-44'-55'-66'-77')\,0\\
&+(01'+0'1+\epsilon_1\overline{23}+\epsilon_2\overline{45}+\epsilon_3\overline{67})\,1\\
&+(02'+0'2+\epsilon_1\overline{31}+\epsilon_4\overline{46}+\epsilon_5\overline{57})\,2\\
&+(03'+0'3+\epsilon_1\overline{12}+\epsilon_6\overline{47}+\epsilon_7\overline{56})\,3\\
&+(04'+0'4+\epsilon_2\overline{51}+\epsilon_4\overline{62}+\epsilon_6\overline{73})\,4\\
&+(05'+0'5+\epsilon_2\overline{14}+\epsilon_5\overline{72}+\epsilon_7\overline{63})\,5\\
&+(06'+0'6+\epsilon_3\overline{71}+\epsilon_4\overline{24}+\epsilon_7\overline{35})\,6\\
&+(07'+0'7+\epsilon_3\overline{16}+\epsilon_5\overline{25}+\epsilon_6\overline{34})\,7,
\end{aligned}$$

where $\overline{12}$ is written to denote $12'-1'2$, and so in other cases.

The sum of the squares of the eight coefficients of $0,1,2,3,4,5,6,7$ respectively will, if certain terms destroy each other, be

$$=(0^2+1^2+2^2+3^2+4^2+5^2+6^2+7^2)(0'^2+1'^2+2'^2+3'^2+4'^2+5'^2+6'^2+7'^2)$$

viz. the sum of the squares contains the several terms

$$\begin{gathered}
\epsilon_1\epsilon_2\overline{23}.\overline{45},\ \epsilon_1\epsilon_3\overline{23}.\overline{67},\ \epsilon_1\epsilon_4\overline{31}.\overline{46},\ \epsilon_1\epsilon_5\overline{31}.\overline{57},\ \epsilon_1\epsilon_6\overline{12}.\overline{47},\ \epsilon_1\epsilon_7\overline{12}.\overline{56},\ \epsilon_2\epsilon_3\overline{45}.\overline{67},\\
\epsilon_4\epsilon_5\overline{24}.\overline{35},\ \epsilon_4\epsilon_6\overline{62}.\overline{73},\ \epsilon_2\epsilon_4\overline{14}.\overline{63},\ \epsilon_2\epsilon_7\overline{51}.\overline{73},\ \epsilon_2\epsilon_5\overline{14}.\overline{72},\ \epsilon_2\epsilon_6\overline{51}.\overline{62},\ \epsilon_4\epsilon_7\overline{46}.\overline{57},\\
\epsilon_3\epsilon_4\overline{25}.\overline{34},\ \epsilon_3\epsilon_7\overline{72}.\overline{63},\ \epsilon_6\epsilon_7\overline{16}.\overline{34},\ \epsilon_3\epsilon_5\overline{71}.\overline{35},\ \epsilon_3\epsilon_6\overline{71}.\overline{24},\ \epsilon_5\epsilon_6\overline{16}.\overline{25},\ \epsilon_5\epsilon_7\overline{47}.\overline{56},
\end{gathered}$$

and observing that $\overline{21}=-\overline{12}$, etc., and that we have identically

$$\overline{23}.\overline{45}+\overline{24}.\overline{53}+\overline{25}.\overline{34}=\text{zero},\quad \text{etc.},$$

then the three terms of each column will vanish, provided a proper relation exists between the ϵ's: viz. the conditions which we thus obtain are

$$\begin{gathered}
\epsilon_1\epsilon_2=\epsilon_4\epsilon_6=\epsilon_5\epsilon_7,\\
\epsilon_1\epsilon_3=\epsilon_3\epsilon_6=\epsilon_2\epsilon_7,\\
\epsilon_1\epsilon_5=\epsilon_3\epsilon_7=\epsilon_2\epsilon_6,\\
\epsilon_1\epsilon_6=\epsilon_3\epsilon_5=\epsilon_3\epsilon_4,\\
\epsilon_1\epsilon_7=-\epsilon_2\epsilon_4=\epsilon_3\epsilon_5,\\
\epsilon_2\epsilon_3=-\epsilon_4\epsilon_6=\epsilon_4\epsilon_7.
\end{gathered}$$

We may without loss of generality assume $\epsilon_1=\epsilon_2=\epsilon_3=+$; the equations then become

$$\begin{gathered}
+=-\epsilon_4\epsilon_7=\epsilon_5\epsilon_6,\\
+=-\epsilon_4\epsilon_6=\epsilon_5\epsilon_7,\\
+=-\epsilon_4\epsilon_5=\epsilon_6\epsilon_7,\\
\epsilon_4=\epsilon_5=\epsilon_6,\\
\epsilon_5=\epsilon_7=\epsilon_6,\\
\epsilon_6=\epsilon_5=-\epsilon_4,\\
\epsilon_7=-\epsilon_4=\epsilon_5,
\end{gathered}$$

and writing $\theta = \pm$ at pleasure, these are all satisfied if $-\epsilon_4 = \epsilon_5 = \epsilon_6 = \epsilon_7 = \theta$. The terms written down all disappear, and the sum of the squares of the eight coefficients thus becomes equal to the product of two sums each of them of eight squares, viz. this is the case if $\epsilon_1 = \epsilon_2 = \epsilon_3 = +$, $-\epsilon_4 = \epsilon_5 = \epsilon_6 = \epsilon_7 = \theta$, θ being $= \pm$ at pleasure: the resulting system of imaginaries may be said to be an 8-square system.

We may inquire whether the system is associative; for this purpose, supposing in the first instance that the ϵ's remain arbitrary, we form the complete system of the values of the triplets 12.3, 1.23, etc., (read the top line $12.3 = -\epsilon_1 0$, $1.23 = -\epsilon_1 0$, the next line $12.4 = \epsilon_1\epsilon_6 7$, $1.24 = \epsilon_2\epsilon_4 7$, and so in other cases):

$$\begin{array}{lll}
12.3 = 1.23 = & -\epsilon_1 & -\epsilon_1 \cdot 0 \\
12.4 = 1.24 = & \epsilon_1\epsilon_6 & \epsilon_2\epsilon_4 \cdot 7 \\
12.5 = 1.25 = & \epsilon_1\epsilon_7 & -\epsilon_2\epsilon_5 \cdot 6 \\
12.6 = 1.26 = & -\epsilon_1\epsilon_7 & -\epsilon_4\epsilon_2 \cdot 5 \\
12.7 = 1.27 = & -\epsilon_1\epsilon_6 & \epsilon_3\epsilon_2 \cdot 4 \\
13.4 = 1.34 = & -\epsilon_1\epsilon_4 & -\epsilon_3\epsilon_6 \cdot 6 \\
13.5 = 1.35 = & -\epsilon_1\epsilon_5 & \epsilon_3\epsilon_7 \cdot 7 \\
13.6 = 1.36 = & \epsilon_1\epsilon_4 & \epsilon_3\epsilon_7 \cdot 4 \\
13.7 = 1.37 = & \epsilon_1\epsilon_5 & -\epsilon_3\epsilon_6 \cdot 5 \\
14.5 = 1.45 = & -\epsilon_2 & -\epsilon_2 \cdot 0 \\
14.6 = 1.46 = & \epsilon_2\epsilon_4 & \epsilon_1\epsilon_4 \cdot 3 \\
14.7 = 1.47 = & \epsilon_2\epsilon_6 & -\epsilon_1\epsilon_6 \cdot 2 \\
15.6 = 1.56 = & -\epsilon_2\epsilon_4 & -\epsilon_1\epsilon_7 \cdot 2 \\
15.7 = 1.57 = & -\epsilon_2\epsilon_5 & \epsilon_1\epsilon_5 \cdot 3 \\
16.7 = 1.67 = & -\epsilon_3 & -\epsilon_3 \cdot 0 \\
23.4 = 2.34 = & \epsilon_1\epsilon_6 & -\epsilon_4\epsilon_6 \cdot 5 \\
23.5 = 2.35 = & -\epsilon_1\epsilon_7 & -\epsilon_4\epsilon_7 \cdot 4 \\
23.6 = 2.36 = & \epsilon_1\epsilon_4 & -\epsilon_5\epsilon_7 \cdot 7 \\
23.7 = 2.37 = & -\epsilon_1\epsilon_5 & -\epsilon_4\epsilon_5 \cdot 6 \\
24.5 = 2.45 = & -\epsilon_4\epsilon_6 & -\epsilon_5\epsilon_7 \cdot 3 \\
24.6 = 2.46 = & -\epsilon_4 & -\epsilon_4 \cdot 0 \\
24.7 = 2.47 = & \epsilon_4\epsilon_6 & \epsilon_1\epsilon_6 \cdot 1 \\
25.6 = 2.56 = & -\epsilon_4\epsilon_5 & \epsilon_1\epsilon_7 \cdot 1 \\
25.7 = 2.57 = & -\epsilon_5 & -\epsilon_5 \cdot 0 \\
26.7 = 2.67 = & -\epsilon_4\epsilon_6 & -\epsilon_1\epsilon_3 \cdot 3 \\
34.5 = 3.45 = & -\epsilon_6\epsilon_7 & \epsilon_1\epsilon_2 \cdot 2 \\
34.6 = 3.46 = & -\epsilon_6\epsilon_7 & -\epsilon_1\epsilon_4 \cdot 1 \\
34.7 = 3.47 = & -\epsilon_6 & -\epsilon_6 \cdot 0 \\
35.6 = 3.56 = & -\epsilon_7 & -\epsilon_7 \cdot 0 \\
35.7 = 3.57 = & \epsilon_5\epsilon_7 & -\epsilon_1\epsilon_5 \cdot 1 \\
36.7 = 3.67 = & -\epsilon_6\epsilon_7 & \epsilon_1\epsilon_3 \cdot 2 \\
45.6 = 4.56 = & \epsilon_2\epsilon_3 & -\epsilon_6\epsilon_7 \cdot 7 \\
45.7 = 4.57 = & -\epsilon_2\epsilon_3 & -\epsilon_4\epsilon_5 \cdot 6 \\
46.7 = 4.67 = & -\epsilon_4\epsilon_6 & -\epsilon_2\epsilon_3 \cdot 5 \\
56.7 = 5.67 = & -\epsilon_5\epsilon_7 & \epsilon_2\epsilon_6 \cdot 4.
\end{array}$$

Write as before $\epsilon_1 = \epsilon_2 = \epsilon_3 = +$; then, disregarding the lines (such as the first line) which contain the symbol 0, and writing down only the signs as given in the third and fourth columns, these are

$$\begin{array}{cc}
\epsilon_6 & \epsilon_4 \\
\epsilon_7 & -\epsilon_5 \\
-\epsilon_7 & -\epsilon_4 \\
-\epsilon_6 & \epsilon_5 \\
-\epsilon_4 & -\epsilon_6 \\
-\epsilon_5 & \epsilon_7 \\
\epsilon_4 & \epsilon_7 \\
\epsilon_5 & -\epsilon_6 \\
\epsilon_7 & \epsilon_4 \\
\epsilon_5 & -\epsilon_6 \\
-\epsilon_4 & -\epsilon_7 \\
-\epsilon_5 & \epsilon_5 \\
+ & -\epsilon_6\epsilon_7 \\
 & -\epsilon_4\epsilon_5 \\
+ & -\epsilon_5\epsilon_7 \\
- & -\epsilon_4\epsilon_6 \\
-\epsilon_4\epsilon_7 & - \\
\epsilon_4 & \epsilon_6 \\
-\epsilon_5 & \epsilon_7 \\
-\epsilon_4\epsilon_6 & - \\
-\epsilon_5\epsilon_6 & + \\
-\epsilon_6 & -\epsilon_4 \\
\epsilon_7 & -\epsilon_5 \\
-\epsilon_5\epsilon_7 & + \\
+ & -\epsilon_6\epsilon_7 \\
- & -\epsilon_4\epsilon_5 \\
-\epsilon_4\epsilon_5 & - \\
-\epsilon_6\epsilon_7 & +
\end{array}$$

We hence see at once that the pairs of signs in the two columns respectively cannot be made identical: to make them so, we should have $\epsilon_6 = \epsilon_4$, $\epsilon_7 = -\epsilon_4$, $\epsilon_7 = \epsilon_4$, that is, $\epsilon_4 = \epsilon_6 = \epsilon_7 = -\epsilon_5$, which is inconsistent with the last equation of the system $-\epsilon_6\epsilon_7 = +$.

Hence the imaginaries $1, 2, 3, 4, 5, 6, 7$, as defined by the original conditions, are not in any case associative.

If we have $\epsilon_1 = \epsilon_2 = \epsilon_3 = +$ and also $-\epsilon_4 = \epsilon_5 = \epsilon_6 = \epsilon_7 = \theta$, that is, if the imaginaries belong to the 8-square formula, then it is at once seen that each pair consists of two opposite signs; that is, for the several triads 123, 145, 167, 246, 257, 347, 356 used for the definition of the imaginaries, the associative property holds good, $12.3 = 1.23$, etc.; but for each of the remaining twenty-eight triads, the two terms *are equal but of opposite signs*, viz. $12.4 = -1.24$, etc.; so that the product 124 of any such three symbols has no determinate meaning.

Baltimore, March 5th, 1882.

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TABLES FOR THE BINARY SEXTIC.

The Leading Coefficients of the First 18 of the 26 Covariants.

[From the *American Journal of Mathematics*, vol. iv. (1881), pp. 379–384.]

Including the sextic itself, the number of covariants of the binary sextic is = 26, as shown in the table p. 296 of Clebsch's *Theorie der binären algebraischen Formen*, Leipzig, 1872; viz. this is

	Order					
Deg.	2	4	6	8	10	12
1			f			
2	A		i		H	
3		l		p	$(i, f)_1$	T
4	B		$(i, f)_2$	$(f, l)_1$	$(p, f)_1$	$(B, i)_1$
5			$(i, l)_3$	$(i, l)_2$	$(H, l)_1$	
6	A_2		$(p, l)_2$			
7		$(l, p)_4$	$(l, l)_3$			
8		$(i, l^2)_4$				
9			$((l, l)_2, l)_4$			
10	$(l, l^2)_6$	$(l, l^2)_5$				
12		$(l \cdot i)_6$				
15	$((l, l)_2, l)_6$					

Or, using the capital letters $A, B, \ldots, Z$ to denote the 26 covariants in the same order, the table is

	2	4	6	8	10	12
1			A			
2	B		C		D	
3		E		F	$G, = (A, C)^2$	H
4	I		$J, = (A, E)^2$	$K, = (A, E)^3$	$L, = (D, C)^2$	
5		$M, = (C, E)^2$	$N, = (C, E)^3$	$O, = (D, E)^2$		
6	P		$R, = (C, E^2)^4$			
7		$S, = (A, E^2)^2$	$T, = (A, E^2)^3$			
8		$U, = (C, E^2)^4$				
9			$V, = (C, E^2)^4$			
10	$W, = (A, E^3)^6$	$X, = (A, E^3)^5$				
12		$Y, = (C, E^3)^6$				
15	$Z, = (C, E^3)^6$					

A is the sextic.
B is Salmon's A, p. 202.
I , , B, p. 203.
P is Salmon's C, p. 204.
W, , , C, p. 207.
Z, , E, p. 253.
The references are to Salmon's *Higher Algebra*, 2nd Ed., 1866.

In the present short paper I give the leading coefficients of the first 18 covariants, A to R (some of these are of course known values, but it is convenient to include them): for the next four covariants S, T, U, V, the leading coefficients depend upon the coefficients of A, C, G and E^2, viz. writing

$$A = (a, b, c, d, e, f, g \mid x, y)^6,$$

$$E^2 = (0, \tfrac{1}{2}\beta, \tfrac{1}{4}\gamma, \tfrac{1}{6}\delta, \tfrac{1}{4}\epsilon, \tfrac{1}{2}\zeta \mid x, y)^6,$$

$$C = (\alpha', \tfrac{1}{2}\beta', \tfrac{1}{4}\gamma', \tfrac{1}{6}\delta', \tfrac{1}{4}\epsilon', \ldots \mid x, y)^6,$$

$$G = (\alpha'', \tfrac{1}{2}\beta'', \tfrac{1}{4}\gamma'', \tfrac{1}{6}\delta'', \tfrac{1}{4}\epsilon'', \ldots \mid x, y)^6,$$

we have

$$\begin{aligned} S,\ \text{Coeff. } x^4 &= a\epsilon - b\delta + c\gamma - d\beta + e\alpha, \\ T,\ \ ,\ \ ,\ \ ,\ x^6 &= a\delta - 2b\gamma + 3c\beta - 4d\alpha, \\ U,\ \ ,\ \ ,\ \ ,\ x^4 &= 2\alpha'\delta - \beta'\gamma + \delta'\beta - 2\epsilon'\alpha, \\ V,\ \ ,\ \ ,\ \ ,\ x^6 &= 280\alpha''\epsilon - 35\beta''\delta + 10\gamma''\gamma - 20\delta''\beta + 24\epsilon''\alpha. \end{aligned}$$

Similarly the invariant W and the leading coefficients of X, Y depend on the coefficients of A, G and E^2; and the invariant Z depends on the coefficients of G and E^2.

But these two invariants W and Z have been already calculated; viz. as already mentioned, W is Salmon's invariant D, and Z his invariant E, given each of them in the second edition of his *Higher Algebra* (but not reproduced in the third edition): on account of the great length of these expressions, it has been thought that it was not expedient to give them here.

For the reason appearing above, I have added the expressions for the remaining coefficients of C, E, O.

Leading coefficients of A, B, C, D, E, F, G, H.

$A,\ x^6$	$B,\ x^0$	$C,\ x^6$	$D,\ x^{10}$	$E,\ x^4$	$F,\ x^8$	$G,\ x^{10}$	$H,\ x^{12}$
$a \quad +1$	$ag+1$	$ae+1$	$ac+1$	$aeg+1$	$ace+1$	$a^2g \ +1$	a^2b+1
	$bf-6$	a^2bd-4	a^2b^2-1	$df-3$	d^2-1	$abe-5$	$abc-3$
	c^2+15	c^3+3		e^2+2	a^2b^2e-1	$cd+2$	a^2b^2+2
	d^2-10			$a^2b^2d^2-1$	$bcd+2$	a^2b^2d+8	
				$bcf+3$	c^2-1	bc^2-6	
				$bde-1$			
				c^2e-3			
				cd^2+2			

$I,\ x^0$	$J,\ x^4$	$K,\ x^6$	$L,\ x^{10}$	$M,\ x^6$	$N,\ x^8$	$O,\ x^6$
ace^2+1	$adg+1$	a^2dg+1	a^2bf+1	a^2bf+1	a^2cg-1	a^2cdg
af^2-1	$abef-10$	$aef-1$	$de-1$	$dfg-6$	$deg+1$	$cef-1$
d^2f-1	$cdf+4$	$abcg-3$	$aby-1$	eg^2+8	df^2+3	$df-3$
$dej+2$	ce^2+16	$bdf-2$	$bce-2$	cf^2-3	$ef-3$	b^2-2
e^2-1	d^2e-12	be^2+5	bd^2+4	a^2bg-1	a^2bfg+1	a^2b^2dg
a^2b^2g-1	a^2b^2df+16	$bf+9$	c^2d-1	$bcfg+6$	$bceg+2$	b^2f+1
$bf+1$	b^2e+9	$cde-17$	a^2b^2e+3	$bdeg-34$	bcf^2+3	b^2c^2
$bdg+2$	$bcf-12$	d^2+8	b^2d-6	bcg^2+48	bd^2g-4	$bcdf-14$
$bce-2$	$bcdg-76$	a^2b^2g+2	$bcg+3$	$bef-18$	$bdef-12$	bce^2+11
$bdg-2$	bd^2+48	b^2f-6		c^3g+45	be^2+15	bd^2e+1
bd^2+2	bd^2+48	b^3de+2		c^2df-45	$bef+15$	c^2f+9
c^3e+1	c^3e-32	b^2ce+6		c^2dg+1	c^2dg+9	c^2e-14
c^2d+2	c^2d-32	bcd^2-4		$cdef+78$	$cdf+4$	cd^2+6
c^2d+1				ce^2-36	cde^2-21	a^2b^2f
cd^2e-3				df^2-48	d^3e+8	b^2g+8
d^3+1				c^3d^2+28	a^2b^2cg-3	b^2e^2-9
				a^2b^3ceg	b^3cd+6	b^2cf-6
				b^2d^2g+64	b^2cf+9	$b^2cde+16$
				$b^2def-144$	b^2df+32	b^2d^2-8
				b^2e^2+81	b^2eg-39	bc^3e-3
				b^2cf-96	b^2eg-3	bc^2d^2+2
				b^2c^3+108	b^2fg-3	
				$bcdg+96$	bc^3d-66	
					bc^2f+18	
				$bcde^2-120$	$bcde+76$	
				bd^2e+16	bd^3-32	
				c^4g+36	c^4f+27	
				c^3df-72	c^3de-45	
				c^3e^2-27	c^2d^3+20	
				c^2d^2e+96		
				cd^4-32		

Leading coefficients of P, Q, R.

$P,\ x^2$	$Q,\ x^6$	$R,\ x^6$
$a^2d^2g^2+1$	$a^2dg^2\quad -1$	$+1$
$defy-6$	$efg\quad +9$	$+9$
df^2+4	$f^2\quad -8$	$+3$
$f^3\quad +4$	a^2bcg^2+3	-24
$g^2y\quad -3$	$bdyg\ -24$	-15
ey^2-3	$be^2g\ -45$	-66
$a\,bcdg^2-6$	$bef\quad +66$	$-2,\ +3$
$bcefg+18$	$cyf\quad +5,\ +48$	-3
bcf^2-12	$cdeg\quad +5,\ +48$	-12
bd^2fg+12	$cdf\quad +6,\ -12$	-12
bde^2g-18	$cef\ -7$	-51
$bey\quad +6$	$d^3\quad -3$	-16
$cy\quad +4$	$d^2ef\ -3$	$+36$
c^2e^2g-24	$d^2\quad +4$	-8
c^2fl^2-18	$a^2by^2g^2\ -2$	-2
$c^2ef\ +30$	$b^2g^2\ +4$	$+12$
cd^2eg+54	$b^2deg\ -5$	$+192$
cdf^2-12	$b^2df\ -6$	-48
$cdeg^2-42$	$b^2e^2\quad +7$	-144
ce^3+12	$bc^2eg\ -5$	-159
$dy\quad -20$	$bcd^2f\ -6$	$+18$
$d^2ef\ +24$	$bcd^2g\ +7$	-48
$d^2\quad -8$	$bcdef\ -16$	$+24$
$a^2b^2c^2g+4$	$bce^2\ +23$	$+279$
b^2d^2-12	$bdy\quad +30$	-48
b^2f^3+8	$bd^2y\ -33$	-84
b^2d^2f-3	$c^2dg\quad -1$	$+42$
b^2d^2+30	$c^2ef\quad +36$	$+153$
$b^2cef-24$	$c^2dy\quad -37$	-36
b^2d^2eg-12	$c^2d^2\quad -53$	-399
$b^2c^2y^2-24$	$cd^3e\ +79$	$+312$
b^2de^2y+60	$d^5\quad -24$	-64
b^2c^4-27	$a^2b^3y\ -2$	
bcy^2d+6	$b^2ceg\ +5$	
$bc^2deg-42$	$b^2cf^2\ +6$	
bc^2df^2+60	$b^2c^2g\ +2$	-224
bc^2ey-30	$b^2def\ +22$	$+144$
bcd^2e^2+24	$b^2e^3\ -27$	$+54$
$bcdy-84$	$b^2c^2dg\ -8$	$+336$
bcd^2e+66	$b^2c^2ef\ -39$	-108
$bdy\quad +24$	$b^2cdy\quad -50$	$+384$
bd^2e^2-24	$b^2cde^2\ +107$	-684
$c^3ey\ +12$	$bc^3d^2e\ -22$	$+144$
$cy\quad -27$	$bc^3f\ +3$	-126
cf^2-8	$bc^3dy\quad +84$	-648
$c^3def+66$	$bc^2d^3\quad -21$	$+432$
c^3e^3-8	$bc^2d^2h\ -102$	$+564$
c^3dy-24	$bcd^4\quad +44$	-288
c^3d^2-39	$c^4f\quad -27$	$+270$
cd^4e+36	$c^4de\ +45$	-450
d^6-8	$c^3d^3\ -20$	$+200$

Remaining Coefficients of C, E, O.

C		E	
x^6y^0	xy	x^2y	x^2y^2
$af+2$	$adg+1$	a^2g+1	$aeg-7$
$be-6$	$aef-1$	a^2f+2	af^2-14
$cd+4$	$bcg-1$	$ace-19$	$bdg+28$
	$bdf-8$	ad^2+8	$bef+42$
x^0y^2	be^2+9	b^2c-6	c^2g+14
$ag+1$	c^2f+9	$bcd+44$	$cdf-168$
$cc-9$	$cde-17$	b^3-30	ce^2+105
d^2+8	d^3+8		
x^2y^2	x^0y^3	xy^4	x^3y
$bg+1$	$aeg+1$	$abg+7$	$afg-7$
$cf-6$	af^2-1	$aef-14$	$beg+14$
$de+4$	$bdg-3$	$ade-14$	bf^2
	$bef+3$	b^2f	$cdg+14$
x^3y^0	c^2g+2	$bce-21$	$cef+21$
$eg+1$	$cdf-1$	bd^2+112	d^2f-112
$df-4$	ce^2-3	c^2d-70	de^2+70
e^2+3	d^2e+2		
x^4y^0	x^3y^1	x^3y^1	x^4y^1
		$acg+7$	$adg-1$
		$adf-28$	b^2g-2
		ae^2-14	$ceg+19$
		$bg+14$	cf^2+6
		$bcf-42$	d^2g-8
		$bde+168$	$def-44$
		c^2e-105	e^3+30
	x^2y^3	x^2y^4	
	adg	bg^2-1	
	$aef-35$	$efg+5$	
	$bcg+35$	$deg-2$	
	bdf	df^2-8	
	be^2+105	$ef+6$	
	c^2f-105		

Note.—In the tables on this page, a has been treated like the other letters; on the preceding pages, the powers of a have been suppressed except in the first of every series of terms containing a common power of a.

The final result is that we have the values of the invariants B, I, P, W, Z and the leading coefficients of the covariants A, C, D, E, F, G, H, J, K, L, M, N, O, Q, R: also the means of calculating the leading coefficients of the remaining covariants S, T, U, V, X, Y.

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regarded as a singly infinite system of points, of lines, or of planes; secondly, the surface, which may be regarded as a doubly infinite system of points or of planes, and also as a special triply infinite system of lines (viz. the tangent-lines of the surface are a special complex): as distinct particular cases of the former figure, we have the plane curve and the cone; and as a particular case of the latter figure, the ruled surface or singly infinite system of lines; we have besides the congruence, or doubly infinite system of lines, and the complex, or triply infinite system of lines. But, even if in solid geometry we attend only to the curve and the surface, there are crowds of theories which have scarcely any analogues in plane geometry. The relation of a curve to the various surfaces which can be drawn through it, or of a surface to the various curves that can be drawn upon it, is different in kind from that which in plane geometry most nearly corresponds to it, the relation of a system of points to the curves through them, or of a curve to the points upon it. In particular, there is nothing in plane geometry corresponding to the theory of the curves of curvature of a surface. To the single theorem of plane geometry, a right line is the shortest distance between two points, there correspond in solid geometry two extensive and difficult theories—that of the geodesic lines upon a given surface, and that of the surface of minimum area for any given boundary. Again, in solid geometry we have the interesting and difficult question of the representation of a curve by means of equations; it is not every curve, but only a curve which is the complete intersection of two surfaces, which can be properly represented by two equations $(x,\ y,\ z,\ w)^m = 0$, $(x,\ y,\ z,\ w)^n = 0$, in quadriplanar coordinates; and in regard to this question, which may also be regarded as that of the classification of curves in space, we have quite recently three elaborate memoirs by Nöther, Halphen, and Valentiner respectively.

In n-dimensional geometry, only isolated questions have been considered. The field is simply too wide; the comparison with each other of the two cases of plane geometry and solid geometry is enough to show how the complexity and difficulty of the theory would increase with each successive dimension.

In Transcendental Analysis, or the Theory of Functions, we have all that has been done in the present century with regard to the general theory of the function of an imaginary variable by Gauss, Cauchy, Puiseux, Briot, Bouquet, Liouville, Riemann, Fuchs, Weierstrass, and others. The fundamental idea of the geometrical representation of an imaginary variable $x + iy$, by means of the point having x, y for its coordinates, belongs, as I mentioned, to Gauss; of this I have already spoken at some length. The notion has been applied to differential equations; in the modern point of view, the problem in regard to a given differential equation is, not so much to reduce the differential equation to quadratures, as to determine from it directly the course of the integrals for all positions of the point representing the independent variable: in particular, the differential equation of the second order leading to the hypergeometric series $F(\alpha, \beta, \gamma, x)$ has been treated in this manner, with the most interesting results; the function so determined for all values of the parameters (α, β, γ) is thus becoming a known function. I would here also refer to the new notion in this part of analysis introduced by Weierstrass—that of the one-valued integer function, as defined by an

[57–2]

infinite series of ascending powers, convergent for all finite values, real or imaginary, of the variable x or $1/(x-c)$, and so having the one essential singular point $x=\infty$ or $x=c$, as the case may be: the memoir is published in the Berlin *Abhandlungen*, 1876.

But it is not only general theory: I have to speak of the various special functions to which the theory has been applied, or say the various known functions.

For a long time the only known transcendental functions were the circular functions—sine, cosine, &c.; the logarithm, i.e. for analytical purposes the hyperbolic logarithm to the base e; and, as implied therein, the exponential function e^x. More completely stated, the group comprises the direct circular functions sin, cos, &c.; the inverse circular functions $\sin^{-1}$ or arc sin, &c.; the exponential function, exp.; and the inverse exponential, or logarithmic, function, log.

Passing over the very important Eulerian integral of the second kind or gamma-function, the theory of which has quite recently given rise to some very interesting developments—and omitting to mention at all various functions of minor importance—we come (1811–1829) to the very wide groups, the elliptic functions and the single theta-functions. I give the interval of date so as to include Legendre's two systematic works, the *Exercices de Calcul Intégral* (1811–1816) and the *Théorie des Fonctions Elliptiques* (1825–1828); also Jacobi's *Fundamenta nova theoriae Functionum Ellipticarum* (1829), calling to mind that many of Jacobi's results were obtained simultaneously by Abel. I remark that Legendre started from the consideration of the integrals depending on a radical $\sqrt{X}$, the square root of a rational and integral quartic function of a variable x; for this he substituted a radical $\Delta\phi, = \sqrt{(1-k^2\sin^2\phi)}$, and he arrived at his three kinds of elliptic integrals $F\phi$, $E\phi$, $\Pi\phi$, depending on the argument or amplitude ϕ, the modulus k, and also the last of them on a parameter n; the F function is properly an inverse function, and in place of it Abel and Jacobi each of them introduced the direct functions corresponding to the circular functions sine and cosine, Abel's functions called by him ϕ, f, F, and Jacobi's functions sinam, cosam, Δam, or as they are also written sn, cn, dn. Jacobi, moreover, in the development of his theory of transformation obtained a multitude of formulae containing q, a transcendental function of the modulus defined by the equation $q = e^{-\pi K'/K}$, and he was also led by it to consider the two new functions H, Θ, which (taken each separately with two different arguments) are in fact the four functions called elsewhere by him Θ_1, Θ_2, Θ_3, Θ_4; these are the so-called theta-functions, or, when the distinction is necessary, the single theta-functions. Finally, Jacobi using the transformation $\sin\phi = \operatorname{sinam} u$, expressed Legendre's integrals of the second and third kinds as integrals depending on the new variable u, denoting them by means of the letters Z, Π, and connecting them with his own functions H and Θ; and the elliptic functions sn, cn, dn are expressed with these, or say with Θ_1, Θ_2, Θ_3, Θ_4, as fractions having a common denominator.

It may be convenient to mention that Hermite in 1858, introducing into the theory in place of q the new variable ω connected with it by the equation $q = e^{i\pi\omega}$, was led to consider three functions $\varphi\omega$, $\psi\omega$, $\chi\omega$, depending on and slightly differing from the three differences of the functions Θ for the three values $\frac{K}{3}$, $\frac{2K}{3}$, $K - \frac{K}{3}$ of the argument; these functions $\varphi\omega$, $\psi\omega$, $\chi\omega$ are connected with the modulus k by very simple relations (they are in fact, except as to constant factors, the fourth roots of k, k', and kk'); and they are functions which play an important part in the new theory of equations.

The hyperelliptic integrals which succeed to the elliptic ones present themselves in the Abelian theory, but were probably first considered by Jacobi: he showed that the existing theory of elliptic functions led to the consideration of the integrals

$$\int \frac{dx}{\sqrt{X}}, \quad \int \frac{x\,dx}{\sqrt{X}},$$

where X is now a rational and integral sextic function of x; these are, in the simplest case, the so-called hyperelliptic integrals, and the corresponding inverse functions are the hyperelliptic functions; only here, presenting themselves as inverse functions of integrals depending on the same radical $\sqrt{X}$, we have, not single but pairs of functions of two variables; and the case is generally that of p functions of p variables.

The general theory of the Abelian functions was first considered by Riemann in his memoir "Theorie der Abel'schen Functionen," *Crelle*, t. LIV. (1857); and we have also the great memoir by Clebsch and Gordan, *Theorie der Abelschen Functionen* (Leipzig, 1866). The theta-functions for any value whatever of p have been considered by Riemann, and also by Weierstrass in a memoir published in the *Berlin Monatsbericht*, 1869; but the theory has been chiefly developed for the next succeeding case $p = 2$, that of the double theta-functions, in connexion with hyperelliptic integrals, by various writers including Göpel, Rosenhain, Borchardt, Weber, Königsberger, Brioschi, Cayley, and others.

I would mention also the linear differential equations with rational algebraical coefficients, especially the equation of the second order treated by Fuchs and others; and as bearing thereon the theory of the so-called automorphic functions, treated of by Klein and Poincaré.

I pass now to the Theory of Numbers. The first leading question is that of the prime numbers, the determination of which is a very interesting one, although it does not seem at present to give rise to any general theory of importance. The next leading question is that of the residues of powers in regard to a prime modulus; this question is included in the still wider question of the theory of forms, viz. given a homogeneous quadratic function $ax^2 + 2bxy + cy^2$, of two variables x, y (called a form), and a prime modulus p, the question is to determine the values which this can take to the modulus p.

We have also the theory of complex numbers $a + bi$ ($i = \sqrt{(-1)}$); and Gauss considered the corresponding theory of the complex numbers $a + b\rho$ where ρ is an imaginary cube root of unity ($\rho^2 + \rho + 1 = 0$); these are the theories used in his demonstrations of the laws of quadratic and cubic reciprocity.

But the great extension is by Kummer in his memoir "Theorie der idealen Primfactoren der complexen Zahlen," *Berl. Abh.*, 1856. We have first the prime number p and its pth roots of unity, 1, η, η^2, $\ldots$, η^{p-1}, satisfying the equation $\eta^p - 1 = 0$, or breaking this up into the equation $\eta - 1 = 0$, with the single root $\eta = 1$, and the equation $1 + \eta + \eta^2 + \cdots + \eta^{p-1} = 0$, with the $p - 1$ roots η, η^2, $\ldots$, η^{p-1}; then if η be a root of this last equation, the complex numbers $a + a_1\eta + a_2\eta^2 + \cdots + a_{p-1}\eta^{p-1}$ (with a, a_1, $\ldots$, a_{p-1} positive or negative ordinary integers, including zero) are the numbers considered in his theory.

But more generally Kummer considered the so-called "periods" of the pth roots of unity, viz. the prime number p being supposed of the form $p = \theta f + 1$, dividing the $p - 1$ roots η, η^2, $\ldots$, η^{p-1} into θ sets each containing f roots, each such set has for its sum a so-called period η_1, η_2, $\ldots$, η_θ, and these periods are by themselves the roots of an equation of the order θ; and if η_1, η_2, $\ldots$, η_θ are taken to be a sum of roots, of the equation $x^n - 1 = 0$, then the complex numbers considered are the numbers of the form $a_1\eta_1 + a_2\eta_2 + \cdots + a_\theta\eta_\theta$ (a_1, a_2, $\ldots$, a_θ positive or negative ordinary integers, including zero): f may be $= 1$, and the theory for the periods thus includes that for the single roots.

We have thus a new and very general theory, including within itself that of the complex numbers $a + bi$ and $a + b\rho$. But a new phenomenon presents itself; for these special forms the properties in regard to prime numbers corresponded precisely with those for real numbers; a non-prime number was in one way only a product of prime factors; the power of a prime number has only factors which are lower powers of the same prime number: for instance, if p be a prime number, then, excluding the obvious decomposition $p \cdot p^2$, we cannot have $p^3 =$ a product of two factors A, B. In the general case this is not so, but the exception first presents itself for the number 23; in the theory of the numbers composed with the 23rd roots of unity, we have prime numbers p, such that $p^3 = AB$. To restore the theorem, it is necessary to establish the notion of ideal numbers; a prime number p is by definition not the product of two actual numbers, but in the example just referred to the number p is the product of two ideal numbers having for their cubes the two actual numbers A, B, respectively, and we thus have $p^3 = AB$. It is, I think, in this way that we most easily get some notion of the meaning of an ideal number, but the mode of treatment (in Kummer's great memoir, "Ueber die Zerlegung der aus Wurzeln der Einheit gebildeten complexen Zahlen in ihre Primfactoren," *Crelle*, t. XXXV. 1847) is a much more refined one; an ideal number, without ever being isolated, is made to manifest itself in the properties of the prime number of which it is a factor, and without reference to the theorem afterwards arrived at, that there is always some power of the ideal number which is an actual number. In the still later developments of the Theory of Numbers by Dedekind, the units, or incommensurables, are the roots of any irreducible equation having for its coefficients ordinary integer numbers, and with the coefficient unity for the highest power of x. The question arises, What is the analogue of a whole number? Thus, for the very simple case of the equation $x^2 + 3 = 0$, we have as a whole number the apparently fractional form $\frac{1}{2}(1 + i\sqrt{3})$ which is the imaginary cube root of unity, the ρ of Eisenstein's theory. We have, moreover, the (as far as appears) wholly distinct complex theory of the numbers composed with the congruence-imaginaries of Galois: viz. these are imaginary numbers assumed to satisfy a congruence which is not satisfied by any real number; for instance, the congruence $x^2 \equiv -2 \pmod{5}$ has no real root, but we assume an imaginary root i, the other root is then $= -i$, and we then consider the system of complex numbers $a+bi \pmod{5}$, viz. we have thus the 5^2 numbers obtained by giving to each of the numbers a, b, the values $0, 1, 2, 3, 4$, successively. And so in general, the consideration of an irreducible congruence $F(x) \equiv 0 \pmod{p}$ of the order n, to any prime modulus p, gives rise to an imaginary congruence root i, and to complex numbers of the form $a + bi + ci^2 + \cdots + fi^{n-1}$, where a, b, c, $\ldots$, &c., are ordinary integers each $= 0, 1, 2, \ldots, p - 1$.

As regards the theory of forms, we have in the ordinary theory, in addition to the binary and ternary quadratic forms, which have been very thoroughly studied, the quaternary and higher quadratic forms (to these last belong, as very particular cases, the theories of the representation of a number as a sum of four, five or more squares), and also binary cubic and quartic forms, and ternary cubic forms, in regard to all of which something has been done; the binary quadratic forms have been studied in the theory of the complex numbers $a + bi$.

A seemingly isolated question in the Theory of Numbers, the demonstration of Fermat's theorem of the impossibility for any exponent λ greater than 3, of the equation $x^\lambda + y^\lambda = z^\lambda$, has given rise to investigations of very great interest and difficulty.

Outside of ordinary mathematics, we have some theories which must be referred to: algebraical, geometrical, logical. It is, as in many other cases, difficult to draw the line; we do in ordinary mathematics use symbols not denoting quantities, which we nevertheless combine in the way of addition and multiplication, $a + b$, and ab, and which may be such as not to obey the commutative law $ab = ba$: in particular, this is or may be so in regard to symbols of operation; and it could hardly be said that any development whatever of the theory of such symbols of operation did not belong to ordinary algebra. But I do separate from ordinary mathematics the system of multiple algebra or linear associative algebra, developed in the valuable memoir by the late Benjamin Peirce, *Linear Associative Algebra* (1870, reprinted 1881 in the *American Journal of Mathematics*, vol. IV., with notes and addenda by his son, C. S. Peirce); we here consider symbols A, B, &c. which are linear functions of a determinate number of letters or units with i, j, k, l, &c., coefficients which are ordinary analytical magnitudes, real or imaginary, viz. the coefficients are in general of the form $x + iy$, where

i is the before-mentioned imaginary or $\sqrt{(-1)}$ of ordinary analysis. The letters i, j, &c., are such that every binary combination i^2, ij, ji, &c., (the ij being in general not $= ji$), is equal to a linear function of the letters, but under the restriction of satisfying the associative law: viz. for each combination of three letters $ij \cdot k$ is $= i \cdot jk$, so that there is a determinate and unique product of three or more letters; or, what is the same thing, the laws of combination of the units i, j, k, are defined by a multiplication table giving the values of i^2, ij, ji, &c.; the original units may be replaced by linear functions of these units, so as to give rise, for the units finally adopted, to a multiplication table of the most simple form; and it is very remarkable, how frequently in these simplified forms we have nilpotent or idempotent symbols ($i^2 = 0$, or $i^2 = i$, as the case may be), and symbols i, j, such that $ij = ji = 0$; and consequently how simple are the forms of the multiplication tables which define the several systems respectively.

I have spoken of this multiple algebra before referring to various geometrical theories of earlier date, because I consider it as the general analytical basis, and the true basis, of these theories. I do not realise to myself directly the notions of the addition or multiplication of two lines, areas, rotations, forces, or other geometrical, kinematical, or mechanical entities; and I would formulate a general theory as follows: consider any such entity as determined by the proper number of parameters a, b, c (for instance, in the case of a finite line given in magnitude and position, these might be the length, the coordinates of one end, and the direction-cosines of the line considered as drawn from this end); and represent it by or connect it with the linear function $ai + bj + ck + \&c.$, formed with these parameters as coefficients, and with a given set of units, i, j, k, &c. Conversely, any such linear function represents an entity of the kind in question. Two given entities are represented by two linear functions; the sum of these is a like linear function representing an entity of the same kind, which may be regarded as the sum of the two entities; and the product of them (taken in a determined order, when the order is material) is an entity of the same kind, which may be regarded as the product (in the same order) of the two entities. We thus establish by definition the notion of the sum of the two entities, and that of the product (in a determinate order, when the order is material) of the two entities. The value of the theory in regard to any kind of entity would of course depend on the choice of a system of units, i, j, k,..., with such laws of combination as would give a geometrical or kinematical or mechanical significance to the notions of the sum and product as thus defined.

Among the geometrical theories referred to, we have a theory (that of Argand, Warren, and Peacock) of imaginaries in plane geometry; Sir W. R. Hamilton's very valuable and important theory of Quaternions; the theories developed in Grassmann's *Ausdehnungslehre*, 1841 and 1862; Clifford's theory of Biquaternions; and recent extensions of Grassmann's theory to non-Euclidian space, by Mr Homersham Cox. These different theories have of course been developed, not in anywise from the point of view in which I have been considering them, but from the points of view of their several authors respectively.

The literal symbols x, y, &c., used in Boole's *Laws of Thought* (1854) to represent things as subjects of our conceptions, are symbols obeying the laws of algebraic combination (the distributive, commutative, and associative laws) but which are such that for any one of them, say x, we have $x - x^2 = 0$, this equation not implying (as in ordinary algebra it would do) either $x = 0$ or else $x = 1$. In the latter part of the work relating to the Theory of Probabilities, there is a difficulty in making out the precise meaning of the symbols; and the remarkable theory there developed has, it seems to me, passed out of notice, without having been properly discussed. A paper by the same author, "Of Propositions numerically definite" (*Camb. Phil. Trans.* 1869), is also on the border-land of logic and mathematics. It would be out of place to consider other systems of mathematical logic, but I will just mention that Mr C. S. Peirce in his "Algebra of Logic," *American Math. Journal*, vol. III., establishes a notation for relative terms, and that these present themselves in connexion with the systems of units of the linear associative algebra.

Connected with logic, but primarily mathematical and of the highest importance, we have Schubert's *Abzählende Geometrie* (1878). The general question is, How many curves or other figures are there which satisfy given conditions? for example, How many conics are there which touch each of five given conics? The class of questions in regard to the conic was first considered by Chasles, and we have his beautiful theory of the characteristics μ, ν, of the conics which satisfy four given conditions; questions relating to cubics and quartics were afterwards considered by Maillard and Zeuthen; and in the work just referred to the theory has become a very wide one. The noticeable point is that the symbols used by Schubert are in the first instance, not numbers, but mere logical symbols: for example, a letter g denotes the condition that a line shall cut a given line; g^2 that it shall cut each of two given lines; and so in other cases; and these logical symbols are combined together by algebraical laws: they first acquire a numerical signification when the number of conditions becomes equal to the number of parameters upon which the figure in question depends.

In all that I have last said in regard to theories outside of ordinary mathematics, I have been still speaking on the text of the vast extent of modern mathematics. In conclusion I would say that mathematics have steadily advanced from the time of the Greek geometers. Nothing is lost or wasted; the achievements of Euclid, Archimedes, and Apollonius are as admirable now as they were in their own days. Descartes' method of coordinates is a possession for ever. But mathematics have never been cultivated more zealously and diligently, or with greater success, than in this century—in the last half of it, or at the present time: the advances made have been enormous, the actual field is boundless, the future full of hope. In regard to pure mathematics we may most confidently say:

> Yet I doubt not through the ages one increasing purpose runs,
> And the thoughts of men are widened with the process of the suns.

[58–2]

. . . .

785.

CURVE.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. VI. (1877), pp. 716—728.]

THIS subject is treated here from an historical point of view, for the purpose of showing how the different leading ideas in the theory were successively arrived at and developed.

A curve is a line, or continuous singly infinite system of points. We consider in the first instance, and chiefly, a plane curve described according to a law. Such a curve may be regarded geometrically as actually described, or kinematically as in course of description by the motion of a point; in the former point of view, it is the locus of all the points which satisfy a given condition; in the latter, it is the locus of a point moving subject to a given condition. Thus the most simple and earliest known curve, the circle, is the locus of all the points at a given distance from a fixed centre, or else the locus of a point moving so as to be always at a given distance from a fixed centre. (The straight line and the point are not for the moment regarded as curves.)

Next to the circle we have the conic sections, the invention of them attributed to Plato (who lived 430 to 347 B.C.); the original definition of them as the sections of a cone was by the Greek geometers who studied them soon replaced by a proper definition *in plano* like that for the circle, viz. a conic section (or as we now say a "conic") is the locus of a point such that its distance from a given point, the focus, is in a given ratio to its (perpendicular) distance from a given line, the directrix; or it is the locus of a point which moves so as always to satisfy the foregoing condition. Similarly any other property might be used as a definition; an ellipse is the locus of a point such that the sum of its distances from two fixed points (the foci) is constant, &c., &c.

The Greek geometers invented other curves; in particular, the "conchoid," which is the locus of a point such that its distance from a given line, measured along the line drawn through it to a fixed point, is constant; and the "cissoid" which is the locus of a point such that its distance from a fixed point is always equal to the intercept (on the line through the fixed point) between a circle passing through the fixed point and the tangent to the circle at the point opposite to the fixed point. Obviously the number of such geometrical or kinematical definitions is infinite. In a machine of any kind, each point describes a curve; a simple but important instance is the "three-bar curve," or locus of a point in or rigidly connected with a bar pivotted on to two other bars which rotate about fixed centres respectively. Every curve thus arbitrarily defined has its own properties; and there was not any principle of classification.

The principle of classification first presented itself in the *Géométrie* of Descartes (1637). The idea was to represent any curve whatever by means of a relation between the coordinates $(x,\ y)$ of a point of the curve, or say to represent the curve by means of its equation.

Descartes takes two lines xx', yy', called axes of coordinates, intersecting at a point O called the origin (the axes are usually at right angles to each other, and for the present they are considered as being so); and he determines the position of a point P by means of its distances OM (or NP) $= x$, and MP (or ON) $= y$, from these two axes respectively; where x is regarded as positive or negative according as it is in the sense Ox or Ox' from O; and similarly y as positive or negative according as it is in the sense Oy or Oy' from O; or, what is the same thing,

In the quadrant xy, or N.E., we have			x	y
			$+$	$+$
„	$x'y$	„ N.W.,	$-$	$+$
„	xy'	„ S.E.,	$+$	$-$
„	$x'y'$	„ S.W.,	$-$	$-$

Any relation whatever between $(x,\ y)$ determines a curve, and conversely every curve whatever is determined by a relation between $(x,\ y)$.

Observe that the distinctive feature is in the exclusive use of such determination of a curve by means of its equation. The Greek geometers were perfectly familiar with the property of an ellipse which in the Cartesian notation is $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, the equation of the curve; but it was as one of a number of properties, and in no wise selected out of the others for the characteristic property of the curve[1].

We obtain from the equation the notion of an algebraical or geometrical as opposed to a transcendental curve, viz. an algebraical or geometrical curve is a curve having an equation $F(x,\ y) = 0$, where $F(x,\ y)$ is a rational and integral algebraical function of the coordinates $(x,\ y)$; and in what follows we attend throughout (unless the contrary is stated) only to such curves. The equation is sometimes given, and may conveniently be used, in an irrational form, but we always imagine it reduced to the foregoing rational and integral form, and regard this as the equation of the curve. And we have hence the notion of a curve of a given order, viz. the order of the curve is equal to that of the term or terms of highest

[1]There is no exercise more profitable for a student than that of tracing a curve from its equation, or say rather that of so tracing a considerable number of curves. And he should make the equations for himself. The equation should be in the first instance a purely numerical one, where y is given or can be found as an explicit function of x; here, by giving different numerical values to x, the corresponding values of y may be found; and a sufficient number of points being thus determined, the curve is traced by drawing a continuous line through these points. The next step should be to consider an equation involving literal coefficients; thus, after such curves as $y = x^2$, $y = x(x-1)(x-2)$, $y = (x-1)\sqrt{x-2}$, &c., he should proceed to trace such curves as $y = (x-a)(x-b)(x-c)$, $y = (x-a)\sqrt{x-b}$, &c., and endeavour to ascertain for what different relations of equality or inequality between the coefficients the curve will assume essentially or notably distinct forms. The purely numerical equations will present instances of nodes, cusps, inflexions, double tangents, asymptotes, &c.—specialities which he should be familiar with before he has to consider their general theory. And he may then consider an equation such that neither coordinate can be expressed as an explicit function of the other of them (practically, an equation such as $x^3 + y^3 - 3xy = 0$, which requires the solution of a cubic equation, belongs to this class); the problem of tracing the curve here frequently requires special methods, and it may easily be such as to require and serve as an exercise for the powers of an advanced algebraist.

order in the coordinates $(x,\ y)$ conjointly in the equation of the curve; for instance, $xy - 1 = 0$ is a curve of the second order.

It is to be noticed here that the axes of coordinates may be any two lines at right angles to each other whatever; and that the equation of a curve will be different according to the selection of the axes of coordinates; but the order is independent of the axes, and has a determinate value for any given curve.

We hence divide curves according to their order, viz. a curve is of the first order, second order, third order, &c., according as it is represented by an equation of the first order, $ax + by + c = 0$, or say $(\ast\!\!\between x,\ y,\ 1) = 0$; or by an equation of the second order; $ax^2 + 2hxy + by^2 + 2fy + 2gx + c = 0$, say $(\ast\!\!\between x,\ y,\ 1)^2 = 0$; or by an equation of the third order, &c.; or, what is the same thing, according as the equation is linear, quadric, cubic, &c.

A curve of the first order is a right line; and conversely every right line is a curve of the first order.

A curve of the second order is a conic, or as it is also called a quadric; and conversely every conic, or quadric, is a curve of the second order.

A curve of the third order is called a cubic; one of the fourth order a quartic; and so on.

A curve of the order m has for its equation $(\ast\!\!\between x,\ y,\ 1)^m = 0$; and when the coefficients of the function are arbitrary, the curve is said to be the general curve of the order m. The number of coefficients is $\frac{1}{2}(m+1)(m+2)$; but there is no loss of generality if the equation be divided by one coefficient so as to reduce the coefficient of the corresponding term to unity, hence the number of coefficients may be reckoned as $\frac{1}{2}(m+1)(m+2)-1$, that is, $\frac{1}{2}m(m+3)$; and a curve of the order m may be made to satisfy this number of conditions; for example, to pass through $\frac{1}{2}m(m+3)$ points.

It is to be remarked that an equation may break up; thus a quadric equation may be $(ax + by + c)(a'x + b'y + c') = 0$, breaking up into the two equations $ax + by + c = 0$, $a'x + b'y + c' = 0$, viz. the original equation is satisfied if either of these is satisfied. Each of these last equations represents a curve of the first order, or right line; and the original equation represents this pair of lines, viz. the pair of lines is considered as a quadric curve. But it is an improper quadric curve; and in speaking of curves of the second or any other given order, we frequently imply that the curve is a proper curve represented by an equation which does not break up.

The intersections of two curves are obtained by combining their equations; viz. the elimination from the two equations of y (or x) gives for x (or y) an equation of a certain order, say the resultant equation; and then to each value of x (or y) satisfying this equation there corresponds in general a single value of y (or x), and consequently a single point of intersection; the number of intersections is thus equal to the order of the resultant equation in x (or y).

Supposing that the two curves are of the orders m, n, respectively, then the order of the resultant equation is in general and at most $= mn$; in particular, if the curve of the order n is an arbitrary line ($n = 1$), then the order of the resultant equation is $= m$; and the curve of the order m meets therefore the line in m points. But the resultant equation may have all or any of its roots imaginary, and it is thus not always that there are m real intersections.

The notion of imaginary intersections, thus presenting itself, through algebra, in geometry, must be accepted in geometry—and it in fact plays an all-important part in modern geometry. As in algebra we say that an equation of the mth order has m roots, viz. we state this generally without in the first instance, or it may be without ever, distinguishing whether these are real or imaginary; so in geometry we say that a curve of the mth order is met by an arbitrary line in m points, or rather we thus, through algebra, obtain the proper geometrical definition of a curve of the mth order, as a curve which is met by an arbitrary line in m points (that is, of course, in m, and not more than m, points).

The theorem of the m intersections has been stated in regard to an arbitrary line; in fact, for particular lines the resultant equation may be or appear to be of an order less than m; for instance, taking $m = 2$, if the hyperbola $xy - 1 = 0$ be cut by the line $y = \beta$, the resultant equation in x is $\beta x - 1 = 0$, and there is apparently only the intersection $(x = 1/\beta,\ y = \beta)$; but the theorem is, in fact, true for every line whatever: a curve of the order m meets every line whatever in precisely m points. We have, in the case just referred to, to take account of a point at infinity on the line $y = \beta$; the two intersections are the point $(x = 1/\beta,\ y = \beta)$, and the point at infinity on the line $y = \beta$.

It is moreover to be noticed that the points at infinity may be all or any of them imaginary, and that the points of intersection, whether finite or at infinity, real or imaginary, may coincide two or more of them together, and have to be counted accordingly; to support the theorem in its universality, it is necessary to take account of these various circumstances.

The foregoing notion of a point at infinity is a very important one in modern geometry; and we have also to consider the paradoxical statement that in plane geometry, or say as regards the plane, infinity is a right line. This admits of an easy illustration in solid geometry. If with a given centre of projection, by drawing from it lines to every point of a given line, we project the given line on a given plane, the projection is a line, i.e., this projection is the intersection of the given plane with the plane through the centre and the given line. Say the projection is always a line, then if the figure is such that the two planes are parallel, the projection is the intersection of the given plane by a parallel plane, or it is the system of points at infinity on the given plane, that is, these points at infinity are regarded as situate on a given line, the line infinity of the given plane[2].

Reverting to the purely plane theory, infinity is a line, related like any other right line to the curve, and thus intersecting it in m points, real or imaginary, distinct or coincident.

Descartes in the *Géométrie* defined and considered the remarkable curves called after him ovals of Descartes, or simply Cartesians, which will be again referred to. The next important work, founded

[2]More generally, in solid geometry infinity is a plane,—its intersection with any given plane being the right line which is the infinity of this given plane.

on the *Géométrie*, was Sir Isaac Newton's *Enumeratio linearum tertii ordinis* (1706), establishing a classification of cubic curves founded chiefly on the nature of their infinite branches, which was in some details completed by Stirling, Murdoch, and Cramer; the work contains also the remarkable theorem (to be again referred to), that there are five kinds of cubic curves giving by their projections every cubic curve whatever.

Various properties of curves in general, and of cubic curves, are established in Maclaurin's memoir, "De linearum geometricarum proprietatibus generalibus Tractatus" (posthumous, say 1746, published in the 6th edition of his *Algebra*). We have in it a particular kind of correspondence of two points on a cubic curve, viz. two points correspond to each other when the tangents at the two points again meet the cubic in the same point.

The *Géométrie Descriptive* by Monge was written in the year 1794 or 1795 (7th edition, Paris, 1847), and in it we find stated, in plano with regard to the circle, and in three dimensions with regard to a surface of the second order, the fundamental theorem of reciprocal polars, viz. "Given a surface of the second order and a circumscribed conic surface which touches it ... then if the conic surface moves so that its summit is always in the same plane, the plane of the curve of contact passes always through the same point." The theorem is here referred to partly on account of its bearing on the theory of imaginaries in geometry. It is, in Brianchon's memoir "Sur les surfaces du second degré" (*Jour. Polyt.*, t. VI., 1806), shown how for any given position of the summit the plane of contact is determined, or reciprocally; say the plane XY is determined when the point P is given, or reciprocally; and it is noticed that when P is situate in the interior of the surface the plane XY does not cut the surface; that is, we have a real plane XY intersecting the surface in the imaginary curve of contact of the imaginary circumscribed cone having for its summit a given real point P inside the surface.

Stating the theorem in regard to a conic, we have a real point P (called the pole) and a real line XY (called the polar), the line joining the two (real or imaginary) points of contact of the (real or imaginary) tangents drawn from the point to the conic; and the theorem is that when the point describes a line the line passes through a point, this line and point being polar and pole to each other. The term "pole" was first used by Servois, and "polar" by Gergonne (*Gerg.*, t. I. and III., 1810–13); and from the theorem we have the method of reciprocal polars for the transformation of geometrical theorems, used already by Brianchon (in the memoir above referred to) for the demonstration of the theorem called by his name, and in a similar manner by various writers in the earlier volumes of Gergonne. We are here concerned with the method less in itself than as leading to the general notion of duality. And, bearing in a somewhat similar manner also on the theory of imaginaries in geometry (but the notion presents itself in a more explicit form), there is the memoir by Gaultier, on the graphical construction of circles and spheres (*Jour. Polyt.*, t. IX., 1813). The well-known theorem as to radical axes may be stated as follows. Consider two circles partially drawn so that it does not appear whether the circles, if completed, would or would not intersect in real points, say two arcs of circles; then we can, by means of a third circle drawn so as to intersect in two real points each of the two arcs, determine a right line, which, if the complete circles intersect in two real points, passes through the points, and which is on this account regarded as a line passing through two (real or imaginary) points of intersection of the two circles. The construction in fact is, join the two points in which the third circle meets the first arc, and join also the two points in which the third circle meets the second arc, and from the point of intersection of the two joining lines, let fall a perpendicular on the line joining the centre of the two circles; this perpendicular (considered as an indefinite line) is what Gaultier terms the "radical axis of the two circles"; it is a line determined by a real construction and itself always real; and by what precedes it is the line joining two (real or imaginary, as the case may be) intersections of the given circles.

The intersections which lie on the radical axis are two out of the four intersections of the two circles. The question as to the remaining two intersections did not present itself to Gaultier, but it is answered in Poncelet's *Traité des propriétés projectives* (1822), where we find (p. 49) the statement, "deux cercles placés arbitrairement sur un plan... ont idéalement deux points imaginaires communs à l'infini"; that is, a circle qua curve of the second order is met by the line infinity in two points; but, more than this, they are the same two points for any circle whatever. The points in question have since been called (it is believed first by Dr Salmon) the circular points at infinity, or they may be called the circular points; these are also frequently spoken of as the points I, J; and we have thus the circle characterized as a conic which passes through the two circular points at infinity; the number of conditions thus imposed upon the conic is $= 2$, and there remain three arbitrary constants, which is the right number for the circle. Poncelet throughout his work makes continual use of the foregoing theories of imaginaries and infinity, and also of the before-mentioned theory of reciprocal polars.

Poncelet's two memoirs "Sur les centres des moyennes harmoniques," and "Sur la théorie générale des polaires réciproques," although presented to the Paris Academy in 1824 were only published (*Crelle*, t. III. and IV., 1828, 1829), subsequent to the memoir by Gergonne, "Considérations philosophiques sur les élémens de la science de l'étendue" (*Gerg.*, t. XVI., 1825–26). In this memoir by Gergonne, the theory of duality is very clearly and explicitly stated; for instance, we find "dans la géométrie plane, à chaque théorème il en répond nécessairement un autre qui s'en déduit en échangeant simplement entre eux les deux mots *points* et *droites*; tandis que dans la géométrie de l'espace ce sont les mots *points* et *plans* qu'il faut échanger entre eux pour passer d'un théorème à son corrélatif"; and the plan is introduced of printing correlative theorems, opposite to each other, in two columns. There was a reclamation as to priority by Poncelet in the *Bulletin Universel* reprinted with remarks by Gergonne (*Gerg.*, t. XIX., 1827), and followed by a short paper by Gergonne, "Rectifications de quelques théorèmes, &c.," which is important as first introducing the word *class*. We find in it explicitly the two correlative definitions: "a plane curve is said to be of the mth degree (order) when it has with a line m real or ideal intersections," and "a plane curve is said to be of the mth class when from any point of its plane there can be drawn to it m real or ideal tangents."

It may be remarked that in Poncelet's memoir on reciprocal polars, above referred to, we have the theorem that the number of tangents from a point to a curve of the order m, or say the class of the curve, is in general and at most $= m(m-1)$, and that he mentions that this number is subject to reduction when the curve has double points or cusps.

The theorem of duality as regards plane figures may be thus stated: two figures may correspond to each other in such manner that to each point and line in either figure there corresponds in the other figure a line and point respectively. It is to be understood that the theorem extends to all points or lines, drawn or not drawn; thus if in the first figure there are any number of points on a line drawn or not drawn, the corresponding lines in the second figure, produced if necessary, must meet in a point. And we thus see how the theorem extends to curves, their points and tangents: if there is in the first figure a curve of the order m, any line meets it in m points; and hence from the corresponding point in the second figure there must be to the corresponding curve m tangents; that is, the corresponding curve must be of the class m.

Trilinear coordinates (to be again referred to) were first used by Bobillier in the memoir, "Essai sur un nouveau mode de recherche des propriétés de l'étendue" (*Gerg.*, t. XVIII., 1827–28). It is convenient to use these rather than Cartesian coordinates. We represent a curve of the order m by an equation $(*\between x,\ y,\ z)^m = 0$, the function on the left-hand being a homogeneous rational and integral function of the order m of the three coordinates $(x,\ y,\ z)$; clearly the number of constants is the same as for the equation $(*\between x,\ y,\ 1)^m$ in Cartesian coordinates.

The theory of duality is considered and developed, but chiefly in regard to its metrical applications, by Chasles in the "Mémoire de géométrie sur deux principes généraux de la science, la dualité et l'homographie," which forms a sequel to the "Aperçu historique sur l'origine et le développement des méthodes en géométrie" (*Mém. de Brux.*, t. XI., 1837).

We now come to Plücker; his "six equations" were given in a short memoir in *Crelle* (1842) preceding his great work, the *Theorie der algebraischen Curven* (1844).

Plücker first gave a scientific dual definition of a curve, viz. "A curve is a locus generated by a point, and enveloped by a line—the point moving continuously along the line, while the line rotates continuously about the point"; the point is a point (*ineunt*) of the curve, the line is a tangent of the curve.

And, assuming the above theory of geometrical imaginaries, a curve such that m of its points are situate in an arbitrary line is said to be of the order m; a curve such that n of its tangents pass through an arbitrary point is said to be of the class n; as already appearing, this notion of the order and the class of a curve is, however, due to Gergonne. Thus the line is a curve of the order 1 and the class 0; and corresponding dually thereto, we have the point as a curve of the order 0 and the class 1.

Plücker moreover imagined a system of line-coordinates (tangential coordinates). The Cartesian coordinates $(x,\ y)$ and trilinear coordinates $(x,\ y,\ z)$ are point-coordinates for determining the position of a point; the new coordinates, say $(\xi,\ \eta,\ \zeta)$, are line-coordinates for determining the position of a line. It is possible, and (not so much for any application thereof as in order to more fully establish the analogy between the two kinds of coordinates) important, to give independent quantitative definitions of the two kinds of coordinates; but we may also derive the notion of line-coordinates from that of point-coordinates; viz. taking $\xi x + \eta y + \zeta z = 0$ to be the equation of a line, we say that $(\xi,\ \eta,\ \zeta)$ are the line-coordinates of this line. A linear relation $a\xi + b\eta + c\zeta = 0$ between these coordinates determines a point, viz. the point whose point-coordinates are $(a,\ b,\ c)$; in fact, the equation in question $a\xi + b\eta + c\zeta = 0$ expresses that the equation $\xi x + \eta y + \zeta z = 0$, where $(x,\ y,\ z)$ are current point-coordinates, is satisfied on writing therein $x,\ y,\ z = a,\ b,\ c$; or that the line in question passes through the point $(a,\ b,\ c)$. Thus $(\xi,\ \eta,\ \zeta)$ are the line-coordinates of any line whatever; but when these, instead of being absolutely arbitrary, are subject to the restriction $a\xi + b\eta + c\zeta = 0$, this obliges the line to pass through a point $(a,\ b,\ c)$; and the last-mentioned equation $a\xi + b\eta + c\zeta = 0$ is considered as the line-equation of this point.

A line has only a point-equation, and a point has only a line-equation; but any other curve has a point-equation and also a line-equation; the point-equation $(\ast\!\!\!\{x,\ y,\ z)^m = 0$ is the relation which is satisfied by the point-coordinates $(x,\ y,\ z)$ of each point of the curve; and similarly the line-equation $(\ast\!\!\!\{\xi,\ \eta,\ \zeta)^n = 0$ is the relation which is satisfied by the line-coordinates $(\xi,\ \eta,\ \zeta)$ of each line (tangent) of the curve.

There is in analytical geometry little occasion for any explicit use of line-coordinates; but the theory is very important; it serves to show that, in demonstrating by point-coordinates any purely descriptive theorem whatever, we demonstrate the correlative theorem; that is, we do not demonstrate the one theorem, and then (as by the method of reciprocal polars) deduce from it the other, but we do at one and the same time demonstrate the two theorems; our $(x,\ y,\ z)$ instead of meaning point-coordinates may mean line-coordinates, and the demonstration is then in every step of it a demonstration of the correlative theorem.

The above dual generation explains the nature of the singularities of a plane curve. The ordinary singularities, arranged according to a cross division, are

	Proper.	Improper.
Point-singularities {	1. The stationary point, cusp, or spinode;	2. The double point, or n
Line-singularities {	3. The stationary tangent, or inflexion;	4. The double tangent;

arising as follows:

1. The cusp: the point as it travels along the line may come to rest, and then reverse the direction of its motion.

3. The stationary tangent: the line may in the course of its rotation come to rest, and then reverse the direction of its rotation.

2. The node: the point may in the course of its motion come to coincide with a former position of the point, the two positions of the line not in general coinciding.

4. The double tangent: the line may in the course of its motion come to coincide with a former position of the line, the two positions of the point not in general coinciding.

It may be remarked that we cannot with a real point and line obtain the node with two imaginary tangents (conjugate or isolated point, or acnode), nor again the real double tangent with two imaginary points of contact; but this is of little consequence, since in the general theory the distinction between real and imaginary is not attended to.

The singularities (1) and (3) have been termed proper singularities, and (2) and (4) improper; in each of the first-mentioned cases there is a real singularity, or peculiarity in the motion; in the other two cases there is not; in (2) there is not when the point is first at the node, or when it is secondly at the node, any peculiarity in the motion; the singularity consists in the point coming twice into the same position; and so in (4) the singularity is in the line coming twice into the same position. Moreover (1) and (2) are, the former a proper singularity, and the latter an improper singularity, as regards the motion of the point; and similarly (3) and (4) are, the former a proper singularity, and the latter an improper singularity, as regards the motion of the line.

But as regards the representation of a curve by an equation, the case is very different.

First, if the equation be in point-coordinates, (3) and (4) are in a sense not singularities at all. The curve $(\ast\!\!\!\!\lbrack x,\ y,\ z)^m = 0$, or general curve of the order m, has double tangents and inflexions; (2) presents itself as a singularity, for the equations $\partial_x(\ast\!\!\!\!\lbrack x,\ y,\ z)^m = 0$, $\partial_y(\ast\!\!\!\!\lbrack x,\ y,\ z)^m = 0$, $\partial_z(\ast\!\!\!\!\lbrack x,\ y,\ z)^m = 0$, implying $(\ast\!\!\!\!\lbrack x,\ y,\ z)^m = 0$, are not in general satisfied by any values $(a,\ b,\ c)$ whatever of $(x,\ y,\ z)$, but if such values exist, then the point $(a,\ b,\ c)$ is a node or double point; and (1) presents itself as a further singularity or sub-case of (2), a cusp being a double point for which the two tangents become coincident.

In line-coordinates all is reversed: (1) and (2) are not singularities; (3) presents itself as a sub-case of (4).

The theory of compound singularities will be referred to further on.

In regard to the ordinary singularities, we have

m, the order,
n, the class,
δ, the number of double points,
κ, the number of cusps,
τ, the number of double tangents,
ι, the number of inflexions;

and this being so, Plücker's "six equations" are

$$
\begin{aligned}
&(1) \quad n = m(m-1) - 2\delta - 3\kappa,\\
&(2) \quad \iota = 3m(m-2) - 6\delta - 8\kappa,\\
&(3) \quad \tau = \tfrac{1}{2}m(m-2)(m^2-9) - (m^2-m-6)(2\delta+3\kappa)\\
&\qquad\qquad + 2\delta(\delta-1) + 6\delta\kappa + \tfrac{9}{2}\kappa(\kappa-1),\\
&(4) \quad m = n(n-1) - 2\tau - 3\iota,\\
&(5) \quad \kappa = 3n(n-2) - 6\tau - 8\iota,\\
&(6) \quad \delta = \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2\tau+3\iota)\\
&\qquad\qquad + 2\tau(\tau-1) + 6\tau\iota + \tfrac{9}{2}\iota(\iota-1).
\end{aligned}
$$

It is easy to derive the further forms

$$
\begin{aligned}
&(7) \quad \iota - \kappa = 3(n - m),\\
&(8) \quad 2(\tau - \delta) = (n - m)(n + m - 9),\\
&(9) \quad \tfrac{1}{2}m(m + 3) - \delta - 2\kappa = \tfrac{1}{2}n(n + 3) - \tau - 2\iota,\\
&(10) \quad \tfrac{1}{2}(m - 1)(m - 2) - \delta - \kappa = \tfrac{1}{2}(n - 1)(n - 2) - \tau - \iota,\\
&(11,\ 12) \quad m^2 - 2\delta - 3\kappa = n^2 - 2\tau - 3\iota,\ = m + n,
\end{aligned}
$$

the whole system being equivalent to three equations only: and it may be added that, using a to denote the equal quantities $3m + \iota$ and $3n + \kappa$, everything may be expressed in terms of m, n, a. We have

$$
\begin{aligned}
\kappa &= a - 3n,\\
\iota &= a - 3m,\\
2\delta &= m^2 - m + 8n - 3a,\\
2\tau &= n^2 - n + 8m - 3a.
\end{aligned}
$$

It is implied in Plücker's theorem that, m, n, δ, κ, τ, ι signifying as above in regard to any curve, then in regard to the reciprocal curve n, m, τ, ι, δ, κ, will have the same significations, viz. for the reciprocal curve these letters denote respectively the order, class, number of nodes, cusps, double tangents, and inflexions.

The expression $\frac{1}{2}m(m+3) - \delta - 2\kappa$ is that of the number of the disposable constants in a curve of the order m with δ nodes and κ cusps (in fact that there shall be a node is 1 condition, a cusp 2 conditions): and the equation (9) thus expresses that the curve and its reciprocal contain each of them the same number of disposable constants.

For a curve of the order m, the expression $\frac{1}{2}(m-1)(m-2)-\delta-\kappa$ is termed the "deficiency" (as to this more hereafter); the equation (10) expresses therefore that the curve and its reciprocal have each of them the same deficiency.

The relations $m^2 - 2\delta - 3\kappa = n^2 - 2\tau - 3\iota, = m + n$, present themselves in the theory of envelopes, as will appear further on.

With regard to the demonstration of Plücker's equations it is to be remarked that we are not able to write down the equation in point-coordinates of a curve of the order m, having the given numbers δ and κ of nodes and cusps. We can only use the general equation $(*\!\!\![x,\ y,\ z)^m = 0$, say for shortness $u = 0$, of a curve of the mth order, which equation, so long as the coefficients remain arbitrary, represents a curve without nodes or cusps. Seeking then, for this curve, the values n, ι, τ of the class, number of inflexions, and number of double tangents—first, as regards the class, this is equal to the number of tangents which can be drawn to the curve from an arbitrary point, or what is the same thing, it is equal to the number of the points of contact of these tangents. The points of contact are found as the intersections of the curve $u = 0$ by a curve depending on the position of the arbitrary point, and called the "first polar" of this point; the order of the first polar is $= m - 1$, and the number of intersections is thus $= m(m-1)$. But it can be shown, analytically or geometrically, that if the given curve has a node, the first polar passes through this node, which therefore counts as two intersections; and that if the curve has a cusp, the first polar passes through the cusp, touching the curve there, and hence the cusp counts as three intersections. But, as is evident, the node or cusp is not a point of contact of a proper tangent from the arbitrary point; we have, therefore, for a node a diminution 2, and for a cusp a diminution 3, in the number of the intersections; and thus, for a curve with δ nodes and κ cusps, there is a diminution $2\delta + 3\kappa$, and the value of n is $n = m(m-1) - 2\delta - 3\kappa$.

Secondly, as to the inflexions, the process is a similar one; it can be shown that the inflexions are the intersections of the curve by a derivative curve called (after Hesse, who first considered it) the Hessian, defined geometrically as the locus of a point such that its conic polar in regard to the curve breaks up into a pair of lines, and which has an equation $H = 0$, where H is the determinant formed with the second differential coefficients of u in regard to the variables $(x,\ y,\ z)$; $H = 0$ is thus a curve of the order $3(m-2)$, and the number of inflexions is $= 3m(m-2)$. But if the given curve has a node, then not only the Hessian passes through the node, but it has there a node the two branches at which touch respectively the two branches of the curve, and the node thus counts as six intersections; so if the curve has a cusp, then the Hessian not only passes through the cusp, but it has there a cusp through which it again passes, that is, there is a cuspidal branch touching the cuspidal branch of the curve, and besides a simple branch passing through the cusp, and hence the cusp counts as eight intersections. The node or cusp is not an inflexion, and we have thus for a node a diminution 6, and for a cusp a diminution 8, in the number of the intersections; hence for a curve with δ nodes and κ cusps, the diminution is $= 6\delta + 8\kappa$, and the number of inflexions is $\iota = 3m(m-2) - 6\delta - 8\kappa$.

Thirdly, for the double tangents; the points of contact of these are obtained as the intersections of the curve by a curve $\Pi = 0$, which has not as yet been geometrically defined, but which is found analytically to be of the order $(m-2)(m^2-9)$; the number of intersections is thus $= m(m-2)(m^2-9)$; but if the given curve has a node then there is a diminution $4(m^2 - m - 6)$, and if it has a cusp then there is a diminution $= 6(m^2 - m - 6)$, where, however, it is to be noticed that the factor $(m^2 - m - 6)$ is in the case of a curve having only a node or only a cusp the number of the tangents which can be drawn from the node or cusp to the curve, and is used as denoting the number of these tangents, and ceases to be the correct expression if the number of nodes and cusps is greater than unity. Hence, in the case of a curve which has δ nodes and κ cusps, the apparent diminution $2(m^2 - m - 6)(2\delta + 3\kappa)$ is too great, and it has in fact to be diminished by $2\{2\delta(\delta - 1) + 6\delta\kappa + \frac{9}{2}\kappa(\kappa - 1)\}$, or the half thereof is 4 for each pair of nodes, 6 for each combination of a node and cusp, and 9 for each pair of cusps. We have thus finally an expression for $2\tau, = m(m - 2)(m^2 - 9) - \&c.$; or dividing the whole by 2, we have the expression for τ given by the third of Plücker's equations.

It is obvious that we cannot by consideration of the equation $u = 0$ in point-coordinates obtain the remaining three of Plücker's equations; they might be obtained in a precisely analogous manner by means of the equation $v = 0$ in line-coordinates, but they follow at once from the principle of duality, viz. they are obtained by the mere interchange of m, δ, κ with n, τ, ι respectively.

To complete Plücker's theory it is necessary to take account of compound singularities; it might be possible, but it is at any rate difficult, to effect this by considering the curve as in course of description by the point moving along the rotating line; and it seems easier to consider the compound singularity as arising from the variation of an actually described curve with ordinary singularities. The most simple case is when three double points come into coincidence, thereby giving rise to a triple point; and a somewhat more complicated one is when we have a cusp of the second kind, or node-cusp arising from the coincidence of a node, a cusp, an inflexion, and a double tangent, as shown in the annexed figure, which represents the singularities as on the point of coalescing. The general conclusion (see Cayley, *Quart. Math. Jour.* t. VII., 1866, [374], "On the higher singularities of a plane curve") is that every singularity whatever may be considered as compounded of ordinary singularities, say we have a singularity $= \delta'$ nodes, κ' cusps, τ' double tangents, and ι' inflexions. So that, in fact, Plücker's equations properly understood apply to a curve with any singularities whatever.

By means of Plücker's equations we may form a table

m	n	δ	κ	τ	ι
0	1	—	—	—	—
1	0	—	—	—	—
2	2	—	—	—	—
3	6	—	—	—	9
3	4	1	—	—	3
3	3	—	1	—	1
4	12	—	—	28	24
4	10	1	—	16	18
4	9	—	1	10	16
4	8	2	—	8	12
4	7	1	1	4	10
4	6	—	2	1	8
4	6	3	—	4	6
4	5	2	1	2	4
4	4	1	2	1	2
4	3	—	3	1	—

The table is arranged according to the value of m; and we have $m = 0$, $n = 1$, the point; $m = 1$, $n = 0$, the line; $m = 2$, $n = 2$, the conic; of $m = 3$, the cubic, there are three cases, the class being 6, 4, or 3, according as the curve is without singularities, or as it has 1 node, or 1 cusp; and so of $m = 4$, the quartic, there are nine cases, where observe that in two of them the class is $= 6$, the reduction of class arising from two cusps or else from three nodes. The nine cases may be also grouped together into four, according as the number of nodes and cusps $(\delta + \kappa)$ is $= 0$, 1, 2, or 3.

The cases may be divided into sub-cases, by the consideration of compound singularities; thus when $m = 4$, $n = 6$, $\delta = 3$, the three nodes may be all distinct, which is the general case, or two of them may unite together into the singularity called a tacnode, or all three may unite together into a triple point, or else into an oscnode.

We may further consider the inflexions and double tangents, as well in general as in regard to cubic and quartic curves.

The expression for the number of inflexions $3m(m - 2)$ for a curve of the order m was obtained analytically by Plücker, but the theory was first given in a complete form by Hesse in the two papers "Ueber die Elimination, u.s.w.," and "Ueber die Wendepuncte der Curven dritter Ordnung" (*Crelle*, t. XXVIII., 1844); in the latter of these the points of inflexion are obtained as the intersections of the curve $u = 0$ with the Hessian, or curve $\Delta = 0$, where Δ is the determinant formed with the second derived functions of u. We have in the Hessian the first instance of a covariant of a ternary form. The whole theory of the inflexions of a cubic curve is discussed in a very interesting manner by means of the canonical form of the equation $x^3 + y^3 + z^3 + 6lxyz = 0$; and in particular a proof is given of Plücker's theorem that the nine points of inflexion of a cubic curve lie by threes in twelve lines.

It may be noticed that the nine inflexions of a cubic curve are three real, six imaginary; the three real inflexions lie in a line, as was known to Newton and Maclaurin. For an acnodal cubic the six imaginary inflexions disappear, and there remain three real inflexions lying in a line. For a crunodal cubic, the six inflexions which disappear are two of them real, the other four imaginary, and there remain two imaginary inflexions and one real inflexion. For a cuspidal cubic the six imaginary inflexions and two of the real inflexions disappear, and there remains one real inflexion.

A quartic curve has 24 inflexions; it was conjectured by Salmon, and has been verified recently by Zeuthen, that at most 8 of these are real.

The expression $\frac{1}{2}m(m-2)(m^2-9)$ for the number of double tangents of a curve of the order m was obtained by Plücker only as a consequence of his first, second, fourth, and fifth equations. An investigation by means of the curve $\Pi = 0$, which by its intersections with the given curve determines the points of contact of the double tangents, is indicated by Cayley, "Recherches sur l'élimination et la théorie des courbes," (*Crelle*, t. XXXIV., 1847), [53]; and in part carried out by Hesse in the memoir "Ueber Curven dritter Ordnung" (*Crelle*, t. XXXVI., 1848). A better process was indicated by Salmon in the "Note on the double tangents to plane curves," *Phil. Mag.* 1858; considering the $m-2$ points in which any tangent to the curve again meets the curve, he showed how to form the equation of a curve of the order $(m-2)$, giving by its intersection with the tangent the points in question; making the tangent touch this curve of the order $(m-2)$, it will be a double tangent of the original curve. See Cayley, "On the Double Tangents of a Plane Curve," (*Phil. Trans.* t. CXLVIII., 1859), [260], and Dersch (*Math. Ann.* t. VII., 1874). The solution is still in so far incomplete that we have no properties of the curve $\Pi = 0$, to distinguish one such curve from the several other curves which pass through the points of contact of the double tangents.

A quartic curve has 28 double tangents, their points of contact determined as the intersections of the curve by a curve $\Pi = 0$ of the order 14, the equation of which in a very elegant form was first obtained by Hesse (1849). Investigations in regard to them are given by Plücker in the *Theorie der algebraischen Curven*, and in two memoirs by Hesse and Steiner (*Crelle*, t. XLV., 1855), in respect to the triads of double tangents which have their points of contact on a conic, and other like relations. It was assumed by Plücker that the number of real double tangents might be 28, 16, 8, 4, or 0, but Zeuthen has recently found that the last case does not exist.

The Hessian Δ has just been spoken of as a covariant of the form u; the notion of invariants and covariants belongs rather to the form u than to the curve $u = 0$ represented by means of this form; and the theory may be very briefly referred to. A curve $u = 0$ may have some invariantive property, viz. a property independent of the particular axes of coordinates used in the representation of the curve by its equation; for instance, the curve may have a node, and in order to this, a relation, say $A = 0$, must exist between the coefficients of the equation; supposing the axes of coordinates altered, so that the equation becomes $u' = 0$, and writing $A' = 0$ for the relation between the new coefficients, then the relations $A = 0$, $A' = 0$, as two different expressions of the same geometrical property, must each of them imply the other; this can only be the case when A, A' are functions differing only by a constant factor, or say, when A is an invariant of u. If, however, the geometrical property requires two or more relations between the coefficients, say $A = 0$, $B = 0$, &c., then we must have between the new coefficients the like relations, $A' = 0$, $B' = 0$, &c., and the two systems of equations must each of them imply the other; when this is so, the system of equations, $A = 0$, $B = 0$, &c., is said to be invariantive, but it does not follow that A, B, &c., are of necessity invariants of u. Similarly, if we have a curve $U = 0$ derived from the curve $u = 0$ in a manner independent of the particular axes of coordinates, then from the transformed equation $u' = 0$ deriving in like manner the curve $U' = 0$, the two equations $U = 0$, $U' = 0$ must each of them imply the other; and when this is so, U will be a covariant of u. The case is less frequent, but it may arise, that there are covariant systems $U = 0$, $V = 0$, &c., and $U' = 0$, $V' = 0$, &c., each implying the other, but where the functions U, V, &c., are not of necessity covariants of u.

The theory of the invariants and covariants of a ternary cubic function u has been studied in detail, and brought into connexion with the cubic curve $u = 0$; but the theory of the invariants and covariants for the next succeeding case, the ternary quartic function, is still very incomplete.

In further illustration of the Plückerian dual generation of a curve, we may consider the question of the envelope of a variable curve. The notion is very probably older, but it is at any rate to be found in Lagrange's *Théorie des fonctions analytiques* (1798); it is there remarked that the equation obtained by the elimination of the parameter a from an equation $f(x,\ y,\ a) = 0$ and the derived equation in respect to a is a curve, the envelope of the series of curves represented by the equation $f(x,\ y,\ a) = 0$ in question. To develope the theory, consider the curve corresponding to any particular value of the parameter; this has with the consecutive curve (or curve belonging to the consecutive value of the parameter) a certain number of intersections, and of common tangents, which may be considered as the tangents at the intersections; and the so-called envelope is the curve which is at the same time generated by the points of intersection and enveloped by the common tangents; we have thus a dual generation. But the question needs to be further examined. Suppose that in general the variable curve is of the order m with δ nodes and κ cusps, and therefore of the class n with τ double tangents and ι inflexions, m, n, δ, κ, τ, ι being connected by the Plückerian equations; the number of nodes or cusps may be greater for particular values of the parameter, but this is a speciality which may be here disregarded. Considering the variable curve corresponding to a given value of the parameter, or say simply the variable curve, the consecutive curve has then also δ and κ nodes and cusps, consecutive to those of the variable curve; and it is easy to see that among the intersections of the two curves we have the nodes each counting twice, and the cusps each counting three times; the number of the remaining $= m^2 - 2\delta - 3\kappa$ intersections is

$m^2-2\delta-3\kappa$. Similarly among the common tangents of the two curves we have the double tangents each counting twice, and the stationary tangents each counting three times, and the number of the remaining common tangents is $= n^2-2\tau-3\iota$ $(= m^2-2\delta-3\kappa$, inasmuch as each of these numbers is as was seen $= m+n)$. At any one of the $m^2-2\delta-3\kappa$ points the variable curve and the consecutive curve have tangents distinct from yet infinitesimally near to each other, and each of these two tangents is also infinitesimally near to one of the $n^2-2\tau-3\iota$ common tangents of the two curves; whence, attending only to the variable curve, and considering the consecutive curve as coming into actual coincidence with it, the $n^2-2\tau-3\iota$ common tangents are the tangents to the variable curve at the $m^2-2\delta-3\kappa$ points respectively, and the envelope is at the same time generated by the $m^2-2\delta-3\kappa$ points, and enveloped by the $n^2-2\tau-3\iota$ tangents; we have thus a dual generation of the envelope, which only differs from Plücker's dual generation, in that in place of a single point and tangent we have the group of $m^2-2\delta-3\kappa$ points and $n^2-2\tau-3\iota$ tangents.

The parameter which determines the variable curve may be given as a point upon a given curve, or say as a parametric point; that is, to the different positions of the parametric point on the given curve correspond the different variable curves, and the nature of the envelope will thus depend on that of the given curve; we have thus the envelope as a derivative curve of the given curve. Many well-known derivative curves present themselves in this manner; thus the variable curve may be the normal (or line at right angles to the tangent) at any point of the given curve; the intersection of the consecutive normals is the centre of curvature, and we have the evolute as at once the locus of the centre of curvature and the envelope of the normal. It may be added that the given curve is one of a series of curves, each cutting the several normals at right angles. Any one of these is a "parallel" of the given curve; and it can be obtained as the envelope of a circle of constant radius having its centre on the given curve.

We have in like manner, as derivatives of a given curve, the caustic, catacaustic, or diacaustic, as the case may be, and the secondary caustic, or curve cutting at right angles the reflected or refracted rays.

We have in much that precedes disregarded, or at least been indifferent to, reality; it is only thus that the conception of a curve of the mth order, as one which is met by every right line in m points, is arrived at; and the curve itself, and the line which cuts it, although both are tacitly assumed to be real, may perfectly well be imaginary. For real figures we have the general theorem that imaginary intersections, &c., present themselves in conjugate pairs: hence, in particular, that a curve of an even order is met by a line in an even number (which may be $= 0$) of points; a curve of an odd order in an odd number of points, hence in one point at least; it will be seen further on that the theorem may be generalized in a remarkable manner. Again, when there is in question only one pair of points or lines, these, if coincident, must be real; thus, a line meets a cubic curve in three points, one of them real, the other two real or imaginary; but if two of the intersections coincide they must be real, and we have a line cutting a cubic in one real point and touching it in another real point. It may be remarked that this is a limit separating the two cases where the intersections are all real, and where they are one real, two imaginary.

Considering always real curves, we obtain the notion of a branch; any portion capable of description by the continuous motion of a point is a branch; and a curve consists of one or more branches. Thus the curve of the first order or right line consists of one branch; but in curves of the second order, or conics, the ellipse and the parabola consist each of one branch, the hyperbola of two branches. A branch is either re-entrant, or it extends both ways to infinity, and in this case, we may regard it as consisting of two legs (*crura*, Newton), each extending one way to infinity, but without any definite separation. The branch, whether re-entrant or infinite, may have a cusp or cusps, or it may cut itself or another branch, thus having or giving rise to crunodes; an acnode is a branch by itself,—it may be considered as an indefinitely small re-entrant branch. A branch may have inflexions and double tangents, or there may be double tangents which touch two distinct branches; there are also double tangents with imaginary points of contact, which are thus lines having no visible connexion with the curve. A re-entrant branch not cutting itself may be everywhere convex, and it is then properly said to be an oval; but the term oval may be used more generally for any re-entrant branch not cutting itself; and we may thus speak of a once indented, twice indented oval, &c., or even of a cuspidate oval. Other descriptive names for ovals and re-entrant branches cutting themselves may be used when required; thus, in the last-mentioned case a simple form is that of a figure of eight; such a form may break up into two ovals, or into a doubly indented oval or hour-glass. A form which presents itself is when two ovals, one inside the other, unite, so as to give rise to a crunode—in default of a better name this may be called, after the curve of that name, a *limaçon.* Names may also be used for the different forms of infinite branches, but we have first to consider the distinction of hyperbolic and parabolic. The leg of an infinite branch may have at the extremity a tangent; this is an asymptote of the curve, and the leg is then hyperbolic; or the leg may tend to a fixed direction, but so that the tangent goes further and further off to infinity, and the leg is then parabolic; a branch may thus be hyperbolic or parabolic as to its two legs; or it may be hyperbolic as to one leg, and parabolic as to the other. The epithets hyperbolic and parabolic are of course derived from the conics hyperbola and parabola respectively. The nature of the two kinds of branches is best understood by considering them as projections, in the same way as we in effect consider the hyperbola and the parabola as projections of the ellipse. If a line Ω cut an arc aa', so that the two segments ab, ba' lie on opposite sides of the line, then projecting the figure so that the line Ω goes off to infinity, the tangent at b is projected into the asymptote, and the arc ab is projected into a hyperbolic leg touching the asymptote at one extremity; the arc ba' will at the same time be projected into a hyperbolic leg touching the same asymptote at the other extremity (and on the opposite side), but so that the two hyperbolic legs may or may not belong to one and the same branch. And we thus see that the two hyperbolic legs belong to a simple intersection of the curve by the line infinity. Next, if the line Ω touch at b the arc aa', so that the two portions ab', ba lie on the same side of the line Ω, then projecting the figure as before, the tangent at b, that is, the line Ω itself, is projected to infinity; the arc ab is projected into a parabolic leg, and at the same time the arc ba' is projected into a parabolic leg, having at infinity the same direction as the other leg, but so that the two legs may or may not belong to the same branch. And we thus see that the two parabolic legs represent a contact of the line infinity with the curve,—the point of contact being of course the point at infinity determined by the common direction of the two legs. It will readily be understood how the like considerations apply to other cases,—for instance, if the line Ω is a tangent at an inflexion, passes through a crunode, or touches one of the branches of a crunode, &c.; thus, if the line Ω passes through a crunode we have pairs of hyperbolic legs belonging to two parallel asymptotes. The foregoing considerations also show (what is very important) how different branches are connected together at infinity, and lead to the notion of a complete branch, or circuit.

The two legs of a hyperbolic branch may belong to different asymptotes, and in this case we have the forms which Newton calls inscribed, circumscribed, ambigene, &c.; or they may belong to the same asymptote, and in this case we have the serpentine form, where the branch cuts the asymptote, so as to touch it at its two extremities on opposite sides, or the conchoidal form, where it touches the asymptote on the same side. The two legs of a parabolic branch may converge to ultimate parallelism, as in the conic parabola, or diverge to ultimate parallelism, as in the semi-cubical parabola $y^2 = x^3$, and the branch is said to be convergent, or divergent, accordingly; or they may tend to parallelism in opposite senses, as in the cubical parabola $y = x^3$. As mentioned with regard to a branch generally, an infinite branch of any kind may have cusps, or, by cutting itself or another branch, may have or give rise to a crunode, &c.

We may now consider the various forms of cubic curves, as appearing by Newton's *Enumeratio*, and by the figures belonging thereto. The species are reckoned as 72, which are numbered accordingly 1 to 72; but to these should be added 10^a, 13^a, 22^a, and 22^b. It is not intended here to consider the division into species, nor even completely that into genera, but only to explain the principle of classification. It may be remarked generally that there are at most three infinite branches, and that there may besides be a re-entrant branch or oval.

The genera may be arranged as follows:

1, 2, 3, 4 redundant hyperbolas,
5, 6 defective hyperbolas,
7, 8 parabolic hyperbolas,
9 hyperbolisms of hyperbola,
10 „ „ ellipse,
11 „ „ parabola,
12 trident curve,
13 divergent parabolas,
14 cubic parabola

and, thus arranged, they correspond to the different relations of the line infinity to the curve. First, if the three intersections by the line infinity are all distinct, we have the hyperbolas; if the points are real, the redundant hyperbolas, with three hyperbolic branches; but if only one of them is real, the defective hyperbolas, with one hyperbolic branch. Secondly, if two of the intersections coincide, say if the line infinity meets the curve in a onefold point and a twofold point, both of them real, then there is always one asymptote: the line infinity may at the twofold point touch the curve, and we have the parabolic hyperbolas; or the twofold point may be a singular point, viz. a crunode giving the hyperbolisms of the hyperbola; an acnode, giving the hyperbolisms of the ellipse; or a cusp, giving the hyperbolisms of the parabola. As regards the so-called hyperbolisms, observe that (besides the single asymptote) we have in the case of those of the hyperbola two parallel asymptotes; in the case of those of the ellipse the two parallel asymptotes become imaginary, that is, they disappear, and in the case of those of the parabola they become coincident, that is, there is here an ordinary asymptote, and a special asymptote answering to a cusp at infinity. Thirdly, the three intersections by the line infinity may be coincident and real; or say we have a threefold point: this may be an inflexion, a crunode, or a cusp, that is, the line infinity may be a tangent at an inflexion, and we have the divergent parabolas; a tangent at a crunode to one branch, and we have the trident curve; or lastly, a tangent at a cusp, and we have the cubical parabola.

It is to be remarked that the classification mixes together non-singular and singular curves, in fact, the five kinds presently referred to: thus the hyperbolas and the divergent parabolas include curves of every kind, the separation being made in the species; the hyperbolisms of the hyperbola and ellipse, and the trident curve, are nodal; the hyperbolisms of the parabola, and the cubical parabola, are cuspidal. The divergent parabolas are of five species which respectively belong to and determine the five kinds of cubic curves; Newton gives (in two short paragraphs without any development) the remarkable theorem that the five divergent parabolas by their shadows generate and exhibit all the cubic curves.

The five divergent parabolas are curves each of them symmetrical with regard to an axis. There are two non-singular kinds, the one with, the other without, an oval, but each of them has an infinite (as Newton describes it) campaniform branch; this cuts the axis at right angles, being at first convex, but ultimately concave, towards the axis, the two legs continually tending to become at right angles to the axis. The oval may unite itself with the infinite branch, or it may dwindle into a point, and we have the crunodal and the acnodal forms respectively; or if simultaneously the oval dwindles into a point and unites itself to the infinite branch, we have the cuspidal form. Drawing a line to cut any one of these curves and projecting the line to infinity, it would not be difficult to show how the line should be drawn in order to obtain a curve of any given species. We have herein a better principle of classification; considering cubic curves, in the first instance, according to singularities, the curves are non-singular, nodal (viz. crunodal or acnodal), or cuspidal; and we see further that there are two kinds of non-singular curves, the complex and the simplex. There is thus a complete division into the five kinds, the complex, simplex, crunodal, acnodal, and cuspidal. Each singular kind presents itself as a limit separating two kinds of inferior singularity; the cuspidal separates the crunodal and the acnodal, and these last separate from each other the complex and the simplex.

The whole question is discussed very fully and ably by Möbius in the memoir "Ueber die Grundformen der Linien dritter Ordnung" (*Abh. der K. Sachs. Ges. zu Leipzig*, t. I., 1852; *Ges. Werke*, t. I.). The author considers not only plane curves, but also cones, or, what is almost the same thing, the spherical curves which are their sections by a concentric sphere. Stated in regard to the cone, we have there the fundamental theorem that there are two different kinds of sheets: viz. the single sheet, not separated into two parts by the vertex (an instance is afforded by the plane considered as a cone of the first order generated by the motion of a line about a point), and the double or twin-pair sheet, separated into two parts by the vertex (as in the cone of the second order). And it then appears that there are two kinds of non-singular cubic cones, viz. the simplex, consisting of a single sheet, and the complex, consisting of a single sheet and a twin-pair sheet; and we thence obtain (as for cubic curves) the crunodal, the acnodal, and the cuspidal kinds of cubic cones. It may be mentioned that the single sheet is a sort of wavy form, having upon it three lines of inflexion, and which is met by any plane through the vertex in one or in three lines; the twin-pair sheet has no lines of inflexion, and resembles in its form a cone on an oval base.

In general a cone consists of one or more single or twin-pair sheets, and if we consider the section of the cone by a plane, the curve consists of one or more complete branches, or say circuits, each of them the section of one sheet of the cone;

thus, a cone of the second order is one twin-pair sheet, and any section of it is one circuit composed, it may be, of two branches. But although we thus arrive by projection at the notion of a circuit, it is not necessary to go out of the plane, and we may (with Zeuthen, using the shorter term *circuit* for his *complete branch*) define a circuit as any portion (of a curve) capable of description by the continuous motion of a point, it being understood that a passage through infinity is permitted. And we then say that a curve consists of one or more circuits; thus the right line, or curve of the first order, consists of one circuit; a curve of the second order consists of one circuit; a cubic curve consists of one circuit or else of two circuits.

A circuit is met by any right line always in an even number, or always in an odd number, of points, and it is said to be an even circuit or an odd circuit accordingly; the right line is an odd circuit, the conic an even circuit. And we have then the theorem, two odd circuits intersect in an odd number of points; an odd and an even circuit, or two even circuits, in an even number of points. An even circuit not cutting itself divides the plane into two parts, the one called the internal part, incapable of containing any odd circuit, the other called the external part, capable of containing an odd circuit.

We may now state in a more convenient form the fundamental distinction of the kinds of cubic curve. A non-singular cubic is simplex, consisting of one odd circuit, or it is complex, consisting of one odd circuit and one even circuit. It may be added that there are on the odd circuit three inflexions, but on the even circuit no inflexion; it hence also appears that from any point of the odd circuit there can be drawn to the odd circuit two tangents, and to the even circuit (if any) two tangents, but that from a point of the even circuit there cannot be drawn (either to the odd or the even circuit) any real tangent; consequently, in a simplex curve the number of tangents from any point is two; but in a complex curve the number is four, or none, four if the point is on the odd circuit, none if it is on the even circuit. It at once appears from inspection of the figure of a non-singular cubic curve, which is the odd and which the even circuit. The singular kinds arise as before; in the crunodal and the cuspidal kinds the whole curve is an odd circuit, but in the acnodal kind the acnode must be regarded as an even circuit.

The analogous question of the classification of quartics (in particular non-singular quartics and nodal quartics) is considered in Zeuthen's memoir "Sur les différentes formes des courbes planes du quatrième ordre" (*Math. Ann.* t. VII., 1874). A non-singular quartic has only even circuits; it has at most four circuits external to each other, or two circuits one internal to the other, and in this last case the internal circuit has no double tangents or inflexions. A very remarkable theorem is established as to the double tangents of such a quartic: distinguishing as a double tangent of the first kind a real double tangent which either twice touches the same circuit, or else touches the curve in two imaginary points, the number of the double tangents of the first kind of a non-singular quartic is $= 4$; it follows that the quartic has at most 8 real inflexions. The forms of the non-singular quartics are very numerous, but it is not necessary to go further into the question.

We may consider in relation to a curve, not only the line infinity, but also the circular points at infinity; assuming the curve to be real, these present themselves always conjointly; thus a circle is a conic passing through the two circular points, and is thereby distinguished from other conics. Similarly a cubic through the two circular points is termed a circular cubic; a quartic through the two points is termed a circular quartic, and if it passes twice through each of them, that is, has each of them for a node, it is termed a bicircular quartic. Such a quartic is of course binodal ($m = 4$, $\delta = 2$, $\kappa = 0$); it has not in general, but it may have, a third node, or a cusp. Or again, we may have a quartic curve having a cusp at each of the circular points: such a curve is a "Cartesian," it being a complete definition of the Cartesian to say that it is a bicuspidal quartic curve ($m = 4$, $\delta = 0$, $\kappa = 2$), having a cusp at each of the circular points. The circular cubic and the bicircular quartic, together with the Cartesian (being in one point of view a particular case thereof), are interesting curves which have been much studied, generally, and in reference to their focal properties.

The points called foci presented themselves in the theory of the conic, and were well known to the Greek geometers, but the general notion of a focus was first established by Plücker, in the memoir "Ueber solche Puncte die bei Curven einer höheren Ordnung den Brennpuncten der Kegelschnitte entsprechen," (*Crelle*, t. X., 1833). We may from each of the circular points draw tangents to a given curve; the intersection of two such tangents (belonging of course to the two circular points respectively) is a focus. There will be from each circular point λ tangents (λ a number depending on the class of the curve and its relation to the line infinity and the circular points, $= 2$ for the general conic, 1 for the parabola, 2 for a circular cubic or a bicircular quartic, &c.); the λ tangents from the one circular point and those from the other circular point intersect in λ real foci (viz. each of these is the only real point on each of the tangents through it), and in $\lambda^2 - \lambda$ imaginary foci; each pair of real foci determines a pair of imaginary foci (the so-called antipoints of the two real foci), and the $\frac{1}{2}\lambda(\lambda - 1)$ pairs of real foci thus determine the $\lambda^2 - \lambda$ imaginary foci. There are in some cases points termed centres, or singular or multiple foci (the nomenclature is unsettled), which are the intersections of improper tangents from the two circular points respectively; thus, in the circular cubic, the tangents to the curve at the two circular points respectively (or two imaginary asymptotes of the curve) meet in a centre.

The notions of distance and of lines at right angles are connected with the circular points; and almost every construction of a curve by means of lines of a determinate length, or at right angles to each other, and (as such) mechanical constructions by means of linkwork, give rise to curves passing the same definite number of times through the two circular points respectively, or say to circular curves, and in which the fixed centres of the construction present themselves as ordinary, or as singular, foci. Thus the general curve of three-bar motion (or locus of the vertex of a triangle, the other two vertices whereof move on fixed circles) is a tricircular sextic, having besides three nodes ($m = 6$, $\delta = 3 + 3 + 3, = 9$), and having the centres of the fixed circles each for a singular focus; there is a third singular focus, and we have thus the remarkable theorem (due to Mr S. Roberts) of the triple generation of the curve by means of the three several pairs of singular foci.

Again, the normal, qua line at right angles to the tangent, is connected with the circular points, and these accordingly present themselves in the before-mentioned theories of evolutes and parallel curves.

We have several recent theories which depend on the notion of correspondence: two points whether in the same plane or in different planes, or on the same curve or in different curves, may determine each other in such wise that to any given position of the first point there correspond α' positions of the second point, and to any given position of the second point α positions of the first point; the two points have then an $(\alpha,\ \alpha')$ correspondence; and if α, α' are each $= 1$, then the two points have a $(1,\ 1)$ or rational correspondence. Connecting with each theory the author's name, the theories in question are—Riemann, the rational transformation of a plane curve; Cremona, the rational transformation of a plane; and Chasles, correspondence of points on the same curve, and united points. The theory first referred to, with the resulting notion of *Geschlecht*, or deficiency, is more than the other two an essential part of the theory of curves, but they will all be considered.

Riemann's results are contained in the memoirs on "Theorie der Abel'schen Functionen," (*Crelle*, t. LIV., 1857); and we have next Clebsch, "Ueber die Singularitäten algebraischer Curven," (*Crelle*, t. LXV., 1865), and Cayley, "On the Transformation of Plane Curves," (*Proc. Lond. Math. Soc.* t. I., 1865, [384]). The fundamental notion of the rational transformation is as follows:

Taking u, X, Y, Z to be rational and integral functions (X, Y, Z all of the same order) of the coordinates $(x,\ y,\ z)$, and u', X', Y', Z' rational and integral functions (X', Y', Z' all of the same order) of the coordinates $(x',\ y',\ z')$, we transform a given curve $u = 0$, by the equations $x' : y' : z' = X : Y : Z$, thereby obtaining a transformed curve $u' = 0$, and a converse set of equations $x : y : z = X' : Y' : Z'$; viz. assuming that this is so, the point $(x,\ y,\ z)$ on the curve $u = 0$ and the point $(x',\ y',\ z')$ on the curve $u' = 0$ will be points having a $(1,\ 1)$ correspondence. To show how this is, observe that to a given point $(x,\ y,\ z)$ on the curve $u = 0$ there corresponds a single point $(x',\ y',\ z')$ determined by the equations $x' : y' : z' = X : Y : Z$; from these equations and the equation $u = 0$ eliminating x, y, z

we obtain the equation $u' = 0$ of the transformed curve. To a given point $(x',\ y',\ z')$ not on the curve $u' = 0$ there corresponds, not a single point, but the system of points $(x,\ y,\ z)$ given by the equations $x' : y' : z' = X : Y : Z$, viz. regarding $x',\ y',\ z'$ as constants (and to fix the ideas, assuming that the curves $X = 0$, $Y = 0$, $Z = 0$ have no common intersections), these are the points of intersection of the curves $X : Y : Z = x' : y' : z'$, but no one of these points is situate on the curve $u = 0$. If, however, the point $(x',\ y',\ z')$ is situate on the curve $u' = 0$, then one point of the system of points in question is situate on the curve $u = 0$, that is, to a given point of the curve $u' = 0$ there corresponds a single point of the curve $u = 0$; and hence also this point must be given by a system of equations such as $x : y : z = X' : Y' : Z'$.

It is an old and easily proved theorem that, for a curve of the order m, the number $\delta + \kappa$ of nodes and cusps is at most $= \frac{1}{2}(m-1)(m-2)$; for a given curve the deficiency of the actual number of nodes and cusps below this maximum number, viz.

$\frac{1}{2}(m-1)(m-2)-\delta-\kappa$, is the "*Geschlecht*," or "deficiency," of the curve, say this is $= D$. When $D = 0$, the curve is said to be unicursal, when $= 1$, bicursal, and so on.

The general theorem is that two curves corresponding rationally to each other have the same deficiency. In particular, a curve and its reciprocal have this rational or $(1, 1)$ correspondence, and it has been already seen that a curve and its reciprocal have the same deficiency.

A curve of a given order can in general be rationally transformed into a curve of a lower order; thus a curve of any order for which $D = 0$, that is, a unicursal curve, can be transformed into a line; a curve of any order having the deficiency 1 or 2 can be rationally transformed into a curve of the order $D + 2$, deficiency D; and a curve of any order deficiency $=$ or > 3 can be rationally transformed into a curve of the order $D + 3$, deficiency D.

Taking x', y', z' as coordinates of a point of the transformed curve, and in its equation writing $x' : y' : z' = 1 : \theta : \phi$, we have ϕ a certain irrational function of θ, and the theorem is that the coordinates x, y, z of any point of the given curve can be expressed as proportional to rational and integral functions of θ, ϕ, that is, of θ and a certain irrational function of θ.

In particular, if $D = 0$, that is, if the given curve be unicursal, the transformed curve is a line, ϕ is a mere linear function of θ, and the theorem is that the coordinates x, y, z of a point of the unicursal curve can be expressed as proportional to rational and integral functions of θ; it is easy to see that for a given curve of the order m, these functions of θ must be of the same order m.

If $D = 1$, then the transformed curve is a cubic; it can be shown that in a cubic, the axes of coordinates being properly chosen, ϕ can be expressed as the square root of a quartic function of θ; and the theorem is that the coordinates x, y, z of a point of the bicursal curve can be expressed as proportional to rational and integral functions of θ, and of the square root of a quartic function of θ.

And so if $D = 2$, then the transformed curve is a nodal quartic; ϕ can be expressed as the square root of a sextic function of θ, and the theorem is, that the coordinates x, y, z of a point of the tricursal curve can be expressed as proportional to rational and integral functions of θ, and of the square root of a sextic function of θ. But when $D = 3$, we have no longer the like law, viz. ϕ is not expressible as the square root of an octic function of θ.

Observe that the radical, square root of a quartic function, is connected with the theory of elliptic functions, and the radical, square root of a sextic function, with that of the first kind of Abelian functions, but that the next kind of Abelian functions does not depend on the radical, square root of an octic function.

It is a form of the theorem for the case $D = 1$, that the coordinates x, y, z of a point of the bicursal curve, or in particular the coordinates of a point of the cubic, can be expressed as proportional to rational and integral functions of the elliptic functions $\operatorname{sn} u$, $\operatorname{cn} u$, $\operatorname{dn} u$; in fact, taking the radical to be $\sqrt{(1 - \theta^2 \cdot 1 - k^2\theta^2)}$, and writing

$\theta = \operatorname{sn} u$, the radical becomes $= \operatorname{cn} u \cdot \operatorname{dn} u$; and we have expressions of the form in question.

It will be observed that the equations $x' : y' : z' = X : Y : Z$ before-mentioned do not of themselves lead to the other system of equations $x : y : z = X' : Y' : Z'$, and thus that the theory does not in anywise establish a $(1,\ 1)$ correspondence between the points $(x,\ y,\ z)$ and $(x',\ y',\ z')$ of two planes or of the same plane; this is the correspondence of Cremona's theory.

In this theory, given in the memoirs "Sulle trasformazioni geometriche delle figure piane," *Mem. di Bologna*, t. II. (1863), and t. V. (1865), we have a system of equations $x' : y' : z' = X : Y : Z$ which does lead to a system $x : y : z = X' : Y' : Z'$, where, as before, X, Y, Z denote rational and integral functions, all of the same order, of the coordinates x, y, z, and X', Y', Z' rational and integral functions, all of the same order, of the coordinates x', y', z', and there is thus a $(1,\ 1)$ correspondence given by these equations between the two points $(x,\ y,\ z)$ and $(x',\ y',\ z')$. To explain this, observe that starting from the equations $x' : y' : z' = X : Y : Z$, to a given point $(x,\ y,\ z)$ there corresponds one point $(x',\ y',\ z')$, but that if n be the order of the functions X, Y, Z, then to a given point x', y', z' there would, if the curves $X = 0$, $Y = 0$, $Z = 0$ had no common intersections, correspond n^2 points $(x,\ y,\ z)$. If, however, the functions are such that the curves $X = 0$, $Y = 0$, $Z = 0$ have k common intersections, then among the n^2 points are included these k points, which are fixed points independent of the point $(x',\ y',\ z')$; so that, disregarding these fixed points, the number of points $(x,\ y,\ z)$ corresponding to the given point $(x',\ y',\ z')$ is $= n^2 - k$; and in particular if $k = n^2 - 1$, then we have one corresponding point; and hence the original system of equations $x' : y' : z' = X : Y : Z$ must lead to the equivalent system $x : y : z = X' : Y' : Z'$; and in this system by the like reasoning the functions must be such that the curves $X' = 0$, $Y' = 0$, $Z' = 0$ have $n'^2 - 1$ common intersections. The most simple example is in the two systems of equations $x' : y' : z' = yz : zx : xy$ and $x : y : z = y'z' : z'x' : x'y'$; where $yz = 0$, $zx = 0$, $xy = 0$ are conics (pairs of lines) having three common intersections, and where obviously either system of equations leads to the other system. In the case where X, Y, Z are of an order exceeding 2, the required number $n^2 - 1$ of common intersections can only occur by reason of common multiple points on the three curves; and assuming that the curves $X = 0$, $Y = 0$, $Z = 0$ have $a_1 + a_2 + a_3 + \cdots + a_{n-1}$ common intersections, where the a_1 points are ordinary points, the a_2 points are double points, the a_3 points are triple points, &c., on each curve, we have the condition

$$a_1 + 4a_2 + 9a_3 + \cdots + (n-1)^2 a_{n-1} = n^2 - 1;$$

but to this must be joined the condition

$$a_1 + 3a_2 + 6a_3 + \cdots + \tfrac{1}{2}(n-1)(n-2)\, a_{n-1} = \tfrac{1}{2}n(n+3) - 2,$$

(without which the transformation would be illusory); and the conclusion is that a_1, a_2, ..., a_{n-1} may be any numbers satisfying these two equations. It may be added that the two equations together give

$$a_2 + 3a_3 + \cdots + \tfrac{1}{2}(n-1)(n-2)\, a_{n-1} = \tfrac{1}{2}(n-1)(n-2),$$

which expresses that the curves $X = 0$, $Y = 0$, $Z = 0$ are unicursal. The transformation may be applied to any curve $u = 0$, which is thus rationally transformed into a curve $u' = 0$, by a rational transformation such as is considered in Riemann's theory; hence the two curves have the same deficiency.

Coming next to Chasles, the principle of correspondence is established and used by him in a series of memoirs relating to the conics which satisfy given conditions, and to other geometrical questions, contained in the *Comptes Rendus*, t. LVIII. et seq. (1864 to the present time). The theorem of united points in regard to points in a right line was given in a paper, June–July 1864, and it was extended to unicursal curves in a paper of the same series (March 1866), "Sur les courbes planes ou à double courbure dont les points peuvent se déterminer individuellement—application du principe de correspondance dans la théorie de ces courbes."

The theorem is as follows: if in a unicursal curve two points have an $(\alpha,\ \beta)$ correspondence, then the number of united points (or points each corresponding to itself) is $= \alpha + \beta$. In fact, in a unicursal curve the coordinates of a point are given as proportional to rational and integral functions of a parameter, so that any point of the curve is determined uniquely by means of this parameter; that is, to each point of the curve corresponds one value of the parameter, and to each value of the parameter one point on the curve; and the $(\alpha,\ \beta)$ correspondence between the two points is given by an equation of the form $(*[\theta,\ 1)^{\alpha}(\phi,\ 1)^{\beta} = 0$ between their parameters θ and ϕ; at a united point $\phi = \theta$, and the value of θ is given by an equation of the order $\alpha + \beta$. The extension to curves of any given deficiency D was made in the memoir of Cayley, "On the correspondence of two points on a curve," *Proc. Lond. Math. Soc.* t. I. (1866), [385], viz. taking P, P' as the corresponding points in an $(\alpha,\ \alpha')$ correspondence on a curve of deficiency D, and supposing that when P is given the corresponding points P' are found as the intersections of the curve by a curve containing the coordinates of P as parameters, and having with the given curve k intersections at the point P, then the number of united points is $a = \alpha + \alpha' + 2kD$; and more generally, if the curve Θ intersect the given curve in a set of points P' each p times, a set of points Q' each q times, &c., in such manner that the points $(P,\ P')$, the points $(P,\ Q')$, &c., are pairs of points corresponding to each other according to distinct laws; then if $(P,\ P')$ are points having an $(\alpha,\ \alpha')$ correspondence with a number $= a$ of united points, $(P,\ Q')$ points having a $(\beta,\ \beta')$ correspondence with a number $= b$ of united points, and so on, the theorem is that we have

$$p(a - \alpha - \alpha') + q(b - \beta - \beta') + \cdots = 2kD.$$

The principle of correspondence, or say rather the theorem of united points, is a most powerful instrument of investigation, which may be used in place of analysis for the determination of the number of solutions of almost every geometrical problem. We can by means of it investigate the class of a curve, number of inflexions, &c.—in fact, Plücker's equations; but it is necessary to take account of special solutions; thus, in one of the most simple instances, in finding the class of a curve, the cusps present themselves as special solutions.

Imagine a curve of order m, deficiency D, and let the corresponding points P, P' be such that the line joining them passes through a given point O; this is an $(m-1,\ m-1)$ correspondence, and the value of k is $=1$, hence the number of united points is $=2m-2+2D$; the united points are the points of contact of the tangents from O and (as special solutions) the cusps, and we have thus the relation $n+\kappa=2m-2+2D$; or, writing $D=\frac{1}{2}(m-1)(m-2)-\delta-\kappa$, this is $n=m(m-1)-2\delta-3\kappa$, which is right.

The principle in its original form as applying to a right line was used throughout by Chasles in the investigations on the number of the conics which satisfy given conditions, and on the number of solutions of very many other geometrical problems.

There is one application of the theory of the $(\alpha,\ \alpha')$ correspondence between two planes which it is proper to notice.

Imagine a curve, real or imaginary, represented by an equation (involving, it may be, imaginary coefficients) between the Cartesian coordinates u, u'; then, writing $u=x+iy$, $u'=x'+iy'$, the equation determines real values of $(x,\ y)$, and of $(x',\ y')$, corresponding to any given real values of $(x',\ y')$ and $(x,\ y)$ respectively; that is, it establishes a real correspondence (not of course a rational one) between the points $(x,\ y)$ and $(x',\ y')$; for example in the imaginary circle $u^2+u'^2=(a+bi)^2$, the correspondence is given by the two equations $x^2-y^2+x'^2-y'^2=a^2-b^2$, $xy+x'y'=ab$. We have thus a means of geometrical representation for the portions, as well imaginary as real, of any real or imaginary curve. Considerations such as these have been used for determining the series of values of the independent variable, and the irrational functions thereof in the theory of Abelian integrals, but the theory seems to be worthy of further investigation.

The researches of Chasles (*Comptes Rendus*, t. LVIII., 1864, et seq.) refer to the conics which satisfy given conditions. There is an earlier paper by De Jonquières, "Théorèmes généraux concernant les courbes géométriques planes d'un ordre quelconque," *Liouv.* t. VI. (1861), which establishes the notion of a system of curves (of any order) of the index N, viz. considering the curves of the order n which satisfy $\frac{1}{2}n(n+3)-1$ conditions, then the index N is the number of these curves which pass through a given arbitrary point.

But Chasles in the first of his papers (February 1864), considering the conics which satisfy four conditions, establishes the notion of the two characteristics (μ, ν) of such a system of conics, viz. μ is the number of the conics which pass through a given arbitrary point, and ν is the number of the conics which touch a given arbitrary line. And he gives the theorem, a system of conics satisfying four conditions, and having the characteristics (μ, ν) contains $2\nu - \mu$ line-pairs (that is, conics, each of them a pair of lines), and $2\mu - \nu$ point-pairs (that is, conics, each of them a pair of points, *coniques infiniment aplaties*), which is a fundamental one in the theory. The characteristics of the system can be determined when it is known how many there are of these two kinds of degenerate conics in the system, and how often each is to be counted. It was thus that Zeuthen (in the paper *Nyt Bydrag*, "Contribution to the Theory of Systems of Conics which satisfy four Conditions," Copenhagen, 1865, translated with an addition in the *Nouvelles Annales*) solved the question of finding the characteristics of the systems of conics which satisfy four conditions of contact with a given curve or curves; and this led to the solution of the further problem of finding the number of the conics which satisfy five conditions of contact with a given curve or curves (Cayley, *Comptes Rendus*, t. LXIII., 1866, [377]), and "On the Curves which satisfy given Conditions" (*Phil. Trans.* t. CLVIII., 1868, [406]).

It may be remarked that although, as a process of investigation, it is very convenient to seek for the characteristics of a system of conics satisfying 4 conditions, yet what is really determined is in every case the number of the conics which satisfy 5 conditions; the characteristics of the system $(4p)$ of the conics which pass through 4 points are $(5p)$, $(4p,\ 1l)$, the number of the conics which pass through 5 points, and which pass through 4 points and touch 1 line: and so in other cases. Similarly as regards cubics, or curves of any other order: a cubic depends on 9 constants, and the elementary problems are to find the number of the cubics $(9p)$, $(8p,\ 1l)$, &c., which pass through 9 points, pass through 8 points and touch 1 line, &c.; but it is in the investigation convenient to seek for the characteristics of the systems of cubics $(8p)$, &c., which satisfy 8 instead of 9 conditions.

The elementary problems in regard to cubics are solved very completely by Maillard in his *Thèse, Recherche des caractéristiques des systèmes élémentaires des courbes planes du troisième ordre* (Paris, 1871). Thus, considering the several cases of a cubic

	No. of consts.
1. With a given cusp	5,
2. „ cusp on given line	6,
3. „ cusp	7,
4. „ a given node	6,
5. „ node on given line	7,
6. „ node	8,
7. non-singular	9,

he determines in every case the characteristics $(\mu,\ \nu)$ of the corresponding systems of cubics $(4p)$, $(3p,\ 1l)$, &c. The same problems, or most of them, and also the elementary problems in regard to quartics are solved by Zeuthen, who in the elaborate memoir "Almindelige Egenskaber, &c.," Danish Academy, t. X. (1873), considers the problem in reference to curves of any order, and applies his results to cubic and quartic curves.

The methods of Maillard and Zeuthen are substantially identical; in each case the question considered is that of finding the characteristics (μ, ν) of a system of curves by consideration of the special or degenerate forms of the curves included in the system. The quantities which have to be considered are very numerous. Zeuthen in the case of curves of any given order establishes between the characteristics μ, ν, and 18 other quantities, in all 20 quantities, a set of 24 equations (equivalent to 23 independent equations), involving (besides the 20 quantities) other quantities relating to the various forms of the degenerate curves, which supplementary terms he determines, partially for curves of any order, but completely only for quartic curves. It is in the discussion and complete enumeration of the special or degenerate forms of the curves, and of the supplementary terms to which they give rise, that the great difficulty of the question seems to consist; it would appear that the 24 equations are a complete system, and that (subject to a proper determination of the supplementary terms) they contain the solution of the general problem.

The remarks which follow have reference to the analytical theory of the degenerate curves which present themselves in the foregoing problem of the curves which satisfy given conditions.

A curve represented by an equation in point-coordinates may break up: thus if P_1, P_2, ... be rational and integral functions of the coordinates (x, y, z) of the orders m_1, m_2, ... respectively, we have the curve $P_1^{a_1} P_2^{a_2} \cdots = 0$, of the order $m = a_1 m_1 + a_2 m_2 + \ldots$, composed of the curve $P_1 = 0$ taken a_1 times, the curve $P_2 = 0$ taken a_2 times, &c.

Instead of the equation $P_1^{a_1} P_2^{a_2} \cdots = 0$, we may start with an equation $u = 0$, where u is a function of the order m containing a parameter θ, and for a particular value say $\theta = 0$, of the parameter reducing itself to $P_1^{a_1} P_2^{a_2} \ldots$. Supposing θ indefinitely small, we have what may be called the penultimate curve, and when $\theta = 0$ the ultimate curve. Regarding the ultimate curve as derived from a given penultimate curve, we connect with the ultimate curve, and consider as belonging to it, certain points called "summits" on the component curves $P_1 = 0$, $P_2 = 0$, respectively; a summit S is a point such that, drawing from an arbitrary point O the tangents to the penultimate curve, we have OS as the limit of one of these tangents. The ultimate curve together with its summits may be regarded as a degenerate form of the curve $u = 0$. Observe that the positions of the summits depend on the penultimate curve $u = 0$, viz. on the values of the coefficients in the terms multiplied by θ, θ^2, ...; they are thus in some measure arbitrary points as regards the ultimate curve $P_1^{a_1} P_2^{a_2} \cdots = 0$.

It may be added that we have summits only on the component curves $P_i = 0$, of a multiplicity $a_i > 1$; the number of summits on such a curve is in general $= (a_i^2 - a_i) m_i^2$. Thus assuming that the penultimate curve is without nodes or cusps, the number of the tangents to it is $= m^2 - m = (a_1 m_1 + a_2 m_2 + \ldots)^2 - (a_1 m_1 + a_2 m_2 + \ldots)$, taking $P_1 = 0$ to have δ_1 nodes and κ_1 cusps, and therefore its class n_1 to be $= m_1^2 - m_1 - 2\delta_1 - 3\kappa_1$, &c., the expression for the number of tangents to the penultimate curve is

$$\begin{aligned}&= (a_1^2 - a_1)\, m_1^2 + (a_2^2 - a_2)\, m_2^2 + \dots \\ &\quad + 2a_1a_2m_1m_2 + \dots + a_1(n_1 + 2\delta_1 + 3\kappa_1) + a_2(n_2 + 2\delta_2 + 3\kappa_2) + \dots\end{aligned}$$

where a term $2a_1a_2m_1m_2$ indicates tangents which are in the limit the lines drawn to the intersections of the curves $P_1 = 0$, $P_2 = 0$, each line $2a_1a_2$ times; a term $a_1(n_1 + 2\delta_1 + 3\kappa_1)$ tangents which are in the limit the proper tangents to $P_1 = 0$ each a_1 times, the lines to its nodes each $2a_1$ times, and the lines to its cusps each $3a_1$ times; the remaining terms $(a_1^2 - a_1)m_1^2 + (a_2^2 - a_2)m_2^2 + \dots$ indicate tangents which are in the limit the lines drawn to the several summits, that is, we have $= (a_i^2 - a_i)m_i^2$ summits on the curve $P_i = 0$.

There is of course a precisely similar theory as regards line-coordinates; taking Π_1, Π_2, &c., to be rational and integral functions of the coordinates $(\xi,\ \eta,\ \zeta)$, we connect with the ultimate curve $= \Pi_1^{a_1}\Pi_2^{a_2} \cdots = 0$, and consider as belonging to it certain lines, which for the moment may be called "axes," tangents to the component curves

$\Pi_1 = 0$, $\Pi_2 = 0$ respectively. Considering an equation in point-coordinates, we may have among the component curves right lines; and, if in order to put these in evidence, we take the equation to be $L_1^{a_1} \dots P_1^{b_1} \dots = 0$, where $L_1 = 0$ is a right line, $P_1 = 0$ a curve of the second or any higher order, then the curve will contain as part of itself summits not exhibited in this equation, but the corresponding line-equation will be $\Lambda_1^{a_1} \dots \Pi_1^{b_1} \dots = 0$, where $\Lambda_1 = 0, \dots$ are the equations of the summits in question, $\Pi_1 = 0$, &c., are the line-equations corresponding to the several point-equations $P_1 = 0$, &c.; and this curve will contain as part of itself axes not exhibited by this equation, but which are the lines $L_1 = 0, \dots$ of the equation in point-coordinates.

In conclusion a little may be said as to curves of double curvature, otherwise twisted curves, or curves in space. The analytical theory by Cartesian coordinates was first considered by Clairaut, *Recherches sur les courbes à double courbure* (Paris, 1731). Such a curve may be considered as described by a point, moving in a line which at the same time rotates about the point in a plane which at the same time rotates about the line; the point is a point, the line a tangent, and the plane an osculating plane, of the curve; moreover the line is a generating line, and the plane a tangent plane, of a developable surface or torse, having the curve for its edge of regression. Analogous to the order and class of a plane curve we have the order, rank, and class, of the system (assumed to be a geometrical one), viz. if an arbitrary plane contains m points, an arbitrary line meets r lines, and an arbitrary point lies in n planes, of the system, then m, r, n are the order, rank, and class respectively. The system has singularities, and there exist between m, r, n and the numbers of the several singularities equations analogous to Plücker's equations for a plane curve.

It is a leading point in the theory that a curve in space cannot in general be represented by means of two equations $U = 0$, $V = 0$; the two equations represent surfaces, intersecting in a curve; but there are curves which are not the complete intersection of any two surfaces; thus we have the cubic in space, or skew cubic, which is the residual intersection of two quadric surfaces which have a line in common; the equations $U = 0$, $V = 0$ of the two quadric surfaces represent the cubic curve, not by itself, but together with the line.

. . . .

786.

EQUATION.

[From the *Encyclopædia Britannica*, Ninth Edition, vol. VIII. (1878), pp. 497—509.]

The present article includes Determinant and Theory of Equations; and it may be proper to explain the relation to each other of the two subjects. Theory of Equations is used in its ordinary conventional sense to denote the theory of a single equation of any order in one unknown quantity; that is, it does not include the theory of a system or systems of equations of any order between any number of unknown quantities. Such systems occur very frequently in analytical geometry and other parts of mathematics, but they are hardly as yet the subject-matter of a distinct theory; and even Elimination, the transition-process for passing from a system of any number of equations involving the same number of unknown quantities to a single equation in one unknown quantity, hardly belongs to the Theory of Equations in the above restricted sense. But there is one case of a system of equations which precedes the Theory of Equations, and indeed presents itself at the outset of algebra, that of a system of simple (or linear) equations. Such a system gives rise to the function called a Determinant, and it is by means of these functions that the solution of the equations is effected. We have thus the subject Determinant as nearly equivalent to (but somewhat more extensive than) that of a system of linear equations; and we have the other subject, Theory of Equations, used in the restricted sense above referred to, and as not including Elimination.

Determinant.

1. A sketch of the history of determinants is given under [the Article] Algebra; it thereby appears that the algebraical function called a determinant presents itself in the solution of a system of simple equations, and we have herein a natural source of the theory. Thus, considering the equations

$$\begin{aligned} ax + by + cz &= d, \\ a'x + b'y + c'z &= d', \\ a''x + b''y + c''z &= d'', \end{aligned}$$

and proceeding to solve them by the so-called method of cross multiplication, we multiply the equations by factors selected in such a manner that, upon adding the results, the whole coefficient of y becomes $= 0$ and the whole coefficient of z becomes $= 0$; the factors in question are $b'c'' - b''c'$, $b''c - bc''$, $bc' - b'c$ (values which, as at once seen, have the desired property); we thus obtain an equation which contains on the left-hand side only a multiple of x, and on the right-hand side a constant term; the coefficient of x has the value

$$a(b'c'' - b''c') + a'(b''c - bc'') + a''(bc' - b'c),$$

and this function, represented in the form

$$\begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix},$$

is said to be a determinant; or, the number of elements being 3^2, it is called a determinant of the third order. It is to be noticed that the resulting equation is

$$x \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} = \begin{vmatrix} d, & b, & c \\ d', & b', & c' \\ d'', & b'', & c'' \end{vmatrix},$$

where the expression on the right-hand side is the like function with d, d', d'' in place of a, a', a'' respectively, and is of course also a determinant. Moreover, the functions $b'c'' - b''c'$, $b''c - bc''$, $bc' - b'c$ used in the process are themselves the determinants of the second order

$$\begin{vmatrix} b', & c' \\ b'', & c'' \end{vmatrix}, \quad \begin{vmatrix} b'', & c'' \\ b, & c \end{vmatrix}, \quad \begin{vmatrix} b, & c \\ b', & c' \end{vmatrix}.$$

We have herein the suggestion of the rule for the derivation of the determinants of the orders 1, 2, 3, 4, &c., each from the preceding one, viz. we have

$$
\begin{aligned}
|a| &= a,\\
\begin{vmatrix} a, & b \\ a', & b' \end{vmatrix} &= a\,|b'| - a'\,|b|,\\
\begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} &= a \begin{vmatrix} b', & c' \\ b'', & c'' \end{vmatrix} - a' \begin{vmatrix} b, & c \\ b'', & c'' \end{vmatrix} + a'' \begin{vmatrix} b, & c \\ b', & c' \end{vmatrix},
\end{aligned}
$$

and so on, the terms being all + for a determinant of an odd order, but alternately + and − for a determinant of an even order.

[62–2]

2. It is easy, by induction, to arrive at the general results:—

A determinant of the order n is the sum of the $1 \cdot 2 \cdot 3 \ldots n$ products which can be formed with n elements out of n^2 elements arranged in the form of a square, no two of the n elements being in the same line or in the same column, and each such product having the coefficient $\pm$ unity.

The products in question may be obtained by permuting in every possible manner the columns (or the lines) of the determinant, and then taking for the factors the n elements in the dexter diagonal. And we thence derive the rule for the signs, viz. considering the primitive arrangement of the columns as positive, then an arrangement obtained therefrom by a single interchange (inversion, or derangement) of two columns is regarded as negative; and so in general an arrangement is positive or negative according as it is derived from the primitive arrangement by an even or an odd number of interchanges. This implies the theorem that a given arrangement can be derived from the primitive arrangement only by an odd number, or else only by an even number of interchanges,—a theorem the verification of which may be easily obtained from the theorem (in fact, a particular case of the general one), an arrangement can be derived from itself only by an even number of interchanges. And this being so, each product has the sign belonging to the corresponding arrangement of the columns; in particular, a determinant contains with the sign $+$ the product of the elements in its dexter diagonal. It is to be observed that the rule gives as many positive as negative arrangements, the number of each being $= \frac{1}{2} \cdot 1 \cdot 2 \ldots n$.

The rule of signs may be expressed in a different form. Giving to the columns in the primitive arrangement the numbers $1, 2, 3, \ldots, n$, to obtain the sign belonging to any other arrangement we take, as often as a lower number succeeds a higher one, the sign $-$, and, compounding together all these minus signs, obtain the proper sign, $+$ or $-$ as the case may be.

Thus, for three columns, it appears by either rule that 123, 231, 312 are positive; 132, 321, 213 are negative; and the developed expression of the foregoing determinant of the third order is

$$= ab'c'' - ab''c' + a'b''c - a'bc'' + a''bc' - a''b'c.$$

3. It further appears that a determinant is a linear function[3] of the elements of each column thereof, and also a linear function of the elements of each line thereof; moreover, that the determinant retains the same value, only its sign being altered, when any two columns are interchanged, or when any two lines are interchanged; more generally, when the columns are permuted in any manner, or when the lines are permuted in any manner, the determinant retains its original value, with the sign + or − according as the new arrangement (considered as derived from the primitive arrangement) is positive or negative according to the foregoing rule of signs.

[3]The expression, a linear function, is here used in its narrowest sense, a linear function without constant term; what is meant is, that the determinant is in regard to the elements a, a', a'', $\dots$ of any column or line thereof, a function of the form $Aa + A'a' + A''a'' + \dots$, without any term independent of a, a', a'', $\dots$.

It at once follows that, if two columns are identical, or if two lines are identical, the value of the determinant is $= 0$. It may be added that, if the lines are converted into columns, and the columns into lines, in such a way as to leave the dexter diagonal unaltered, the value of the determinant is unaltered; the determinant is in this case said to be transposed.

4. By what precedes it appears that there exists a function of the n^2 elements, linear as regards the terms of each column (or say, for shortness, linear as to each column), and such that only the sign is altered when any two columns are interchanged; these properties completely determine the function, except as to a common factor which may multiply all the terms. If, to get rid of this arbitrary common factor, we assume that the product of the elements in the dexter diagonal has the coefficient $+1$, we have a complete definition of the determinant; and it is interesting to show how from these properties, assumed for the definition of the determinant, it at once appears that the determinant is a function serving for the solution of a system of linear equations. Observe that the properties show at once that if any column is 0 (that is, if the elements in the column are each $= 0$), then the determinant is $= 0$; and further that, if any two columns are identical, then the determinant is $= 0$.

5. Reverting to the system of linear equations written down at the beginning of this article, consider the determinant

$$\begin{vmatrix} ax+by+cz-d, & b, & c \\ a'x+b'y+c'z-d', & b', & c' \\ a''x+b''y+c''z-d'', & b'', & c'' \end{vmatrix};$$

it appears that this is

$$= x\begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} + y\begin{vmatrix} b, & b, & c \\ b', & b', & c' \\ b'', & b'', & c'' \end{vmatrix} + z\begin{vmatrix} c, & b, & c \\ c', & b', & c' \\ c'', & b'', & c'' \end{vmatrix} - \begin{vmatrix} d, & b, & c \\ d', & b', & c' \\ d'', & b'', & c'' \end{vmatrix};$$

viz. the second and the third terms each vanishing, it is

$$= x\begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} - \begin{vmatrix} d, & b, & c \\ d', & b', & c' \\ d'', & b'', & c'' \end{vmatrix}.$$

But if the linear equations hold good, then the first column of the original determinant is $= 0$, and therefore the determinant itself is $= 0$; that is, the linear equations give

$$x \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} - \begin{vmatrix} d, & b, & c \\ d', & b', & c' \\ d'', & b'', & c'' \end{vmatrix} = 0;$$

which is the result obtained above.

We might in a similar way find the values of y and z, but there is a more symmetrical process. Join to the original equations the new equation

$$\alpha x + \beta y + \gamma z = \delta;$$

a like process shows that, the equations being satisfied, we have

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & -\delta \\ a, & b, & c, & d \\ a', & b', & c', & d' \\ a'', & b'', & c'', & d'' \end{vmatrix} = 0;$$

or, as this may be written,

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} \cdot \delta - \begin{vmatrix} a, & b, & c, & d \\ a', & b', & c', & d' \\ a'', & b'', & c'', & d'' \end{vmatrix} = 0;$$

which, considering δ as standing herein for its value $\alpha x + \beta y + \gamma z$, is a consequence of the original equations only. We have thus an expression for $\alpha x + \beta y + \gamma z$, an arbitrary linear function of the unknown quantities x, y, z; and by comparing the coefficients of α, β, γ on the two sides respectively, we have the values of x, y, z; in fact, these quantities, each multiplied by

$$\begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix},$$

are in the first instance obtained in the forms

$$\begin{vmatrix} a, & b, & c, & d \\ a', & b', & c', & d' \\ a'', & b'', & c'', & d'' \end{vmatrix} \text{ (the columns being taken in appropriate orders),}$$

but these are

$$\begin{vmatrix} b, & c, & d \\ b', & c', & d' \\ b'', & c'', & d'' \end{vmatrix}, \quad \begin{vmatrix} d, & c, & a \\ d', & c', & a' \\ d'', & c'', & a'' \end{vmatrix}, \quad \begin{vmatrix} d, & a, & b \\ d', & a', & b' \\ d'', & a'', & b'' \end{vmatrix}$$

or, what is the same thing,

$$\begin{vmatrix} d, & b, & c \\ d', & b', & c' \\ d'', & b'', & c'' \end{vmatrix}, \quad \begin{vmatrix} a, & d, & c \\ a', & d', & c' \\ a'', & d'', & c'' \end{vmatrix}, \quad \begin{vmatrix} a, & b, & d \\ a', & b', & d' \\ a'', & b'', & d'' \end{vmatrix}$$

respectively.

6. *Multiplication of two determinants of the same order.*—The theorem is obtained very easily from the last preceding definition of a determinant. It is most simply expressed thus

$$\begin{vmatrix} (a,b,c)(\alpha,\alpha',\alpha''), & (a,b,c)(\beta,\beta',\beta''), & (a,b,c)(\gamma,\gamma',\gamma'') \\ (a',b',c')(\alpha,\alpha',\alpha''), & (a',b',c')(\beta,\beta',\beta''), & (a',b',c')(\gamma,\gamma',\gamma'') \\ (a'',b'',c'')(\alpha,\alpha',\alpha''), & (a'',b'',c'')(\beta,\beta',\beta''), & (a'',b'',c'')(\gamma,\gamma',\gamma'') \end{vmatrix} = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix} \cdot \begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix},$$

where the expression on the left side stands for a determinant, the terms of the first line being $(a,b,c)(\alpha,\alpha',\alpha'')$, that is, $a\alpha+b\alpha'+c\alpha''$, $(a,b,c)(\beta,\beta',\beta'')$, that is, $a\beta+b\beta'+c\beta''$, $(a,b,c)(\gamma,\gamma',\gamma'')$, that is, $a\gamma+b\gamma'+c\gamma''$; and similarly the terms in the second and third lines are the like functions with (a',b',c') and (a'',b'',c'') respectively.

There is an apparently arbitrary transposition of lines and columns; the result would hold good if on the left-hand side we had written (α,β,γ), (α',β',γ'), $(\alpha'',\beta'',\gamma'')$, or what is the same thing, if on the right-hand side we had transposed the second determinant; and either of these changes would, it might be thought, increase the elegance of the form, but, for a reason which need not be explained[4], the form actually adopted is the preferable one.

To indicate the method of proof, observe that the determinant on the left-hand side, qua linear function of its columns, may be broken up into a sum of ($3^3 =$) 27 determinants, each of which is either of some such form as

$$\alpha\beta\gamma' \begin{vmatrix} a, & a, & b \\ a', & a', & b' \\ a'', & a'', & b'' \end{vmatrix},$$

where the term $\alpha\beta\gamma'$ is not a term of the $\alpha\beta\gamma$-determinant, and its coefficient (as a determinant with two identical columns) vanishes; or else it is of a form such as

$$\alpha\beta'\gamma'' \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix};$$

that is, every term which does not vanish contains as a factor the *abc*-determinant last written down; the sum of all other factors

[4]The reason is the connexion with the corresponding theorem for the multiplication of two matrices.

$\pm\alpha\beta'\gamma''$ is the $\alpha\beta\gamma$-determinant of the formula; and the final result then is, that the determinant on the left-hand side is equal to the product on the right-hand side of the formula.

7. *Decomposition of a determinant into complementary determinants.*—Consider, for simplicity, a determinant of the fifth order, $5 = 2 + 3$, and let the top two lines be

$$\begin{array}{l} a,\ b,\ c,\ d,\ e \\ a',\ b',\ c',\ d',\ e'; \end{array}$$

then, if we consider how these elements enter into the determinant, it is at once seen that they enter only through the determinants of the second order $\begin{vmatrix} a, & b \\ a', & b' \end{vmatrix}$, &c., which can be formed by selecting any two columns at pleasure. Moreover, representing the remaining three lines by

$$\begin{array}{l} a'',\ b'',\ c'',\ d'',\ e'' \\ a''',\ b''',\ c''',\ d''',\ e''' \\ a'''',\ b'''',\ c'''',\ d'''',\ e'''', \end{array}$$

it is further seen that the factor which multiplies the determinant formed with any two columns of the first set is the determinant of the third order formed with the complementary three columns of the second set; and it thus appears that the determinant of the fifth order is a sum of all the products of the form

$$\begin{vmatrix} a, & b \\ a', & b' \end{vmatrix} \cdot \begin{vmatrix} c'', & d'', & e'' \\ c''', & d''', & e''' \\ c'''', & d'''', & e'''' \end{vmatrix},$$

the sign $+$ being in each case such that the sign of the term $\pm ab' \cdot c''d'''e''''$ obtained from the diagonal elements of the component determinants may be the actual sign of this term in the determinant of the fifth order; for the product written down the sign is obviously $+$.

Observe that for a determinant of the nth order, taking the decomposition to be $1 + (n - 1)$, we fall back upon the equations given at the commencement, in order to show the genesis of a determinant.

8. Any determinant $\begin{vmatrix} a, & b \\ a', & b' \end{vmatrix}$ formed out of the elements of the original determinant, by selecting the lines and columns at pleasure, is termed a minor of the original determinant; and when the number of lines and columns, or order of the determinant, is $n - 1$, then such determinant is called a first minor; the number of the first minors is $= n^2$, the first minors, in fact, corresponding to the several elements of the determinant—that is, the coefficient therein of any term whatever is the corresponding first minor. The first minors,

each divided by the determinant itself, form a system of elements inverse to the elements of the determinant.

A determinant is symmetrical when every two elements symmetrically situated in regard to the dexter diagonal are equal to each other; if they are equal and opposite (that is, if the sum of the two elements be $= 0$), this relation not extending to the diagonal elements themselves, which remain arbitrary, then the determinant is skew; but if the relation does extend to the diagonal terms (that is, if these are each $= 0$), then the determinant is skew symmetrical; thus the determinants

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix}, \quad \begin{vmatrix} a, & -\nu, & \mu \\ \nu, & b, & -\lambda \\ -\mu, & \lambda, & c \end{vmatrix}, \quad \begin{vmatrix} 0, & \nu, & -\mu \\ -\nu, & 0, & \lambda \\ \mu, & -\lambda, & 0 \end{vmatrix}$$

are respectively symmetrical, skew, and skew symmetrical.

The theory admits of very extensive algebraic developments, and applications in algebraical geometry and other parts of mathematics; but the fundamental properties of the functions may fairly be considered as included in what precedes.

Theory of Equations.

9. In the subject "Theory of Equations," the term equation is used to denote an equation of the form $x^n - p_1 x^{n-1} + \dots + p_n = 0$, where p_1, p_2, ..., p_n are regarded as known, and x as a quantity to be determined; for shortness, the equation is written $f(x) = 0$.

The equation may be *numerical*; that is, the coefficients p_1, p_2, ..., p_n are then numbers,—understanding by number a quantity of the form $\alpha + \beta i$ (α and β having any positive or negative real values whatever, or say each of these is regarded as susceptible of continuous variation from an indefinitely large negative to an indefinitely large positive value), and i denoting $\sqrt{(-1)}$.

Or the equation may be *algebraical*; that is, the coefficients are not then restricted to denote, or are not explicitly considered as denoting, numbers.

I. We consider first numerical equations. (Real theory, 10 to 14; Imaginary theory, 15 to 18.)

10. Postponing all consideration of imaginaries, we take in the first instance the coefficients to be real, and attend only to the real roots (if any); that is, p_1, p_2, ..., p_n are real positive or negative quantities, and a root α, if it exists, is a positive or negative quantity such that $\alpha^n - p_1\alpha^{n-1} + \dots + p_n = 0$, or say, $f(\alpha) = 0$. The fundamental theorems are given in the article Algebra, sections X., XIII., XIV.; but there are various points in the theory which require further development.

It is very useful to consider the curve $y = f(x)$, or, what would come to the same, the curve $Ay = f(x)$,—but it is better to retain the first-mentioned form of equation, drawing, if need be, the ordinate y on a reduced scale. For instance, if the given equation be $x^3 - 6x^2 + 11x - 6{\cdot}06 = 0$,[5] then the curve $y = x^3 - 6x^2 + 11x - 6{\cdot}06$ is as shown in the figure at page 501, without any reduction of scale for

[5]The coefficients were selected so that the roots might be nearly 1, 2, 3.

the ordinate. It is clear that, in general, y is a continuous one-valued function of x, finite for every finite value of x, but becoming infinite when x is infinite; i.e. assuming throughout that the coefficient of x^n is $+1$, then when $x = +\infty$, $y = +\infty$; but when $x = -\infty$, then $y = +\infty$ or $-\infty$, according as n is even or odd; the curve cuts any line whatever, and in particular it cuts the axis of x, in at most n points; and the value of x, at any point of intersection with the axis, is a root of the equation $f(x) = 0$.

If β, α are any two values of x ($\alpha > \beta$, that is, α nearer $+\infty$), then if $f(\beta)$, $f(\alpha)$ have opposite signs, the curve cuts the axis an odd number of times, and therefore at least once, between the points $x = \beta$, $x = \alpha$; but if $f(\beta)$, $f(\alpha)$ have the same sign, then between these points the curve cuts the axis an even number of times, or it may be not at all. That is, $f(\beta)$, $f(\alpha)$ having opposite signs, there are between the limits β, α an odd number of real roots, and therefore at least one real root; but $f(\beta)$, $f(\alpha)$ having the same sign, there are between these limits an even number of real roots, or it may be there is no real root. In particular, by giving to β, α the values $-\infty$, $+\infty$ (or, what is the same thing, any two values sufficiently near to these values respectively) it appears that an equation of an odd order has always an odd number of real roots, and therefore at least one real root; but that an equation of an even order has an even number of real roots, or it may be no real root.

If α be such that for $x =$ or $> \alpha$ (that is, x nearer to $+\infty$) $f(x)$ is always $+$, and β be such that for $x =$ or $< \beta$ (that is, x nearer to $-\infty$) $f(x)$ is always $-$, then the real roots (if any) lie between these limits $x = \beta$, $x = \alpha$; and it is easy to find by trial such two limits including between them all the real roots (if any).

11. Suppose that the positive value δ is an inferior limit to the difference between two real roots of the equation; or rather (since the foregoing expression would imply the existence of real roots) suppose that there are not two real roots such that their difference taken positively is $=$ or $< \delta$; then, γ being any value whatever, there is clearly at most one real root between the limits γ and $\gamma + \delta$; and by what precedes there is such real root or there is not such real root, according as $f(\gamma)$, $f(\gamma + \delta)$ have opposite signs or have the same sign. And by dividing in this manner the interval β to α into intervals each of which is $=$ or $< \delta$, we should not only ascertain the number of the real roots (if any), but we should also separate the real roots, that is, find for each of them limits γ, $\gamma + \delta$ between which there lies this one, and only this one, real root.

In particular cases it is frequently possible to ascertain the number of the real roots, and to effect their separation by trial or otherwise, without much difficulty; but the foregoing was the general process as employed by Lagrange even in the second edition (1808) of the *Traité de la résolution des Equations Numériques*[6]; the determination of the limit δ had to be effected by means of the "equation of differences" or equation of the order $\frac{1}{2}n(n-1)$, the roots of which are the squares of the differences of the roots of the

[6] The third edition (1826) is a reproduction of that of 1808; the first edition has the date 1798, but a large part of the contents is taken from memoirs of 1767–68 and 1770–71.

given equation, and the process is a cumbrous and unsatisfactory one.

12. The great step was effected by Sturm's theorem (1835)—viz. here starting from the function $f(x)$, and its first derived function $f'(x)$, we have (by a process which is a slight modification of that for obtaining the greatest common measure of these two functions) to form a series of functions

$$f(x),\ f'(x),\ f_2(x),\ \ldots,\ f_n(x)$$

of the degrees n, $n-1$, $n-2$, …, 0 respectively, the last term $f_n(x)$ being thus an absolute constant. These lead to the immediate determination of the number of real roots (if any) between any two given limits α, β; viz. supposing $\alpha > \beta$ (that is, α nearer to $+\infty$), then substituting successively these two values in the series of functions, and attending only to the signs of the resulting values, the number of the changes of sign lost in passing from β to α is the required number of real roots between the two limits. In particular, taking β, $\alpha = -\infty$, $+\infty$ respectively, the signs of the several functions depend merely on the signs of the terms which contain the highest powers of x, and are seen by inspection, and the theorem thus gives at once the whole number of real roots.

And although theoretically, in order to complete by a finite number of operations the separation of the real roots, we still need to know the value of the before-mentioned limit δ; yet in any given case the separation may be effected by a limited number of repetitions of the process. The practical difficulty is when two or more roots are very near to each other. Suppose, for instance, that the theorem shows that there are two roots between 0 and 10; by giving to x the values $1, 2, 3, \ldots$ successively, it might appear that the two roots were between 5 and 6; then again that they were between $5 \cdot 3$ and $5 \cdot 4$, then between $5 \cdot 34$ and $5 \cdot 35$, and so on until we arrive at a separation; say it appears that between $5 \cdot 346$ and $5 \cdot 347$ there is one root, and between $5 \cdot 348$ and $5 \cdot 349$ the other root. But in the case in question δ would have a very small value, such as ·002, and even supposing this value known, the direct application of the first-mentioned process would be still more laborious.

13. Supposing the separation once effected, the determination of the single real root which lies between the two given limits may be effected to any required degree of approximation either by the processes of Horner and Lagrange (which are in principle a carrying out of the method of Sturm's theorem), or by the process of Newton, as perfected by Fourier (which requires to be separately considered).

First as to Horner and Lagrange. We know that between the limits β, α there lies one, and only one, real root of the equation; $f(\beta)$ and $f(\alpha)$ have therefore opposite signs. Suppose any intermediate value is θ; in order to determine by Sturm's theorem whether the root lies between β, θ, or between θ, α, it would be quite unnecessary to calculate the signs of $f(\theta)$, $f'(\theta)$, $f_2(\theta)$, $\ldots$; only the sign of $f(\theta)$ is required: for, if this has the same sign as $f(\beta)$, then the root is between θ, α; if the same sign as $f(\alpha)$, then the root is between β, θ. We want to make θ increase from the inferior limit β, at which $f(\theta)$ has the sign of $f(\beta)$, so long as $f(\theta)$ retains this sign, and then to a value for which it assumes the opposite sign; we have thus two nearer limits of the required root, and the process may be repeated indefinitely.

Horner's method (1819) gives the root as a decimal, figure by figure; thus, if the equation be known to have one real root between 0 and 10, it is in effect shown—say that 5 is too small (that is, the root is between 5 and 6); next that $5 \cdot 4$ is too small (that is, the root is between $5 \cdot 4$ and $5 \cdot 5$); and so on to any number of decimals. Each figure is obtained, not by the successive trial of all the figures which precede it, but (as in the ordinary process of the extraction of a square root, which is in fact Horner's process applied to this particular case) it is given presumptively as the first figure of a quotient; such value may be too large, and then the next inferior integer must be tried instead of it, or it may require to be further diminished. And it is to be remarked that the process not only gives the approximate value α of the root, but (as in the extraction of a square root) it includes the calculation of the function $f(\alpha)$ which should be, and approximately is, $= 0$. The arrangement of the calculations is very elegant, and forms an integral part of the actual method. It is to be observed that after a certain number of decimal places have been obtained, a good many more can be found by a mere division. It is in the progress tacitly assumed that the roots have been first separated.

Lagrange's method (1767) gives the root as a continued fraction $a + \cfrac{1}{b + \cfrac{1}{c + \dots}}$, where a is a positive or negative integer (which may be $= 0$), but b, c, $\dots$ are positive integers. Suppose the roots have been separated; then (by trial if need be of consecutive integer values) the limits may be made to be consecutive integer numbers: say they are a, $a + 1$; the value of x is therefore $= a + \dfrac{1}{y}$, where y is positive and greater than 1; from the given equation for x, writing therein $x = a + \dfrac{1}{y}$, we form an equation of the same order for y, and this equation will have one, and only one, positive root greater than 1; hence finding for it the limits b, $b + 1$ (where b is $=$ or > 1), we have $y = b + \dfrac{1}{z}$, where z is positive and greater than 1; and so on; that is, we thus obtain the successive denominators b, c, d, $\dots$ of the continued fraction. The method is theoretically very elegant, but the disadvantage is that it gives the result in the form of a continued fraction, which for the most part must ultimately be converted into a decimal. There is one advantage in the method, that a commensurable root (that is, a root equal to a rational fraction) is found accurately, since, when such root exists, the continued fraction terminates.

14. Newton's method (1711), as perfected by Fourier (1831), may be roughly stated as follows. If $x = \gamma$ be an approximate value of any root, and $\gamma + h$ the correct value, then $f(\gamma + h) = 0$, that is,

$$f(\gamma) + hf'(\gamma) + \tfrac{1}{1\cdot 2}h^2 f''(\gamma) + \dots = 0;$$

and then, if h be so small that the terms after the second may be neglected, $f(\gamma) + hf'(\gamma) = 0$, that is, $h = -\dfrac{f(\gamma)}{f'(\gamma)}$, or the new approximate value is $\gamma' = \gamma - \dfrac{f(\gamma)}{f'(\gamma)}$, and so on, as often as we please. It will be observed that so far nothing has been assumed as to the separation of the roots, or even as to the existence of a real root; γ has been taken as the approximate value of a root, but no precise meaning has been attached to this expression. The question arises, what are the conditions to be satisfied by γ in order that the process may by successive repetitions actually lead to a certain real root of the equation; or say that, γ being an approximate value of a certain real root, the new value $\gamma - \dfrac{f(\gamma)}{f'(\gamma)}$ may be a more approximate value.

Referring to the figure, it is easy to see that, if OC represent the assumed value γ, then, drawing the ordinate CP to meet the curve in P, and the tangent PO' to meet the axis in C', we shall have OC' as the new approximate value of the root. But observe that there is here a real root OX, and that the curve beyond [the article ends on page 500 mid-discussion; continues in the next chunk].

[*The discussion of Newton's method, Fourier's conditions, and the imaginary theory continues at page 501; see the volume's next chunk (pages 501–550).*]